

15.8.11 DCMI data register (DCMI_DR)

Address offset: 0x28

Reset value: 0x0000 0x0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Byte3								Byte2								Byte1								Byte0							
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:24 Data byte 3

Bits 23:16 Data byte 2

Bits 15:8 Data byte 1

Bits 7:0 Data byte 0

The digital camera Interface packages all the received data in 32-bit format before requesting a DMA transfer. A 4-word deep FIFO is available to leave enough time for DMA transfers and avoid DMA overrun conditions.

15.8.12 DCMI register map

[Table 88](#) summarizes the DCMI registers.

Table 88. DCMI register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	DCMI_CR	Reserved																	ENABLE	Reserved	EDM		FCR		VSPOL	HSPOL	PCKPOL	ESS	JPEG	CROP	CM	CAPTURE	
	Reset value																		0		0	0	0	0	0	0	0	0	0	0	0	0	
0x04	DCMI_SR	Reserved																								FNE		VSYNC	HSYNC				
	Reset value																									0	0	0					
0x08	DCMI_RIS	Reserved																								LINE_RIS	VSYNC_RIS	ERR_RIS	OVR_RIS	FRAME_RIS			
	Reset value																									0	0	0	0	0			
0x0C	DCMI_IER	Reserved																								LINE_IE	VSYNC_IE	ERR_IE	OVR_IE	FRAME_IE			
	Reset value																									0	0	0	0	0			
0x10	DCMI_MIS	Reserved																								LINE_MIS	VSYNC_MIS	ERR_MIS	OVR_MIS	FRAME_MIS			
	Reset value																									0	0	0	0	0			

Table 88. DCMI register map and reset values (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x14	DCMI_ICR	Reserved																												LINE_ISC	VSYNC_ISC	ERR_ISC	OVR_ISC	FRAME_ISC
	Reset value																													0	0	0	0	0
0x18	DCMI_ESCR	FEC								LEC								LSC								FSC								
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x1C	DCMI_ESUR	FEU								LEU								LSU								FSU								
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x20	DCMI_CWSTR_T	Reserved	VST[12:0]														Reserved	HOFFCNT[13:0]																
	Reset value		0	0	0	0	0	0	0	0	0	0	0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x24	DCMI_CWSIZE	Reserved	VLINE13:0]														Reserved	CAPCNT[13:0]																
	Reset value		0	0	0	0	0	0	0	0	0	0	0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x28	DCMI_DR	Byte3								Byte2								Byte1								Byte0								
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.

16 LCD-TFT controller (LTDC)

This section applies only to STM32F429xx/439xx devices.

16.1 Introduction

The LCD-TFT (Liquid Crystal Display - Thin Film Transistor) display controller provides a parallel digital RGB (Red, Green, Blue) and signals for horizontal, vertical synchronisation, Pixel Clock and Data Enable as output to interface directly to a variety of LCD and TFT panels.

16.2 LTDC main features

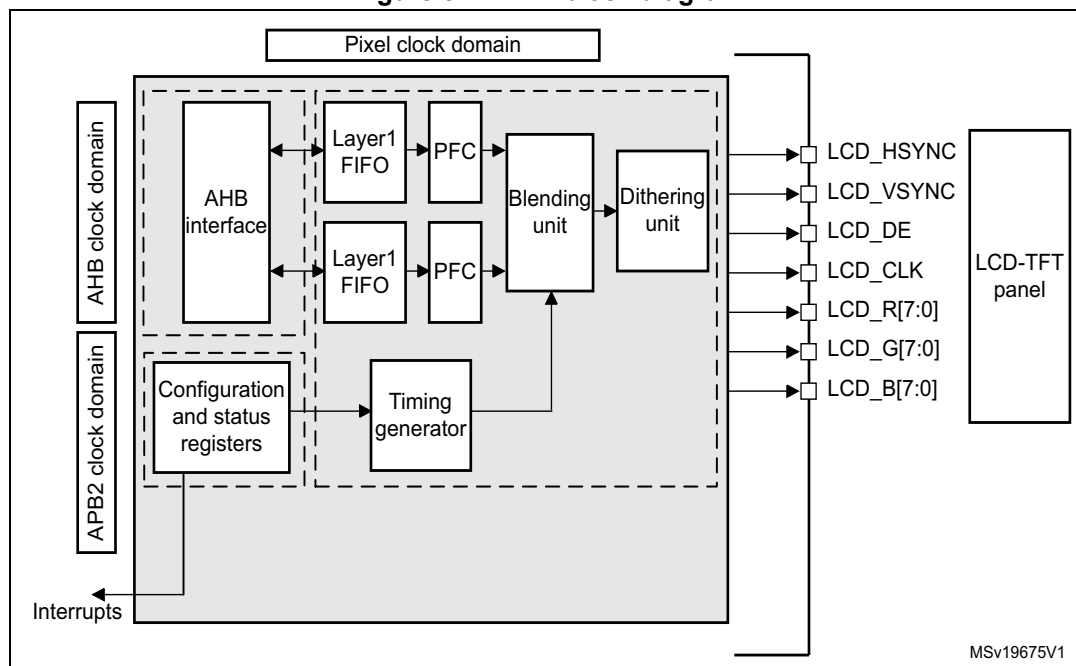
- 24-bit RGB Parallel Pixel Output; 8 bits-per-pixel (RGB888)
- 2 display layers with dedicated FIFO (64x32-bit)
- Color Look-Up Table (CLUT) up to 256 color (256x24-bit) per layer
- Supports up to XGA (1024x768) resolution
- Programmable timings for different display panels
- Programmable Background color
- Programmable polarity for HSync, VSync and Data Enable
- Up to 8 Input color formats selectable per layer
 - ARGB8888
 - RGB888
 - RGB565
 - ARGB1555
 - ARGB4444
 - L8 (8-bit Luminance or CLUT)
 - AL44 (4-bit alpha + 4-bit luminance)
 - AL88 (8-bit alpha + 8-bit luminance)
- Pseudo-random dithering output for low bits per channel
 - Dither width 2-bits for Red, Green, Blue
- Flexible blending between two layers using alpha value (per pixel or constant)
- Color Keying (transparency color)
- Programmable Window position and size
- Supports thin film transistor (TFT) color displays
- AHB master interface with burst of 16 words
- Up to 4 programmable interrupt events

16.3 LTDC functional description

16.3.1 LTDC block diagram

The block diagram of the LTDC is shown in [Figure 81: LTDC block diagram](#).

Figure 81. LTDC block diagram



Layer FIFO: One FIFO 64x32 bit per layer.

PFC: Pixel Format Converter performing the pixel format conversion from the selected input pixel format of a layer to words.

AHB interface: For data transfer from memories to the FIFO.

Blending, Dithering unit and Timings Generator: Refer to [Section 16.4.1](#) and [Section 16.4.2](#).

16.3.2 LTDC reset and clocks

The LCD-TFT controller peripheral uses 3 clock domains:

- The AHB clock domain (HCLK) is used for data transfer from the memories to the Layer FIFO and frame buffer configuration register
- The APB2 clock domain (PCLK2) is used for global configuration register and interrupt registers
- The Pixel Clock domain (LCD_CLK) is used to generate LCD-TFT interface signals, pixel data generation and layer configuration. The LCD_CLK output should be configured following the panel requirements. The LCD_CLK is configured through the PLLSAI (refer to RCC section).

[Table 89](#) summarizes the clock domain for each register.

Table 89. LTDC registers versus clock domain

LTDC registers	Clock domain
LTDC_LxCR	HCLK
LTDC_LxCFBAR	
LTDC_LxCFBLR	
LTDC_LxCFBLNR	
LTDC_SRCR	PCLK2
LTDC_IER	
LTDC_ISR	
LTDC_ICR	
LTDC_SSCR	Pixel Clock (LCD_CLK)
LTDC_BPCR	
LTDC_AWCR	
LTDC_TWCR	
LTDC_GCR	
LTDC_BCCR	
LTDC_LIPCR	
LTDC_CPSR	
LTDC_CDSR	
LTDC_LxWHPCR	
LTDC_LxWVPCR	
LTDC_LxCKCR	
LTDC_LxPFCR	
LTDC_LxCACR	
LTDC_LxDCCR	
LTDC_LxBFCR	
LTDC_LxCLUTWR	

Care must be taken when accessing the LTDC registers since the APB2 bus is stalling when the following operations are ongoing:

- Register write access and update for 6 xPCKL2 period + 5x LCD_CLK period (5x HCLK period for register on AHB clock domain)
- Register read access for 7xPCKL2 period + 5x LCD_CLK period (5x HCLK period for register on AHB clock domain).

For registers on PCLK2 clock domain, APB2 bus is stalling during the register write access for 6 xPCKL2 period and 7xPCKL2 period for read access.

The LCD controller can be reset by setting the corresponding bit in the RCC_APB2RSTR register. It resets the three clock domains.

16.3.3 LCD-TFT pins and signal interface

The Table below summarizes the LTDC signal interface:

Table 90. LCD-TFT pins and signal interface

LCD-TFT signals	I/O	Description
LCD_CLK	O	Clock Output
LCD_HSYNC	O	Horizontal Synchronization
LCD_VSYNC	O	Vertical Synchronization
LCD_DE	O	Not Data Enable
LCD_R[7:0]	O	Data: 8-bit Red data
LCD_G[7:0]	O	Data: 8-bit Green data
LCD_B[7:0]	O	Data: 8-bit Blue data

The LCD-TFT controller pins must be configured by the user application. The unused pins can be used for other purposes.

For LTDC outputs up to 24-bit (RGB888), if less than 8bpp are used to output for example RGB565 or RGB666 to interface on 16b-bit or 18-bit displays, the RGB display data lines must be connected to the MSB of the LCD-TFT controller RGB data lines. As an example, in the case of an LCD-TFT controller interfacing with a RGB565 16-bit display, the LCD display R[4:0], G[5:0] and B[4:0] data lines pins must be connected to LCD-TFT controller LCD_R[7:3], LCD_G[7:2] and LCD_B[7:3].

16.4 LTDC programmable parameters

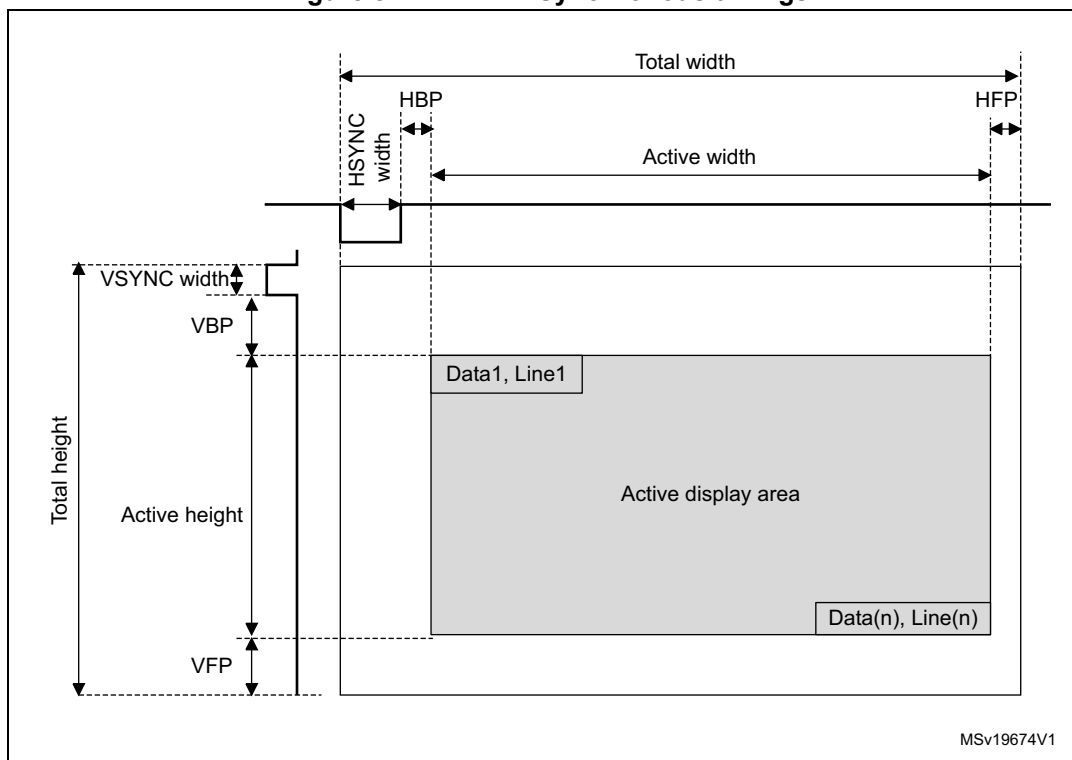
The LCD-TFT controller provides flexible configurable parameters. It can be enabled or disabled through the **LTDC_GCR** register.

16.4.1 LTDC Global configuration parameters

Synchronous Timings:

[Figure 82](#) presents the configurable timing parameters generated by the Synchronous Timings Generator block presented in the block diagram [Figure 81](#). It generates the Horizontal and Vertical Synchronization timings panel signals, the Pixel Clock and the Data Enable signals.

Figure 82. LCD-TFT Synchronous timings



Note: The HBP and HFP are respectively the Horizontal back porch and front porch period.
The VBP and the VFP are respectively the Vertical back porch and front porch period.

The LCD-TFT programmable synchronous timings are:

- HSYNC and VSYNC Width: Horizontal and Vertical Synchronization width configured by programming a value of **HSYNC Width - 1** and **VSYNC Width - 1** in the **LTDC_SSCR** register.
- HBP and VBP: Horizontal and Vertical Synchronization back porch width configured by programming the accumulated value **HSYNC Width + HBP - 1** and the accumulated value **VSYNC Width + VBP - 1** in the **LTDC_BPCR** register.
- Active Width and Active Height: The Active Width and Active Height are configured by programming the accumulated value **HSYNC Width + HBP + Active Width - 1** and the accumulated value **VSYNC Width + VBP + Active Height - 1** in the **LTDC_AWCR** register (only up to 1024x768 is supported).
- Total Width: The Total width is configured by programming the accumulated value **HSYNC Width + HBP + Active Width + HFP - 1** in the **LTDC_TWCR** register. The HFP is the Horizontal front porch period.
- Total Height: The Total Height is configured by programming the accumulated value **VSYNC Height + VBP + Active Height + VFP - 1** in the **LTDC_TWCR** register. The VFP is the Vertical front porch period.

Note: When the LTDC is enabled, the timings generated start with X/Y=0/0 position as the first horizontal synchronization pixel in the vertical synchronization area and following the back porch, active data display area and the front porch.

When the LTDC is disabled, the timing generator block is reset to $X = \text{Total Width} - 1$, $Y = \text{Total Height} - 1$ and held the last pixel before the vertical synchronization phase and the FIFO are flushed. Therefore only blanking data is output continuously.

Example of Synchronous timings configuration

TFT-LCD timings (should be extracted from Panel datasheet):

- Horizontal and Vertical Synchronization width: 0x8 pixels and 0x4 lines
- Horizontal and Vertical back porch: 0x7 pixels and 0x2 lines
- Active Width and Active Height: 0x280 pixels, 0x1E0 lines (640x480)
- Horizontal front porch: 0x6 pixels
- Vertical front porch: 0x2 lines

The programmed values in the LTDC timings registers are:

- **LTDC_SSCR** register: to be programmed to 0x00070003. (HSW[11:0] is 0x7 and VSH[10:0] is 0x3)
- **LTDC_BPCR** register: to be programmed to 0x000E0005. (AHBP[11:0] is 0xE(0x8 + 0x6) and AVBP[10:0] is 0x5(0x4 + 0x1))
- **LTDC_AWCR** register: to be programmed to 0x028E01E5. (AAW[11:0] is 0x28E(0x8 + 0x7 + 0x27F) and AAH[10:0] is 0x1E5(0x4 + 0x2 + 0x1DF))
- **LTDC_TWCR** register: to be programmed to 0x00000294. (TOTALW[11:0] is 0x294(0x8 + 0x7 + 0x280 + 0x5))
- **LTDC_THCR** register: to be programmed to 0x000001E7. (TOTALH[10:0] is 0x1E7(0x4 + 0x2 + 0x1E0 + 1))

Programmable polarity

The Horizontal and Vertical Synchronization, Data Enable and Pixel Clock output signals polarity can be programmed to active high or active low through the **LTDC_GCR** register.

Background Color

A constant background color (RGB888) can be programmed through the **LTDC_BCCR** register. It is used for blending with the bottom layer.

Dithering

The Dithering pseudo-random technique using an LFSR is used to add a small random value (threshold) to each pixel color channel (R, G or B) value, thus rounding up the MSB in some cases when displaying a 24-bit data on 18-bit display. Thus the Dithering technique is used to round data which is different from one frame to the other.

The Dither pseudo-random technique is the same as comparing LSBs against a threshold value and adding a 1 to the MSB part only, if the LSB part is $\geq$ the threshold. The LSBs are typically dropped once dithering was applied.

The width of the added pseudo-random value is 2 bits for each color channel; 2 bits for Red, 2 bits for Green and 2 bits for Blue.

Once the LCD-TFT controller is enabled, the LFSR starts running with the first active pixel and it is kept running even during blanking periods and when dithering is switched off. If the LTDC is disabled, the LFSR is reset.

The Dithering can be switched On and Off on the fly through the **LTDC_GCR** register.

Reload Shadow registers

Some configuration registers are shadowed. The shadow registers values can be reloaded immediately to the active registers when writing to these registers or at the beginning of the vertical blanking period following the configuration in the **LTDC_SRCR** register. If the immediate reload configuration is selected, the reload should be only activated when all new registers have been written.

The shadow registers should not be modified again before the reload has been done. Reading from the shadow registers returns the actual active value. The new written value can only be read after the reload has taken place.

A register reload interrupt can be generated if enabled in the **LTDC_IER** register.

The shadowed registers are all the Layer 1 and Layer 2 registers except the **LTDC_LxCLUTWR** register.

Interrupt generation event

Refer to [Section 16.5: LTDC interrupts](#) for interrupt configuration.

16.4.2 Layer programmable parameters

Up to two layers can be enabled, disabled and configured separately. The layer display order is fixed and it is bottom up. If two layers are enabled, the Layer2 is the top displayed window.

Windowing

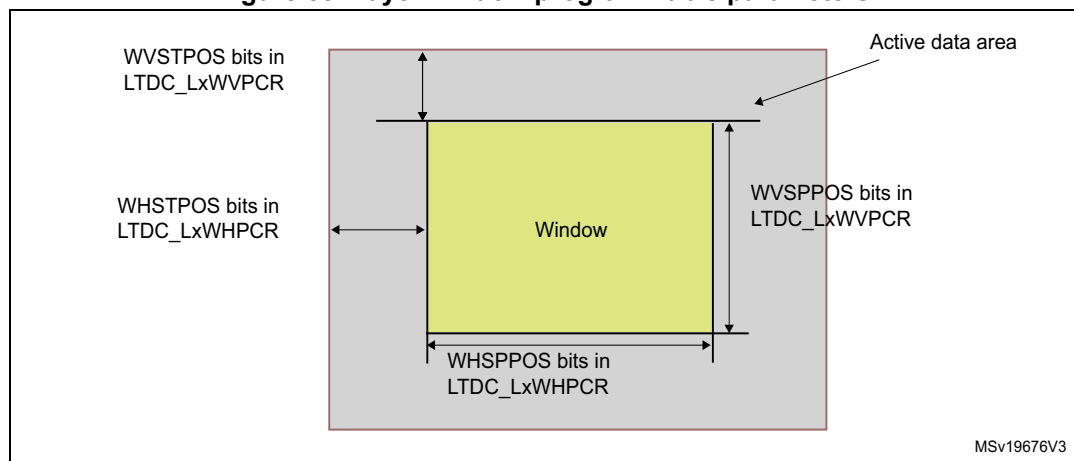
Every layer can be positioned and resized and it must be inside the Active Display area.

The window position and size are configured through the top-left and bottom-right X/Y positions and the Internal timing generator which includes the synchronous, back porch size and the active data area. Refer to **LTDC_LxWHPCR** and **LTDC_WVPCR** registers.

The programmable layer position and size defines the first/last visible pixel of a line and the first/last visible line in the window. It allows to display either the full image frame or only a part of the image frame. Refer to [Figure 83](#)

- The first and the last visible pixel in the layer are set by configuring the WHSTPOS[11:0] and WHSPPOS[11:0] in the **LTDC_LxWHPCR** register.
- The first and the last visible lines in the layer are set by configuring the WVSTPOS[10:0] and WVSPPOS[10:0] in the **LTDC_LxWVPCR** register.

Figure 83. Layer window programmable parameters:



Pixel input Format

The programmable pixel format is used for the data stored in the frame buffer of a layer.

Up to 8 input pixel formats can be configured for every layer through the **LTDC_LxPFCR** register

The pixel data is read from the frame buffer and then transformed to the internal 8888 (ARGB) format as follows:

- Components which have a width of less than 8 bits get expanded to 8 bits by bit replication. The selected bit range is concatenated multiple times until it is longer than 8 bits. Of the resulting vector, the 8 MSB bits are chosen. Example: 5 bits of an RGB565 red channel become (bit positions): 43210432 (the 3 LSBs are filled with the 3 MSBs of the 5 bits)

The figure below describes the pixel data mapping depending on the selected format.

Table 91. Pixel Data mapping versus Color Format

ARGB8888			
@+3 $A_x[7:0]$	@+2 $R_x[7:0]$	@+1 $G_x[7:0]$	@ $B_x[7:0]$
@+7 $A_{x+1}[7:0]$	@+6 $R_{x+1}[7:0]$	@+5 $G_{x+1}[7:0]$	@+4 $B_{x+1}[7:0]$
RGB888			
@+3 $B_{x+1}[7:0]$	@+2 $R_x[7:0]$	@+1 $G_x[7:0]$	@ $B_x[7:0]$
@+7 $G_{x+2}[7:0]$	@+6 $B_{x+2}[7:0]$	@+5 $R_{x+1}[7:0]$	@+4 $G_{x+1}[7:0]$
RGB565			
@+3 $R_{x+1}[4:0] \ G_{x+1}[5:3]$	@+2 $G_{x+1}[2:0] \ B_{x+1}[4:0]$	@+1 $R_x[4:0] \ G_x[5:3]$	@ $G_x[2:0] \ B_x[4:0]$

Table 91. Pixel Data mapping versus Color Format (continued)

ARGB8888			
@+7 R _{x+3} [4:0] G _{x+3} [5:3]	@+6 G _{x+3} [2:0] B _{x+3} [4:0]	@+5 R _{x+2} [4:0] G _{x+2} [5:3]	@+4 G _{x+2} [2:0] B _{x+2} [4:0]
ARGB1555			
@+3 A _{x+1} [0]R _{x+1} [4:0] G _{x+1} [4:3]	@+2 G _{x+1} [2:0] B _{x+1} [4:0]	@+1 A _x [0] R _x [4:0] G _x [4:3]	@ G _x [2:0] B _x [4:0]
@+7 A _{x+3} [0]R _{x+3} [4:0] G _{x+3} [4:3]	@+6 G _{x+3} [2:0] B _{x+3} [4:0]	@+5 A _{x+2} [0]R _{x+2} [4:0]G _{x+2} [4:3]	@+4 G _{x+2} [2:0] B _{x+2} [4:0]
ARGB4444			
@+3 A _{x+1} [3:0]R _{x+1} [3:0]	@+2 G _{x+1} [3:0] B _{x+1} [3:0]	@+1 A _x [3:0] R _x [3:0]	@ G _x [3:0] B _x [3:0]
@+7 A _{x+3} [3:0]R _{x+3} [3:0]	@+6 G _{x+3} [3:0] B _{x+3} [3:0]	@+5 A _{x+2} [3:0]R _{x+2} [3:0]	@+4 G _{x+2} [3:0] B _{x+2} [3:0]
L8			
@+3 L _{x+3} [7:0]	@+2 L _{x+2} [7:0]	@+1 L _{x+1} [7:0]	@ L _x [7:0]
@+7 L _{x+7} [7:0]	@+6 L _{x+6} [7:0]	@+5 L _{x+5} [7:0]	@+4 L _{x+4} [7:0]
AL44			
@+3 A _{x+3} [3:0] L _{x+3} [3:0]	@+2 A _{x+2} [3:0] L _{x+2} [3:0]	@+1 A _{x+1} [3:0] L _{x+1} [3:0]	@ A _x [3:0] L _x [3:0]
@+7 A _{x+7} [3:0] L _{x+7} [3:0]	@+6 A _{x+6} [3:0] L _{x+6} [3:0]	@+5 A _{x+5} [3:0] L _{x+5} [3:0]	@+4 A _{x+4} [3:0] L _{x+4} [3:0]
AL88			
@+3 A _{x+1} [7:0]	@+2 L _{x+1} [7:0]	@+1 A _x [7:0]	@ L _x [7:0]
@+7 A _{x+3} [7:0]	@+6 L _{x+3} [7:0]	@+5 A _{x+2} [7:0]	@+4 L _{x+2} [7:0]

Color Look-Up Table (CLUT)

The CLUT can be enabled at run-time for every layer through the **LTDC_LxCR** register and it is only useful in case of indexed color when using the L8, AL44 and AL88 input pixel format.

First, the CLUT has to be loaded with the R, G and B values that replace the original R, G, B values of that pixel (indexed color). Each color (RGB value) has its own address which is the position within the CLUT.

The R, G and B values and their own respective address are programmed through the **LTDC_LxCLUTWR** register.

- In case of L8 and AL88 input pixel format, the CLUT has to be loaded by 256 colors. The address of each color is configured in the CLUTADD bits in the **LTDC_LxCLUTWR** register.
- In case of AL44 input pixel format, the CLUT has to be only loaded by 16 colors. The address of each color must be filled by replicating the 4-bit L channel to 8-bit as follows:
 - L0 (indexed color 0), at address 0x00
 - L1, at address 0x11
 - L2, at address 0x22
 -
 - L15, at address 0xFF

Color Frame Buffer Address

Every Layer has a start address for the color frame buffer configured through the **LTDC_LxCFBAR** register.

When a layer is enabled, the data is fetched from the Color Frame Buffer.

Color Frame Buffer Length

Every layer has a total line length setting for the color frame buffer in bytes and a number of lines in the frame buffer configurable in the **LTDC_LxCFBLR** and **LTDC_LxCFBLNR** register respectively.

The line length and the number of lines settings are used to stop the prefetching of data to the layer FIFO at the end of the frame buffer.

- If it is set to less bytes than required, a FIFO underrun interrupt is generated if it has been previously enabled.
- If it is set to more bytes than actually required, the useless data read from the FIFO is discarded. The useless data is not displayed.

Color Frame Buffer Pitch

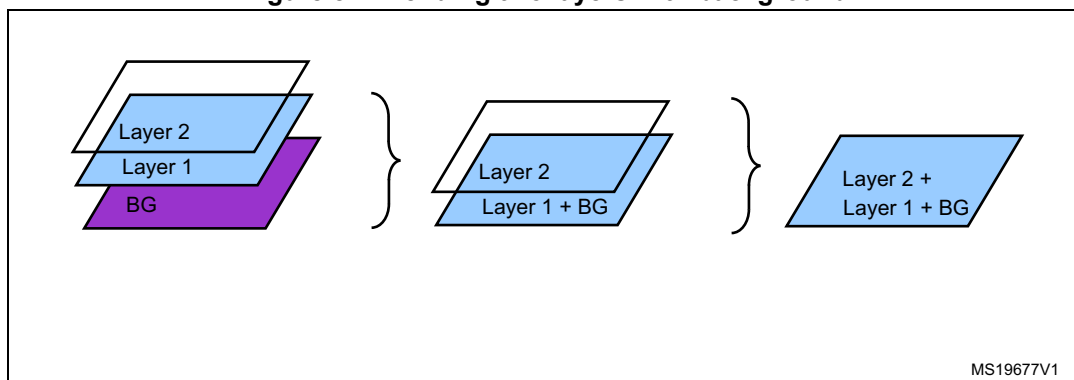
Every layer has a configurable pitch for the color frame buffer, which is the distance between the start of one line and the beginning of the next line in bytes. It is configured through the **LTDC_LxCFBLR** register.

Layer Blending

The blending is always active and the two layers can be blended following the blending factors configured through the **LTDC_LxBFGR** register.

The blending order is fixed and it is bottom up. If two layers are enabled, first the Layer1 is blended with the Background color, then the Layer2 is blended with the result of blended color of Layer1 and the background. Refer to [Figure 84](#).

Figure 84. Blending two layers with background



Default color

Every layer can have a default color in the format ARGB which is used outside the defined layer window or when a layer is disabled.

The default color is configured through the **LTDC_LxDCCR** register.

The blending is always performed between the two layers even when a layer is disabled. To avoid displaying the default color when a layer is disabled, keep the blending factors of this layer in the LTDC_LxBFCR register to their reset value.

Color Keying

A color key (RGB) can be configured to be representative for a transparent pixel.

If the Color Keying is enabled, the current pixels (after format conversion and before blending) are compared to the color key. If they match for the programmed RGB value, all channels (ARGB) of that pixel are set to 0.

The Color Key value can be configured and used at run-time to replace the pixel RGB value.

The Color Keying is enabled through the **LTDC_LxCKCR** register.

16.5 LTDC interrupts

The LTDC provides four maskable interrupts logically ORed to two interrupt vectors.

The interrupt sources can be enabled or disabled separately through the **LTDC_IER** register. Setting the appropriate mask bit to 1 enables the corresponding interrupt.

The two interrupts are generated on the following events:

- Line interrupt: generated when a programmed line is reached. The line interrupt position is programmed in the LTDC_LIPCR register
- Register Reload interrupt: generated when the shadow registers reload was performed during the vertical blanking period
- FIFO Underrun interrupt: generated when a pixel is requested from an empty layer FIFO
- Transfer Error interrupt: generated when an AHB bus error occurs during data transfer

Those interrupts events are connected to the NVIC controller as described in the figure below.

Figure 85. Interrupt events

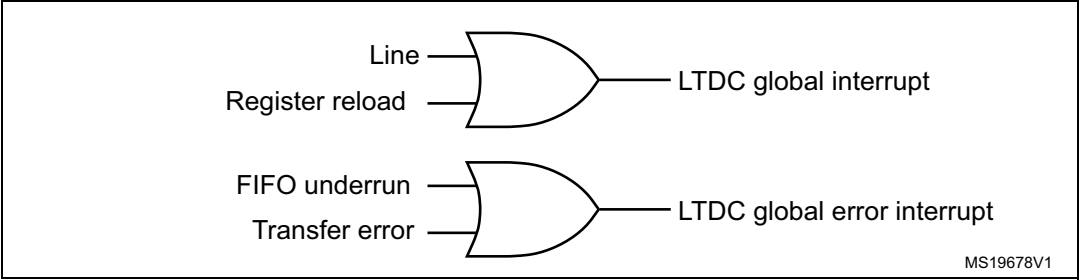


Table 92. LTDC interrupt requests

Interrupt event	Event flag	Enable Control bit
Line	LIF	LIE
Register Reload	RRIF	RRIEN
FIFO Underrun	FUDERRIF	FUDERRIE
Transfer Error	TERRIF	TERRIE

16.6 LTDC programming procedure

- Enable the LTDC clock in the RCC register
- Configure the required Pixel clock following the panel datasheet
- Configure the Synchronous timings: VSYNC, HSYNC, Vertical and Horizontal back porch, active data area and the front porch timings following the panel datasheet as described in the [Section 16.4.1: LTDC Global configuration parameters](#)
- Configure the synchronous signals and clock polarity in the **LTDC_GCR** register
- If needed, configure the background color in the **LTDC_BCCR** register
- Configure the needed interrupts in the **LTDC_IER** and **LTDC_LIPCR** register
- Configure the Layer1/2 parameters by programming:
 - The Layer window horizontal and vertical position in the **LTDC_LxWHPCR** and **LTDC_WVPCR** registers. The layer window must be in the active data area.
 - The pixel input format in the **LTDC_LxPFCR** register
 - The color frame buffer start address in the **LTDC_LxCFBAR** register
 - The line length and pitch of the color frame buffer in the **LTDC_LxCFBLR** register
 - The number of lines of the color frame buffer in the **LTDC_LxCFBLNR** register
 - if needed, load the CLUT with the RGB values and its address in the **LTDC_LxCLUTWR** register
 - If needed, configure the default color and the blending factors respectively in the **LTDC_LxDCCR** and **LTDC_LxBFCR** registers
- Enable Layer1/2 and if needed the CLUT in the **LTDC_LxCR** register
- If needed, dithering and color keying can be enabled respectively in the **LTDC_GCR** and **LTDC_LxCKCR** registers. It can be also enabled on the fly.
- Reload the shadow registers to active register through the **LTDC_SRCR** register.
- Enable the LCD-TFT controller in the **LTDC_GCR** register.
- All layer parameters can be modified on the fly except the CLUT. The new configuration has to be either reloaded immediately or during vertical blanking period by configuring the **LTDC_SRCR** register.

Note: All layer's registers are shadowed. Once a register is written, it should not be modified again before the reload has been done. Thus, a new write to the same register overrides the previous configuration if not yet reloaded.

16.7 LTDC registers

16.7.1 LTDC Synchronization Size Configuration Register (LTDC_SSCR)

This register defines the number of Horizontal Synchronization pixels minus 1 and the number of Vertical Synchronization lines minus 1. Refer to [Figure 82](#) and [Section 16.4: LTDC programmable parameters](#) for an example of configuration.

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				HSW[11:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				VSH[10:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:28 Reserved, must be kept at reset value

Bits 27:16 **HSW[11:0]**: Horizontal Synchronization Width (in units of pixel clock period)
These bits define the number of Horizontal Synchronization pixel minus 1.

Bits 15:11 Reserved, must be kept at reset value

Bits 10:0 **VSH[10:0]**: Vertical Synchronization Height (in units of horizontal scan line)
These bits define the vertical Synchronization height minus 1. It represents the number of horizontal synchronization lines.

16.7.2 LTDC Back Porch Configuration Register (LTDC_BPCR)

This register defines the accumulated number of Horizontal Synchronization and back porch pixels minus 1 (**HSYNC Width + HBP - 1**) and the accumulated number of Vertical Synchronization and back porch lines minus 1 (**VSYNC Height + VBP - 1**). Refer to [Figure 82](#) and [Section 16.4: LTDC programmable parameters](#) for an example of configuration.

Address offset: 0x0C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				AHBP[11:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				AVBP[10:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:28 Reserved, must be kept at reset value

Bits 27:16 **AHBP[11:0]**: Accumulated Horizontal back porch (in units of pixel clock period)

These bits define the Accumulated Horizontal back porch width which includes the Horizontal Synchronization and Horizontal back porch pixels minus 1.

The Horizontal back porch is the period between Horizontal Synchronization going inactive and the start of the active display part of the next scan line.

Bits 15:11 Reserved, must be kept at reset value

Bits 10:0 **AVBP[10:0]**: Accumulated Vertical back porch (in units of horizontal scan line)

These bits define the accumulated Vertical back porch width which includes the Vertical Synchronization and Vertical back porch lines minus 1.

The Vertical back porch is the number of horizontal scan lines at a start of frame to the start of the first active scan line of the next frame.

16.7.3 LTDC Active Width Configuration Register (LTDC_AWCR)

This register defines the accumulated number of Horizontal Synchronization, back porch and Active pixels minus 1 (**HSYNC width + HBP + Active Width - 1**) and the accumulated number of Vertical Synchronization, back porch lines and Active lines minus 1 (**VSYNC Height+ BVBP + Active Height - 1**). Refer to [Figure 82](#) and [Section 16.4: LTDC programmable parameters](#) for an example of configuration.

Address offset: 0x10

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				AAW[11:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				AAH[10:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:28 Reserved, must be kept at reset value

Bits 27:16 **AAW[11:0]**: Accumulated Active Width (in units of pixel clock period)

These bits define the Accumulated Active Width which includes the Horizontal Synchronization, Horizontal back porch and Active pixels minus 1.

The Active Width is the number of pixels in active display area of the panel scan line. The maximum Active Width supported is 0x400.

Bits 15:11 Reserved, must be kept at reset value

Bits 10:0 **AAH[10:0]**: Accumulated Active Height (in units of horizontal scan line)

These bits define the Accumulated Height which includes the Vertical Synchronization, Vertical back porch and the Active Height lines minus 1. The Active Height is the number of active lines in the panel. The maximum Active Height supported is 0x300.

16.7.4 LTDC Total Width Configuration Register (LTDC_TWCR)

This register defines the accumulated number of Horizontal Synchronization, back porch, Active and front porch pixels minus 1 (**HSYNC Width + HBP + Active Width + HFP - 1**) and the accumulated number of Vertical Synchronization, back porch lines, Active and Front lines minus 1 (**VSYNC Height+ BVBP + Active Height + VFP - 1**). Refer to [Figure 82](#) and [Section 16.4: LTDC programmable parameters](#) for an example of configuration.

Address offset: 0x14

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				TOTALW[11:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
16	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				TOTALH[10:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:28 Reserved, must be kept at reset value

Bits 27:16 **TOTALW[11:0]**: Total Width (in units of pixel clock period)

These bits defines the accumulated Total Width which includes the Horizontal Synchronization, Horizontal back porch, Active Width and Horizontal front porch pixels minus 1.

Bits 15:11 Reserved, must be kept at reset value

Bits 10:0 **TOTALH[10:0]**: Total Height (in units of horizontal scan line)

These bits defines the accumulated Height which includes the Vertical Synchronization, Vertical back porch, the Active Height and Vertical front porch Height lines minus 1.

16.7.5 LTDC Global Control Register (LTDC_GCR)

This register defines the global configuration of the LCD-TFT controller.

Address offset: 0x18

Reset value: 0x0000 2220

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HSPOL	VSPOL	DEPOL	PCPOL	Reserved											DEN
rw	rw	rw	rw												rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	DRW[2:0]			Reserved	DGW[2:0]			Reserved	DBW[2:0]			Reserved		LTDCEN	
	r	r	r		r	r	r		r	r	r			rw	

- Bit 31 **HSPOL**: Horizontal Synchronization Polarity
This bit is set and cleared by software.
0: Horizontal Synchronization polarity is active low
1: Horizontal Synchronization polarity is active high
- Bit 30 **VSPOL**: Vertical Synchronization Polarity
This bit is set and cleared by software.
0: Vertical Synchronization is active low
1: Vertical Synchronization is active high
- Bit 29 **DEPOL**: Data Enable Polarity
This bit is set and cleared by software.
0: Data Enable polarity is active low
1: Data Enable polarity is active high
- Bit 28 **PCPOL**: Pixel Clock Polarity
This bit is set and cleared by software.
0: input pixel clock
1: inverted input pixel clock
- Bits 27:17 Reserved, must be kept at reset value
- Bit 16 **DEN**: Dither Enable
This bit is set and cleared by software.
0: Dither disable
1: Dither enable
- Bit 15 Reserved, must be kept at reset value
- Bits 14:12 **DRW[2:0]**: Dither Red Width
These bits return the Dither Red Bits
- Bit 11 Reserved, must be kept at reset value
- Bits 10:8 **DGW[2:0]**: Dither Green Width
These bits return the Dither Green Bits
- Bit 7 Reserved, must be kept at reset value
- Bits 6:4 **DBW[2:0]**: Dither Blue Width
These bits return the Dither Blue Bits
- Bits 3:1 Reserved, must be kept at reset value
- Bit 0 **LTDCEN**: LCD-TFT controller enable bit
This bit is set and cleared by software.
0: LTDC disable
1: LTDC enable

16.7.6 LTDC Shadow Reload Configuration Register (LTDC_SRCR)

This register allows to reload either immediately or during the vertical blanking period, the shadow registers values to the active registers. The shadow registers are all Layer1 and Layer2 registers except the LTDC_L1CLUTWR and the LTDC_L2CLUTWR.

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														VBR	IMR
														rw	rw

Bits 31:2 Reserved, must be kept at reset value

Bit 1 **VBR**: Vertical Blanking Reload

This bit is set by software and cleared only by hardware after reload. (it cannot be cleared through register write once it is set)

0: No effect

1: The shadow registers are reloaded during the vertical blanking period (at the beginning of the first line after the Active Display Area)

Bit 0 **IMR**: Immediate Reload

This bit is set by software and cleared only by hardware after reload.

0: No effect

1: The shadow registers are reloaded immediately

Note: The shadow registers read back the active values. Until the reload has been done, the 'old' value is read.

16.7.7 LTDC Background Color Configuration Register (LTDC_BCCR)

This register defines the background color (RGB888).

Address offset: 0x2C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								BCRED[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BCGREEN[7:0]								BCBLUE[7:0]							
rw								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:24 Reserved, must be kept at reset value

Bits 23:16 **BCRED[7:0]**: Background Color Red value
These bits configure the background red value

Bits 15:8 **BCGREEN[7:0]**: Background Color Green value
These bits configure the background green value

Bits 7:0 **BCBLUE[7:0]**: Background Color Blue value
These bits configure the background blue value

16.7.8 LTDC Interrupt Enable Register (LTDC_IER)

This register determines which status flags generate an interrupt request by setting the corresponding bit to 1.

Address offset: 0x34

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												RRIE	TERRIE	FUIE	LIE
												rw	rw	rw	rw

Bits 31:4 Reserved, must be kept at reset value

Bit 3 **RRIE**: Register Reload interrupt enable
This bit is set and cleared by software
0: Register Reload interrupt disable
1: Register Reload interrupt enable

Bit 2 **TERRIE**: Transfer Error Interrupt Enable
This bit is set and cleared by software
0: Transfer Error interrupt disable
1: Transfer Error interrupt enable

Bit 1 **FUIE**: FIFO Underrun Interrupt Enable
This bit is set and cleared by software
0: FIFO Underrun interrupt disable
1: FIFO Underrun Interrupt enable

Bit 0 **LIE**: Line Interrupt Enable
This bit is set and cleared by software
0: Line interrupt disable
1: Line Interrupt enable

16.7.9 LTDC Interrupt Status Register (LTDC_ISR)

This register returns the interrupt status flag

Address offset: 0x38

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												RRIF	TERRIF	FUIF	LIF
												r	r	r	r

Bits 31:24 Reserved, must be kept at reset value

Bit 3 **RRIF**: Register Reload Interrupt Flag

0: No Register Reload interrupt generated

1: Register Reload interrupt generated when a vertical blanking reload occurs (and the first line after the active area is reached)

Bit 2 **TERRIF**: Transfer Error interrupt flag

0: No Transfer Error interrupt generated

1: Transfer Error interrupt generated when a Bus error occurs

Bit 1 **FUIF**: FIFO Underrun Interrupt flag

0: NO FIFO Underrun interrupt generated.

1: A FIFO underrun interrupt is generated, if one of the layer FIFOs is empty and pixel data is read from the FIFO

Bit 0 **LIF**: Line Interrupt flag

0: No Line interrupt generated

1: A Line interrupt is generated, when a programmed line is reached

16.7.10 LTDC Interrupt Clear Register (LTDC_ICR)

Address offset: 0x3C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												CRRIF	CTERRIF	CFUIF	CLIF
												w	w	w	w

Bits 31:24 Reserved, must be kept at reset value

Bit 3 **CRRIF**: Clears Register Reload Interrupt Flag

0: No effect

1: Clears the RRIF flag in the LTDC_ISR register

Bit 2 **CTERRIF**: Clears the Transfer Error Interrupt Flag

0: No effect

1: Clears the TERRIF flag in the LTDC_ISR register.

Bit 1 **CFUIF**: Clears the FIFO Underrun Interrupt flag

0: No effect

1: Clears the FUDERRIF flag in the LTDC_ISR register.

Bit 0 **CLIF**: Clears the Line Interrupt Flag

0: No effect

1: Clears the LIF flag in the LTDC_ISR register.

16.7.11 LTDC Line Interrupt Position Configuration Register (LTDC_LIPCR)

This register defines the position of the line interrupt. The line value to be programmed depends on the timings parameters. Refer to [Figure 82](#).

Address offset: 0x40

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					LIPOS[10:0]										
					r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:11 Reserved, must be kept at reset value

Bits 10:0 **LIPOS[10:0]**: Line Interrupt Position

These bits configure the line interrupt position

16.7.12 LTDC Current Position Status Register (LTDC_CPSR)

Address offset: 0x44

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CXPOS[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CYPOS[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:16: **CXPOS[15:0]**: Current X Position
These bits return the current X position

Bits 15:0 **CYPOS[15:0]**: Current Y Position
These bits return the current Y position

16.7.13 LTDC Current Display Status Register (LTDC_CDSR)

This register returns the status of the current display phase which is controlled by the HSYNC, VSYNC, and Horizontal/Vertical DE signals.

Example: if the current display phase is the vertical synchronization, the VSYNC S bit is set (active high). If the current display phase is the horizontal synchronization, the HSYNC S bit is active high.

Address offset: 0x48

Reset value: 0x0000 000F

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												HSYNC S	VSYNC S	HDES	VDES
												r	r	r	r

Bits 31:24 Reserved, must be kept at reset value

Bit 3 **HSYNC S**: Horizontal Synchronization display Status

0: Active low

1: Active high

Bit 2 **VSYNC S**: Vertical Synchronization display Status

0: Active low

1: Active high

Bit 1 **HDES**: Horizontal Data Enable display Status

0: Active low

1: Active high

Bit 0 **VDES**: Vertical Data Enable display Status

0: Active low

1: Active high

Note: The returned status does not depend on the configured polarity in the **LTDC_GCR** register, instead it returns the current active display phase.

16.7.14 LTDC Layerx Control Register (LTDC_LxCR) (where x=1..2)

Address offset: $0x84 + 0x80 \times (Layerx - 1)$, $Layerx = 1$ or 2

Reset value: $0x0000\ 0000$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved											CLUTEN	Reserved		COLKEN	LEN
											rw			rw	rw

Bits 31:5 Reserved, must be kept at reset value

Bit 4 **CLUTEN**: Color Look-Up Table Enable

This bit is set and cleared by software.

0: Color Look-Up Table disable

1: Color Look-Up Table enable

The CLUT is only meaningful for L8, AL44 and AL88 pixel format. Refer to [Color Look-Up Table \(CLUT\) on page 491](#)

Bit 3 Reserved, must be kept at reset value

Bit 2 Reserved, must be kept at reset value

Bit 1 **COLKEN**: Color Keying Enable

This bit is set and cleared by software.

0: Color Keying disable

1: Color Keying enable

Bit 0 **LEN**: Layer Enable

This bit is set and cleared by software.

0: Layer disable

1: Layer enable

16.7.15 LTDC Layerx Window Horizontal Position Configuration Register (LTDC_LxWHPCR) (where x=1..2)

This register defines the Horizontal Position (first and last pixel) of the layer 1 or 2 window.

The first visible pixel of a line is the programmed value of **AHBP[10:0] bits + 1** in the **LTDC_BPCR** register.

The last visible pixel of a line is the programmed value of **AAW[10:0] bits** in the **LTDC_AWCR** register.

Address offset: $0x88 + 0x80 \times (\text{Layerx} - 1)$, $\text{Layerx} = 1 \text{ or } 2$

Reset value: **0x0000 0000**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				WHSPPOS[11:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				WHSTPOS[11:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:28 Reserved, must be kept at reset value

Bits 27:16 **WHSPPOS[11:0]**: Window Horizontal Stop Position

These bits configure the last visible pixel of a line of the layer window.

The following condition must be respected:

$\text{WHSPPOS}[11:0] \geq \text{AHBP}[10:0] \text{ bits} + 1$ (programmed in LTDC_BPCR register)

Bits 15:12 Reserved, must be kept at reset value

Bits 11:0 **WHSTPOS[11:0]**: Window Horizontal Start Position

These bits configure the first visible pixel of a line of the layer window.

The following condition must be respected:

$\text{WHSTPOS}[11:0] \text{ must be } \leq \text{AAW}[10:0] \text{ bits}$ (programmed in the LTDC_AWCR register).

Example:

The LTDC_BPCR register is configured to 0x000E0005(AHBP[11:0] is 0xE) and the LTDC_AWCR register is configured to 0x028E01E5(AAW[11:0] is 0x28E). To configure the horizontal position of a window size of 630x460, with horizontal start offset of 5 pixels in the Active data area.

1. Layer window first pixel: WHSTPOS[11:0] should be programmed to 0x14 (0xE+1+0x5)
2. Layer window last pixel: WHSPPOS[11:0] should be programmed to 0x28A

16.7.16 LTDC Layerx Window Vertical Position Configuration Register (LTDC_LxWVPCR) (where x=1..2)

This register defines the vertical position (first and last line) of the layer1 or 2 window.

The first visible line of a frame is the programmed value of **AVBP[10:0] bits + 1** in the register **LTDC_BPCR** register.

The last visible line of a frame is the programmed value of **AAH[10:0] bits** in the **LTDC_AWCR** register.

Address offset: $0x8C + 0x80 \times (\text{Layerx} - 1)$, *Layerx* = 1 or 2

Reset value: *0x0000 0000*

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					WVSPPOS[10:0]										
					rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					WVSTPOS[10:0]										
					rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:27 Reserved, must be kept at reset value

Bits 26:16 **WVSPPOS[10:0]**: Window Vertical Stop Position

These bits configures the last visible line of the layer window.

The following condition must be respected:

WVSPPOS[11:0] must be $\geq$ AVBP[10:0] bits + 1 (programmed in LTDC_BPCR register)

Bits 15:11 Reserved, must be kept at reset value

Bits 10:0 **WVSTPOS[10:0]**: Window Vertical Start Position

These bits configure the first visible line of the layer window.

The following condition must be respected:

WVSTPOS[11:0] must be $\leq$ AAH[10:0] bits (programmed in the LTDC_AWCR register)

Example:

The LTDC_BPCR register is configured to 0x000E0005 (AVBP[10:0] is 0x5) and the LTDC_AWCR register is configured to 0x028E01E5 (AAH[10:0] is 0x1E5). To configure the vertical position of a window size of 630x460, with vertical start offset of 8 lines in the Active data area:

1. Layer window first line: WVSTPOS[10:0] should be programmed to 0xE ($0x5 + 1 + 0x8$)
2. Layer window last line: WVSPPOS[10:0] should be programmed to 0x1DA

16.7.17 LTDC Layerx Color Keying Configuration Register (LTDC_LxCKCR) (where x=1..2)

This register defines the color key value (RGB), which is used by the Color Keying.

Address offset: $0x90 + 0x80 \times (Layerx - 1)$, $Layerx = 1$ or 2

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CKRED[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CKGREEN[7:0]								CKBLUE[7:0]							
rw								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:24 Reserved, must be kept at reset value

Bits 23:16 **CKRED[7:0]**: Color Key Red value

Bits 15:8 **CKGREEN[7:0]**: Color Key Green value

Bits 7:0 **CKBLUE[7:0]**: Color Key Blue value

16.7.18 LTDC Layerx Pixel Format Configuration Register (LTDC_LxPFCR) (where x=1..2)

This register defines the pixel format which is used for the stored data in the frame buffer of a layer. The pixel data is read from the frame buffer and then transformed to the internal format 8888 (ARGB).

Address offset: $0x94 + 0x80 \times (Layerx - 1)$, $Layerx = 1$ or 2

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												PF[2:0]			
												rw	rw	rw	

Bits 31:3 Reserved, must be kept at reset value

Bits 2:0 **PF[2:0]**: Pixel Format

These bits configures the Pixel format

000: ARGB8888

001: RGB888

010: RGB565

011: ARGB1555

100: ARGB4444

101: L8 (8-Bit Luminance)

110: AL44 (4-Bit Alpha, 4-Bit Luminance)

111: AL88 (8-Bit Alpha, 8-Bit Luminance)

16.7.19 LTDC Layerx Constant Alpha Configuration Register (LTDC_LxCACR) (where x=1..2)

This register defines the constant alpha value (divided by 255 by Hardware), which is used in the alpha blending. Refer to LTDC_LxBFCR register.

Address offset: $0x98 + 0x80 \times (Layerx - 1)$, $Layerx = 1$ or 2

Reset value: $(Layerx - 1) 0x0000 00FF$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								CONSTA[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value

Bits 7:0 **CONSTA[7:0]**: Constant Alpha

These bits configure the Constant Alpha used for blending. The Constant Alpha is divided by 255 by hardware.

Example: if the programmed Constant Alpha is 0xFF, the Constant Alpha value is $255/255=1$

16.7.20 LTDC Layerx Default Color Configuration Register (LTDC_LxDCCR) (where x=1..2)

This register defines the default color of a layer in the format ARGB. The default color is used outside the defined layer window or when a layer is disabled. The reset value of 0x00000000 defines a transparent black color.

Address offset: $0x9C + 0x80 \times (Layerx - 1)$, $Layerx = 1$ or 2

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DCALPHA[7:0]								DCRED[7:0]							
rw								rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DCGREEN[7:0]								DCBLUE[7:0]							
rw								rw	rw	rw	rw	rw	rw	rw	rw

- Bits 31:24 **DCALPHA[7:0]**: Default Color Alpha
These bits configure the default alpha value
- Bits 23:16 **DCRED[7:0]**: Default Color Red
These bits configure the default red value
- Bits 15:8 **DCGREEN[7:0]**: Default Color Green
These bits configure the default green value
- Bits 7:0 **DCBLUE[7:0]**: Default Color Blue
These bits configure the default blue value



16.7.21 LTDC Layerx Blending Factors Configuration Register (LTDC_LxBFCR) (where x=1..2)

This register defines the blending factors F1 and F2.

The general blending formula is: $BC = BF1 \times C + BF2 \times Cs$

- BC = Blended color
- BF1 = Blend Factor 1
- C = Current layer color
- BF2 = Blend Factor 2
- Cs = subjacent layers blended color

Address offset: $0xA0 + 0x80 \times (Layerx - 1)$, $Layerx = 1$ or 2

Reset value: 0x0000 0607

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					BF1[2:0]			Reserved					BF2[2:0]		
					rw	rw	rw						rw	rw	rw

Bits 31:11 Reserved, must be kept at reset value

Bits 10:8 **BF1[2:0]**: Blending Factor 1

These bits select the blending factor F1

000: Reserved

001: Reserved

010: Reserved

011: Reserved

100: Constant Alpha

101: Reserved

110: Pixel Alpha x Constant Alpha

111: Reserved

Bits 7:3 Reserved, must be kept at reset value

Bits 2:0 **BF2[2:0]**: Blending Factor 2

These bits select the blending factor F2

000: Reserved

001: Reserved

010: Reserved

011: Reserved

100: Reserved

101: 1 - Constant Alpha

110: Reserved

111: 1 - (Pixel Alpha x Constant Alpha)

Note: The Constant Alpha value, is the programmed value in the LxCACR register divided by 255 by hardware.

Example: Only layer1 is enabled, BF1 configured to Constant Alpha

BF2 configured to 1 - Constant Alpha

Constant Alpha: The Constant Alpha programmed in the LxCACR register is 240 (0xF0). Thus, the Constant Alpha value is $240/255 = 0.94$

C: Current Layer Color is 128

Cs: Background color is 48

Layer1 is blended with the background color.

$BC = \text{Constant Alpha} \times C + (1 - \text{Constant Alpha}) \times Cs = 0.94 \times 128 + (1 - 0.94) \times 48 = 123$.

16.7.22 LTDC Layerx Color Frame Buffer Address Register (LTDC_LxCFBAR) (where x=1..2)

This register defines the color frame buffer start address which has to point to the address where the pixel data of the top left pixel of a layer is stored in the frame buffer.

Address offset: $0xAC + 0x80 \times (\text{Layerx} - 1)$, Layerx = 1 or 2

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CFBADD[31:0]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CFBADD[31:0]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:0 **CFBADD[31:0]**: Color Frame Buffer Start Address

These bits defines the color frame buffer start address.

16.7.23 LTDC Layerx Color Frame Buffer Length Register (LTDC_LxCFBLR) (where x=1..2)

This register defines the color frame buffer line length and pitch.

Address offset: $0xB0 + 0x80 \times (\text{Layerx} - 1)$, Layerx = 1 or 2

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			CFBP[12:0]												
			rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			CFBLL[12:0]												
			rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:29 Reserved, must be kept at reset value

Bits 28:16 **CFBP[12:0]**: Color Frame Buffer Pitch in bytes

These bits define the pitch which is the increment from the start of one line of pixels to the start of the next line in bytes.

Bits 15:13 Reserved, must be kept at reset value

Bits 12:0 **CFBLL[12:0]**: Color Frame Buffer Line Length

These bits define the length of one line of pixels in bytes + 3.

The line length is computed as follows: Active high width x number of bytes per pixel + 3.

Example:

- A frame buffer having the format RGB565 (2 bytes per pixel) and a width of 256 pixels (total number of bytes per line is $256 \times 2 = 512$ bytes), where pitch = line length requires a value of 0x02000203 to be written into this register.
- A frame buffer having the format RGB888 (3 bytes per pixel) and a width of 320 pixels (total number of bytes per line is $320 \times 3 = 960$), where pitch = line length requires a value of 0x03C003C3 to be written into this register.

16.7.24 LTDC Layerx ColorFrame Buffer Line Number Register (LTDC_LxCFBLNR) (where x=1..2)

This register defines the number of lines in the color frame buffer.

Address offset: $0xB4 + 0x80 \times (\text{Layerx} - 1)$, Layerx = 1 or 2

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					CFBLNBR[10:0]										
					r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:11 Reserved, must be kept at reset value

Bits 10:0 **CFBLNBR[10:0]**: Frame Buffer Line Number

These bits define the number of lines in the frame buffer which corresponds to the Active high width.

Note: *The number of lines and line length settings define how much data is fetched per frame for every layer. If it is configured to less bytes than required, a FIFO underrun interrupt is generated if enabled.*

The start address and pitch settings on the other hand define the correct start of every line in memory.

16.7.25 LTDC Layerx CLUT Write Register (LTDC_LxCLUTWR) (where x=1..2)

This register defines the CLUT address and the RGB value.

Address offset: $0xC4 + 0x80 \times (\text{Layerx} - 1)$, Layerx = 1 or 2

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CLUTADD[7:0]								RED[7:0]							
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GREEN[7:0]								BLUE[7:0]							
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:24 **CLUTADD[7:0]**: CLUT Address

These bits configure the CLUT address (color position within the CLUT) of each RGB value

Bits 23:16 **RED[7:0]**: Red value

These bits configure the red value

Bits 15:8 **GREEN[7:0]**: Green value

These bits configure the green value

Bits 7:0 **BLUE[7:0]**: Blue value

These bits configure the blue value

Note: The CLUT write register should only be configured during blanking period or if the layer is disabled. The CLUT can be enabled or disabled in the **LTDC_LxCR** register.

The CLUT is only meaningful for L8, AL44 and AL88 pixel format.

16.7.26 LTDC register map

The following table summarizes the LTDC registers. Refer to the register boundary addresses table for the LTDC register base address.

Table 93. LTDC register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0														
0x0008	LTDC_SSCR	Reserved					HSW[9:0]									Reserved					VSH[10:0]																										
	Reset value						0	0	0	0	0	0	0	0	0						0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0							
0x000C	LTDC_BPCR	Reserved					AHBP[11:0]									Reserved					AVBP[10:0]																										
	Reset value						0	0	0	0	0	0	0	0	0						0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0							
0x0010	LTDC_AWCR	Reserved					AAV[11:0]									Reserved					AAH[10:0]																										
	Reset value						0	9	0	0	0	0	0	0	0						0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						
0x0014	LTDC_TWCR	Reserved					TOTALW[11:0]									Reserved					TOTALH[10:0]																										
	Reset value						0	0	0	0	0	0	0	0	0						0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						
0x0018	LTDC_GCR	HSPOL	VSPOL	DEPOL	PCPOL	Reserve										DEN	Reserved	DRW[2:0]			Reserved	DGW[2:0]			Reserved	DBW[2:0]			Reserved	LTDEN																	
	Reset value	0	0	0	0											0		0	1	0		0	1	0		0	1	0			0	1	0														
0x0024	LTDC_SRCR	Reserved																													VBR	IMR															
	Reset value																														0		0														
0x002C	LTDC_BCCR	Reserved									BC[23:0]																																				
	Reset value										0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0							
0x0034	LTDC_IER	Reserved																													RRIE	TERRIE	FUIE	LIE													
	Reset value																														0	0	0	0													
0x0038	LTDC_ISR	Reserved																													RRIF	TERRIF	FUIF	LIF													
	Reset value																														0	0	0	0													
0x003C	LTDC_ICR	Reserved																													CRRIF	CTERRIF	CFUIF	CLIF													
	Reset value																														0	0	0	0													
0x0040	LTDC_LIPCR	Reserved																			LIPOS[10:0]																										
	Reset value																				0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0044	LTDC_CPSR	CXPOS[15:0]															CYPOS[15:0]																														
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0														
0x0048	LTDC_CDSR	Reserved																													HSYNCS	VSYNCS	HDES	VDES													
	Reset value																														1	1	1	1													
0x0084	LTDC_L1CR	Reserved																													CLUTEN	Reserved	COLKEN	LEN													
	Reset value																														0				0	0											
0x0088	LTDC_L1WHPCR	Reserved					WHSPPOS[11:0]									Reserved					WHSTPOS[11:0]																										
	Reset value						0	0	0	0	0	0	0	0	0						0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0							

Table 93. LTDC register map and reset values (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x0134	LTDC_L2CFBLNR	Reserved																				CFBLNBR[10:0]											
	Reset value																					0	0	0	0	0	0	0	0	0	0	0	
0x0144	LTDC_L2CLUTWR	CLUTADD[7:0]								RED[7:0]								GREEN[7:0]								BLUE[7:0]							
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

17 Advanced-control timers (TIM1 and TIM8)

This section applies to the whole STM32F4xx family, unless otherwise specified.

17.1 TIM1 and TIM8 introduction

The advanced-control timers (TIM1 and TIM8) consist of a 16-bit auto-reload counter driven by a programmable prescaler.

It may be used for a variety of purposes, including measuring the pulse lengths of input signals (input capture) or generating output waveforms (output compare, PWM, complementary PWM with dead-time insertion).

Pulse lengths and waveform periods can be modulated from a few microseconds to several milliseconds using the timer prescaler and the RCC clock controller prescalers.

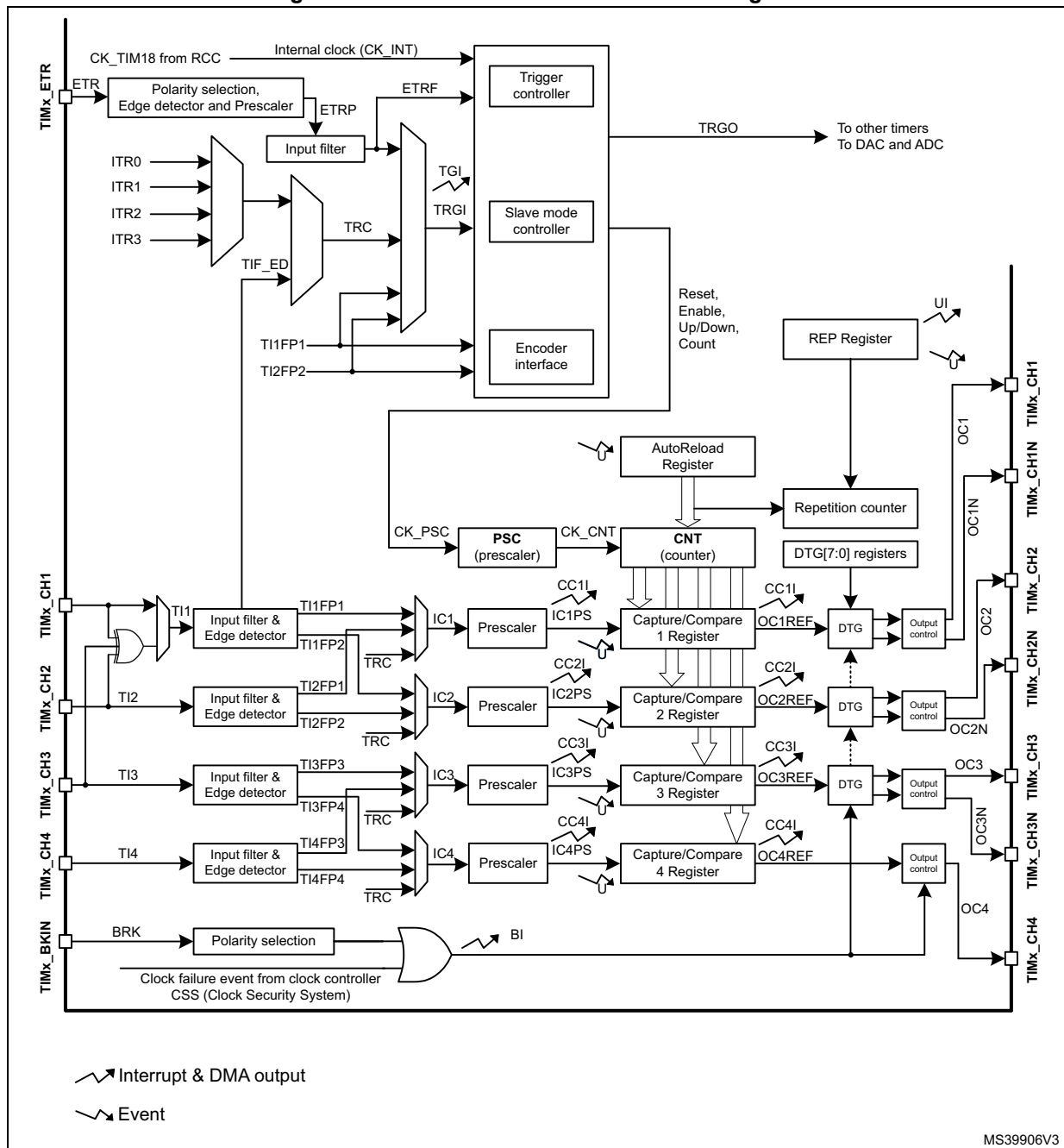
The advanced-control (TIM1 and TIM8) and general-purpose (TIMx) timers are completely independent, and do not share any resources. They can be synchronized together as described in [Section 17.3.20](#).

17.2 TIM1 and TIM8 main features

TIM1 and TIM8 timer features include:

- 16-bit up, down, up/down auto-reload counter.
- 16-bit programmable prescaler allowing dividing (also “on the fly”) the counter clock frequency either by any factor between 1 and 65536.
- Up to 4 independent channels for:
 - Input capture
 - Output compare
 - PWM generation (Edge and Center-aligned mode)
 - One-pulse mode output
- Complementary outputs with programmable dead-time
- Synchronization circuit to control the timer with external signals and to interconnect several timers together.
- Repetition counter to update the timer registers only after a given number of cycles of the counter.
- Break input to put the timer’s output signals in reset state or in a known state.
- Interrupt/DMA generation on the following events:
 - Update: counter overflow/underflow, counter initialization (by software or internal/external trigger)
 - Trigger event (counter start, stop, initialization or count by internal/external trigger)
 - Input capture
 - Output compare
 - Break input
- Supports incremental (quadrature) encoder and hall-sensor circuitry for positioning purposes
- Trigger input for external clock or cycle-by-cycle current management

Figure 86. Advanced-control timer block diagram



17.3 TIM1 and TIM8 functional description

17.3.1 Time-base unit

The main block of the programmable advanced-control timer is a 16-bit counter with its related auto-reload register. The counter can count up, down or both up and down. The counter clock can be divided by a prescaler.

The counter, the auto-reload register and the prescaler register can be written or read by software. This is true even when the counter is running.

The time-base unit includes:

- Counter register (TIMx_CNT)
- Prescaler register (TIMx_PSC)
- Auto-reload register (TIMx_ARR)
- Repetition counter register (TIMx_RCR)

The auto-reload register is preloaded. Writing to or reading from the auto-reload register accesses the preload register. The content of the preload register are transferred into the shadow register permanently or at each update event (UEV), depending on the auto-reload preload enable bit (ARPE) in TIMx_CR1 register. The update event is sent when the counter reaches the overflow (or underflow when downcounting) and if the UDIS bit equals 0 in the TIMx_CR1 register. It can also be generated by software. The generation of the update event is described in detailed for each configuration.

The counter is clocked by the prescaler output CK_CNT, which is enabled only when the counter enable bit (CEN) in TIMx_CR1 register is set (refer also to the slave mode controller description to get more details on counter enabling).

Note that the counter starts counting 1 clock cycle after setting the CEN bit in the TIMx_CR1 register.

Prescaler description

The prescaler can divide the counter clock frequency by any factor between 1 and 65536. It is based on a 16-bit counter controlled through a 16-bit register (in the TIMx_PSC register). It can be changed on the fly as this control register is buffered. The new prescaler ratio is taken into account at the next update event.

Figure 87 and *Figure 88* give some examples of the counter behavior when the prescaler ratio is changed on the fly:

Figure 87. Counter timing diagram with prescaler division change from 1 to 2

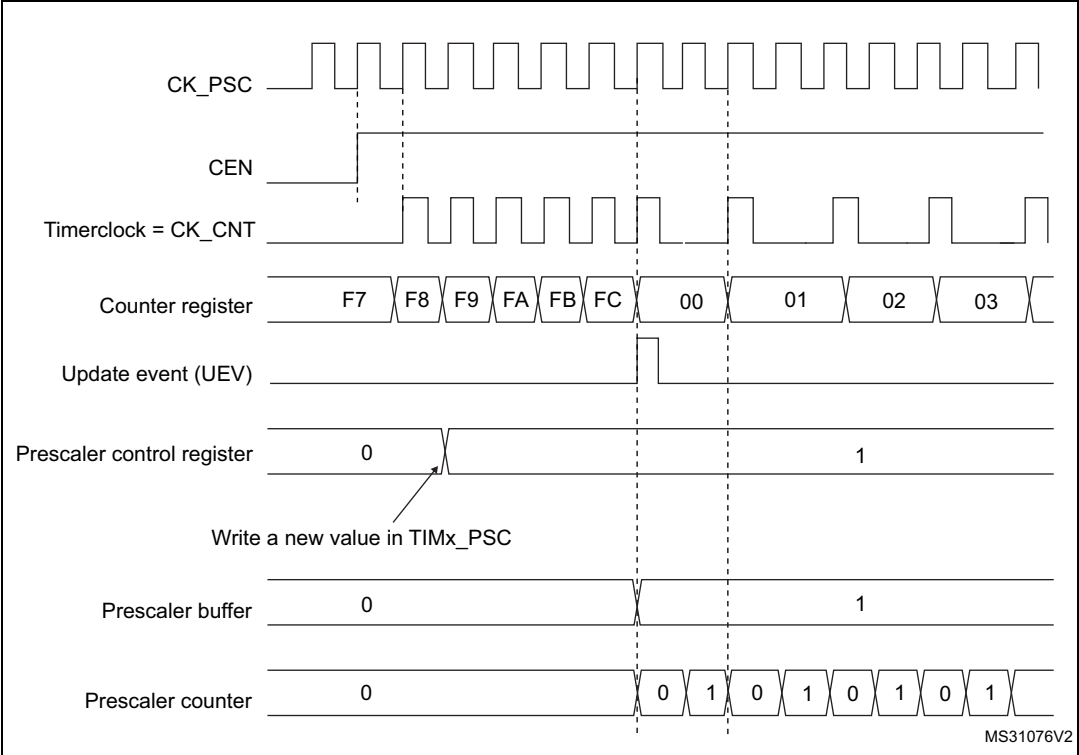
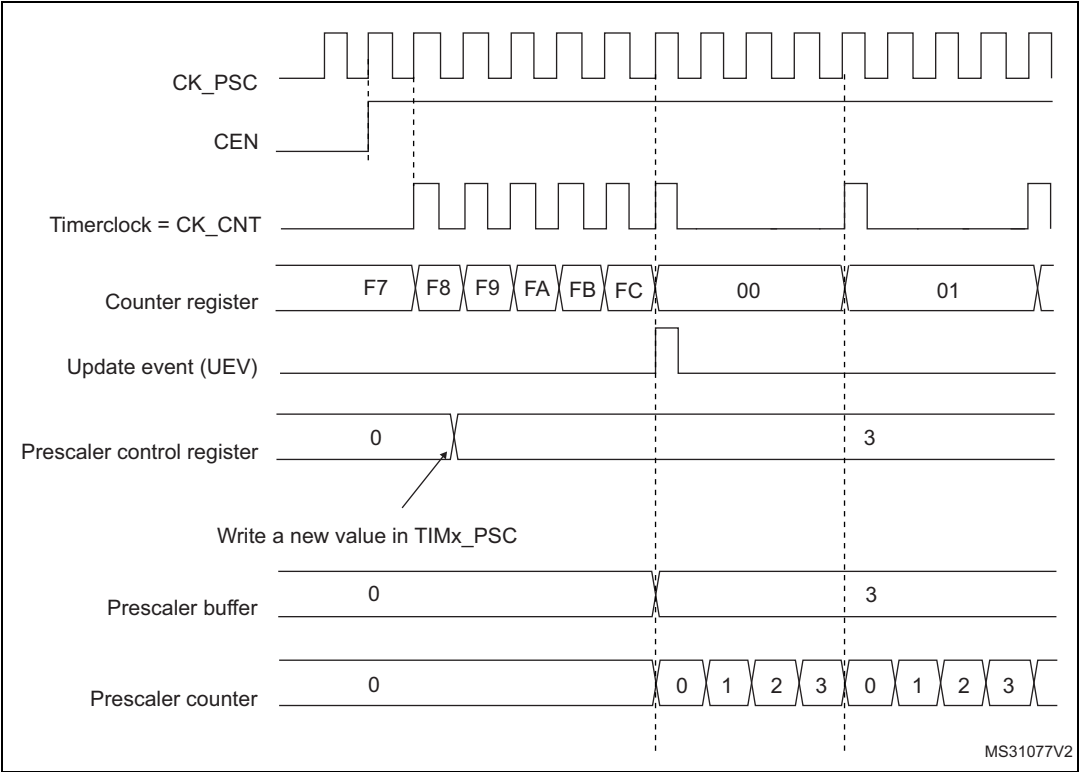


Figure 88. Counter timing diagram with prescaler division change from 1 to 4



17.3.2 Counter modes

Upcounting mode

In upcounting mode, the counter counts from 0 to the auto-reload value (content of the TIMx_ARR register), then restarts from 0 and generates a counter overflow event.

If the repetition counter is used, the update event (UEV) is generated after upcounting is repeated for the number of times programmed in the repetition counter register plus one (TIMx_RCR+1). Else the update event is generated at each counter overflow.

Setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller) also generates an update event.

The UEV event can be disabled by software by setting the UDIS bit in the TIMx_CR1 register. This is to avoid updating the shadow registers while writing new values in the preload registers. Then no update event occurs until the UDIS bit has been written to 0. However, the counter restarts from 0, as well as the counter of the prescaler (but the prescale rate does not change). In addition, if the URS bit (update request selection) in TIMx_CR1 register is set, setting the UG bit generates an update event UEV but without setting the UIF flag (thus no interrupt or DMA request is sent). This is to avoid generating both update and capture interrupts when clearing the counter on the capture event.

When an update event occurs, all the registers are updated and the update flag (UIF bit in TIMx_SR register) is set (depending on the URS bit):

- The repetition counter is reloaded with the content of TIMx_RCR register,
- The auto-reload shadow register is updated with the preload value (TIMx_ARR),
- The buffer of the prescaler is reloaded with the preload value (content of the TIMx_PSC register).

The following figures show some examples of the counter behavior for different clock frequencies when TIMx_ARR=0x36.

Figure 89. Counter timing diagram, internal clock divided by 1

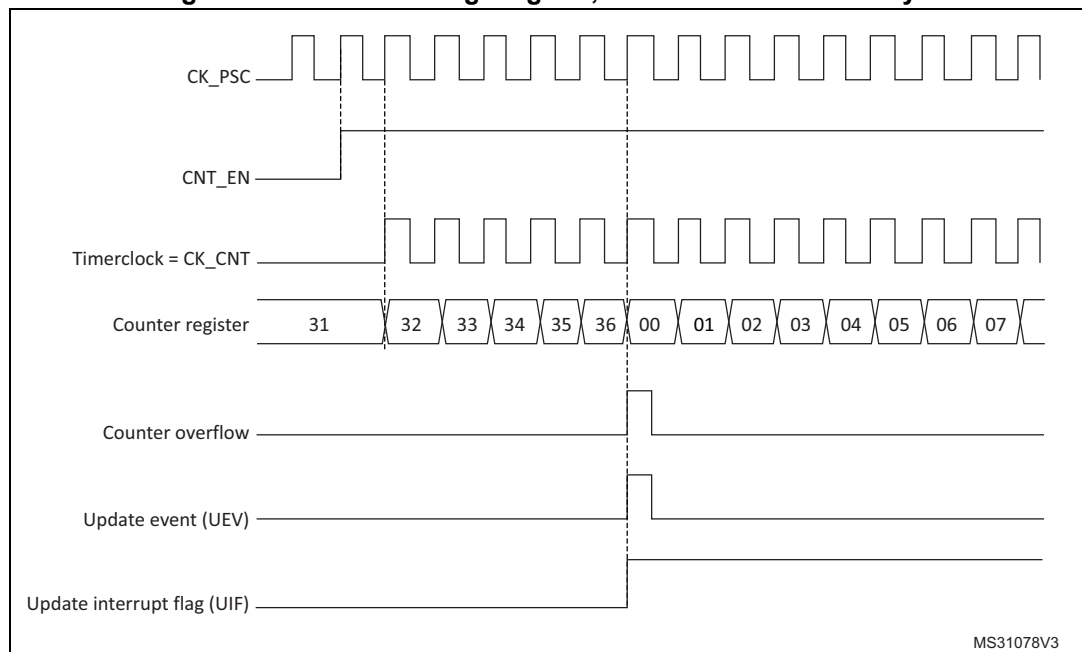


Figure 90. Counter timing diagram, internal clock divided by 2

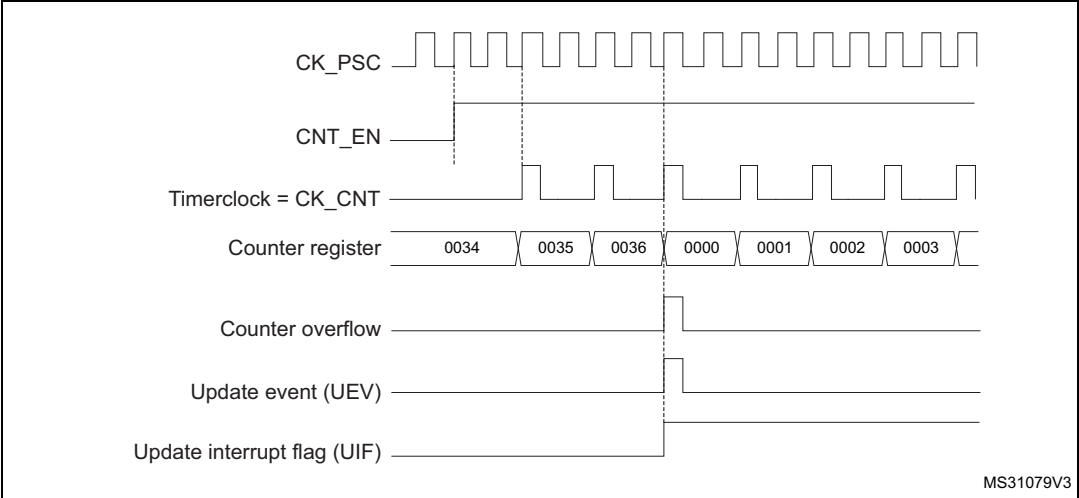


Figure 91. Counter timing diagram, internal clock divided by 4

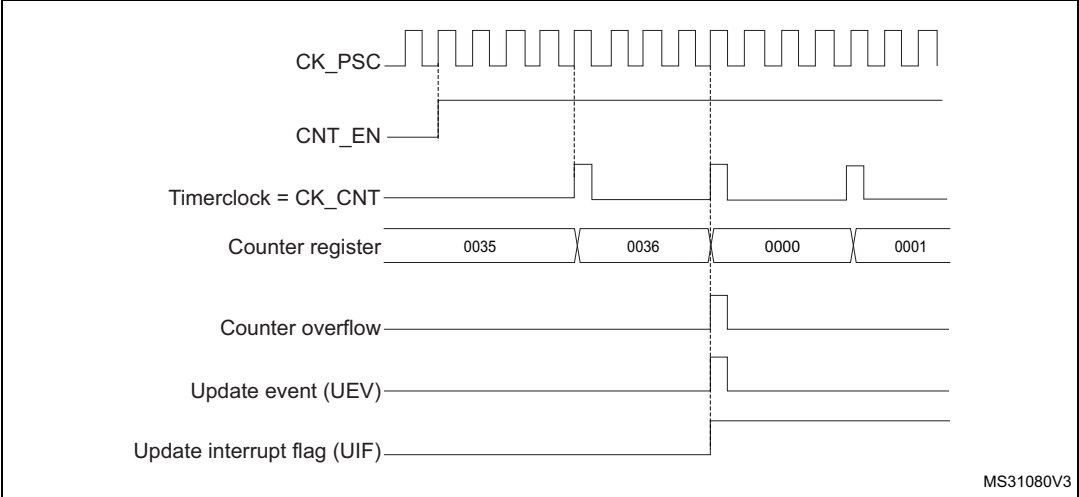


Figure 92. Counter timing diagram, internal clock divided by N

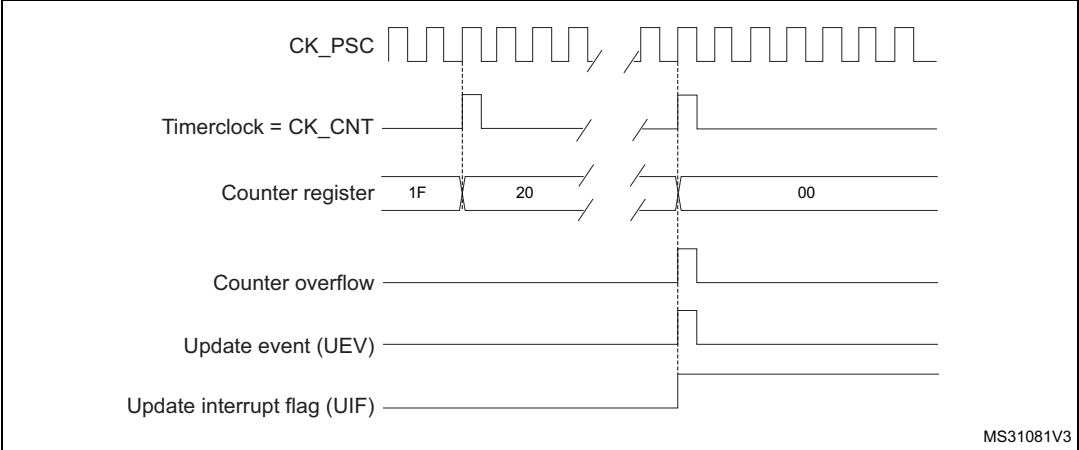


Figure 93. Counter timing diagram, update event when ARPE=0 (TIMx_ARR not preloaded)

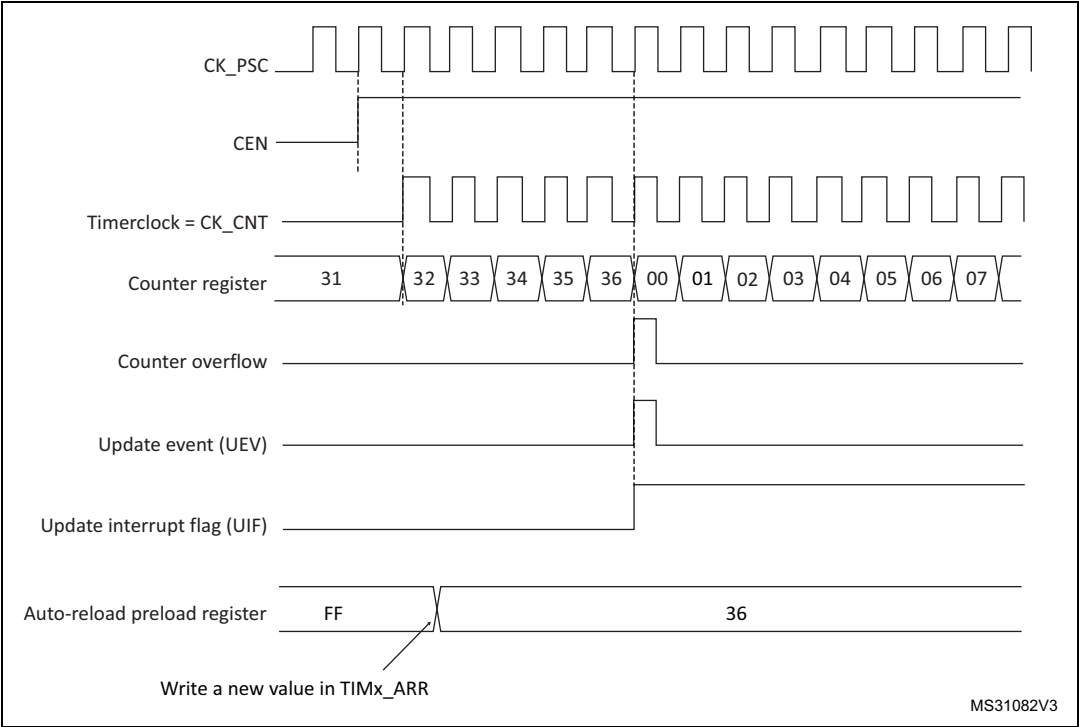
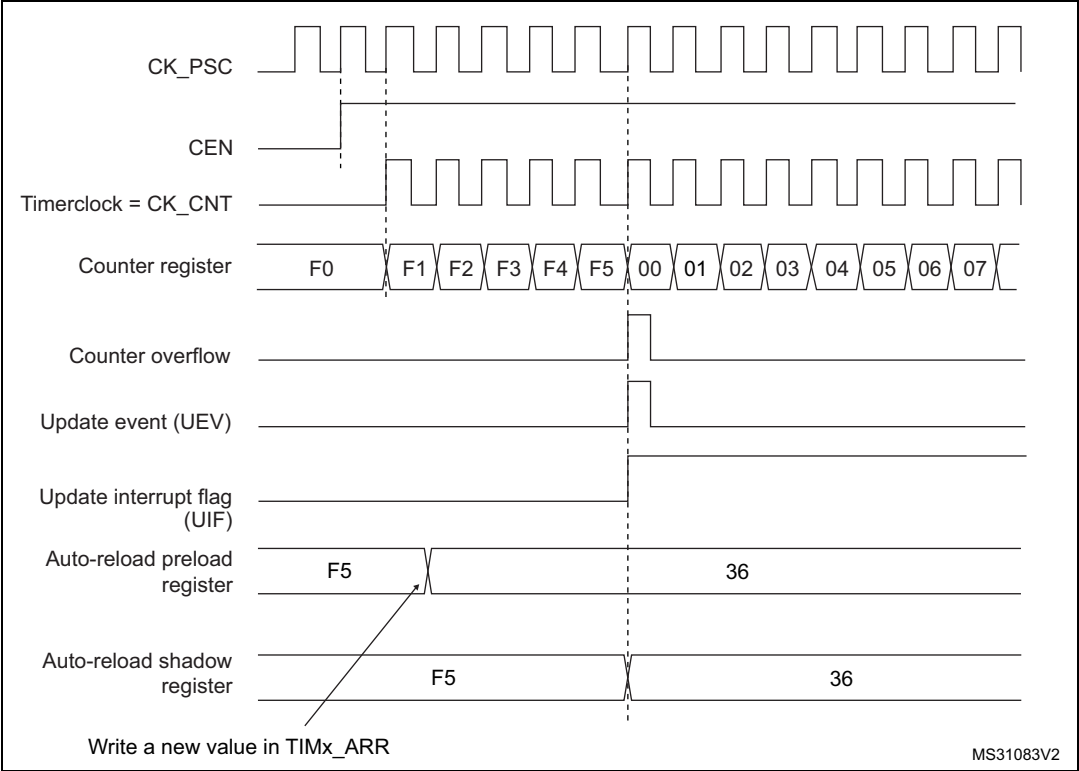


Figure 94. Counter timing diagram, update event when ARPE=1 (TIMx_ARR preloaded)



Downcounting mode

In downcounting mode, the counter counts from the auto-reload value (content of the TIMx_ARR register) down to 0, then restarts from the auto-reload value and generates a counter underflow event.

If the repetition counter is used, the update event (UEV) is generated after downcounting is repeated for the number of times programmed in the repetition counter register plus one (TIMx_RCR+1). Else the update event is generated at each counter underflow.

Setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller) also generates an update event.

The UEV update event can be disabled by software by setting the UDIS bit in TIMx_CR1 register. This is to avoid updating the shadow registers while writing new values in the preload registers. Then no update event occurs until UDIS bit has been written to 0. However, the counter restarts from the current auto-reload value, whereas the counter of the prescaler restarts from 0 (but the prescale rate doesn't change).

In addition, if the URS bit (update request selection) in TIMx_CR1 register is set, setting the UG bit generates an update event UEV but without setting the UIF flag (thus no interrupt or DMA request is sent). This is to avoid generating both update and capture interrupts when clearing the counter on the capture event.

When an update event occurs, all the registers are updated and the update flag (UIF bit in TIMx_SR register) is set (depending on the URS bit):

- The repetition counter is reloaded with the content of TIMx_RCR register
- The buffer of the prescaler is reloaded with the preload value (content of the TIMx_PSC register)
- The auto-reload active register is updated with the preload value (content of the TIMx_ARR register). Note that the auto-reload is updated before the counter is reloaded, so that the next period is the expected one

The following figures show some examples of the counter behavior for different clock frequencies when TIMx_ARR=0x36.

Figure 95. Counter timing diagram, internal clock divided by 1

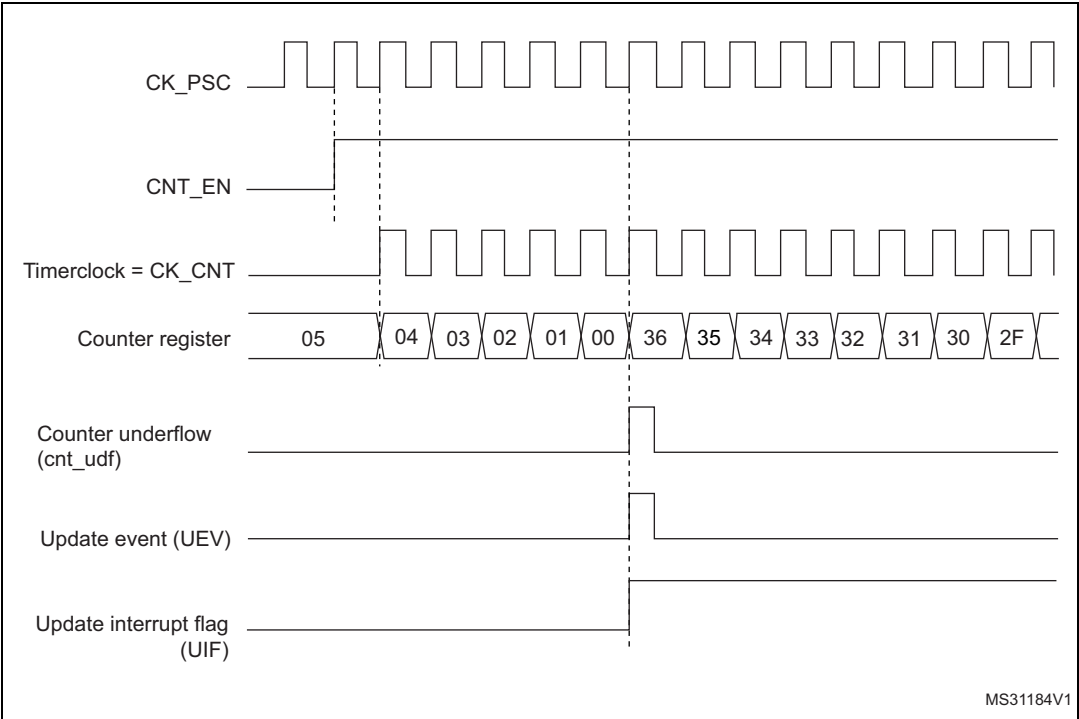


Figure 96. Counter timing diagram, internal clock divided by 2

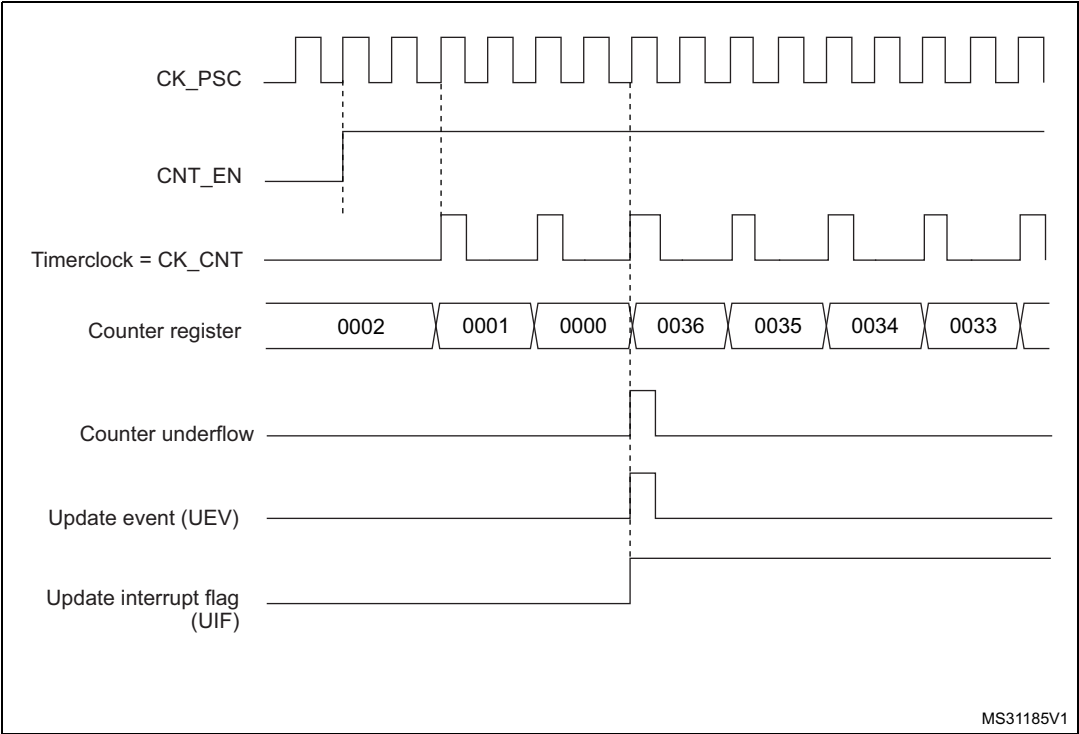


Figure 97. Counter timing diagram, internal clock divided by 4

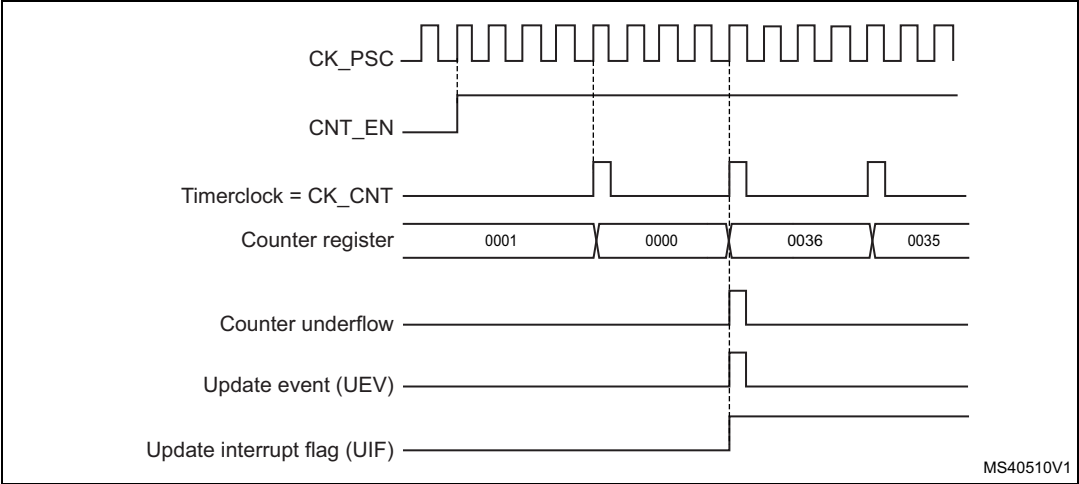


Figure 98. Counter timing diagram, internal clock divided by N

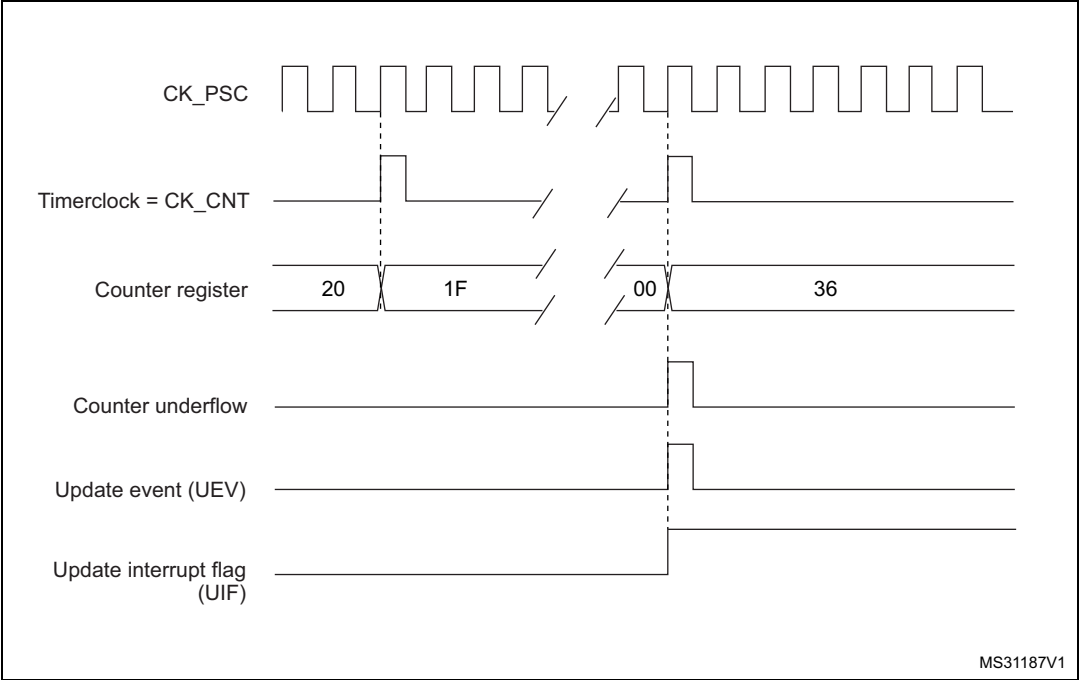
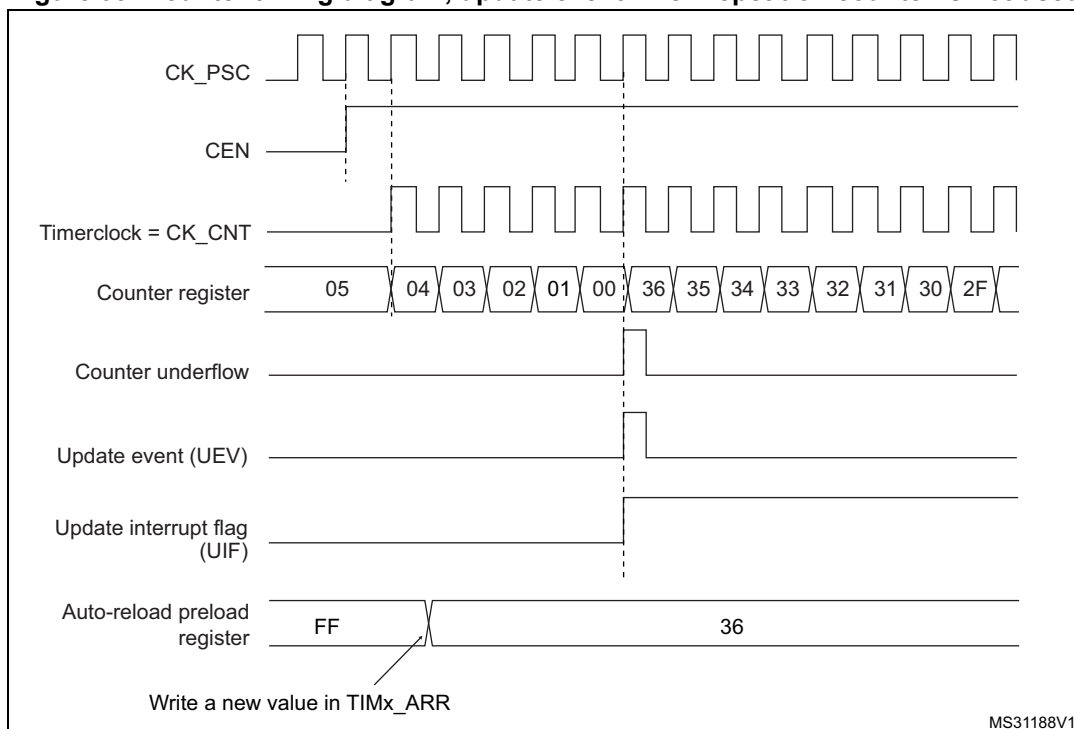


Figure 99. Counter timing diagram, update event when repetition counter is not used**Center-aligned mode (up/down counting)**

In center-aligned mode, the counter counts from 0 to the auto-reload value (content of the TIMx_ARR register) – 1, generates a counter overflow event, then counts from the auto-reload value down to 1 and generates a counter underflow event. Then it restarts counting from 0.

Center-aligned mode is active when the CMS bits in TIMx_CR1 register are not equal to '00'. The Output compare interrupt flag of channels configured in output is set when: the counter counts down (Center aligned mode 1, CMS = "01"), the counter counts up (Center aligned mode 2, CMS = "10") the counter counts up and down (Center aligned mode 3, CMS = "11").

In this mode, the DIR direction bit in the TIMx_CR1 register cannot be written. It is updated by hardware and gives the current direction of the counter.

The update event can be generated at each counter overflow and at each counter underflow or by setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller) also generates an update event. In this case, the counter restarts counting from 0, as well as the counter of the prescaler.

The UEV update event can be disabled by software by setting the UDIS bit in the TIMx_CR1 register. This is to avoid updating the shadow registers while writing new values in the preload registers. Then no update event occurs until UDIS bit has been written to 0. However, the counter continues counting up and down, based on the current auto-reload value.

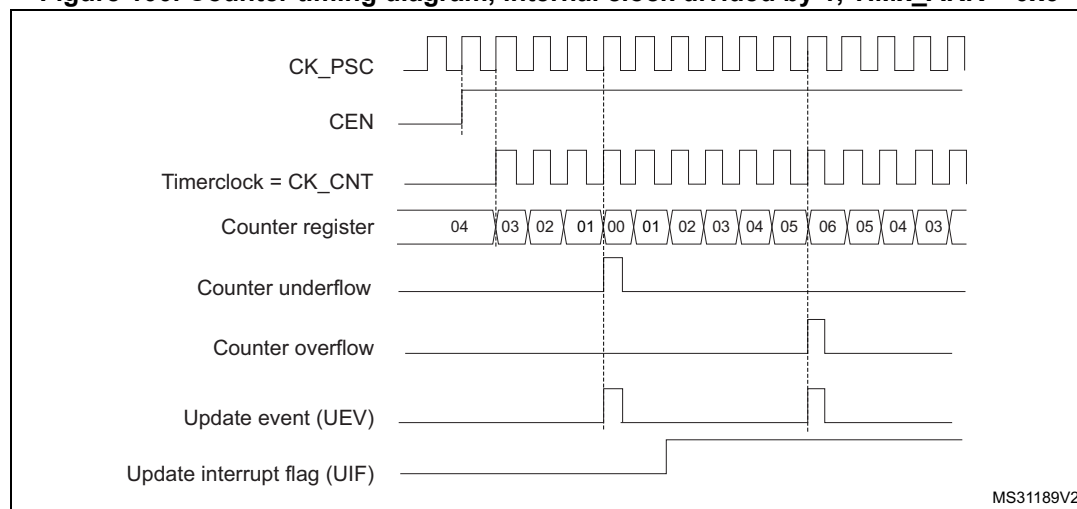
In addition, if the URS bit (update request selection) in TIMx_CR1 register is set, setting the UG bit generates an UEV update event but without setting the UIF flag (thus no interrupt or DMA request is sent). This is to avoid generating both update and capture interrupts when clearing the counter on the capture event.

When an update event occurs, all the registers are updated and the update flag (UIF bit in TIMx_SR register) is set (depending on the URS bit):

- The repetition counter is reloaded with the content of TIMx_RCR register
- The buffer of the prescaler is reloaded with the preload value (content of the TIMx_PSC register)
- The auto-reload active register is updated with the preload value (content of the TIMx_ARR register). Note that if the update source is a counter overflow, the auto-reload is updated before the counter is reloaded, so that the next period is the expected one (the counter is loaded with the new value).

The following figures show some examples of the counter behavior for different clock frequencies.

Figure 100. Counter timing diagram, internal clock divided by 1, TIMx_ARR = 0x6



1. Here, center-aligned mode 1 is used (for more details refer to [Section 17.4: TIM1 and TIM8 registers](#)).

Figure 101. Counter timing diagram, internal clock divided by 2

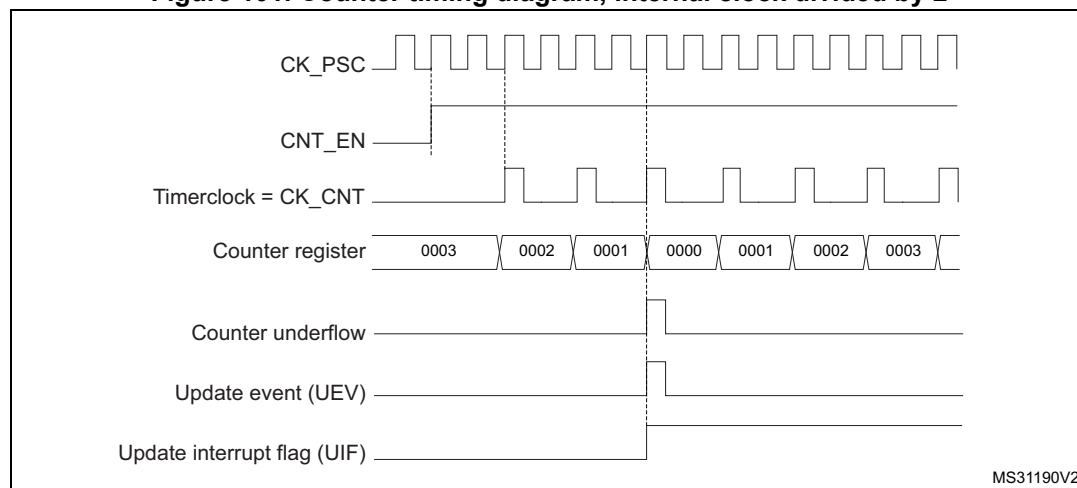
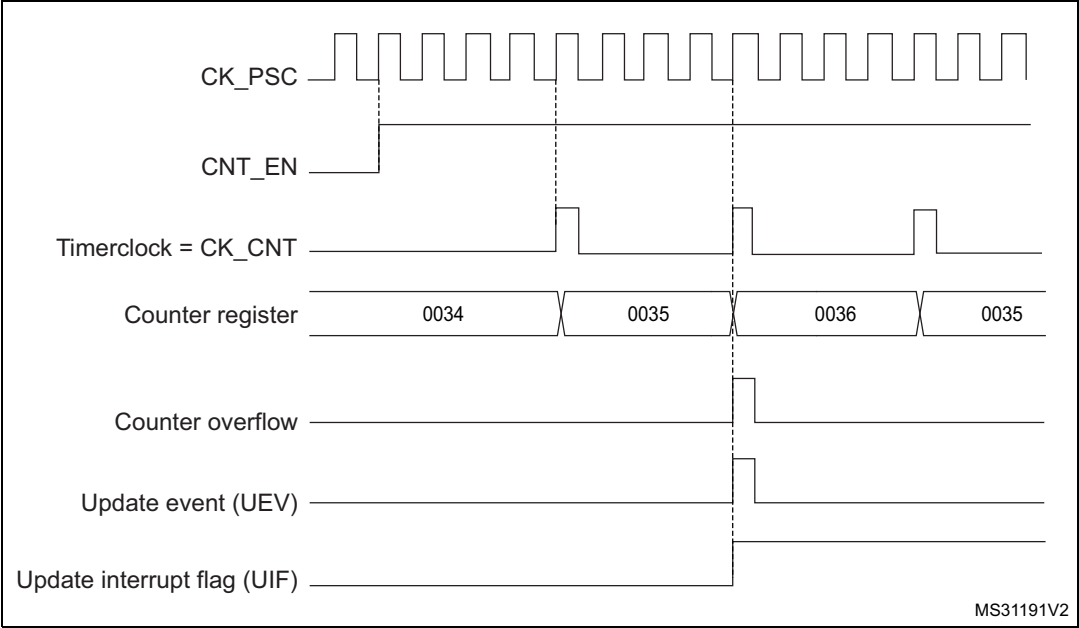


Figure 102. Counter timing diagram, internal clock divided by 4, TIMx_ARR=0x36



1. Center-aligned mode 2 or 3 is used with an UIF on overflow.

Figure 103. Counter timing diagram, internal clock divided by N

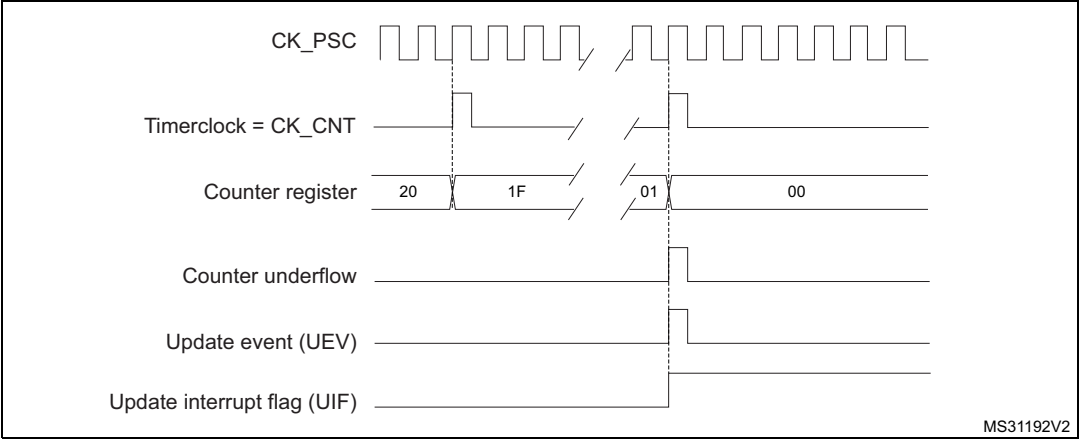
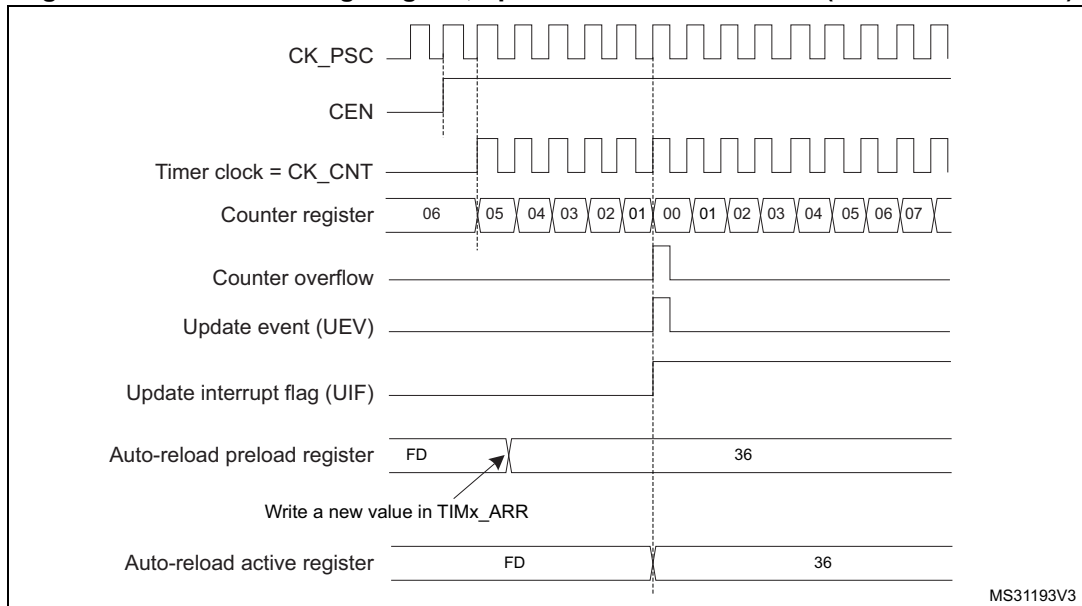
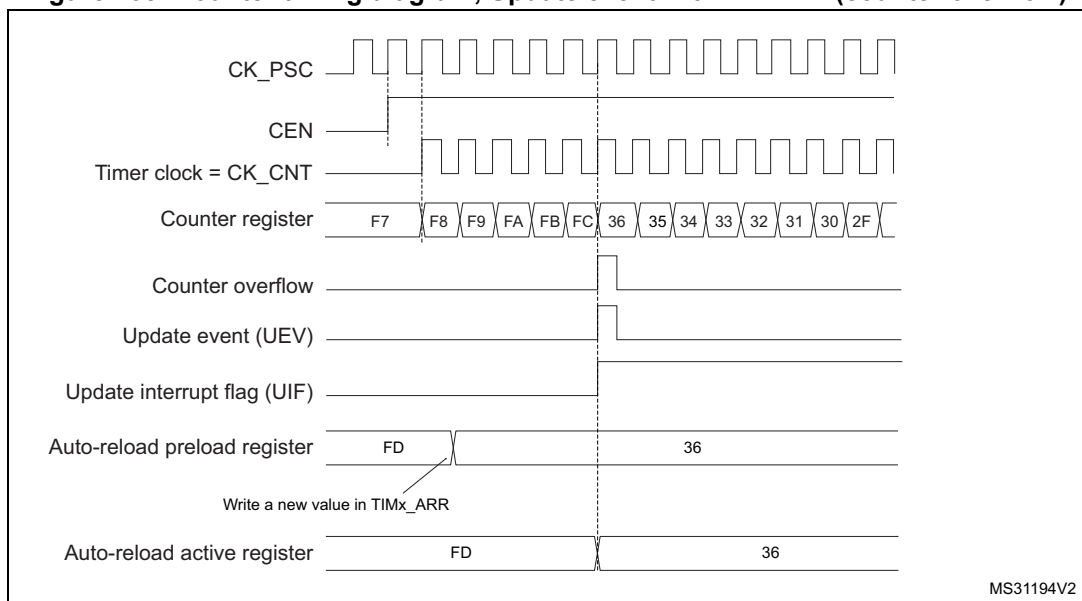


Figure 104. Counter timing diagram, update event with ARPE=1 (counter underflow)**Figure 105. Counter timing diagram, Update event with ARPE=1 (counter overflow)**

17.3.3 Repetition counter

[Section 17.3.1: Time-base unit](#) describes how the update event (UEV) is generated with respect to the counter overflows/underflows. It is actually generated only when the repetition counter has reached zero. This can be useful when generating PWM signals.

This means that data are transferred from the preload registers to the shadow registers (TIMx_ARR auto-reload register, TIMx_PSC prescaler register, but also TIMx_CCRx capture/compare registers in compare mode) every N+1 counter overflows or underflows, where N is the value in the TIMx_RCR repetition counter register.

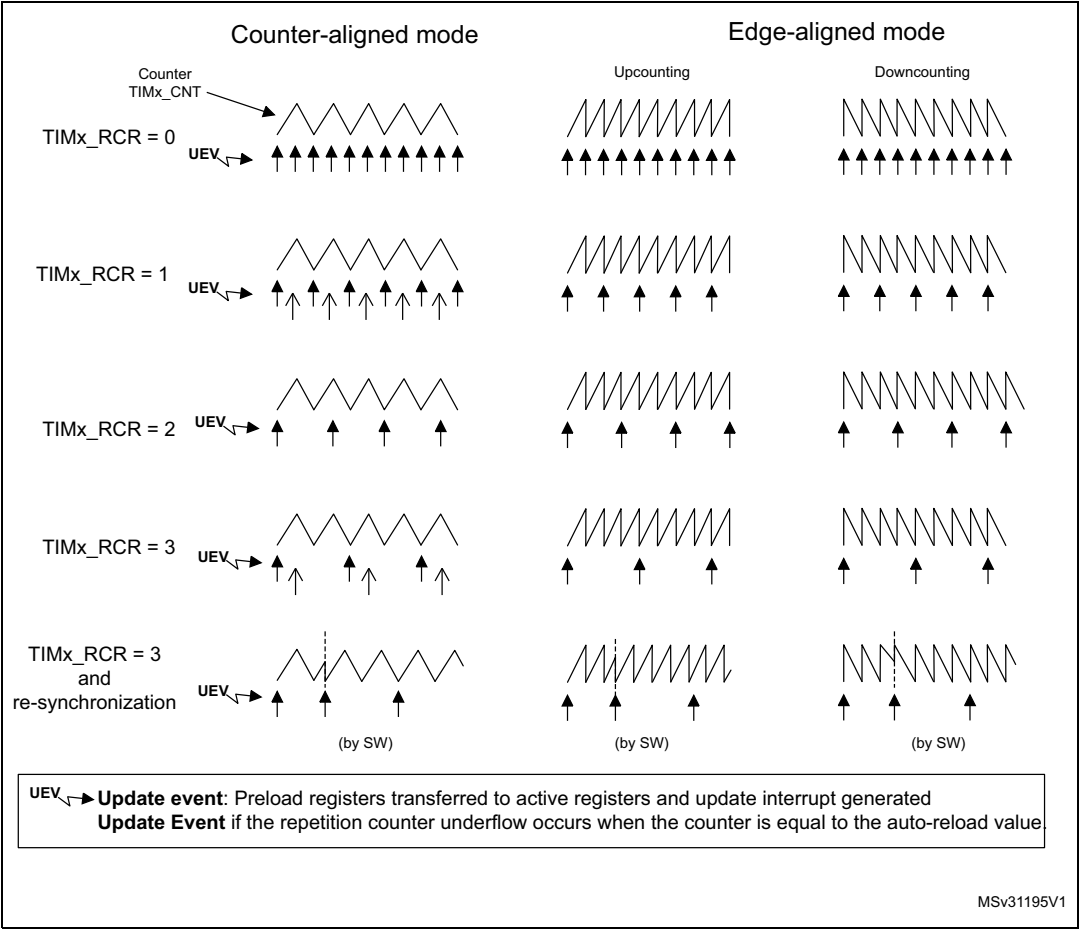
The repetition counter is decremented:

- At each counter overflow in upcounting mode,
- At each counter underflow in downcounting mode,
- At each counter overflow and at each counter underflow in center-aligned mode.
Although this limits the maximum number of repetition to 128 PWM cycles, it makes it possible to update the duty cycle twice per PWM period. When refreshing compare registers only once per PWM period in center-aligned mode, maximum resolution is $2 \times T_{ck}$, due to the symmetry of the pattern.

The repetition counter is an auto-reload type; the repetition rate is maintained as defined by the TIMx_RCR register value (refer to [Figure 106](#)). When the update event is generated by software (by setting the UG bit in TIMx_EGR register) or by hardware through the slave mode controller, it occurs immediately whatever the value of the repetition counter is and the repetition counter is reloaded with the content of the TIMx_RCR register.

In center-aligned mode, for odd values of RCR, the update event occurs either on the overflow or on the underflow depending on when the RCR register was written and when the counter was started. If the RCR was written before starting the counter, the UEV occurs on the overflow. If the RCR was written after starting the counter, the UEV occurs on the underflow. For example for RCR = 3, the UEV is generated on each 4th overflow or underflow event depending on when RCR was written.

Figure 106. Update rate examples depending on mode and TIMx_RCR register settings



17.3.4 Clock selection

The counter clock can be provided by the following clock sources:

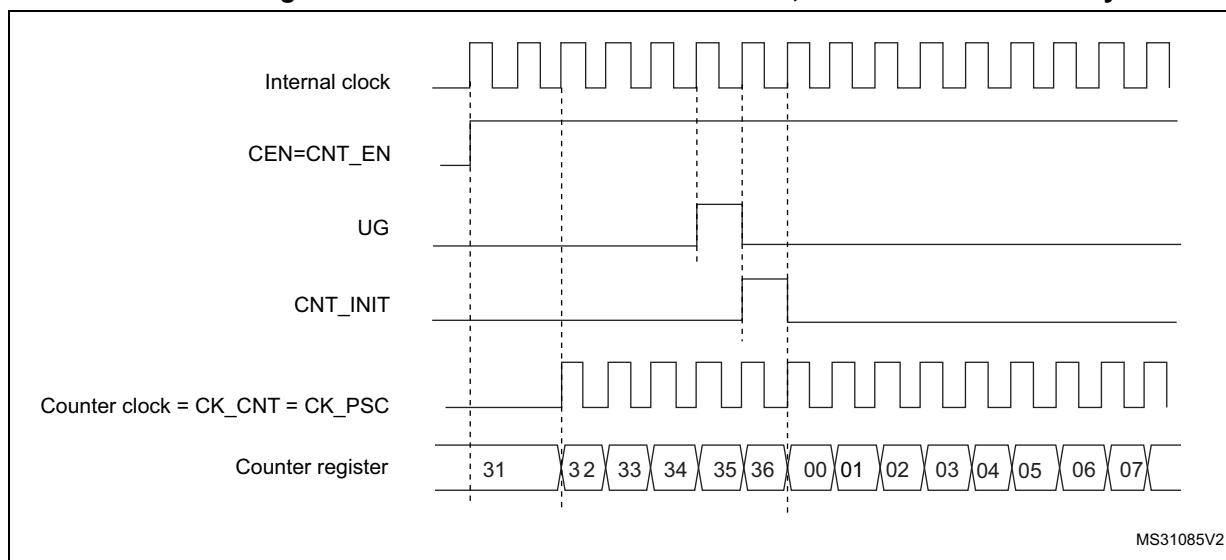
- Internal clock (CK_INT)
- External clock mode1: external input pin
- External clock mode2: external trigger input ETR
- Internal trigger inputs (ITRx): using one timer as prescaler for another timer, for example, the user can configure Timer 1 to act as a prescaler for Timer 2. Refer to [Using one timer as prescaler for another timer](#) for more details.

Internal clock source (CK_INT)

If the slave mode controller is disabled (SMS=000), then the CEN, DIR (in the TIMx_CR1 register) and UG bits (in the TIMx_EGR register) are actual control bits and can be changed only by software (except UG which remains cleared automatically). As soon as the CEN bit is written to 1, the prescaler is clocked by the internal clock CK_INT.

[Figure 107](#) shows the behavior of the control circuit and the upcounter in normal mode, without prescaler.

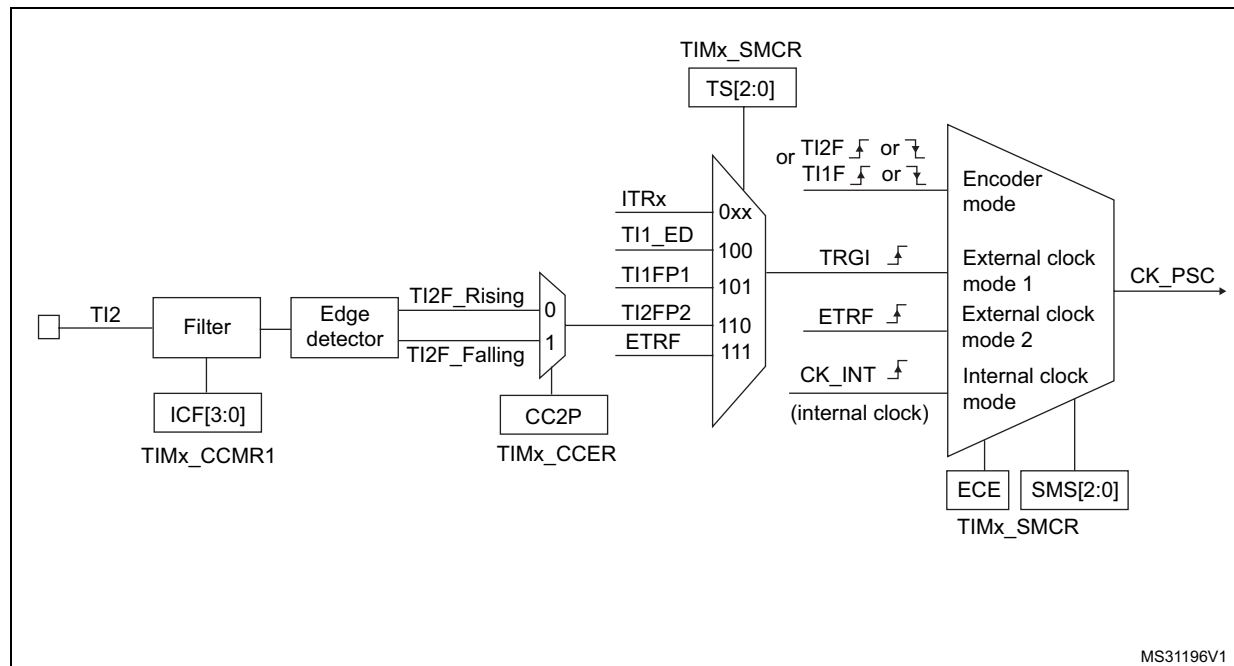
Figure 107. Control circuit in normal mode, internal clock divided by 1



External clock source mode 1

This mode is selected when SMS=111 in the TIMx_SMCR register. The counter can count at each rising or falling edge on a selected input.

Figure 108. TI2 external clock connection example



For example, to configure the upcounter to count in response to a rising edge on the TI2 input, use the following procedure:

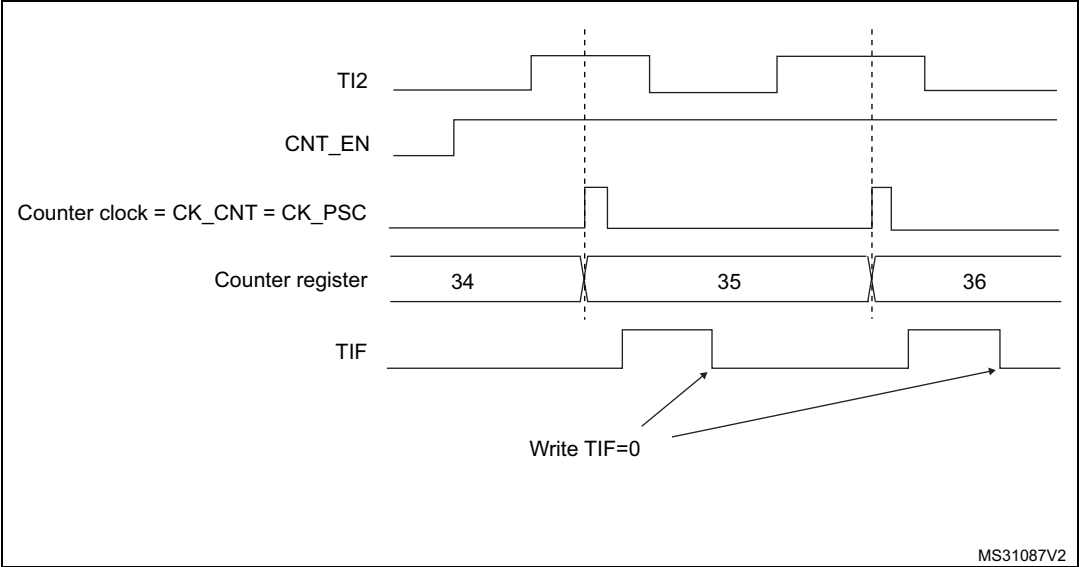
1. Configure channel 2 to detect rising edges on the TI2 input by writing CC2S = '01' in the TIMx_CCMR1 register.
2. Configure the input filter duration by writing the IC2F[3:0] bits in the TIMx_CCMR1 register (if no filter is needed, keep IC2F=0000).
3. Select rising edge polarity by writing CC2P=0 and CC2NP=0 in the TIMx_CCER register.
4. Configure the timer in external clock mode 1 by writing SMS=111 in the TIMx_SMCR register.
5. Select TI2 as the trigger input source by writing TS=110 in the TIMx_SMCR register.
6. Enable the counter by writing CEN=1 in the TIMx_CR1 register.

Note: The capture prescaler is not used for triggering, so the user does not need to configure it.

When a rising edge occurs on TI2, the counter counts once and the TIF flag is set.

The delay between the rising edge on TI2 and the actual clock of the counter is due to the resynchronization circuit on TI2 input.

Figure 109. Control circuit in external clock mode 1



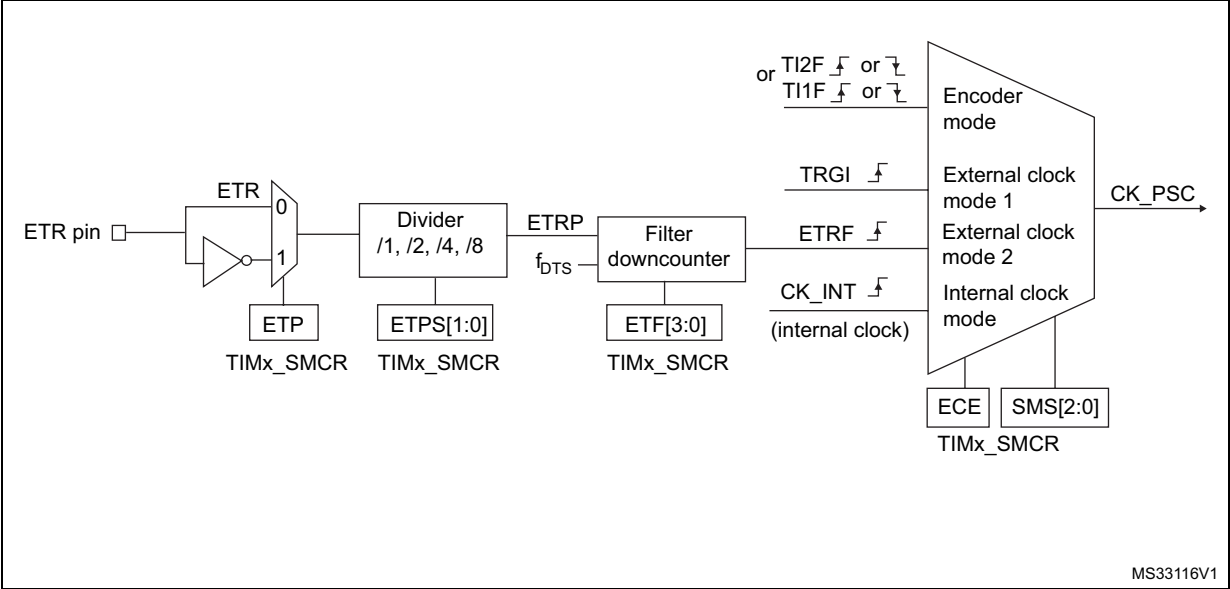
External clock source mode 2

This mode is selected by writing ECE=1 in the TIMx_SMCR register.

The counter can count at each rising or falling edge on the external trigger input ETR.

[Figure 110](#) gives an overview of the external trigger input block.

Figure 110. External trigger input block



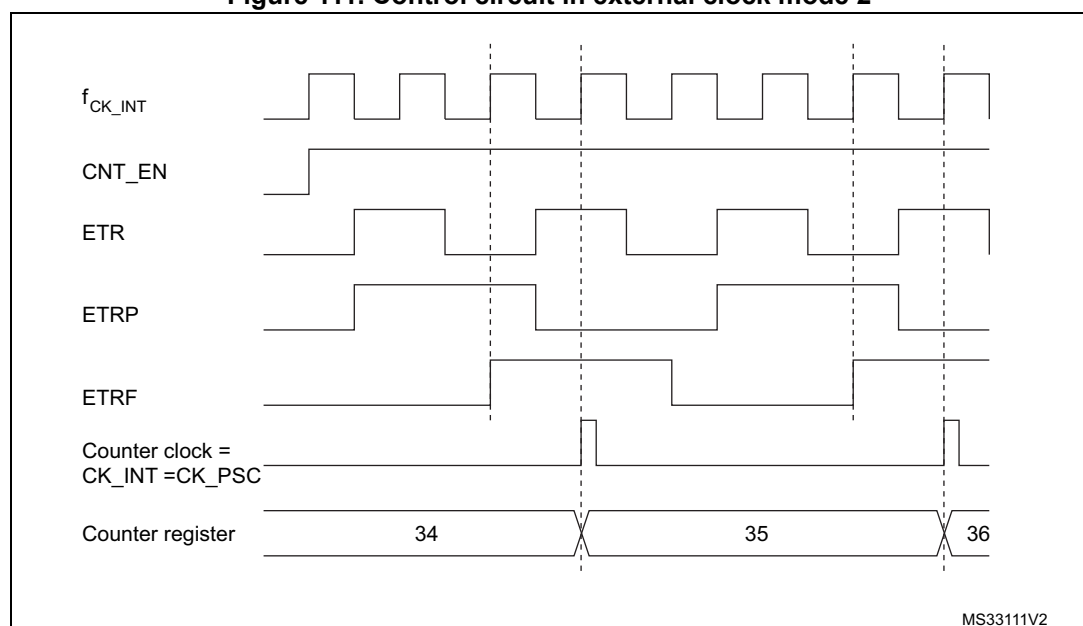
For example, to configure the upcounter to count each 2 rising edges on ETR, use the following procedure:

1. As no filter is needed in this example, write $ETF[3:0]=0000$ in the TIMx_SMCR register.
2. Set the prescaler by writing $ETPS[1:0]=01$ in the TIMx_SMCR register
3. Select rising edge detection on the ETR pin by writing $ETP=0$ in the TIMx_SMCR register
4. Enable external clock mode 2 by writing $ECE=1$ in the TIMx_SMCR register.
5. Enable the counter by writing $CEN=1$ in the TIMx_CR1 register.

The counter counts once each 2 ETR rising edges.

The delay between the rising edge on ETR and the actual clock of the counter is due to the resynchronization circuit on the ETRP signal.

Figure 111. Control circuit in external clock mode 2



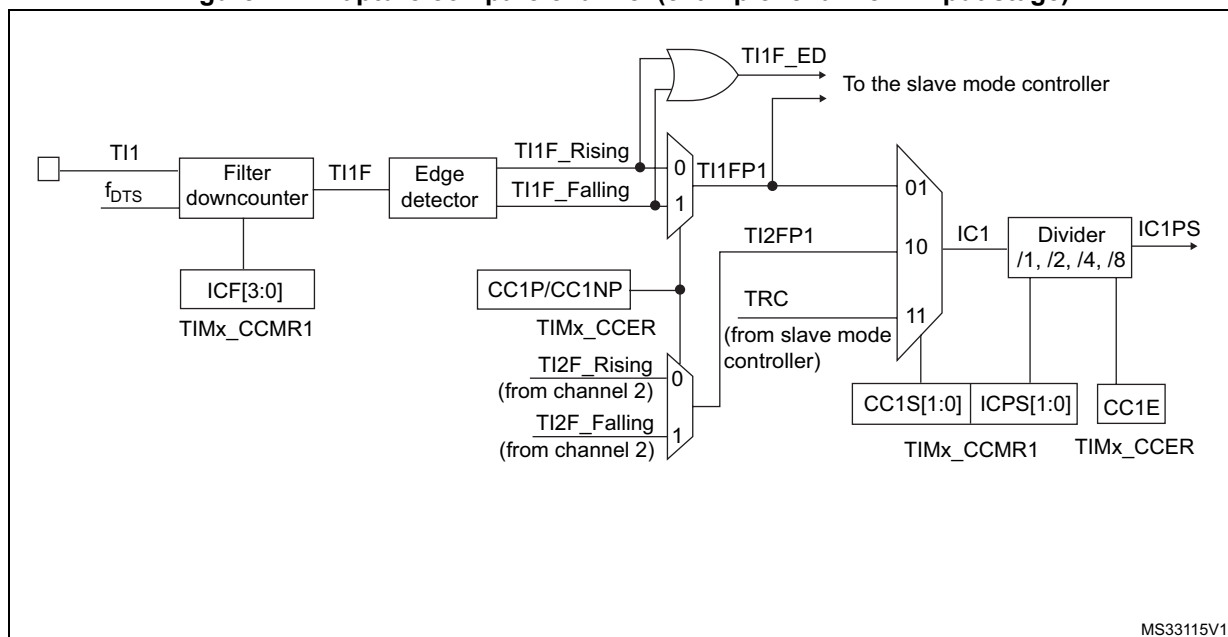
17.3.5 Capture/compare channels

Each Capture/Compare channel is built around a capture/compare register (including a shadow register), a input stage for capture (with digital filter, multiplexing and prescaler) and an output stage (with comparator and output control).

[Figure 112](#) to [Figure 115](#) give an overview of one Capture/Compare channel.

The input stage samples the corresponding TIx input to generate a filtered signal TIxF. Then, an edge detector with polarity selection generates a signal (TIxFPx) which can be used as trigger input by the slave mode controller or as the capture command. It is prescaled before the capture register (ICxPS).

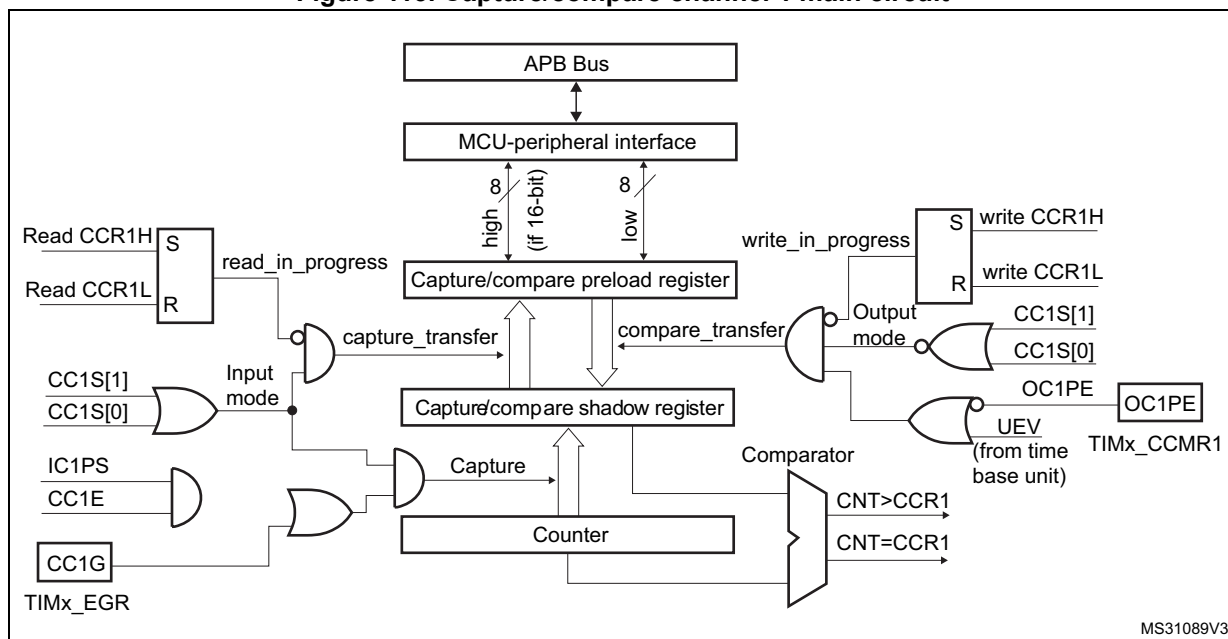
Figure 112. Capture/compare channel (example: channel 1 input stage)



MS33115V1

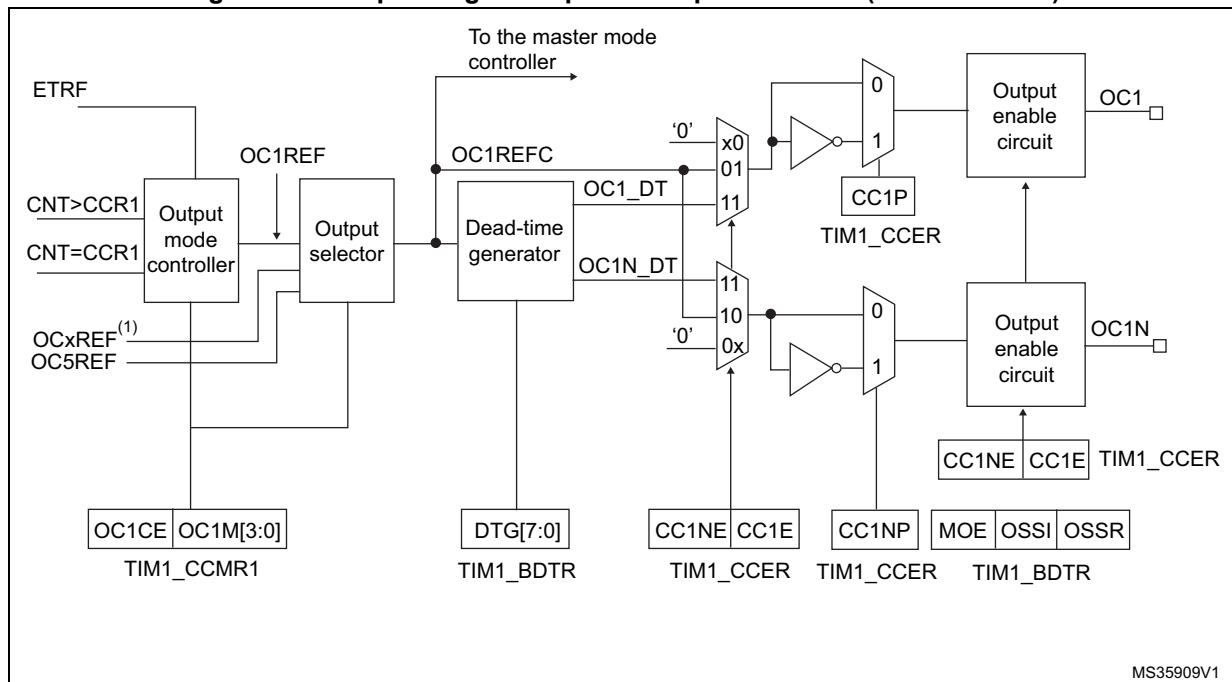
The output stage generates an intermediate waveform that is then used for reference: OCxRef (active high). The polarity acts at the end of the chain.

Figure 113. Capture/compare channel 1 main circuit



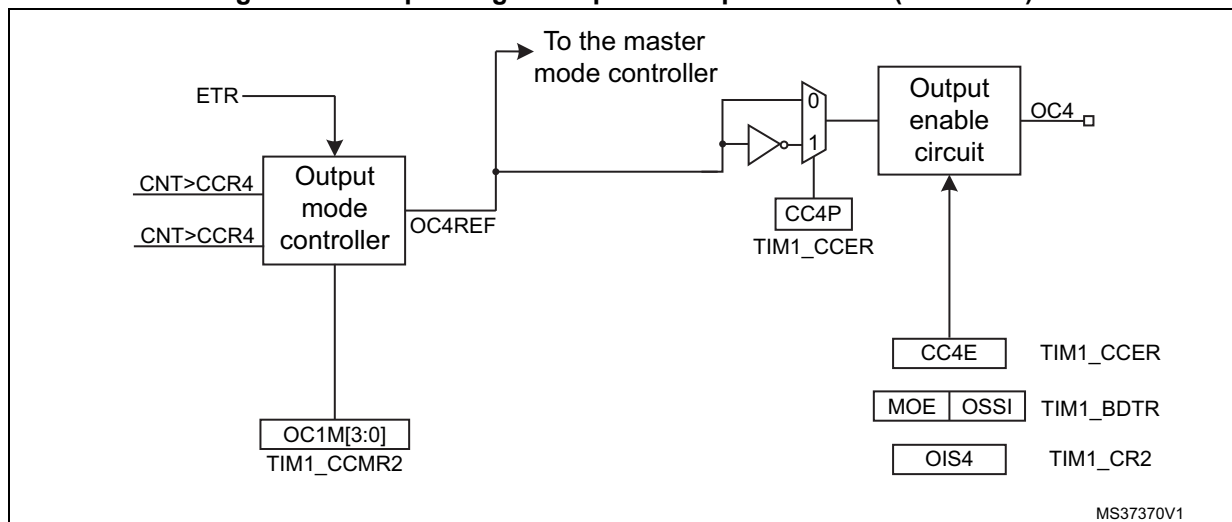
MS31089V3

Figure 114. Output stage of capture/compare channel (channel 1 to 3)



MS35909V1

Figure 115. Output stage of capture/compare channel (channel 4)



MS37370V1

The capture/compare block is made of one preload register and one shadow register. Write and read always access the preload register.

In capture mode, captures are actually done in the shadow register, which is copied into the preload register.

In compare mode, the content of the preload register is copied into the shadow register which is compared to the counter.

17.3.6 Input capture mode

In Input capture mode, the Capture/Compare registers (TIMx_CCRx) are used to latch the value of the counter after a transition detected by the corresponding ICx signal. When a capture occurs, the corresponding CCxIF flag (TIMx_SR register) is set and an interrupt or a DMA request can be sent if they are enabled. If a capture occurs while the CCxIF flag was already high, then the over-capture flag CCxOF (TIMx_SR register) is set. CCxIF can be cleared by software by writing it to '0' or by reading the captured data stored in the TIMx_CCRx register. CCxOF is cleared when written to '0'.

The following example shows how to capture the counter value in TIMx_CCR1 when TI1 input rises. To do this, use the following procedure:

- Select the active input: TIMx_CCR1 must be linked to the TI1 input, so write the CC1S bits to 01 in the TIMx_CCMR1 register. As soon as CC1S becomes different from 00, the channel is configured in input and the TIMx_CCR1 register becomes read-only.
- Program the needed input filter duration with respect to the signal connected to the timer (by programming ICxF bits in the TIMx_CCMRx register if the input is a TIx input). Let's imagine that, when toggling, the input signal is not stable during at most five internal clock cycles. We must program a filter duration longer than these five clock cycles. We can validate a transition on TI1 when 8 consecutive samples with the new level have been detected (sampled at f_{DTS} frequency). Then write IC1F bits to 0011 in the TIMx_CCMR1 register.
- Select the edge of the active transition on the TI1 channel by writing CC1P and CC1NP bits to 0 in the TIMx_CCER register (rising edge in this case).
- Program the input prescaler. In our example, we wish the capture to be performed at each valid transition, so the prescaler is disabled (write IC1PS bits to '00' in the TIMx_CCMR1 register).
- Enable capture from the counter into the capture register by setting the CC1E bit in the TIMx_CCER register.
- If needed, enable the related interrupt request by setting the CC1IE bit in the TIMx_DIER register, and/or the DMA request by setting the CC1DE bit in the TIMx_DIER register.

When an input capture occurs:

- The TIMx_CCR1 register gets the value of the counter on the active transition.
- CC1IF flag is set (interrupt flag). CC1OF is also set if at least two consecutive captures occurred whereas the flag was not cleared.
- An interrupt is generated depending on the CC1IE bit.
- A DMA request is generated depending on the CC1DE bit.

In order to handle the overcapture, it is recommended to read the data before the overcapture flag. This is to avoid missing an overcapture which could happen after reading the flag and before reading the data.

Note: IC interrupt and/or DMA requests can be generated by software by setting the corresponding CCxG bit in the TIMx_EGR register.

forced high (OCxREF is always active high) and OCx get opposite value to CCxP polarity bit.

For example: CCxP=0 (OCx active high) => OCx is forced to high level.

The OCxREF signal can be forced low by writing the OCxM bits to 100 in the TIMx_CCMRx register.

Anyway, the comparison between the TIMx_CCRx shadow register and the counter is still performed and allows the flag to be set. Interrupt and DMA requests can be sent accordingly. This is described in the output compare mode section below.

17.3.9 Output compare mode

This function is used to control an output waveform or indicating when a period of time has elapsed.

When a match is found between the capture/compare register and the counter, the output compare function:

- Assigns the corresponding output pin to a programmable value defined by the output compare mode (OCxM bits in the TIMx_CCMRx register) and the output polarity (CCxP bit in the TIMx_CCER register). The output pin can keep its level (OCxM=000), be set active (OCxM=001), be set inactive (OCxM=010) or can toggle (OCxM=011) on match.
- Sets a flag in the interrupt status register (CCxIF bit in the TIMx_SR register).
- Generates an interrupt if the corresponding interrupt mask is set (CCxIE bit in the TIMx_DIER register).
- Sends a DMA request if the corresponding enable bit is set (CCxDE bit in the TIMx_DIER register, CCDS bit in the TIMx_CR2 register for the DMA request selection).

The TIMx_CCRx registers can be programmed with or without preload registers using the OCxPE bit in the TIMx_CCMRx register.

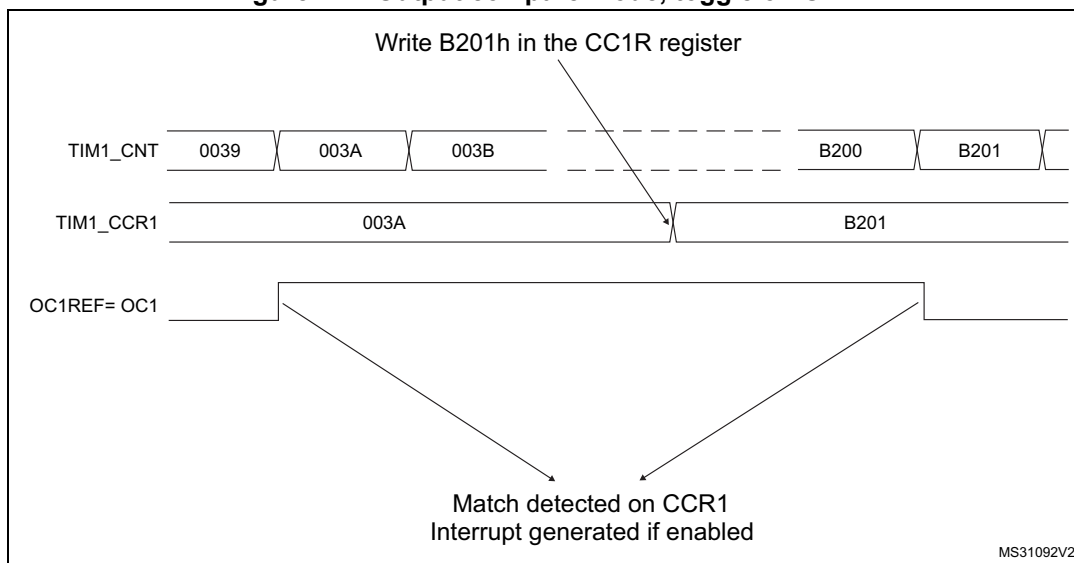
In output compare mode, the update event UEV has no effect on OCxREF and OCx output. The timing resolution is one count of the counter. Output compare mode can also be used to output a single pulse (in One Pulse mode).

Procedure:

1. Select the counter clock (internal, external, prescaler).
2. Write the desired data in the TIMx_ARR and TIMx_CCRx registers.
3. Set the CCxIE bit if an interrupt request is to be generated.
4. Select the output mode. For example:
 - Write OCxM = 011 to toggle OCx output pin when CNT matches CCRx
 - Write OCxPE = 0 to disable preload register
 - Write CCxP = 0 to select active high polarity
 - Write CCxE = 1 to enable the output
5. Enable the counter by setting the CEN bit in the TIMx_CR1 register.

The TIMx_CCRx register can be updated at any time by software to control the output waveform, provided that the preload register is not enabled (OCxPE='0', else TIMx_CCRx shadow register is updated only at the next update event UEV). An example is given in [Figure 117](#).

Figure 117. Output compare mode, toggle on OC1.



17.3.10 PWM mode

Pulse Width Modulation mode allows generating a signal with a frequency determined by the value of the TIMx_ARR register and a duty cycle determined by the value of the TIMx_CCRx register.

The PWM mode can be selected independently on each channel (one PWM per OCx output) by writing '110' (PWM mode 1) or '111' (PWM mode 2) in the OCxM bits in the TIMx_CCMRx register. The corresponding preload register must be enabled by setting the OCxPE bit in the TIMx_CCMRx register, and eventually the auto-reload preload register (in upcounting or center-aligned modes) by setting the ARPE bit in the TIMx_CR1 register.

As the preload registers are transferred to the shadow registers only when an update event occurs, before starting the counter, the user must initialize all the registers by setting the UG bit in the TIMx_EGR register.

OCx polarity is software programmable using the CCxP bit in the TIMx_CCER register. It can be programmed as active high or active low. OCx output is enabled by a combination of the CCxE, CCxNE, MOE, OSSI and OSSR bits (TIMx_CCER and TIMx_BDTR registers). Refer to the TIMx_CCER register description for more details.

In PWM mode (1 or 2), TIMx_CNT and TIMx_CCRx are always compared to determine whether $TIMx_CCRx \leq TIMx_CNT$ or $TIMx_CNT \leq TIMx_CCRx$ (depending on the direction of the counter).

The timer is able to generate PWM in edge-aligned mode or center-aligned mode depending on the CMS bits in the TIMx_CR1 register.

PWM edge-aligned mode

- Upcounting configuration

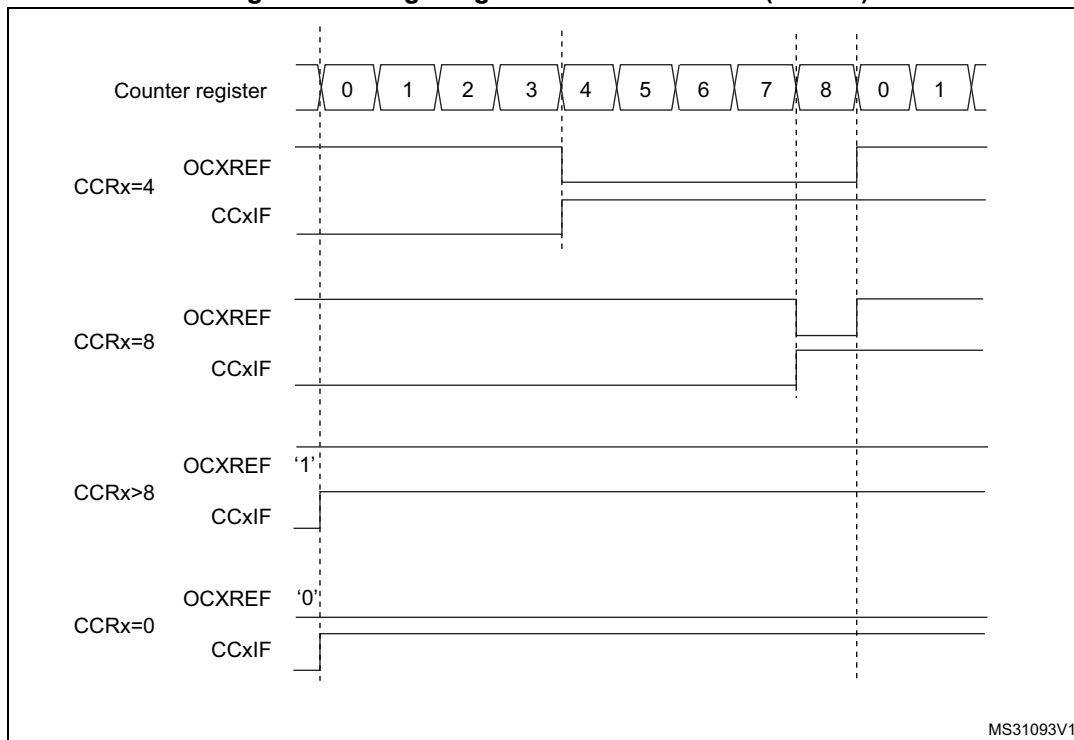
Upcounting is active when the DIR bit in the TIMx_CR1 register is low. Refer to [Upcounting mode](#).

In the following example, we consider PWM mode 1. The reference PWM signal OCxREF is high as long as $TIMx_CNT < TIMx_CCRx$ else it becomes low. If the

compare value in TIMx_CCRx is greater than the auto-reload value (in TIMx_ARR) then OCxREF is held at '1'. If the compare value is 0 then OCxRef is held at '0'.

Figure 118 shows some edge-aligned PWM waveforms in an example where TIMx_ARR=8.

Figure 118. Edge-aligned PWM waveforms (ARR=8)



- **Downcounting configuration**

Downcounting is active when DIR bit in TIMx_CR1 register is high. Refer to [Downcounting mode](#)

In PWM mode 1, the reference signal OCxRef is low as long as $TIMx_CNT > TIMx_CCRx$ else it becomes high. If the compare value in TIMx_CCRx is greater than the auto-reload value in TIMx_ARR, then OCxREF is held at '1'. 0% PWM is not possible in this mode.

PWM center-aligned mode

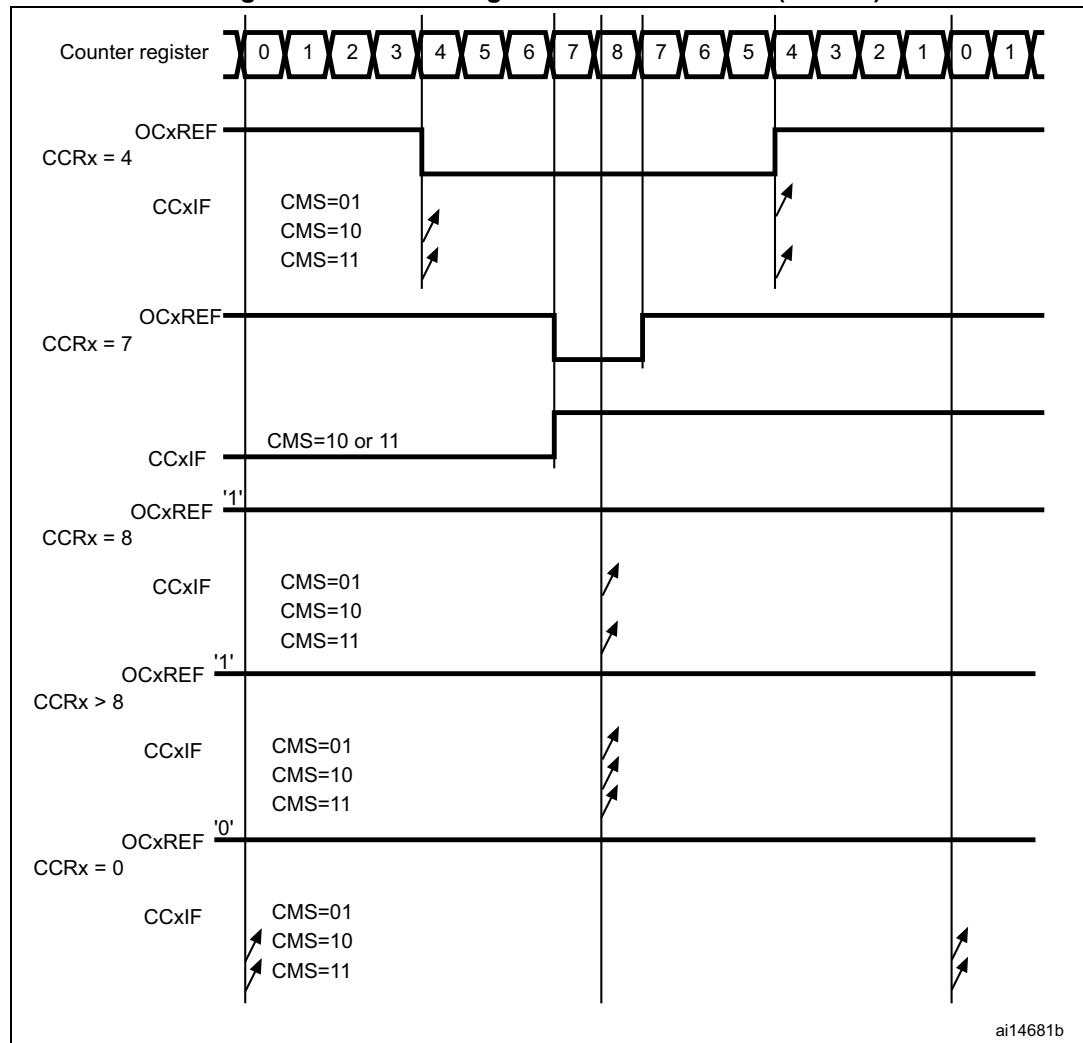
Center-aligned mode is active when the CMS bits in TIMx_CR1 register are different from '00' (all the remaining configurations having the same effect on the OCxRef/OCx signals). The compare flag is set when the counter counts up, when it counts down or both when it counts up and down depending on the CMS bits configuration. The direction bit (DIR) in the

TIMx_CR1 register is updated by hardware and must not be changed by software. Refer to [Center-aligned mode \(up/down counting\)](#).

Figure 119 shows some center-aligned PWM waveforms in an example where:

- TIMx_ARR=8,
- PWM mode is the PWM mode 1,
- The flag is set when the counter counts down corresponding to the center-aligned mode 1 selected for CMS=01 in TIMx_CR1 register.

Figure 119. Center-aligned PWM waveforms (ARR=8)



Hints on using center-aligned mode:

- When starting in center-aligned mode, the current up-down configuration is used. It means that the counter counts up or down depending on the value written in the DIR bit

in the TIMx_CR1 register. Moreover, the DIR and CMS bits must not be changed at the same time by the software.

- Writing to the counter while running in center-aligned mode is not recommended as it can lead to unexpected results. In particular:
 - The direction is not updated if the user writes a value in the counter greater than the auto-reload value (TIMx_CNT > TIMx_ARR). For example, if the counter was counting up, it continues to count up.
 - The direction is updated if the user writes 0 or write the TIMx_ARR value in the counter but no Update Event UEV is generated.
- The safest way to use center-aligned mode is to generate an update by software (setting the UG bit in the TIMx_EGR register) just before starting the counter and not to write the counter while it is running.

17.3.11 Complementary outputs and dead-time insertion

The advanced-control timers (TIM1 and TIM8) can output two complementary signals and manage the switching-off and the switching-on instants of the outputs.

This time is generally known as dead-time and it has to be adjusted depending on the devices connected to the outputs and their characteristics (intrinsic delays of level-shifters, delays due to power switches...)

User can select the polarity of the outputs (main output OCx or complementary OCxN) independently for each output. This is done by writing to the CCxP and CCxNP bits in the TIMx_CCER register.

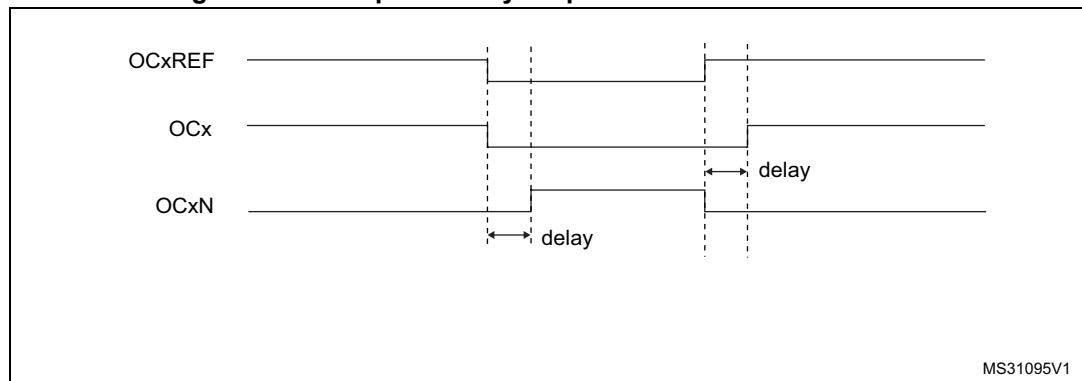
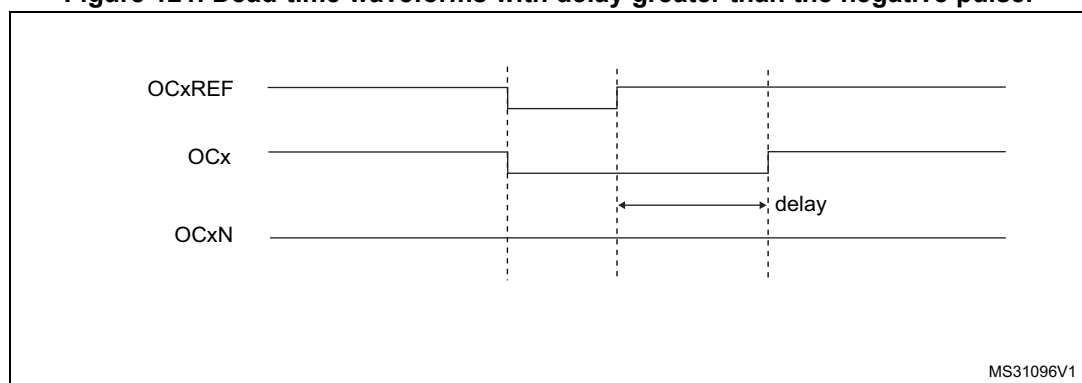
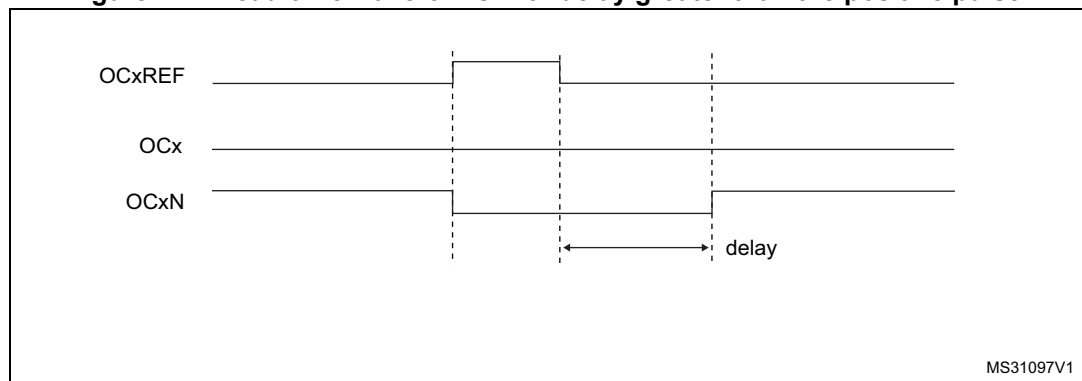
The complementary signals OCx and OCxN are activated by a combination of several control bits: the CCxE and CCxNE bits in the TIMx_CCER register and the MOE, OISx, OISxN, OSSI and OSSR bits in the TIMx_BDTR and TIMx_CR2 registers. Refer to [Table 96](#) for more details. In particular, the dead-time is activated when switching to the IDLE state (MOE falling down to 0).

Dead-time insertion is enabled by setting both CCxE and CCxNE bits, and the MOE bit if the break circuit is present. DTG[7:0] bits of the TIMx_BDTR register are used to control the dead-time generation for all channels. From a reference waveform OCxREF, it generates 2 outputs OCx and OCxN. If OCx and OCxN are active high:

- The OCx output signal is the same as the reference signal except for the rising edge, which is delayed relative to the reference rising edge.
- The OCxN output signal is the opposite of the reference signal except for the rising edge, which is delayed relative to the reference falling edge.

If the delay is greater than the width of the active output (OCx or OCxN) then the corresponding pulse is not generated.

The following figures show the relationships between the output signals of the dead-time generator and the reference signal OCxREF. (we suppose CCxP=0, CCxNP=0, MOE=1, CCxE=1 and CCxNE=1 in these examples)

Figure 120. Complementary output with dead-time insertion.**Figure 121. Dead-time waveforms with delay greater than the negative pulse.****Figure 122. Dead-time waveforms with delay greater than the positive pulse.**

The dead-time delay is the same for each of the channels and is programmable with the DTG bits in the TIMx_BDTR register. Refer to [Section 17.4.18: TIM1 and TIM8 break and dead-time register \(TIMx_BDTR\)](#) for delay calculation.

Re-directing OCxREF to OCx or OCxN

In output mode (forced, output compare or PWM), OCxREF can be re-directed to the OCx output or to OCxN output by configuring the CCxE and CCxNE bits in the TIMx_CCER register.

This allows the user to send a specific waveform (such as PWM or static active level) on one output while the complementary remains at its inactive level. Other possibilities are to

have both outputs at inactive level or both outputs active and complementary with dead-time.

Note: When only OCxN is enabled (CCxE=0, CCxNE=1), it is not complemented and becomes active as soon as OCxREF is high. For example, if CCxNP=0 then OCxN=OCxRef. On the other hand, when both OCx and OCxN are enabled (CCxE=CCxNE=1) OCx becomes active when OCxREF is high whereas OCxN is complemented and becomes active when OCxREF is low.

17.3.12 Using the break function

When using the break function, the output enable signals and inactive levels are modified according to additional control bits (MOE, OSSI and OSSR bits in the TIMx_BDTR register, OISx and OISxN bits in the TIMx_CR2 register). In any case, the OCx and OCxN outputs cannot be set both to active level at a given time. Refer to [Table 96](#) for more details.

The break source can be either the break input pin or a clock failure event, generated by the Clock Security System (CSS), from the Reset Clock Controller. For further information on the Clock Security System, refer to [Section 7.2.7: Clock security system \(CSS\)](#).

When exiting from reset, the break circuit is disabled and the MOE bit is low. User can enable the break function by setting the BKE bit in the TIMx_BDTR register. The break input polarity can be selected by configuring the BKP bit in the same register. BKE and BKP can be modified at the same time. When the BKE and BKP bits are written, a delay of 1 APB clock cycle is applied before the writing is effective. Consequently, it is necessary to wait 1 APB clock period to correctly read back the bit after the write operation.

Because MOE falling edge can be asynchronous, a resynchronization circuit has been inserted between the actual signal (acting on the outputs) and the synchronous control bit (accessed in the TIMx_BDTR register). It results in some delays between the asynchronous and the synchronous signals. In particular, if MOE is written to 1 whereas it was low, a delay (dummy instruction) must be inserted before reading it correctly. This is because the user writes an asynchronous signal, but reads a synchronous signal.

When a break occurs (selected level on the break input):

- The MOE bit is cleared asynchronously, putting the outputs in inactive state, idle state or in reset state (selected by the OSSI bit). This feature functions even if the MCU oscillator is off.
- Each output channel is driven with the level programmed in the OISx bit in the TIMx_CR2 register as soon as MOE=0. If OSSI=0 then the timer releases the enable output else the enable output remains high.
- When complementary outputs are used:
 - The outputs are first put in reset state inactive state (depending on the polarity). This is done asynchronously so that it works even if no clock is provided to the timer.
 - If the timer clock is still present, then the dead-time generator is reactivated in order to drive the outputs with the level programmed in the OISx and OISxN bits after a dead-time. Even in this case, OCx and OCxN cannot be driven to their

active level together. Note that because of the resynchronization on MOE, the dead-time duration is a bit longer than usual (around 2 ck_tim clock cycles).

- If OSSR=0 then the timer releases the enable outputs else the enable outputs remain or become high as soon as one of the CCxE or CCxNE bits is high.
- The break status flag (BIF bit in the TIMx_SR register) is set. An interrupt can be generated if the BIE bit in the TIMx_DIER register is set. A DMA request can be sent if the BDE bit in the TIMx_DIER register is set.
- If the AOE bit in the TIMx_BDTR register is set, the MOE bit is automatically set again at the next update event UEV. This can be used to perform a regulation, for instance. Else, MOE remains low until it is written to '1' again. In this case, it can be used for security and the break input can be connected to an alarm from power drivers, thermal sensors or any security components.

Note: *The break inputs is acting on level. Thus, the MOE cannot be set while the break input is active (neither automatically nor by software). In the meantime, the status flag BIF cannot be cleared.*

The break can be generated by the BRK input which has a programmable polarity and an enable bit BKE in the TIMx_BDTR register.

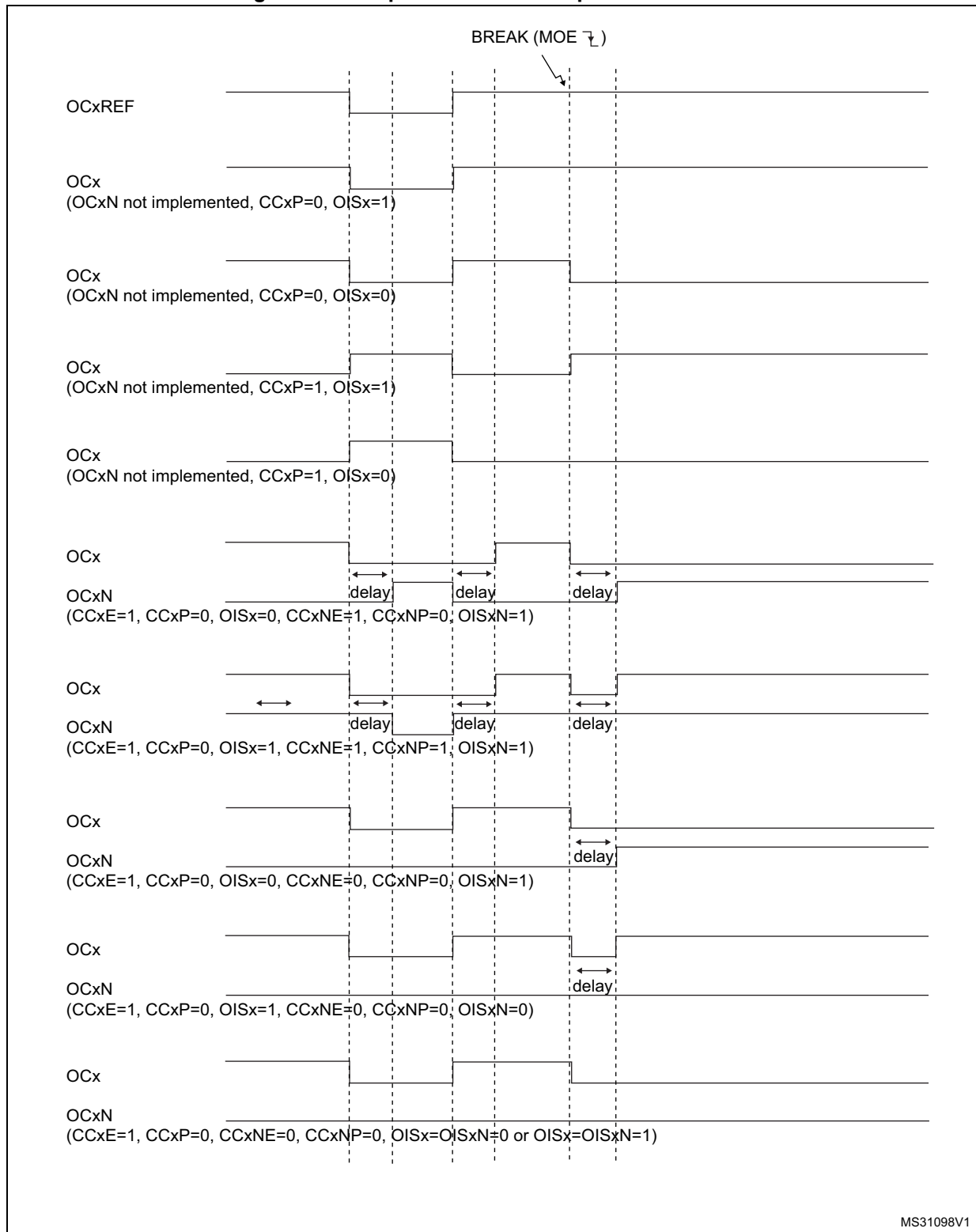
There are two solutions to generate a break:

- By using the BRK input which has a programmable polarity and an enable bit BKE in the TIMx_BDTR register
- By software through the BG bit of the TIMx_EGR register.

In addition to the break input and the output management, a write protection has been implemented inside the break circuit to safeguard the application. It allows freezing the configuration of several parameters (dead-time duration, OCx/OCxN polarities and state when disabled, OCxM configurations, break enable and polarity). The user can choose from three levels of protection selected by the LOCK bits in the TIMx_BDTR register. Refer to [Section 17.4.18: TIM1 and TIM8 break and dead-time register \(TIMx_BDTR\)](#). The LOCK bits can be written only once after an MCU reset.

Figure 123 shows an example of behavior of the outputs in response to a break.

Figure 123. Output behavior in response to a break.



17.3.13 Clearing the OCxREF signal on an external event

The OCxREF signal for a given channel can be driven Low by applying a High level to the ETRF input (OCxCE enable bit of the corresponding TIMx_CCMRx register set to '1'). The OCxREF signal remains Low until the next update event, UEV, occurs.

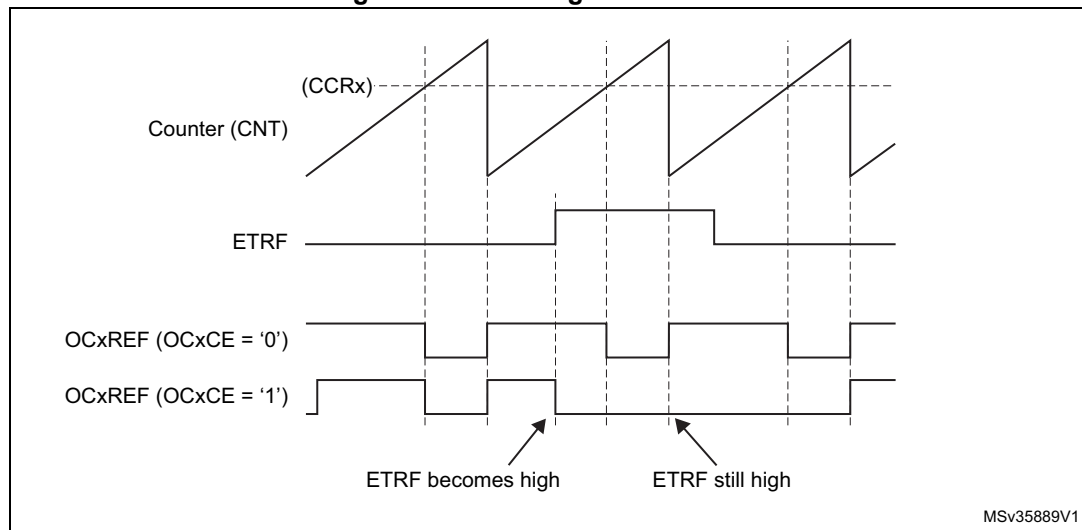
This function can only be used in output compare and PWM modes, and does not work in forced mode.

For example, the ETR signal can be connected to the output of a comparator to be used for current handling. In this case, the ETR must be configured as follow:

1. The External Trigger Prescaler should be kept off: bits ETPS[1:0] of the TIMx_SMCR register set to '00'.
2. The external clock mode 2 must be disabled: bit ECE of the TIMx_SMCR register set to '0'.
3. The External Trigger Polarity (ETP) and the External Trigger Filter (ETF) can be configured according to the user needs.

Figure 124 shows the behavior of the OCxREF signal when the ETRF Input becomes High, for both values of the enable bit OCxCE. In this example, the timer TIMx is programmed in PWM mode.

Figure 124. Clearing TIMx OCxREF



Note: In case of a PWM with a 100% duty cycle (if $CCR_x > ARR$), then OCxREF is enabled again at the next counter overflow.

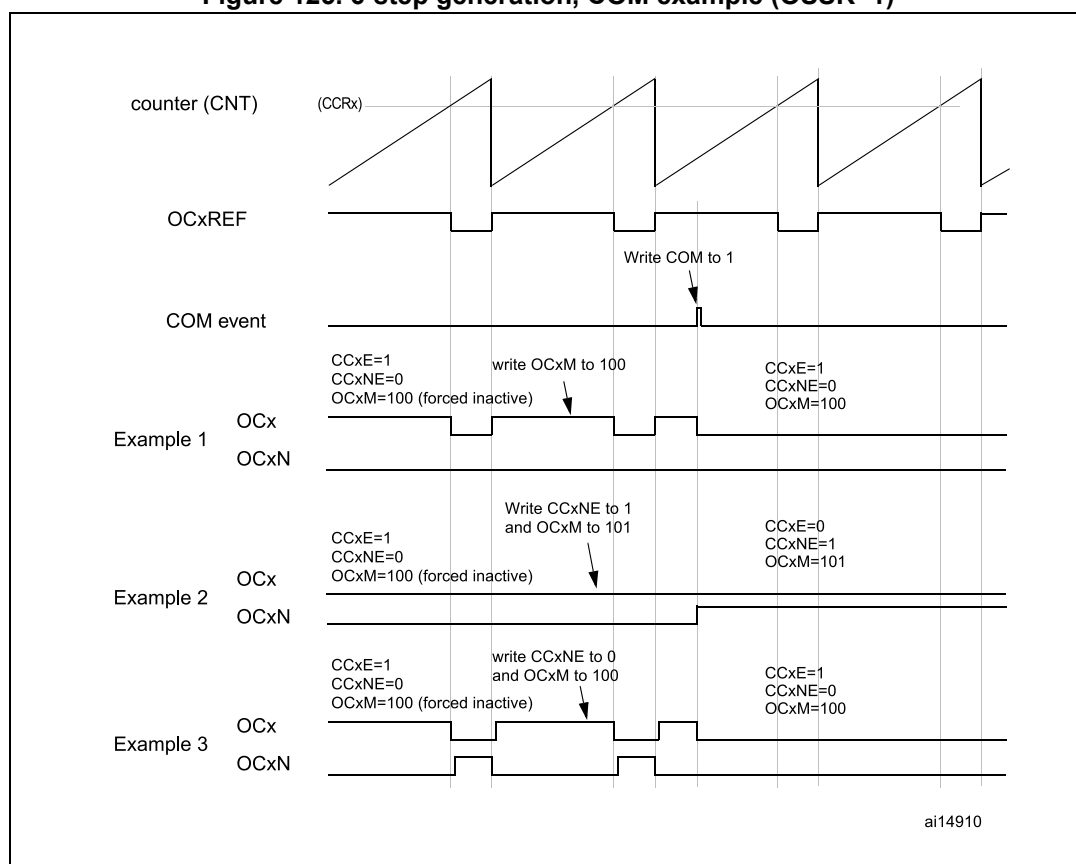
17.3.14 6-step PWM generation

When complementary outputs are used on a channel, preload bits are available on the OCxM, CCxE and CCxNE bits. The preload bits are transferred to the shadow bits at the COM commutation event. The user can thus program in advance the configuration for the next step and change the configuration of all the channels at the same time. COM can be generated by software by setting the COM bit in the TIMx_EGR register or by hardware (on TRGI rising edge).

A flag is set when the COM event occurs (COMIF bit in the TIMx_SR register), which can generate an interrupt (if the COMIE bit is set in the TIMx_DIER register) or a DMA request (if the COMDE bit is set in the TIMx_DIER register).

Figure 125 describes the behavior of the OCx and OCxN outputs when a COM event occurs, in 3 different examples of programmed configurations.

Figure 125. 6-step generation, COM example (OSSR=1)



17.3.15 One-pulse mode

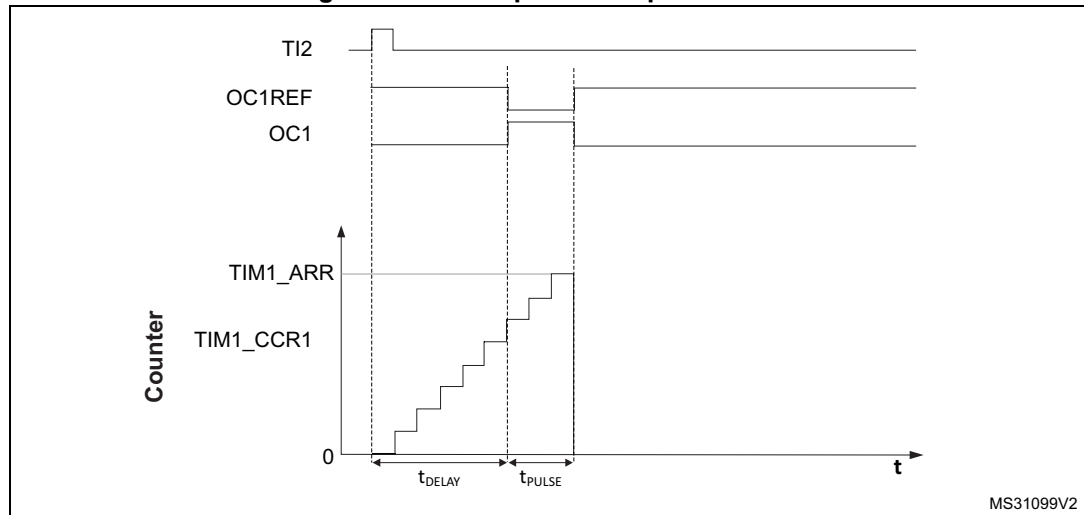
One-pulse mode (OPM) is a particular case of the previous modes. It allows the counter to be started in response to a stimulus and to generate a pulse with a programmable length after a programmable delay.

Starting the counter can be controlled through the slave mode controller. Generating the waveform can be done in output compare mode or PWM mode. Select One-pulse mode by setting the OPM bit in the TIMx_CR1 register. This makes the counter stop automatically at the next update event UEV.

A pulse can be correctly generated only if the compare value is different from the counter initial value. Before starting (when the timer is waiting for the trigger), the configuration must be:

- In upcounting: $CNT < CCRx \leq ARR$ (in particular, $0 < CCRx$)
- In downcounting: $CNT > CCRx$

Figure 126. Example of one pulse mode.



For example the user may want to generate a positive pulse on OC1 with a length of t_{PULSE} and after a delay of t_{DELAY} as soon as a positive edge is detected on the TI2 input pin.

Let's use TI2FP2 as trigger 1:

- Map TI2FP2 to TI2 by writing $CC2S='01'$ in the TIMx_CCMR1 register.
- TI2FP2 must detect a rising edge, write $CC2P='0'$ and $CC2NP='0'$ in the TIMx_CCER register.
- Configure TI2FP2 as trigger for the slave mode controller (TRGI) by writing $TS='110'$ in the TIMx_SMCR register.
- TI2FP2 is used to start the counter by writing SMS to '110' in the TIMx_SMCR register (trigger mode).

The OPM waveform is defined by writing the compare registers (taking into account the clock frequency and the counter prescaler).

- The t_{DELAY} is defined by the value written in the TIMx_CCR1 register.
- The t_{PULSE} is defined by the difference between the auto-reload value and the compare value (TIMx_ARR - TIMx_CCR1).
- Let us say the user wants to build a waveform with a transition from '0' to '1' when a compare match occurs and a transition from '1' to '0' when the counter reaches the auto-reload value. To do this, enable PWM mode 2 by writing OC1M=111 in the TIMx_CCMR1 register. The user can optionally enable the preload registers by writing OC1PE='1' in the TIMx_CCMR1 register and ARPE in the TIMx_CR1 register. In this case the compare value must be written in the TIMx_CCR1 register, the auto-reload value in the TIMx_ARR register, generate an update by setting the UG bit and wait for external trigger event on TI2. CC1P is written to '0' in this example.

In our example, the DIR and CMS bits in the TIMx_CR1 register should be low.

The user only wants one pulse (Single mode), so '1' must be written in the OPM bit in the TIMx_CR1 register to stop the counter at the next update event (when the counter rolls over from the auto-reload value back to 0). When OPM bit in the TIMx_CR1 register is set to '0', so the Repetitive mode is selected.

Particular case: OCx fast enable:

In One-pulse mode, the edge detection on TIx input set the CEN bit which enables the counter. Then the comparison between the counter and the compare value makes the output toggle. But several clock cycles are needed for these operations and it limits the minimum delay $t_{\text{DELAY min}}$ we can get.

If the user wants to output a waveform with the minimum delay, the OCxFE bit in the TIMx_CCMRx register must be set. Then OCxRef (and OCx) are forced in response to the stimulus, without taking in account the comparison. Its new level is the same as if a compare match had occurred. OCxFE acts only if the channel is configured in PWM1 or PWM2 mode.

17.3.16 Encoder interface mode

To select Encoder Interface mode write SMS='001' in the TIMx_SMCR register if the counter is counting on TI2 edges only, SMS='010' if it is counting on TI1 edges only and SMS='011' if it is counting on both TI1 and TI2 edges.

Select the TI1 and TI2 polarity by programming the CC1P and CC2P bits in the TIMx_CCER register. When needed, the user can program the input filter as well. CC1NP and CC2NP must be kept low.

The two inputs TI1 and TI2 are used to interface to an incremental encoder. Refer to [Table 94](#). The counter is clocked by each valid transition on TI1FP1 or TI2FP2 (TI1 and TI2 after input filter and polarity selection, TI1FP1=TI1 if not filtered and not inverted, TI2FP2=TI2 if not filtered and not inverted) assuming that it is enabled (CEN bit in TIMx_CR1 register written to '1'). The sequence of transitions of the two inputs is evaluated and generates count pulses as well as the direction signal. Depending on the sequence the counter counts up or down, the DIR bit in the TIMx_CR1 register is modified by hardware accordingly. The DIR bit is calculated at each transition on any input (TI1 or TI2), whatever the counter is counting on TI1 only, TI2 only or both TI1 and TI2.

Encoder interface mode acts simply as an external clock with direction selection. This means that the counter just counts continuously between 0 and the auto-reload value in the

TIMx_ARR register (0 to ARR or ARR down to 0 depending on the direction). So user must configure TIMx_ARR before starting. In the same way, the capture, compare, prescaler, repetition counter, trigger output features continue to work as normal. Encoder mode and External clock mode 2 are not compatible and must not be selected together.

In this mode, the counter is modified automatically following the speed and the direction of the incremental encoder and its content, therefore, always represents the encoder's position. The count direction correspond to the rotation direction of the connected sensor.

[Table 94](#) summarizes the possible combinations, assuming TI1 and TI2 do not switch at the same time.

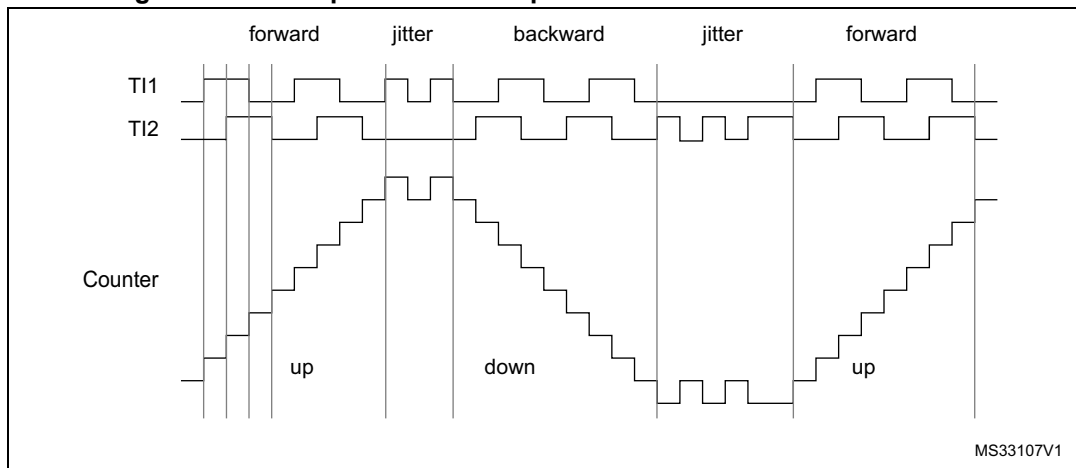
Table 94. Counting direction versus encoder signals

Active edge	Level on opposite signal (TI1FP1 for TI2, TI2FP2 for TI1)	TI1FP1 signal		TI2FP2 signal	
		Rising	Falling	Rising	Falling
Counting on TI1 only	High	Down	Up	No Count	No Count
	Low	Up	Down	No Count	No Count
Counting on TI2 only	High	No Count	No Count	Up	Down
	Low	No Count	No Count	Down	Up
Counting on TI1 and TI2	High	Down	Up	Up	Down
	Low	Up	Down	Down	Up

An external incremental encoder can be connected directly to the MCU without external interface logic. However, comparators are normally be used to convert the encoder's differential outputs to digital signals. This greatly increases noise immunity. The third encoder output which indicate the mechanical zero position, may be connected to an external interrupt input and trigger a counter reset.

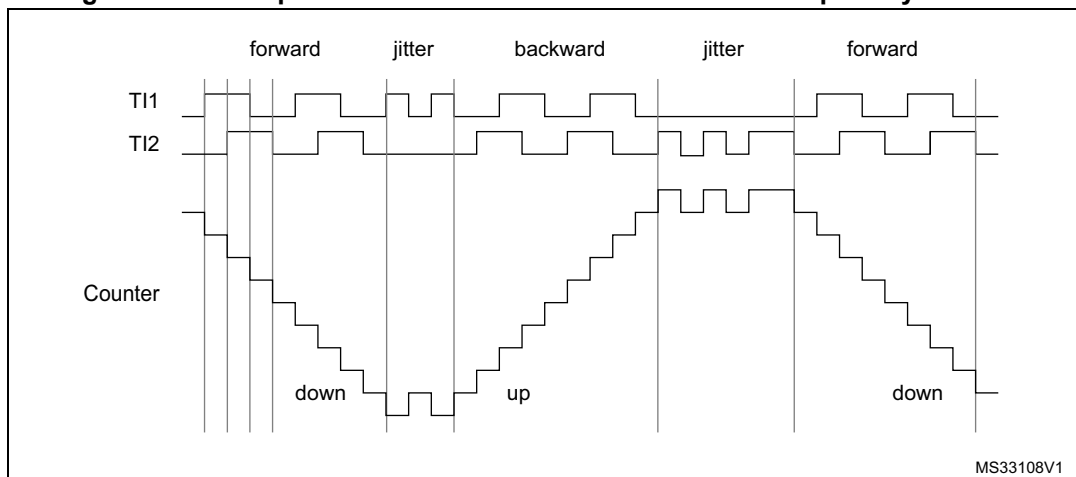
[Figure 127](#) gives an example of counter operation, showing count signal generation and direction control. It also shows how input jitter is compensated where both edges are selected. This might occur if the sensor is positioned near to one of the switching points. For this example we assume that the configuration is the following:

- CC1S='01' (TIMx_CCMR1 register, TI1FP1 mapped on TI1).
- CC2S='01' (TIMx_CCMR1 register, TI1FP2 mapped on TI2).
- CC1P='0', CC1NP='0', and IC1F = '0000' (TIMx_CCER register, TI1FP1 non-inverted, TI1FP1=TI1).
- CC2P='0', CC2NP='0', and IC2F = '0000' (TIMx_CCER register, TI1FP2 non-inverted, TI1FP2= TI2).
- SMS='011' (TIMx_SMCR register, both inputs are active on both rising and falling edges).
- CEN='1' (TIMx_CR1 register, Counter enabled).

Figure 127. Example of counter operation in encoder interface mode.

MS33107V1

Figure 128 gives an example of counter behavior when TI1FP1 polarity is inverted (same configuration as above except CC1P='1').

Figure 128. Example of encoder interface mode with TI1FP1 polarity inverted.

MS33108V1

The timer, when configured in Encoder Interface mode provides information on the sensor's current position. The user can obtain dynamic information (speed, acceleration, deceleration) by measuring the period between two encoder events using a second timer configured in capture mode. The output of the encoder which indicates the mechanical zero can be used for this purpose. Depending on the time between two events, the counter can also be read at regular times. This can be done by latching the counter value into a third input capture register if available (then the capture signal must be periodic and can be generated by another timer). When available, it is also possible to read its value through a DMA request generated by a real-time clock.

17.3.17 Timer input XOR function

The TI1S bit in the TIMx_CR2 register, allows the input filter of channel 1 to be connected to the output of a XOR gate, combining the three input pins TIMx_CH1, TIMx_CH2 and TIMx_CH3.

The XOR output can be used with all the timer input functions such as trigger or input capture. An example of this feature used to interface Hall sensors is given in [Section 17.3.18](#).

17.3.18 Interfacing with Hall sensors

This is done using the advanced-control timers (TIM1 or TIM8) to generate PWM signals to drive the motor and another timer TIMx (TIM2, TIM3, TIM4 or TIM5) referred to as “interfacing timer” in [Figure 129](#). The “interfacing timer” captures the 3 timer input pins (TIMx_CH1, TIMx_CH2, and TIMx_CH3) connected through a XOR to the TI1 input channel (selected by setting the TI1S bit in the TIMx_CR2 register).

The slave mode controller is configured in reset mode; the slave input is TI1F_ED. Thus, each time one of the 3 inputs toggles, the counter restarts counting from 0. This creates a time base triggered by any change on the Hall inputs.

On the “interfacing timer”, capture/compare channel 1 is configured in capture mode, capture signal is TRC (see [Figure 112](#)). The captured value, which corresponds to the time elapsed between 2 changes on the inputs, gives information about motor speed.

The “interfacing timer” can be used in output mode to generate a pulse which changes the configuration of the channels of the advanced-control timer (TIM1 or TIM8) (by triggering a COM event). The TIM1 timer is used to generate PWM signals to drive the motor. To do this, the interfacing timer channel must be programmed so that a positive pulse is generated after a programmed delay (in output compare or PWM mode). This pulse is sent to the advanced-control timer (TIM1 or TIM8) through the TRGO output.

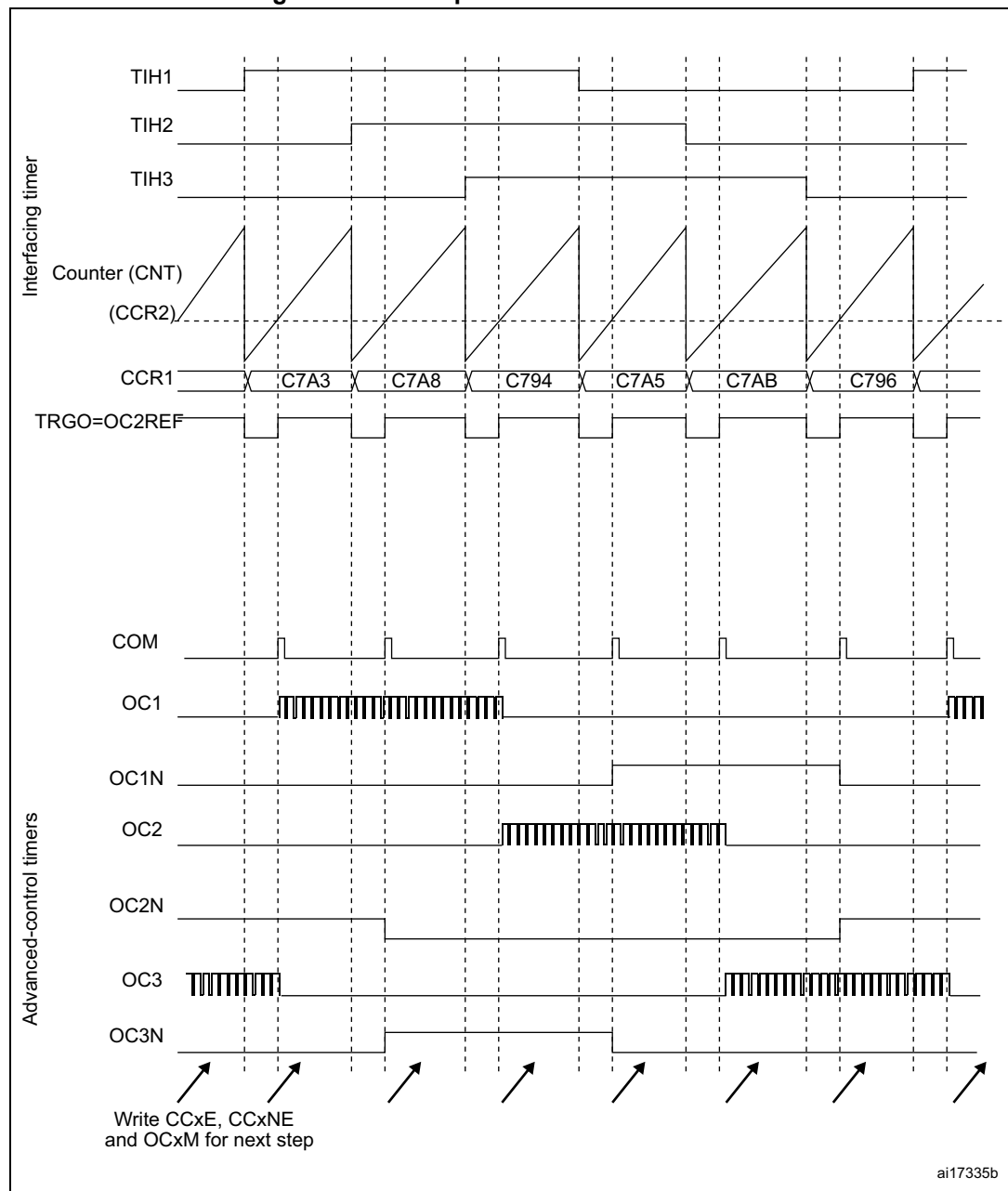
Example: the user wants to change the PWM configuration of the advanced-control timer TIM1 after a programmed delay each time a change occurs on the Hall inputs connected to one of the TIMx timers.

- Configure 3 timer inputs ORed to the TI1 input channel by writing the TI1S bit in the TIMx_CR2 register to ‘1’,
- Program the time base: write the TIMx_ARR to the max value (the counter must be cleared by the TI1 change. Set the prescaler to get a maximum counter period longer than the time between 2 changes on the sensors,
- Program channel 1 in capture mode (TRC selected): write the CC1S bits in the TIMx_CCMR1 register to ‘11’. The user can also program the digital filter if needed,
- Program channel 2 in PWM 2 mode with the desired delay: write the OC2M bits to ‘111’ and the CC2S bits to ‘00’ in the TIMx_CCMR1 register,
- Select OC2REF as trigger output on TRGO: write the MMS bits in the TIMx_CR2 register to ‘101’,

In the advanced-control timer TIM1, the right ITR input must be selected as trigger input, the timer is programmed to generate PWM signals, the capture/compare control signals are preloaded (CCPC=1 in the TIMx_CR2 register) and the COM event is controlled by the trigger input (CCUS=1 in the TIMx_CR2 register). The PWM control bits (CCxE, OCxM) are written after a COM event for the next step (this can be done in an interrupt subroutine generated by the rising edge of OC2REF).

Figure 129 describes this example.

Figure 129. Example of Hall sensor interface



17.3.19 TIMx and external trigger synchronization

The TIMx timer can be synchronized with an external trigger in several modes: Reset mode, Gated mode and Trigger mode.

Slave mode: Reset mode

The counter and its prescaler can be reinitialized in response to an event on a trigger input. Moreover, if the URS bit from the TIMx_CR1 register is low, an update event UEV is generated. Then all the preloaded registers (TIMx_ARR, TIMx_CCRx) are updated.

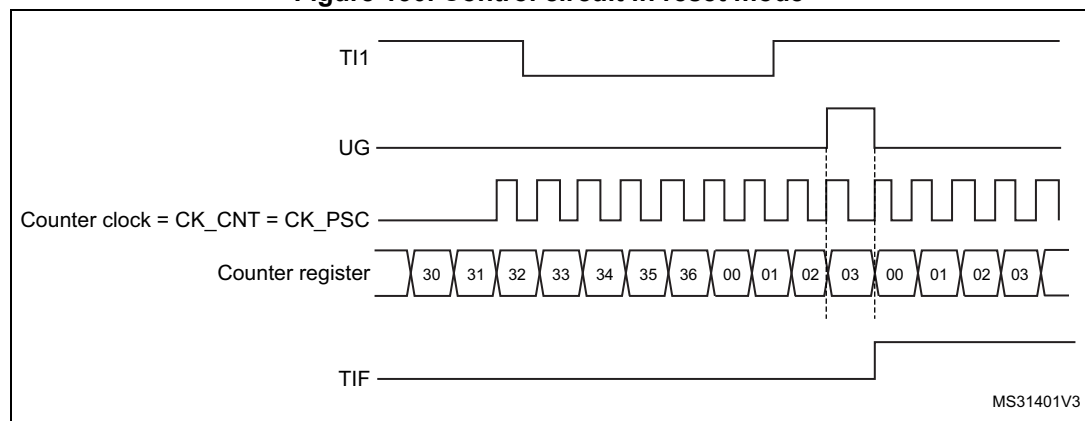
In the following example, the upcounter is cleared in response to a rising edge on TI1 input:

- Configure the channel 1 to detect rising edges on TI1. Configure the input filter duration (in this example, we don't need any filter, so we keep IC1F=0000). The capture prescaler is not used for triggering, so there's no need to configure it. The CC1S bits select the input capture source only, CC1S = 01 in the TIMx_CCMR1 register. Write CC1P=0 and CC1NP='0' in TIMx_CCER register to validate the polarity (and detect rising edges only).
- Configure the timer in reset mode by writing SMS=100 in TIMx_SMCR register. Select TI1 as the input source by writing TS=101 in TIMx_SMCR register.
- Start the counter by writing CEN=1 in the TIMx_CR1 register.

The counter starts counting on the internal clock, then behaves normally until TI1 rising edge. When TI1 rises, the counter is cleared and restarts from 0. In the meantime, the trigger flag is set (TIF bit in the TIMx_SR register) and an interrupt request, or a DMA request can be sent if enabled (depending on the TIE and TDE bits in TIMx_DIER register).

The following figure shows this behavior when the auto-reload register TIMx_ARR=0x36. The delay between the rising edge on TI1 and the actual reset of the counter is due to the resynchronization circuit on TI1 input.

Figure 130. Control circuit in reset mode



Slave mode: Gated mode

The counter can be enabled depending on the level of a selected input.

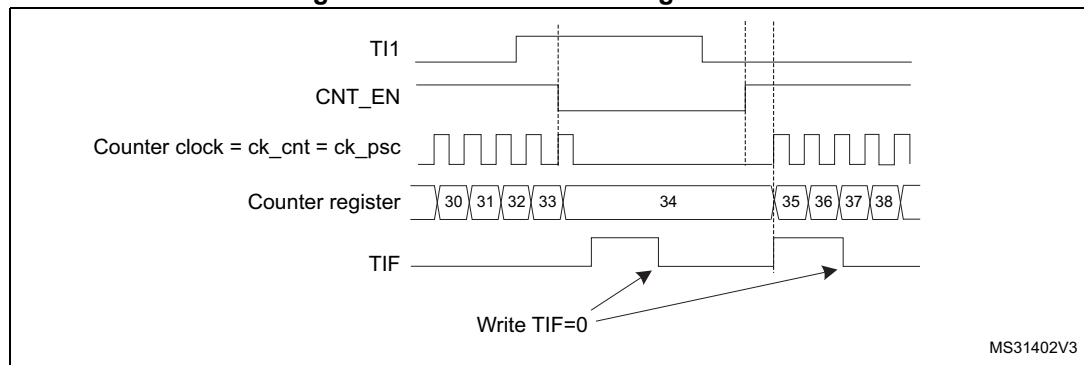
In the following example, the upcounter counts only when TI1 input is low:

- Configure the channel 1 to detect low levels on TI1. Configure the input filter duration (in this example, we don't need any filter, so we keep IC1F=0000). The capture prescaler is not used for triggering, so the user does not need to configure it. The CC1S bits select the input capture source only, CC1S=01 in TIMx_CCMR1 register. Write CC1P=1 and CC1NP='0' in TIMx_CCER register to validate the polarity (and detect low level only).
- Configure the timer in gated mode by writing SMS=101 in TIMx_SMCR register. Select TI1 as the input source by writing TS=101 in TIMx_SMCR register.
- Enable the counter by writing CEN=1 in the TIMx_CR1 register (in gated mode, the counter doesn't start if CEN=0, whatever is the trigger input level).

The counter starts counting on the internal clock as long as TI1 is low and stops as soon as TI1 becomes high. The TIF flag in the TIMx_SR register is set both when the counter starts or stops.

The delay between the rising edge on TI1 and the actual stop of the counter is due to the resynchronization circuit on TI1 input.

Figure 131. Control circuit in gated mode



Slave mode: Trigger mode

The counter can start in response to an event on a selected input.

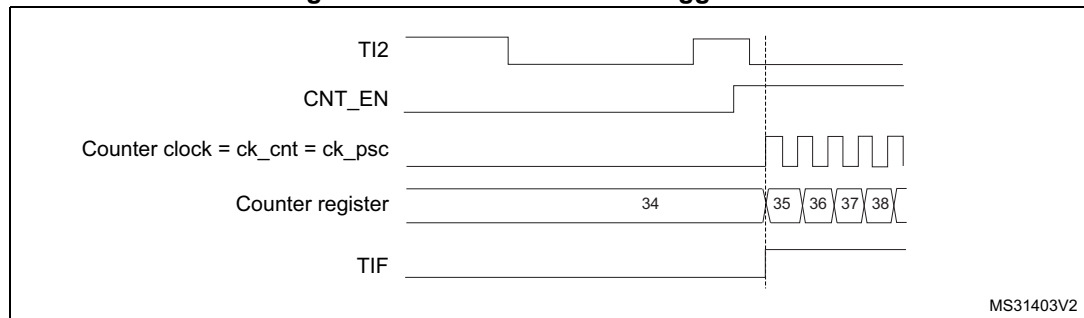
In the following example, the upcounter starts in response to a rising edge on TI2 input:

- Configure the channel 2 to detect rising edges on TI2. Configure the input filter duration (in this example, we don't need any filter, so we keep IC2F=0000). The capture prescaler is not used for triggering, so there's no need to configure it. The CC2S bits are configured to select the input capture source only, CC2S=01 in TIMx_CCMR1 register. Write CC2P=1 and CC2NP=0 in TIMx_CCER register to validate the polarity (and detect low level only).
- Configure the timer in trigger mode by writing SMS=110 in TIMx_SMCR register. Select TI2 as the input source by writing TS=110 in TIMx_SMCR register.

When a rising edge occurs on TI2, the counter starts counting on the internal clock and the TIF flag is set.

The delay between the rising edge on TI2 and the actual start of the counter is due to the resynchronization circuit on TI2 input.

Figure 132. Control circuit in trigger mode



MS31403V2

Slave mode: external clock mode 2 + trigger mode

The external clock mode 2 can be used in addition to another slave mode (except external clock mode 1 and encoder mode). In this case, the ETR signal is used as external clock input, and another input can be selected as trigger input (in reset mode, gated mode or trigger mode). It is recommended not to select ETR as TRGI through the TS bits of TIMx_SMCR register.

In the following example, the upcounter is incremented at each rising edge of the ETR signal as soon as a rising edge of TI1 occurs:

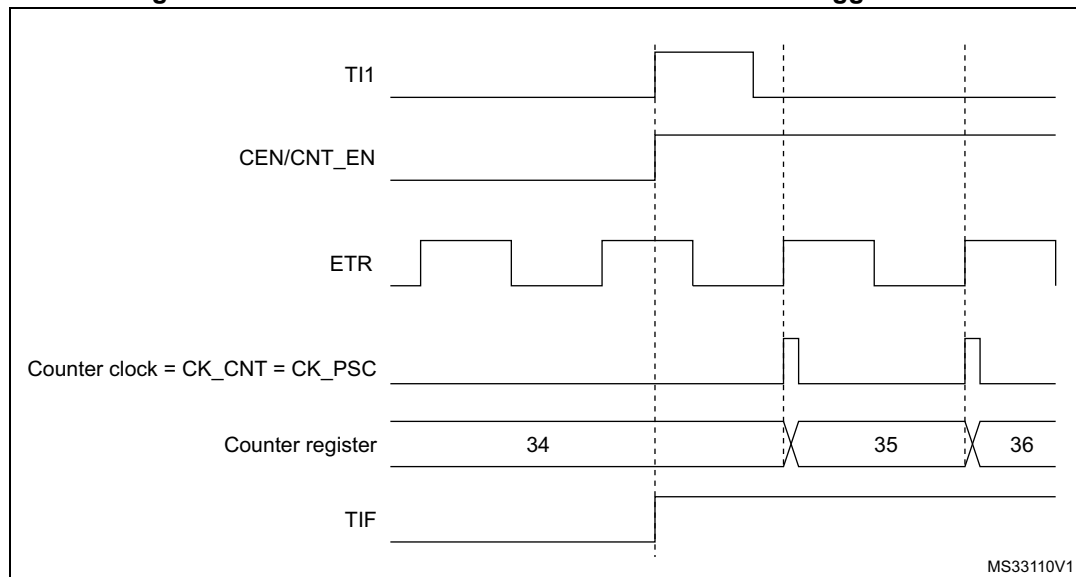
1. Configure the external trigger input circuit by programming the TIMx_SMCR register as follows:
 - ETF = 0000: no filter
 - ETPS = 00: prescaler disabled
 - ETP = 0: detection of rising edges on ETR and ECE=1 to enable the external clock mode 2.
2. Configure the channel 1 as follows, to detect rising edges on TI1:
 - IC1F=0000: no filter.
 - The capture prescaler is not used for triggering and does not need to be configured.
 - CC1S=01 in TIMx_CCMR1 register to select only the input capture source

- CC1P=0 and CC1NP='0' in TIMx_CCER register to validate the polarity (and detect rising edge only).
- 3. Configure the timer in trigger mode by writing SMS=110 in TIMx_SMCR register. Select TI1 as the input source by writing TS=101 in TIMx_SMCR register.

A rising edge on TI1 enables the counter and sets the TIF flag. The counter then counts on ETR rising edges.

The delay between the rising edge of the ETR signal and the actual reset of the counter is due to the resynchronization circuit on ETRP input.

Figure 133. Control circuit in external clock mode 2 + trigger mode



17.3.20 Timer synchronization

The TIM timers are linked together internally for timer synchronization or chaining. Refer to [Section 18.3.15: Timer synchronization](#) for details.

Note: The clock of the slave timer must be enabled prior to receive events from the master timer, and must not be changed on-the-fly while triggers are received from the master timer.

17.3.21 Debug mode

When the microcontroller enters debug mode (Cortex[®]-M4 with FPU core halted), the TIMx counter either continues to work normally or stops, depending on DBG_TIMx_STOP configuration bit in DBG module. For more details, refer to [Section 38.16.2: Debug support for timers, watchdog, bxCAN and I²C](#).

For safety purposes, when the counter is stopped (DBG_TIMx_STOP = 1 in DBGMCU_APBx_FZ register), the outputs are disabled (as if the MOE bit was reset). The outputs can either be forced to an inactive state (OSSI bit = 1), or have their control taken over by the GPIO controller (OSSI bit = 0) to force them to Hi-Z.

17.4 TIM1 and TIM8 registers

Refer to [Section 2.2](#) for a list of abbreviations used in register descriptions.

The peripheral registers must be written by half-words (16 bits) or words (32 bits). Read accesses can be done by bytes (8 bits), half-word (16 bits) or words (32 bits).

17.4.1 TIM1 and TIM8 control register 1 (TIMx_CR1)

Address offset: 0x00

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						CKD[1:0]		ARPE	CMS[1:0]		DIR	OPM	URS	UDIS	CEN
						rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:10 Reserved, must be kept at reset value.

Bits 9:8 **CKD[1:0]**: Clock division

This bit-field indicates the division ratio between the timer clock (CK_INT) frequency and the dead-time and sampling clock (t_{DTS}) used by the dead-time generators and the digital filters (ETR, TIX),

00: $t_{DTS}=t_{CK_INT}$

01: $t_{DTS}=2*t_{CK_INT}$

10: $t_{DTS}=4*t_{CK_INT}$

11: Reserved, do not program this value

Bit 7 **ARPE**: Auto-reload preload enable

0: TIMx_ARR register is not buffered

1: TIMx_ARR register is buffered

Bits 6:5 **CMS[1:0]**: Center-aligned mode selection

00: Edge-aligned mode. The counter counts up or down depending on the direction bit (DIR).

01: Center-aligned mode 1. The counter counts up and down alternatively. Output compare interrupt flags of channels configured in output (CCxS=00 in TIMx_CCMRx register) are set only when the counter is counting down.

10: Center-aligned mode 2. The counter counts up and down alternatively. Output compare interrupt flags of channels configured in output (CCxS=00 in TIMx_CCMRx register) are set only when the counter is counting up.

11: Center-aligned mode 3. The counter counts up and down alternatively. Output compare interrupt flags of channels configured in output (CCxS=00 in TIMx_CCMRx register) are set both when the counter is counting up or down.

Note: It is not allowed to switch from edge-aligned mode to center-aligned mode as long as the counter is enabled (CEN=1)

Bit 4 **DIR**: Direction

0: Counter used as upcounter

1: Counter used as downcounter

Note: This bit is read only when the timer is configured in Center-aligned mode or Encoder mode.

Bit 3 **OPM**: One pulse mode

0: Counter is not stopped at update event

1: Counter stops counting at the next update event (clearing the bit CEN)

Bit 2 URS: Update request source

This bit is set and cleared by software to select the UEV event sources.

0: Any of the following events generate an update interrupt or DMA request if enabled.

These events can be:

- Counter overflow/underflow
- Setting the UG bit
- Update generation through the slave mode controller

1: Only counter overflow/underflow generates an update interrupt or DMA request if enabled.

Bit 1 UDIS: Update disable

This bit is set and cleared by software to enable/disable UEV event generation.

0: UEV enabled. The Update (UEV) event is generated by one of the following events:

- Counter overflow/underflow
- Setting the UG bit
- Update generation through the slave mode controller

Buffered registers are then loaded with their preload values.

1: UEV disabled. The Update event is not generated, shadow registers keep their value (ARR, PSC, CCRx). However the counter and the prescaler are reinitialized if the UG bit is set or if a hardware reset is received from the slave mode controller.

Bit 0 CEN: Counter enable

0: Counter disabled

1: Counter enabled

Note: External clock, gated mode and encoder mode can work only if the CEN bit has been previously set by software. However trigger mode can set the CEN bit automatically by hardware.

17.4.2 TIM1 and TIM8 control register 2 (TIMx_CR2)

Address offset: 0x04

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	OIS4	OIS3N	OIS3	OIS2N	OIS2	OIS1N	OIS1	TI1S	MMS[2:0]			CCDS	CCUS	Res.	CCPC
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw		rw

Bit 15 Reserved, must be kept at reset value.

Bit 14 **OIS4**: Output Idle state 4 (OC4 output)
refer to OIS1 bit

Bit 13 **OIS3N**: Output Idle state 3 (OC3N output)
refer to OIS1N bit

Bit 12 **OIS3**: Output Idle state 3 (OC3 output)
refer to OIS1 bit

Bit 11 **OIS2N**: Output Idle state 2 (OC2N output)
refer to OIS1N bit

Bit 10 **OIS2**: Output Idle state 2 (OC2 output)
refer to OIS1 bit

Bit 9 **OIS1N**: Output Idle state 1 (OC1N output)

0: OC1N=0 after a dead-time when MOE=0

1: OC1N=1 after a dead-time when MOE=0

Note: This bit can not be modified as long as LOCK level 1, 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 8 **OIS1**: Output Idle state 1 (OC1 output)

0: OC1=0 (after a dead-time if OC1N is implemented) when MOE=0

1: OC1=1 (after a dead-time if OC1N is implemented) when MOE=0

Note: This bit can not be modified as long as LOCK level 1, 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 7 **TI1S**: TI1 selection

0: The TIMx_CH1 pin is connected to TI1 input

1: The TIMx_CH1, CH2 and CH3 pins are connected to the TI1 input (XOR combination)

Bits 6:4 **MMS[2:0]**: Master mode selection

These bits allow to select the information to be sent in master mode to slave timers for synchronization (TRGO). The combination is as follows:

000: **Reset** - the UG bit from the TIMx_EGR register is used as trigger output (TRGO). If the reset is generated by the trigger input (slave mode controller configured in reset mode) then the signal on TRGO is delayed compared to the actual reset.

001: **Enable** - the Counter Enable signal CNT_EN is used as trigger output (TRGO). It is useful to start several timers at the same time or to control a window in which a slave timer is enable. The Counter Enable signal is generated by a logic OR between CEN control bit and the trigger input when configured in gated mode. When the Counter Enable signal is controlled by the trigger input, there is a delay on TRGO, except if the master/slave mode is selected (see the MSM bit description in TIMx_SMCR register).

010: **Update** - The update event is selected as trigger output (TRGO). For instance a master timer can then be used as a prescaler for a slave timer.

011: **Compare Pulse** - The trigger output send a positive pulse when the CC1IF flag is to be set (even if it was already high), as soon as a capture or a compare match occurred. (TRGO).

100: **Compare** - OC1REF signal is used as trigger output (TRGO)

101: **Compare** - OC2REF signal is used as trigger output (TRGO)

110: **Compare** - OC3REF signal is used as trigger output (TRGO)

111: **Compare** - OC4REF signal is used as trigger output (TRGO)

Note: The clock of the slave timer and ADC must be enabled prior to receiving events from the master timer, and must not be changed on-the-fly while triggers are received from the master timer.

Bit 3 **CCDS**: Capture/compare DMA selection

0: CCx DMA request sent when CCx event occurs

1: CCx DMA requests sent when update event occurs

Bit 2 **CCUS**: Capture/compare control update selection

0: When capture/compare control bits are preloaded (CCPC=1), they are updated by setting the COMG bit only

1: When capture/compare control bits are preloaded (CCPC=1), they are updated by setting the COMG bit or when an rising edge occurs on TRGI

Note: This bit acts only on channels that have a complementary output.

Bit 1 Reserved, must be kept at reset value.

Bit 0 **CCPC**: Capture/compare preloaded control

0: CCxE, CCxNE and OCxM bits are not preloaded

1: CCxE, CCxNE and OCxM bits are preloaded, after having been written, they are updated only when a commutation event (COM) occurs (COMG bit set or rising edge detected on TRGI, depending on the CCUS bit).

Note: This bit acts only on channels that have a complementary output.

17.4.3 TIM1 and TIM8 slave mode control register (TIMx_SMCR)

Address offset: 0x08

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETP	ECE	ETPS[1:0]		ETF[3:0]				MSM	TS[2:0]			Res.	SMS[2:0]		
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	Res.	rw	rw	rw

Bit 15 **ETP**: External trigger polarity

This bit selects whether ETR or $\overline{ETR}$ is used for trigger operations

0: ETR is non-inverted, active at high level or rising edge.

1: ETR is inverted, active at low level or falling edge.

Bit 14 **ECE**: External clock enable

This bit enables External clock mode 2.

0: External clock mode 2 disabled

1: External clock mode 2 enabled. The counter is clocked by any active edge on the ETRF signal.

Note: **1:** Setting the ECE bit has the same effect as selecting external clock mode 1 with TRGI connected to ETRF (SMS=111 and TS=111).

2: It is possible to simultaneously use external clock mode 2 with the following slave modes: reset mode, gated mode and trigger mode. Nevertheless, TRGI must not be connected to ETRF in this case (TS bits must not be 111).

3: If external clock mode 1 and external clock mode 2 are enabled at the same time, the external clock input is ETRF.

Bits 13:12 **ETPS[1:0]**: External trigger prescaler

External trigger signal ETRP frequency must be at most 1/4 of TIMxCLK frequency. A prescaler can be enabled to reduce ETRP frequency. It is useful when inputting fast external clocks.

00: Prescaler OFF

01: ETRP frequency divided by 2

10: ETRP frequency divided by 4

11: ETRP frequency divided by 8

Bits 11:8 **ETF[3:0]**: External trigger filter

This bit-field then defines the frequency used to sample ETRP signal and the length of the digital filter applied to ETRP. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, sampling is done at f_{DTS}

0001: $f_{SAMPLING}=f_{CK_INT}$, $N=2$

0010: $f_{SAMPLING}=f_{CK_INT}$, $N=4$

0011: $f_{SAMPLING}=f_{CK_INT}$, $N=8$

0100: $f_{SAMPLING}=f_{DTS}/2$, $N=6$

0101: $f_{SAMPLING}=f_{DTS}/2$, $N=8$

0110: $f_{SAMPLING}=f_{DTS}/4$, $N=6$

0111: $f_{SAMPLING}=f_{DTS}/4$, $N=8$

1000: $f_{SAMPLING}=f_{DTS}/8$, $N=6$

1001: $f_{SAMPLING}=f_{DTS}/8$, $N=8$

1010: $f_{SAMPLING}=f_{DTS}/16$, $N=5$

1011: $f_{SAMPLING}=f_{DTS}/16$, $N=6$

1100: $f_{SAMPLING}=f_{DTS}/16$, $N=8$

1101: $f_{SAMPLING}=f_{DTS}/32$, $N=5$

1110: $f_{SAMPLING}=f_{DTS}/32$, $N=6$

1111: $f_{SAMPLING}=f_{DTS}/32$, $N=8$

Bit 7 **MSM**: Master/slave mode

0: No action

1: The effect of an event on the trigger input (TRGI) is delayed to allow a perfect synchronization between the current timer and its slaves (through TRGO). It is useful if we want to synchronize several timers on a single external event.

Bits 6:4 **TS[2:0]**: Trigger selection

This bit-field selects the trigger input to be used to synchronize the counter.

000: Internal Trigger 0 (ITR0)

001: Internal Trigger 1 (ITR1)

010: Internal Trigger 2 (ITR2)

011: Internal Trigger 3 (ITR3)

100: TI1 Edge Detector (TI1F_ED)

101: Filtered Timer Input 1 (TI1FP1)

110: Filtered Timer Input 2 (TI2FP2)

111: External Trigger input (ETRF)

See [Table 95](#) for more details on ITRx meaning for each Timer.

Note: These bits must be changed only when they are not used (e.g. when SMS=000) to avoid wrong edge detections at the transition.

Bit 3 Reserved, must be kept at reset value.

Bits 2:0 **SMS**: Slave mode selection

When external signals are selected the active edge of the trigger signal (TRGI) is linked to the polarity selected on the external input (see Input Control register and Control register description).

000: Slave mode disabled - if CEN = '1' then the prescaler is clocked directly by the internal clock.

001: Encoder mode 1 - Counter counts up/down on TI1FP1 edge depending on TI2FP2 level.

010: Encoder mode 2 - Counter counts up/down on TI2FP2 edge depending on TI1FP1 level.

011: Encoder mode 3 - Counter counts up/down on both TI1FP1 and TI2FP2 edges depending on the level of the other input.

100: Reset mode - Rising edge of the selected trigger input (TRGI) reinitializes the counter and generates an update of the registers.

101: Gated mode - The counter clock is enabled when the trigger input (TRGI) is high. The counter stops (but is not reset) as soon as the trigger becomes low. Both start and stop of the counter are controlled.

110: Trigger mode - The counter starts at a rising edge of the trigger TRGI (but it is not reset). Only the start of the counter is controlled.

111: External Clock mode 1 - Rising edges of the selected trigger (TRGI) clock the counter.

Note: The gated mode must not be used if TI1F_ED is selected as the trigger input (TS='100'). Indeed, TI1F_ED outputs 1 pulse for each transition on TI1F, whereas the gated mode checks the level of the trigger signal.

The clock of the slave timer must be enabled prior to receiving events from the master timer, and must not be changed on-the-fly while triggers are received from the master timer.

Table 95. TIMx Internal trigger connection

Slave TIM	ITR0 (TS = 000)	ITR1 (TS = 001)	ITR2 (TS = 010)	ITR3 (TS = 011)
TIM1	TIM5_TRGO	TIM2_TRGO	TIM3_TRGO	TIM4_TRGO
TIM8	TIM1_TRGO	TIM2_TRGO	TIM4_TRGO	TIM5_TRGO

17.4.4 TIM1 and TIM8 DMA/interrupt enable register (TIMx_DIER)

Address offset: 0x0C

Reset value: 0x0000

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	TDE	COMDE	CC4DE	CC3DE	CC2DE	CC1DE	UDE	BIE	TIE	COMIE	CC4IE	CC3IE	CC2IE	CC1IE	UIE	
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 15 Reserved, must be kept at reset value.

Bit 14 **TDE**: Trigger DMA request enable

0: Trigger DMA request disabled

1: Trigger DMA request enabled

Bit 13 **COMDE**: COM DMA request enable

0: COM DMA request disabled

1: COM DMA request enabled

- Bit 12 **CC4DE**: Capture/Compare 4 DMA request enable
0: CC4 DMA request disabled
1: CC4 DMA request enabled
- Bit 11 **CC3DE**: Capture/Compare 3 DMA request enable
0: CC3 DMA request disabled
1: CC3 DMA request enabled
- Bit 10 **CC2DE**: Capture/Compare 2 DMA request enable
0: CC2 DMA request disabled
1: CC2 DMA request enabled
- Bit 9 **CC1DE**: Capture/Compare 1 DMA request enable
0: CC1 DMA request disabled
1: CC1 DMA request enabled
- Bit 8 **UDE**: Update DMA request enable
0: Update DMA request disabled
1: Update DMA request enabled
- Bit 7 **BIE**: Break interrupt enable
0: Break interrupt disabled
1: Break interrupt enabled
- Bit 6 **TIE**: Trigger interrupt enable
0: Trigger interrupt disabled
1: Trigger interrupt enabled
- Bit 5 **COMIE**: COM interrupt enable
0: COM interrupt disabled
1: COM interrupt enabled
- Bit 4 **CC4IE**: Capture/Compare 4 interrupt enable
0: CC4 interrupt disabled
1: CC4 interrupt enabled
- Bit 3 **CC3IE**: Capture/Compare 3 interrupt enable
0: CC3 interrupt disabled
1: CC3 interrupt enabled
- Bit 2 **CC2IE**: Capture/Compare 2 interrupt enable
0: CC2 interrupt disabled
1: CC2 interrupt enabled
- Bit 1 **CC1IE**: Capture/Compare 1 interrupt enable
0: CC1 interrupt disabled
1: CC1 interrupt enabled
- Bit 0 **UIE**: Update interrupt enable
0: Update interrupt disabled
1: Update interrupt enabled

17.4.5 TIM1 and TIM8 status register (TIMx_SR)

Address offset: 0x10

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			CC4OF	CC3OF	CC2OF	CC1OF	Res.	BIF	TIF	COMIF	CC4IF	CC3IF	CC2IF	CC1IF	UIF
			rc_w0	rc_w0	rc_w0	rc_w0	Res.	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0

Bits 15:13 Reserved, must be kept at reset value.

Bit 12 **CC4OF**: Capture/Compare 4 overcapture flag
refer to CC1OF description

Bit 11 **CC3OF**: Capture/Compare 3 overcapture flag
refer to CC1OF description

Bit 10 **CC2OF**: Capture/Compare 2 overcapture flag
refer to CC1OF description

Bit 9 **CC1OF**: Capture/Compare 1 overcapture flag
This flag is set by hardware only when the corresponding channel is configured in input capture mode. It is cleared by software by writing it to '0'.
0: No overcapture has been detected.
1: The counter value has been captured in TIMx_CCR1 register while CC1IF flag was already set

Bit 8 Reserved, must be kept at reset value.

Bit 7 **BIF**: Break interrupt flag
This flag is set by hardware as soon as the break input goes active. It can be cleared by software if the break input is not active.
0: No break event occurred.
1: An active level has been detected on the break input.

Bit 6 **TIF**: Trigger interrupt flag
This flag is set by hardware on trigger event (active edge detected on TRGI input when the slave mode controller is enabled in all modes but gated mode. It is set when the counter starts or stops when gated mode is selected. It is cleared by software.
0: No trigger event occurred.
1: Trigger interrupt pending.

Bit 5 **COMIF**: COM interrupt flag
This flag is set by hardware on COM event (when Capture/compare Control bits - CCxE, CCxNE, OCxM - have been updated). It is cleared by software.
0: No COM event occurred.
1: COM interrupt pending.

Bit 4 **CC4IF**: Capture/Compare 4 interrupt flag
refer to CC1IF description

Bit 3 **CC3IF**: Capture/Compare 3 interrupt flag
refer to CC1IF description

Bit 2 **CC2IF**: Capture/Compare 2 interrupt flag
refer to CC1IF description

Bit 1 **CC1IF**: Capture/Compare 1 interrupt flag

If channel CC1 is configured as output:

This flag is set by hardware when the counter matches the compare value, with some exception in center-aligned mode (refer to the CMS bits in the TIMx_CR1 register description). It is cleared by software.

0: No match.

1: The content of the counter TIMx_CNT matches the content of the TIMx_CCR1 register. When the contents of TIMx_CCR1 are greater than the contents of TIMx_ARR, the CC1IF bit goes high on the counter overflow (in upcounting and up/down-counting modes) or underflow (in downcounting mode)

If channel CC1 is configured as input:

This bit is set by hardware on a capture. It is cleared by software or by reading the TIMx_CCR1 register.

0: No input capture occurred

1: The counter value has been captured in TIMx_CCR1 register (An edge has been detected on IC1 which matches the selected polarity)

Bit 0 **UIF**: Update interrupt flag

This bit is set by hardware on an update event. It is cleared by software.

0: No update occurred.

1: Update interrupt pending. This bit is set by hardware when the registers are updated:

- At overflow or underflow regarding the repetition counter value (update if repetition counter = 0) and if the UDIS=0 in the TIMx_CR1 register.
- When CNT is reinitialized by software using the UG bit in TIMx_EGR register, if URS=0 and UDIS=0 in the TIMx_CR1 register.
- When CNT is reinitialized by a trigger event (refer to [Section 17.4.3: TIM1 and TIM8 slave mode control register \(TIMx_SMCR\)](#)), if URS=0 and UDIS=0 in the TIMx_CR1 register.

17.4.6 TIM1 and TIM8 event generation register (TIMx_EGR)

Address offset: 0x14

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								BG	TG	COMG	CC4G	CC3G	CC2G	CC1G	UG
								w	w	w	w	w	w	w	w

Bits 15:8 Reserved, must be kept at reset value.

Bit 7 **BG**: Break generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: A break event is generated. MOE bit is cleared and BIF flag is set. Related interrupt or DMA transfer can occur if enabled.

Bit 6 **TG**: Trigger generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: The TIF flag is set in TIMx_SR register. Related interrupt or DMA transfer can occur if enabled.

Bit 5 **COMG**: Capture/Compare control update generation

This bit can be set by software, it is automatically cleared by hardware

0: No action

1: When CCPC bit is set, it allows to update CCxE, CCxNE and OCxM bits

Note: This bit acts only on channels having a complementary output.

Bit 4 **CC4G**: Capture/Compare 4 generation

refer to CC1G description

Bit 3 **CC3G**: Capture/Compare 3 generation

refer to CC1G description

Bit 2 **CC2G**: Capture/Compare 2 generation

refer to CC1G description

Bit 1 **CC1G**: Capture/Compare 1 generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: A capture/compare event is generated on channel 1:

If channel CC1 is configured as output:

CC1IF flag is set, Corresponding interrupt or DMA request is sent if enabled.

If channel CC1 is configured as input:

The current value of the counter is captured in TIMx_CCR1 register. The CC1IF flag is set, the corresponding interrupt or DMA request is sent if enabled. The CC1OF flag is set if the CC1IF flag was already high.

Bit 0 **UG**: Update generation

This bit can be set by software, it is automatically cleared by hardware.

0: No action

1: Reinitialize the counter and generates an update of the registers. Note that the prescaler counter is cleared too (anyway the prescaler ratio is not affected). The counter is cleared if the center-aligned mode is selected or if DIR=0 (upcounting), else it takes the auto-reload value (TIMx_ARR) if DIR=1 (downcounting).

17.4.7 TIM1 and TIM8 capture/compare mode register 1 (TIMx_CCMR1)

Address offset: 0x18

Reset value: 0x0000

The channels can be used in input (capture mode) or in output (compare mode). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits of this register have a different function in input and in output mode. For a given bit, OCxx describes its function when the channel is configured in output, ICxx describes its function when the channel is configured in input. So the user must take care that the same bit can have a different meaning for the input stage and for the output stage.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC2 CE	OC2M[2:0]			OC2 PE	OC2 FE	CC2S[1:0]		OC1 CE	OC1M[2:0]			OC1 PE	OC1 FE	CC1S[1:0]	
IC2F[3:0]			IC2PSC[1:0]		IC1F[3:0]			IC1PSC[1:0]							
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Output compare mode:

Bit 15 **OC2CE**: Output compare 2 clear enable

Bits 14:12 **OC2M[2:0]**: Output compare 2 mode

Bit 11 **OC2PE**: Output compare 2 preload enable

Bit 10 **OC2FE**: Output compare 2 fast enable

Bits 9:8 **CC2S[1:0]**: Capture/Compare 2 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC2 channel is configured as output

01: CC2 channel is configured as input, IC2 is mapped on TI2

10: CC2 channel is configured as input, IC2 is mapped on TI1

11: CC2 channel is configured as input, IC2 is mapped on TRC. This mode is working only if an internal trigger input is selected through the TS bit (TIMx_SMCR register)

Note: CC2S bits are writable only when the channel is OFF (CC2E = '0' in TIMx_CCER).

Bit 7 **OC1CE**: Output compare 1 clear enable

OC1CE: Output compare 1 Clear Enable

0: OC1Ref is not affected by the ETRF Input

1: OC1Ref is cleared as soon as a High level is detected on ETRF input

Bits 6:4 **OC1M**: Output compare 1 mode

These bits define the behavior of the output reference signal OC1REF from which OC1 and OC1N are derived. OC1REF is active high whereas OC1 and OC1N active level depends on CC1P and CC1NP bits.

000: Frozen - The comparison between the output compare register TIMx_CCR1 and the counter TIMx_CNT has no effect on the outputs.(this mode is used to generate a timing base).

001: Set channel 1 to active level on match. OC1REF signal is forced high when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).

010: Set channel 1 to inactive level on match. OC1REF signal is forced low when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).

011: Toggle - OC1REF toggles when TIMx_CNT=TIMx_CCR1.

100: Force inactive level - OC1REF is forced low.

101: Force active level - OC1REF is forced high.

110: PWM mode 1 - In upcounting, channel 1 is active as long as TIMx_CNT<TIMx_CCR1 else inactive. In downcounting, channel 1 is inactive (OC1REF='0') as long as TIMx_CNT>TIMx_CCR1 else active (OC1REF='1').

111: PWM mode 2 - In upcounting, channel 1 is inactive as long as TIMx_CNT<TIMx_CCR1 else active. In downcounting, channel 1 is active as long as TIMx_CNT>TIMx_CCR1 else inactive.

Note: 1: These bits can not be modified as long as LOCK level 3 has been programmed (LOCK bits in TIMx_BDTR register) and CC1S='00' (the channel is configured in output).

2: In PWM mode 1 or 2, the OCREF level changes only when the result of the comparison changes or when the output compare mode switches from "frozen" mode to "PWM" mode.

3: On channels having a complementary output, this bit field is preloaded. If the CCPC bit is set in the TIMx_CR2 register then the OC1M active bits take the new value from the preloaded bits only when a COM event is generated.

Bit 3 **OC1PE**: Output compare 1 preload enable

0: Preload register on TIMx_CCR1 disabled. TIMx_CCR1 can be written at anytime, the new value is taken in account immediately.

1: Preload register on TIMx_CCR1 enabled. Read/Write operations access the preload register. TIMx_CCR1 preload value is loaded in the active register at each update event.

Note: These bits can not be modified as long as LOCK level 3 has been programmed (LOCK bits in TIMx_BDTR register) and CC1S='00' (the channel is configured in output).

Bit 2 **OC1FE**: Output compare 1 fast enable

This bit is used to accelerate the effect of an event on the trigger in input on the CC output.

0: CC1 behaves normally depending on counter and CCR1 values even when the trigger is ON. The minimum delay to activate CC1 output when an edge occurs on the trigger input is 5 clock cycles.

1: An active edge on the trigger input acts like a compare match on CC1 output. Then, OC is set to the compare level independently from the result of the comparison. Delay to sample the trigger input and to activate CC1 output is reduced to 3 clock cycles. OC1FE acts only if the channel is configured in PWM1 or PWM2 mode.

Bits 1:0 **CC1S**: Capture/Compare 1 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output

01: CC1 channel is configured as input, IC1 is mapped on TI1

10: CC1 channel is configured as input, IC1 is mapped on TI2

11: CC1 channel is configured as input, IC1 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC1S bits are writable only when the channel is OFF (CC1E = '0' in TIMx_CCER).

Input capture mode

Bits 15:12 **IC2F**: Input capture 2 filter

Bits 11:10 **IC2PSC[1:0]**: Input capture 2 prescaler

Bits 9:8 **CC2S**: Capture/Compare 2 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC2 channel is configured as output

01: CC2 channel is configured as input, IC2 is mapped on TI2

10: CC2 channel is configured as input, IC2 is mapped on TI1

11: CC2 channel is configured as input, IC2 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC2S bits are writable only when the channel is OFF (CC2E = '0' in TIMx_CCER).

Bits 7:4 **IC1F[3:0]**: Input capture 1 filter

This bit-field defines the frequency used to sample TI1 input and the length of the digital filter applied to TI1. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, sampling is done at f_{DTS}

0001: $f_{SAMPLING}=f_{CK_INT}$, N=2

0010: $f_{SAMPLING}=f_{CK_INT}$, N=4

0011: $f_{SAMPLING}=f_{CK_INT}$, N=8

0100: $f_{SAMPLING}=f_{DTS}/2$, N=6

0101: $f_{SAMPLING}=f_{DTS}/2$, N=8

0110: $f_{SAMPLING}=f_{DTS}/4$, N=6

0111: $f_{SAMPLING}=f_{DTS}/4$, N=8

1000: $f_{SAMPLING}=f_{DTS}/8$, N=6

1001: $f_{SAMPLING}=f_{DTS}/8$, N=8

1010: $f_{SAMPLING}=f_{DTS}/16$, N=5

1011: $f_{SAMPLING}=f_{DTS}/16$, N=6

1100: $f_{SAMPLING}=f_{DTS}/16$, N=8

1101: $f_{SAMPLING}=f_{DTS}/32$, N=5

1110: $f_{SAMPLING}=f_{DTS}/32$, N=6

1111: $f_{SAMPLING}=f_{DTS}/32$, N=8

Bits 3:2 **IC1PSC**: Input capture 1 prescaler

This bit-field defines the ratio of the prescaler acting on CC1 input (IC1).

The prescaler is reset as soon as CC1E='0' (TIMx_CCER register).

00: no prescaler, capture is done each time an edge is detected on the capture input

01: capture is done once every 2 events

10: capture is done once every 4 events

11: capture is done once every 8 events

Bits 1:0 **CC1S**: Capture/Compare 1 Selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output

01: CC1 channel is configured as input, IC1 is mapped on TI1

10: CC1 channel is configured as input, IC1 is mapped on TI2

11: CC1 channel is configured as input, IC1 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC1S bits are writable only when the channel is OFF (CC1E = '0' in TIMx_CCER).

17.4.8 TIM1 and TIM8 capture/compare mode register 2 (TIMx_CCMR2)

Address offset: 0x1C

Reset value: 0x0000

Refer to the above CCMR1 register description.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC4 CE	OC4M[2:0]			OC4 PE	OC4 FE	CC4S[1:0]		OC3 CE.	OC3M[2:0]			OC3 PE	OC3 FE	CC3S[1:0]	
IC4F[3:0]				IC4PSC[1:0]				IC3F[3:0]				IC3PSC[1:0]			
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Output compare mode

Bit 15 **OC4CE**: Output compare 4 clear enable

Bits 14:12 **OC4M**: Output compare 4 mode

Bit 11 **OC4PE**: Output compare 4 preload enable

Bit 10 **OC4FE**: Output compare 4 fast enable

Bits 9:8 **CC4S**: Capture/Compare 4 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC4 channel is configured as output

01: CC4 channel is configured as input, IC4 is mapped on TI4

10: CC4 channel is configured as input, IC4 is mapped on TI3

11: CC4 channel is configured as input, IC4 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC4S bits are writable only when the channel is OFF (CC4E = '0' in TIMx_CCER).

Bit 7 **OC3CE**: Output compare 3 clear enable

Bits 6:4 **OC3M**: Output compare 3 mode

Bit 3 **OC3PE**: Output compare 3 preload enable

Bit 2 **OC3FE**: Output compare 3 fast enable

Bits 1:0 **CC3S**: Capture/Compare 3 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC3 channel is configured as output

01: CC3 channel is configured as input, IC3 is mapped on TI3

10: CC3 channel is configured as input, IC3 is mapped on TI4

11: CC3 channel is configured as input, IC3 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC3S bits are writable only when the channel is OFF (CC3E = '0' in TIMx_CCER).

Input capture mode

Bits 15:12 **IC4F**: Input capture 4 filter

Bits 11:10 **IC4PSC**: Input capture 4 prescaler

Bits 9:8 **CC4S**: Capture/Compare 4 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC4 channel is configured as output

01: CC4 channel is configured as input, IC4 is mapped on TI4

10: CC4 channel is configured as input, IC4 is mapped on TI3

11: CC4 channel is configured as input, IC4 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC4S bits are writable only when the channel is OFF (CC4E = '0' in TIMx_CCER).

Bits 7:4 **IC3F**: Input capture 3 filter

Bits 3:2 **IC3PSC**: Input capture 3 prescaler

Bits 1:0 **CC3S**: Capture/compare 3 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC3 channel is configured as output

01: CC3 channel is configured as input, IC3 is mapped on TI3

10: CC3 channel is configured as input, IC3 is mapped on TI4

11: CC3 channel is configured as input, IC3 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC3S bits are writable only when the channel is OFF (CC3E = '0' in TIMx_CCER).

17.4.9 TIM1 and TIM8 capture/compare enable register (TIMx_CCER)

Address offset: 0x20

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CC4NP	Res.	CC4P	CC4E	CC3NP	CC3NE	CC3P	CC3E	CC2NP	CC2NE	CC2P	CC2E	CC1NP	CC1NE	CC1P	CC1E
rw		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 15 **CC4NP**: Capture/Compare 4 complementary output polarity
refer to CC1NP description

Bit 14 Reserved, must be kept at reset value.

Bit 13 **CC4P**: Capture/Compare 4 output polarity
refer to CC1P description

Bit 12 **CC4E**: Capture/Compare 4 output enable
refer to CC1E description

Bit 11 **CC3NP**: Capture/Compare 3 complementary output polarity
refer to CC1NP description

Bit 10 **CC3NE**: Capture/Compare 3 complementary output enable
refer to CC1NE description

Bit 9 **CC3P**: Capture/Compare 3 output polarity
refer to CC1P description

- Bit 8 **CC3E**: Capture/Compare 3 output enable
refer to CC1E description
- Bit 7 **CC2NP**: Capture/Compare 2 complementary output polarity
refer to CC1NP description
- Bit 6 **CC2NE**: Capture/Compare 2 complementary output enable
refer to CC1NE description
- Bit 5 **CC2P**: Capture/Compare 2 output polarity
refer to CC1P description
- Bit 4 **CC2E**: Capture/Compare 2 output enable
refer to CC1E description
- Bit 3 **CC1NP**: Capture/Compare 1 complementary output polarity
CC1 channel configured as output:
0: OC1N active high.
1: OC1N active low.
CC1 channel configured as input:
This bit is used in conjunction with CC1P to define the polarity of TI1FP1 and TI2FP1. Refer to CC1P description.
- Note: On channels having a complementary output, this bit is preloaded. If the CCPC bit is set in the TIMx_CR2 register then the CC1NP active bit takes the new value from the preloaded bit only when a Commutation event is generated.*
- Note: This bit is not writable as soon as LOCK level 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register) and CC1S="00" (the channel is configured in output).*
- Bit 2 **CC1NE**: Capture/Compare 1 complementary output enable
0: Off - OC1N is not active. OC1N level is then function of MOE, OSSI, OSSR, OIS1, OIS1N and CC1E bits.
1: On - OC1N signal is output on the corresponding output pin depending on MOE, OSSI, OSSR, OIS1, OIS1N and CC1E bits.
- Note: On channels having a complementary output, this bit is preloaded. If the CCPC bit is set in the TIMx_CR2 register then the CC1NE active bit takes the new value from the preloaded bit only when a Commutation event is generated.*

Bit 1 **CC1P**: Capture/Compare 1 output polarity

CC1 channel configured as output:

0: OC1 active high

1: OC1 active low

CC1 channel configured as input:

CC1NP/CC1P bits select the active polarity of TI1FP1 and TI2FP1 for trigger or capture operations.

00: non-inverted/rising edge

The circuit is sensitive to TIxFP1 rising edge (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger operation in gated mode or encoder mode).

01: inverted/falling edge

The circuit is sensitive to TIxFP1 falling edge (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is inverted (trigger operation in gated mode or encoder mode).

10: reserved, do not use this configuration.

11: non-inverted/both edges

The circuit is sensitive to both TIxFP1 rising and falling edges (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger operation in gated mode). This configuration must not be used in encoder mode.

Note: On channels having a complementary output, this bit is preloaded. If the CCPC bit is set in the TIMx_CR2 register then the CC1P active bit takes the new value from the preloaded bit only when a Commutation event is generated.

Note: This bit is not writable as soon as LOCK level 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 0 **CC1E**: Capture/Compare 1 output enable

CC1 channel configured as output:

0: Off - OC1 is not active. OC1 level is then function of MOE, OSS1, OSSR, OIS1, OIS1N and CC1NE bits.

1: On - OC1 signal is output on the corresponding output pin depending on MOE, OSS1, OSSR, OIS1, OIS1N and CC1NE bits.

CC1 channel configured as input:

This bit determines if a capture of the counter value can actually be done into the input capture/compare register 1 (TIMx_CCR1) or not.

0: Capture disabled.

1: Capture enabled.

Note: On channels having a complementary output, this bit is preloaded. If the CCPC bit is set in the TIMx_CR2 register then the CC1E active bit takes the new value from the preloaded bit only when a Commutation event is generated.

Table 96. Output control bits for complementary OCx and OCxN channels with break feature

Control bits					Output states ⁽¹⁾	
MOE bit	OSSI bit	OSSR bit	CCxE bit	CCxNE bit	OCx output state	OCxN output state
1	X	0	0	0	Output Disabled (not driven by the timer), OCx=0, OCx_EN=0	Output Disabled (not driven by the timer), OCxN=0, OCxN_EN=0
		0	0	1	Output Disabled (not driven by the timer), OCx=0, OCx_EN=0	OCxREF + Polarity OCxN=OCxREF xor CCxNP, OCxN_EN=1
		0	1	0	OCxREF + Polarity OCx=OCxREF xor CCxP, OCx_EN=1	Output Disabled (not driven by the timer) OCxN=0, OCxN_EN=0
		0	1	1	OCREF + Polarity + dead-time OCx_EN=1	Complementary to OCREF (not OCREF) + Polarity + dead-time OCxN_EN=1
		1	0	0	Output Disabled (not driven by the timer) OCx=CCxP, OCx_EN=0	Output Disabled (not driven by the timer) OCxN=CCxNP, OCxN_EN=0
		1	0	1	Off-State (output enabled with inactive state) OCx=CCxP, OCx_EN=1	OCxREF + Polarity OCxN=OCxREF xor CCxNP, OCxN_EN=1
		1	1	0	OCxREF + Polarity OCx=OCxREF xor CCxP, OCx_EN=1	Off-State (output enabled with inactive state) OCxN=CCxNP, OCxN_EN=1
		1	1	1	OCREF + Polarity + dead-time OCx_EN=1	Complementary to OCREF (not OCREF) + Polarity + dead-time OCxN_EN=1
0	0	X	X	X	Output disabled (not driven by the timer anymore). The output state is defined by the GPIO controller and can be High, Low or Hi-Z.	
	0		0			
	1		0	1	Off-State (output enabled with inactive state)	
			1	0	Asynchronously: OCx=CCxP, OCx_EN=1, OCxN=CCxNP, OCxN_EN=1	
			1	1	Then if the clock is present: OCx=OISx and OCxN=OISxN after a dead-time, assuming that OISx and OISxN do not correspond to OCx and OCxN both in active state	

1. When both outputs of a channel are not used (CCxE = CCxNE = 0), the OISx, OISxN, CCxP and CCxNP bits must be kept cleared.

Note: The state of the external I/O pins connected to the complementary OCx and OCxN channels depends on the OCx and OCxN channel state and the GPIO registers.

17.4.10 TIM1 and TIM8 counter (TIMx_CNT)

Address offset: 0x24

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **CNT[15:0]**: Counter value**17.4.11 TIM1 and TIM8 prescaler (TIMx_PSC)**

Address offset: 0x28

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **PSC[15:0]**: Prescaler valueThe counter clock frequency (f_{CK_CNT}) is equal to $f_{CK_PSC} / (PSC[15:0] + 1)$.

PSC contains the value to be loaded in the active prescaler register at each update event (including when the counter is cleared through UG bit of TIMx_EGR register or through trigger controller when configured in “reset mode”).

17.4.12 TIM1 and TIM8 auto-reload register (TIMx_ARR)

Address offset: 0x2C

Reset value: 0xFFFF

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **ARR[15:0]**: Auto-reload value

ARR is the value to be loaded in the actual auto-reload register.

Refer to [Section 17.3.1: Time-base unit](#) for more details about ARR update and behavior.

The counter is blocked while the auto-reload value is null.

17.4.13 TIM1 and TIM8 repetition counter register (TIMx_RCR)

Address offset: 0x30

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								REP[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:8 Reserved, must be kept at reset value.

Bits 7:0 **REP[7:0]**: Repetition counter value

These bits allow the user to set-up the update rate of the compare registers (i.e. periodic transfers from preload to active registers) when preload registers are enable, as well as the update interrupt generation rate, if this interrupt is enable.

Each time the REP_CNT related downcounter reaches zero, an update event is generated and it restarts counting from REP value. As REP_CNT is reloaded with REP value only at the repetition update event U_RC, any write to the TIMx_RCR register is not taken in account until the next repetition update event.

It means in PWM mode (REP+1) corresponds to:

- the number of PWM periods in edge-aligned mode
- the number of half PWM period in center-aligned mode.

17.4.14 TIM1 and TIM8 capture/compare register 1 (TIMx_CCR1)

Address offset: 0x34

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1[15:0]															
rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro

Bits 15:0 **CCR1[15:0]**: Capture/Compare 1 value

If channel CC1 is configured as output:

CCR1 is the value to be loaded in the actual capture/compare 1 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR1 register (bit OC1PE). Else the preload value is copied in the active capture/compare 1 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signaled on OC1 output.

If channel CC1 is configured as input:

CCR1 is the counter value transferred by the last input capture 1 event (IC1). The TIMx_CCR1 register is read-only and cannot be programmed.

17.4.15 TIM1 and TIM8 capture/compare register 2 (TIMx_CCR2)

Address offset: 0x38

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR2[15:0]															
rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro

Bits 15:0 **CCR2[15:0]**: Capture/Compare 2 value

If channel CC2 is configured as output:

CCR2 is the value to be loaded in the actual capture/compare 2 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR2 register (bit OC2PE). Else the preload value is copied in the active capture/compare 2 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signalled on OC2 output.

If channel CC2 is configured as input:

CCR2 is the counter value transferred by the last input capture 2 event (IC2). The TIMx_CCR2 register is read-only and cannot be programmed.

17.4.16 TIM1 and TIM8 capture/compare register 3 (TIMx_CCR3)

Address offset: 0x3C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR3[15:0]															
rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro

Bits 15:0 **CCR3[15:0]**: Capture/Compare value

If channel CC3 is configured as output:

CCR3 is the value to be loaded in the actual capture/compare 3 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR3 register (bit OC3PE). Else the preload value is copied in the active capture/compare 3 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signalled on OC3 output.

If channel CC3 is configured as input:

CCR3 is the counter value transferred by the last input capture 3 event (IC3). The TIMx_CCR3 register is read-only and cannot be programmed.

17.4.17 TIM1 and TIM8 capture/compare register 4 (TIMx_CCR4)

Address offset: 0x40

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR4[15:0]															
rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro

Bits 15:0 **CCR4[15:0]**: Capture/Compare value

If channel CC4 is configured as output:

CCR4 is the value to be loaded in the actual capture/compare 4 register (preload value). It is loaded permanently if the preload feature is not selected in the TIMx_CCMR4 register (bit OC4PE). Else the preload value is copied in the active capture/compare 4 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signalled on OC4 output.

If channel CC4 is configured as input:

CCR4 is the counter value transferred by the last input capture 4 event (IC4). The TIMx_CCR3 register is read-only and cannot be programmed.

17.4.18 TIM1 and TIM8 break and dead-time register (TIMx_BDTR)

Address offset: 0x44

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MOE	AOE	BKP	BKE	OSSR	OSSI	LOCK[1:0]		DTG[7:0]							
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Note: As the bits AOE, BKP, BKE, OSSI, OSSR and DTG[7:0] can be write-locked depending on the LOCK configuration, it can be necessary to configure all of them during the first write access to the TIMx_BDTR register.

Bit 15 **MOE**: Main output enable

This bit is cleared asynchronously by hardware as soon as the break input is active. It is set by software or automatically depending on the AOE bit. It is acting only on the channels which are configured in output.

0: OC and OCN outputs are disabled or forced to idle state.

1: OC and OCN outputs are enabled if their respective enable bits are set (CCxE, CCxNE in TIMx_CCER register).

See OC/OCN enable description for more details ([Section 17.4.9: TIM1 and TIM8 capture/compare enable register \(TIMx_CCER\)](#)).

Bit 14 **AOE**: Automatic output enable

0: MOE can be set only by software

1: MOE can be set by software or automatically at the next update event (if the break input is not be active)

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 13 **BKP**: Break polarity

- 0: Break input BRK is active low
- 1: Break input BRK is active high

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Note: Any write operation to this bit takes a delay of 1 APB clock cycle to become effective.

Bit 12 **BKE**: Break enable

- 0: Break inputs (BRK and CSS clock failure event) disabled
- 1: Break inputs (BRK and CSS clock failure event) enabled

Note: This bit cannot be modified when LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Note: Any write operation to this bit takes a delay of 1 APB clock cycle to become effective.

Bit 11 **OSSR**: Off-state selection for Run mode

This bit is used when MOE=1 on channels having a complementary output which are configured as outputs. OSSR is not implemented if no complementary output is implemented in the timer.

See OC/OCN enable description for more details ([Section 17.4.9: TIM1 and TIM8 capture/compare enable register \(TIMx_CCER\)](#)).

- 0: When inactive, OC/OCN outputs are disabled (OC/OCN enable output signal=0).
- 1: When inactive, OC/OCN outputs are enabled with their inactive level as soon as CCxE=1 or CCxNE=1. Then, OC/OCN enable output signal=1

Note: This bit can not be modified as soon as the LOCK level 2 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 10 **OSSI**: Off-state selection for Idle mode

This bit is used when MOE=0 on channels configured as outputs.

See OC/OCN enable description for more details ([Section 17.4.9: TIM1 and TIM8 capture/compare enable register \(TIMx_CCER\)](#)).

- 0: When inactive, OC/OCN outputs are disabled (OC/OCN enable output signal=0).
- 1: When inactive, OC/OCN outputs are forced first with their idle level as soon as CCxE=1 or CCxNE=1. OC/OCN enable output signal=1

Note: This bit can not be modified as soon as the LOCK level 2 has been programmed (LOCK bits in TIMx_BDTR register).

Bits 9:8 **LOCK[1:0]**: Lock configuration

These bits offer a write protection against software errors.

- 00: LOCK OFF - No bit is write protected.
- 01: LOCK Level 1 = DTG bits in TIMx_BDTR register, OISx and OISxN bits in TIMx_CR2 register and BKE/BKP/AOE bits in TIMx_BDTR register can no longer be written.
- 10: LOCK Level 2 = LOCK Level 1 + CC Polarity bits (CCxP/CCxNP bits in TIMx_CCER register, as long as the related channel is configured in output through the CCxS bits) as well as OSSR and OSSI bits can no longer be written.
- 11: LOCK Level 3 = LOCK Level 2 + CC Control bits (OCxM and OCxPE bits in TIMx_CCMRx registers, as long as the related channel is configured in output through the CCxS bits) can no longer be written.

Note: The LOCK bits can be written only once after the reset. Once the TIMx_BDTR register has been written, their content is frozen until the next reset.

Bits 7:0 **DTG[7:0]**: Dead-time generator setup

This bit-field defines the duration of the dead-time inserted between the complementary outputs. DT correspond to this duration.

DTG[7:5]=0xx => $DT = DTG[7:0] \times t_{dtg}$ with $t_{dtg} = t_{DTS}$.

DTG[7:5]=10x => $DT = (64 + DTG[5:0]) \times t_{dtg}$ with $T_{dtg} = 2 \times t_{DTS}$.

DTG[7:5]=110 => $DT = (32 + DTG[4:0]) \times t_{dtg}$ with $T_{dtg} = 8 \times t_{DTS}$.

DTG[7:5]=111 => $DT = (32 + DTG[4:0]) \times t_{dtg}$ with $T_{dtg} = 16 \times t_{DTS}$.

Example if $T_{DTS} = 125\text{ns}$ (8MHz), dead-time possible values are:

0 to 15875 ns by 125 ns steps,

16 us to 31750 ns by 250 ns steps,

32 us to 63us by 1 us steps,

64 us to 126 us by 2 us steps

Note: This bit-field can not be modified as long as LOCK level 1, 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register).

17.4.19 TIM1 and TIM8 DMA control register (TIMx_DCR)

Address offset: 0x48

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			DBL[4:0]					Reserved			DBA[4:0]				
			rw	rw	rw	rw	rw				rw	rw	rw	rw	rw

Bits 15:13 Reserved, must be kept at reset value.

Bits 12:8 **DBL[4:0]**: DMA burst length

This 5-bit vector defines the number of DMA transfers (the timer detects a burst transfer when a read or a write access to the TIMx_DMAR register address is performed).

the TIMx_DMAR address)

00000: 1 transfer

00001: 2 transfers

00010: 3 transfers

...

10001: 18 transfers

Bits 7:5 Reserved, must be kept at reset value.

Bits 4:0 **DBA[4:0]**: DMA base address

This 5-bits vector defines the base-address for DMA transfers (when read/write access are done through the TIMx_DMAR address). DBA is defined as an offset starting from the address of the TIMx_CR1 register.

Example:

00000: TIMx_CR1,

00001: TIMx_CR2,

00010: TIMx_SMCR,

...

Example: Let us consider the following transfer: DBL = 7 transfers and DBA = TIMx_CR1. In this case the transfer is done to/from 7 registers starting from the TIMx_CR1 address.

17.4.20 TIM1 and TIM8 DMA address for full transfer (TIMx_DMAR)

Address offset: 0x4C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DMAB[31:16]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMAB[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **DMAB[31:0]**: DMA register for burst accesses

A read or write operation to the DMAR register accesses the register located at the address
 $(\text{TIMx_CR1 address}) + (\text{DBA} + \text{DMA index}) \times 4$

where TIMx_CR1 address is the address of the control register 1, DBA is the DMA base address configured in TIMx_DCR register, DMA index is automatically controlled by the DMA transfer, and ranges from 0 to DBL (DBL configured in TIMx_DCR).

Example of how to use the DMA burst feature

In this example the timer DMA burst feature is used to update the contents of the CCRx registers (x = 2, 3, 4) with the DMA transferring half words into the CCRx registers.

This is done in the following steps:

1. Configure the corresponding DMA channel as follows:
 - DMA channel peripheral address is the DMAR register address
 - DMA channel memory address is the address of the buffer in the RAM containing the data to be transferred by DMA into CCRx registers.
 - Number of data to transfer = 3 (See note below).
 - Circular mode disabled.
2. Configure the DCR register by configuring the DBA and DBL bit fields as follows:
 DBL = 3 transfers, DBA = 0xE.
3. Enable the TIMx update DMA request (set the UDE bit in the DIER register).
4. Enable TIMx
5. Enable the DMA channel

Note: *This example is for the case where every CCRx register to be updated once. If every CCRx register is to be updated twice for example, the number of data to transfer should be 6. Let's take the example of a buffer in the RAM containing data1, data2, data3, data4, data5 and data6. The data is transferred to the CCRx registers as follows: on the first update DMA request, data1 is transferred to CCR2, data2 is transferred to CCR3, data3 is transferred to CCR4 and on the second update DMA request, data4 is transferred to CCR2, data5 is transferred to CCR3 and data6 is transferred to CCR4.*

17.4.21 TIM1 and TIM8 register map

TIM1 and TIM8 registers are mapped as 16-bit addressable registers as described in the table below:

Table 97. TIM1 and TIM8 register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0x00	TIMx_CR1	Reserved																						CKD [1:0]		ARPE	CMS [1:0]		DIR	OPM	URS	UDIS	CEN		
	Reset value																							0	0	0	0	0	0	0	0				
0x04	TIMx_CR2	Reserved																		OIS4	OIS3N	OIS3	OIS2N	OIS2	OIS1N	OIS1	TI1S	MMS[2:0]		CCDS	CCUS	Reserved	CCPC		
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x08	TIMx_SMCR	Reserved																		ETP	ECE	ETPS [1:0]		ETF[3:0]		MSM	TS[2:0]		Reserved	SMS[2:0]					
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0C	TIMx_DIER	Reserved																		TDE	COMDE	CC4DE	CC3DE	CC2DE	CC1DE	UDE	BIE	TIE	COMIE	CC4IE	CC3IE	CC2IE	CC1IE	UIE	
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x10	TIMx_SR	Reserved																		CC4OF		CC3OF	CC2OF	CC1OF	Reserved	BIF	TIF	COMIF	CC4IF	CC3IF	CC2IF	CC1IF	UIF		
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x14	TIMx_EGR	Reserved																						BG		TG	COMG	CC4G	CC3G	CC2G	CC1G	UG			
	Reset value																							0	0	0	0	0	0	0	0	0	0		
0x18	TIMx_CCMR1 Output compare mode	Reserved																		OC2CE	OC2M [2:0]		OC2PE	OC2FE	CC2S [1:0]		OC1CE	OC1M [2:0]		OC1PE	OC1FE	CC1S [1:0]			
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	TIMx_CCMR1 Input capture mode	Reserved																		IC2F[3:0]			IC2 PSC [1:0]	CC2S [1:0]		IC1F[3:0]			IC1 PSC [1:0]	CC1S [1:0]					
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x1C	TIMx_CCMR2 Output compare mode	Reserved																		OC4CE	OC4M [2:0]		OC4PE	OC4FE	CC4S [1:0]		OC3CE	OC3M [2:0]		OC3PE	OC3FE	CC3S [1:0]			
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	TIMx_CCMR2 Input capture mode	Reserved																		IC4F[3:0]			IC4 PSC [1:0]	CC4S [1:0]		IC3F[3:0]			IC3 PSC [1:0]	CC3S [1:0]					
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x20	TIMx_CCER	Reserved																		CC4NP	Reserved	CC4P	CC4E	CC3NP	CC3NE	CC3P	CC3E	CC2NP	CC2NE	CC2P	CC2E	CC1NP	CC1NE	CC1P	CC1E
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x24	TIMx_CNT	Reserved																		CNT[15:0]															
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x28	TIMx_PSC	Reserved																		PSC[15:0]															
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 97. TIM1 and TIM8 register map and reset values (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x2C	TIMx_ARR	Reserved																ARR[15:0]															
	Reset value																	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0x30	TIMx_RCR	Reserved																							REP[7:0]								
	Reset value																								0	0	0	0	0	0	0	0	0
0x34	TIMx_CCR1	Reserved																CCR1[15:0]															
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x38	TIMx_CCR2	Reserved																CCR2[15:0]															
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x3C	TIMx_CCR3	Reserved																CCR3[15:0]															
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x40	TIMx_CCR4	Reserved																CCR4[15:0]															
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x44	TIMx_BDTR	Reserved																MOE	AOE	BKP	BKE	OSR	OSSI	LOCK [1:0]	DT[7:0]								
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x48	TIMx_DCR	Reserved																				DBL[4:0]				Reserved				DBA[4:0]			
	Reset value																					0	0	0	0					0	0	0	0
0x4C	TIMx_DMAR	DMAB[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.

18 General-purpose timers (TIM2 to TIM5)

This section applies to the whole STM32F4xx family, unless otherwise specified.

18.1 TIM2 to TIM5 introduction

The general-purpose timers consist of a 16-bit or 32-bit auto-reload counter driven by a programmable prescaler.

They may be used for a variety of purposes, including measuring the pulse lengths of input signals (*input capture*) or generating output waveforms (*output compare and PWM*).

Pulse lengths and waveform periods can be modulated from a few microseconds to several milliseconds using the timer prescaler and the RCC clock controller prescalers.

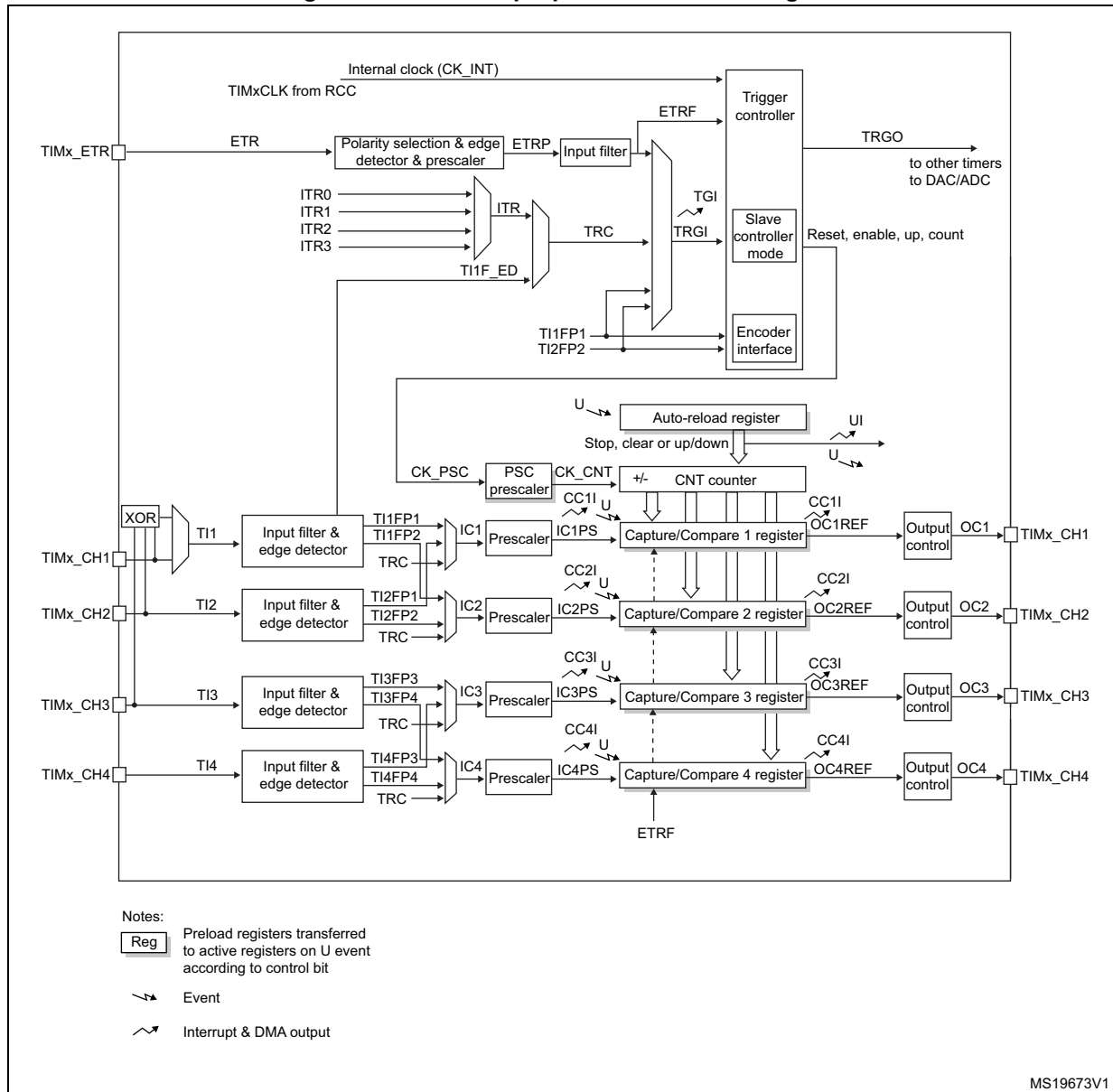
The timers are completely independent, and do not share any resources. They can be synchronized together as described in [Section 18.3.15](#).

18.2 TIM2 to TIM5 main features

General-purpose TIMx timer features include:

- 16-bit (TIM3 and TIM4) or 32-bit (TIM2 and TIM5) up, down, up/down auto-reload counter.
- 16-bit programmable prescaler used to divide (also “on the fly”) the counter clock frequency by any factor between 1 and 65536.
- Up to four independent channels for:
 - Input capture
 - Output compare
 - PWM generation (Edge- and Center-aligned modes)
 - One-pulse mode output
- Synchronization circuit to control the timer with external signals and to interconnect several timers.
- Interrupt/DMA generation on the following events:
 - Update: counter overflow/underflow, counter initialization (by software or internal/external trigger)
 - Trigger event (counter start, stop, initialization or count by internal/external trigger)
 - Input capture
 - Output compare
- Supports incremental (quadrature) encoder and hall-sensor circuitry for positioning purposes
- Trigger input for external clock or cycle-by-cycle current management

Figure 134. General-purpose timer block diagram



18.3 TIM2 to TIM5 functional description

18.3.1 Time-base unit

The main block of the programmable timer is a 16-bit/32-bit counter with its related auto-reload register. The counter can count up. The counter clock can be divided by a prescaler.

The counter, the auto-reload register and the prescaler register can be written or read by software. This is true even when the counter is running.

The time-base unit includes:

- Counter register (TIMx_CNT)
- Prescaler register (TIMx_PSC):
- Auto-Reload register (TIMx_ARR)

The auto-reload register is preloaded. Writing to or reading from the auto-reload register accesses the preload register. The content of the preload register are transferred into the shadow register permanently or at each update event (UEV), depending on the auto-reload preload enable bit (ARPE) in TIMx_CR1 register. The update event is sent when the counter reaches the overflow (or underflow when downcounting) and if the UDIS bit equals 0 in the TIMx_CR1 register. It can also be generated by software. The generation of the update event is described in detail for each configuration.

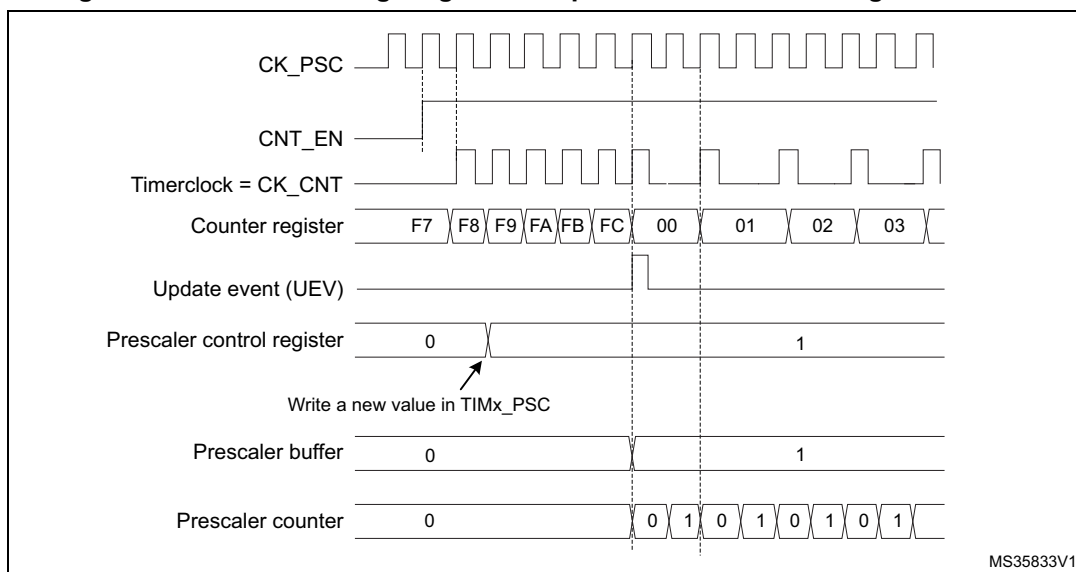
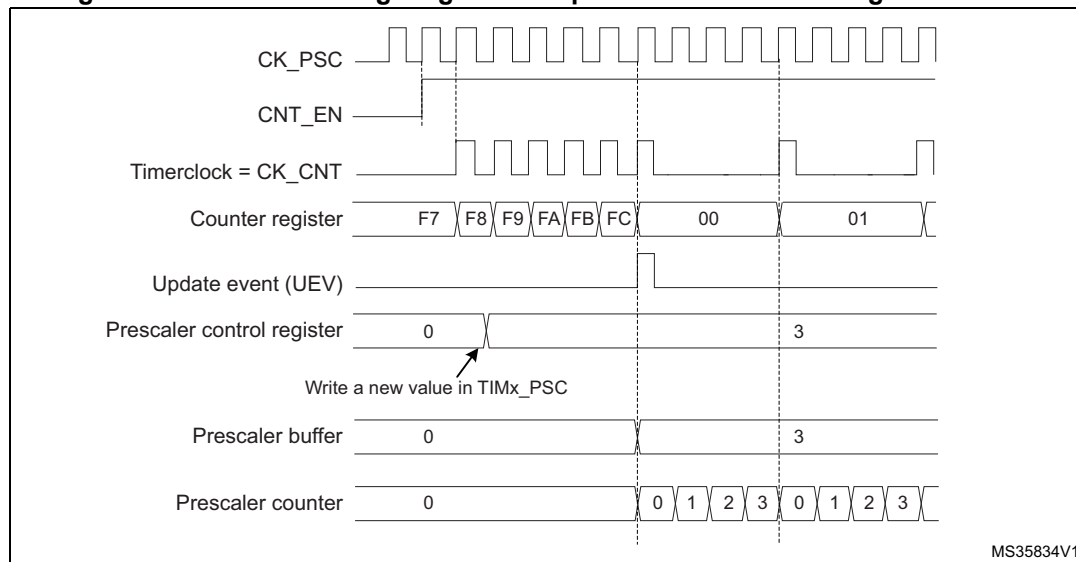
The counter is clocked by the prescaler output CK_CNT, which is enabled only when the counter enable bit (CEN) in TIMx_CR1 register is set (refer also to the slave mode controller description to get more details on counter enabling).

Note that the actual counter enable signal CNT_EN is set 1 clock cycle after CEN.

Prescaler description

The prescaler can divide the counter clock frequency by any factor between 1 and 65536. It is based on a 16-bit counter controlled through a 16-bit/32-bit register (in the TIMx_PSC register). It can be changed on the fly as this control register is buffered. The new prescaler ratio is taken into account at the next update event.

[Figure 135](#) and [Figure 136](#) give some examples of the counter behavior when the prescaler ratio is changed on the fly:

Figure 135. Counter timing diagram with prescaler division change from 1 to 2**Figure 136. Counter timing diagram with prescaler division change from 1 to 4**

18.3.2 Counter modes

Upcounting mode

In upcounting mode, the counter counts from 0 to the auto-reload value (content of the TIMx_ARR register), then restarts from 0 and generates a counter overflow event.

An Update event can be generated at each counter overflow or by setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller).

The UEV event can be disabled by software by setting the UDIS bit in TIMx_CR1 register. This is to avoid updating the shadow registers while writing new values in the preload registers. Then no update event occurs until the UDIS bit has been written to 0. However, the counter restarts from 0, as well as the counter of the prescaler (but the prescale rate

does not change). In addition, if the URS bit (update request selection) in TIMx_CR1 register is set, setting the UG bit generates an update event UEV but without setting the UIF flag (thus no interrupt or DMA request is sent). This is to avoid generating both update and capture interrupts when clearing the counter on the capture event.

When an update event occurs, all the registers are updated and the update flag (UIF bit in TIMx_SR register) is set (depending on the URS bit):

- The buffer of the prescaler is reloaded with the preload value (content of the TIMx_PSC register)
- The auto-reload shadow register is updated with the preload value (TIMx_ARR)

The following figures show some examples of the counter behavior for different clock frequencies when TIMx_ARR=0x36.

Figure 137. Counter timing diagram, internal clock divided by 1

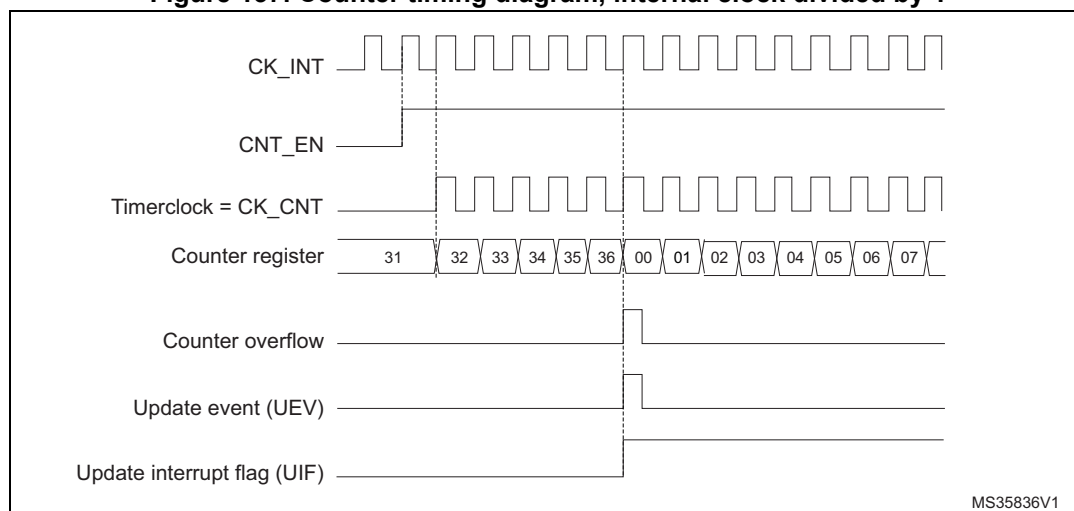


Figure 138. Counter timing diagram, internal clock divided by 2

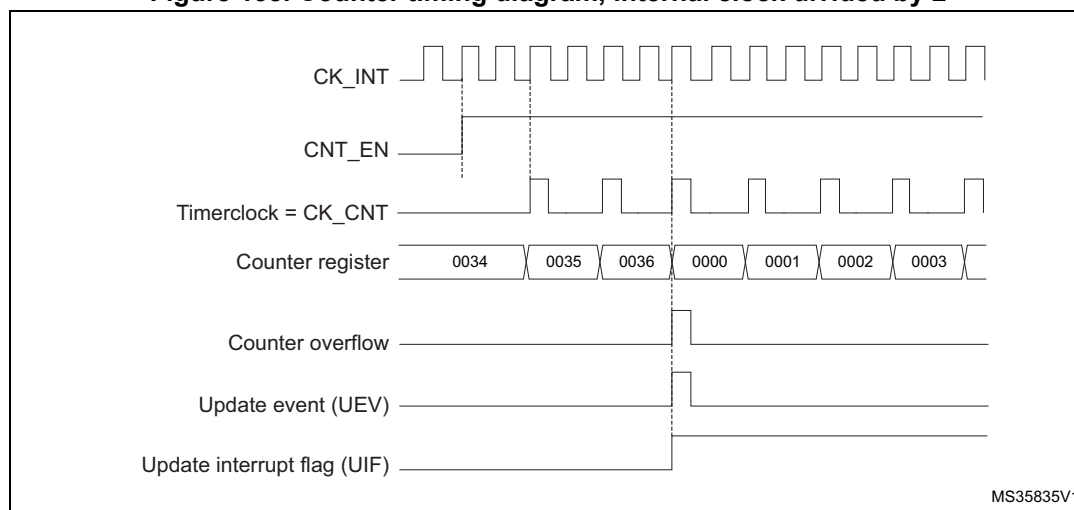


Figure 139. Counter timing diagram, internal clock divided by 4

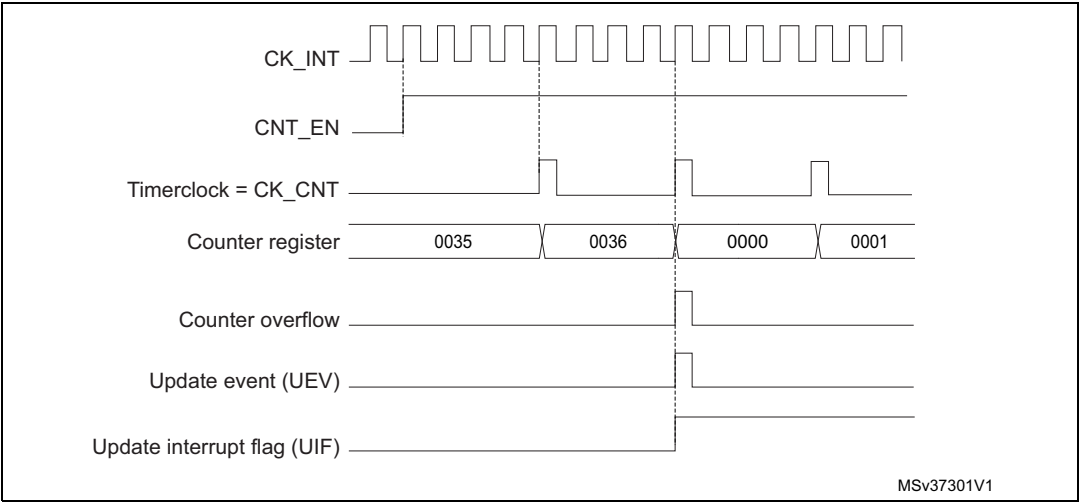


Figure 140. Counter timing diagram, internal clock divided by N

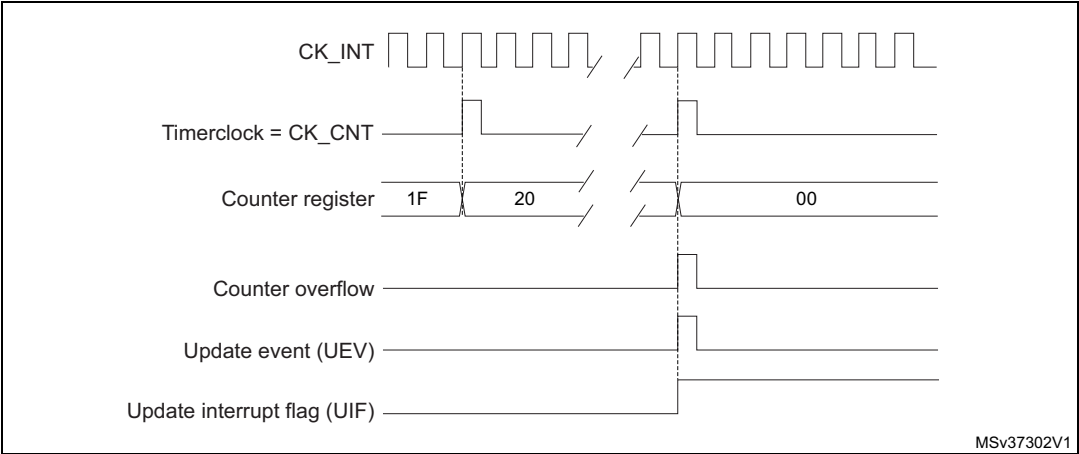
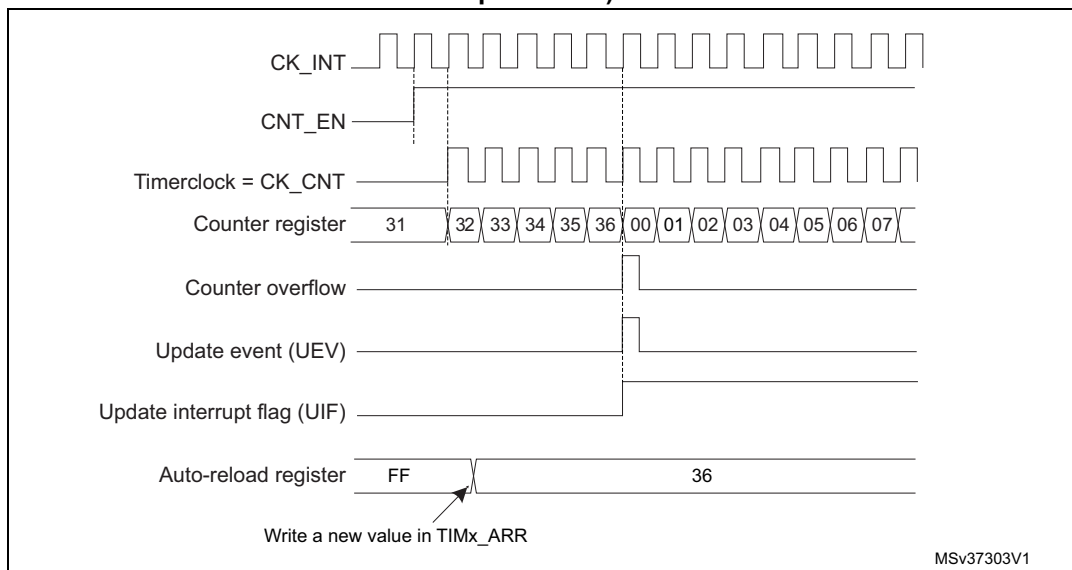
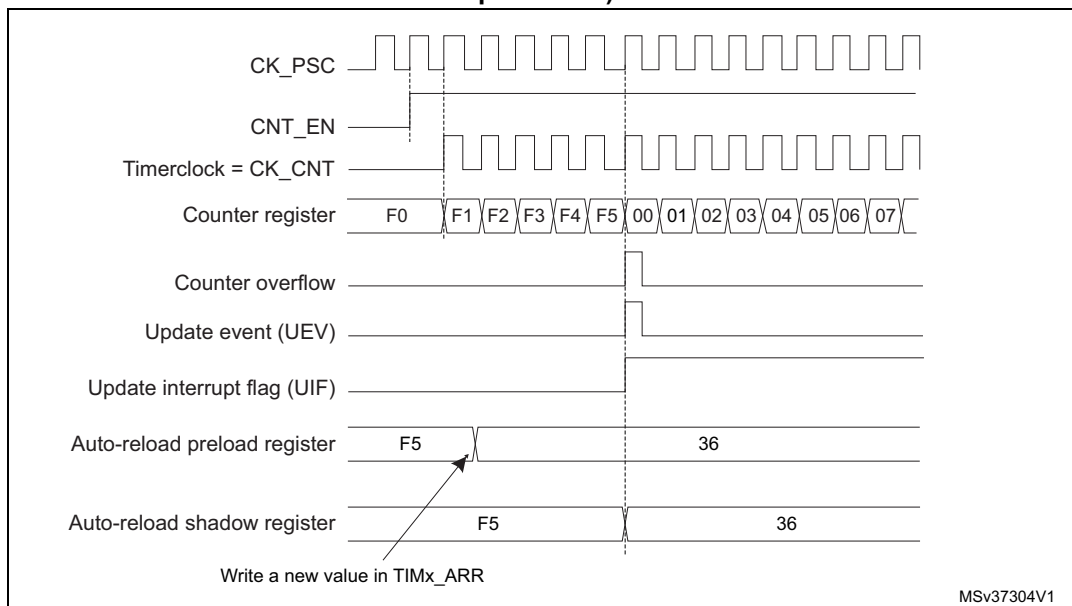


Figure 141. Counter timing diagram, Update event when ARPE=0 (TIMx_ARR not preloaded)**Figure 142. Counter timing diagram, Update event when ARPE=1 (TIMx_ARR preloaded)**

Downcounting mode

In downcounting mode, the counter counts from the auto-reload value (content of the TIMx_ARR register) down to 0, then restarts from the auto-reload value and generates a counter underflow event.

An Update event can be generated at each counter underflow or by setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller)

The UEV update event can be disabled by software by setting the UDIS bit in TIMx_CR1 register. This is to avoid updating the shadow registers while writing new values in the

preload registers. Then no update event occurs until UDIS bit has been written to 0. However, the counter restarts from the current auto-reload value, whereas the counter of the prescaler restarts from 0 (but the prescale rate does not change).

In addition, if the URS bit (update request selection) in TIMx_CR1 register is set, setting the UG bit generates an update event UEV but without setting the UIF flag (thus no interrupt or DMA request is sent). This is to avoid generating both update and capture interrupts when clearing the counter on the capture event.

When an update event occurs, all the registers are updated and the update flag (UIF bit in TIMx_SR register) is set (depending on the URS bit):

- The buffer of the prescaler is reloaded with the preload value (content of the TIMx_PSC register).
- The auto-reload active register is updated with the preload value (content of the TIMx_ARR register). Note that the auto-reload is updated before the counter is reloaded, so that the next period is the expected one.

The following figures show some examples of the counter behavior for different clock frequencies when TIMx_ARR=0x36.

Figure 143. Counter timing diagram, internal clock divided by 1

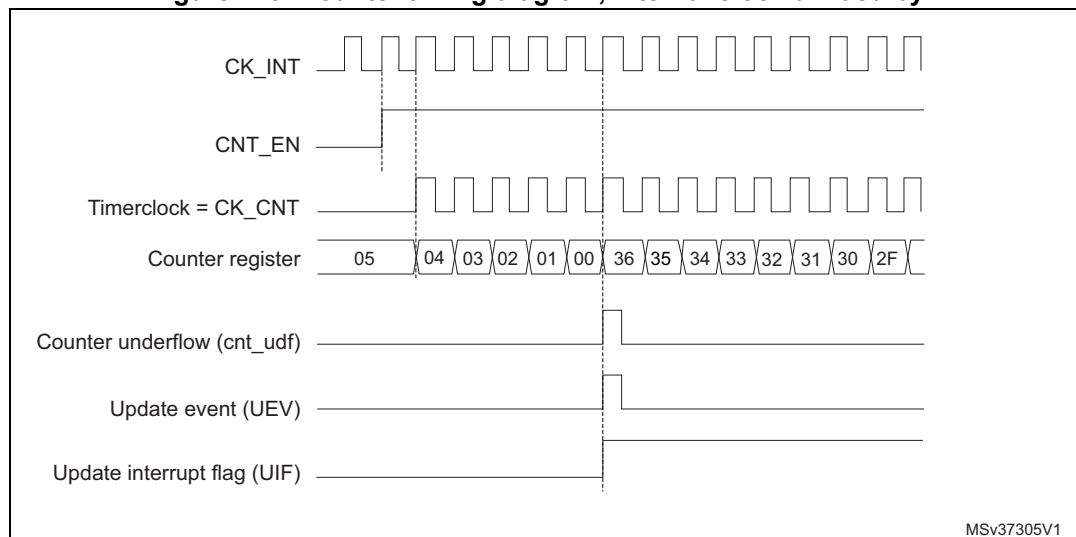


Figure 144. Counter timing diagram, internal clock divided by 2

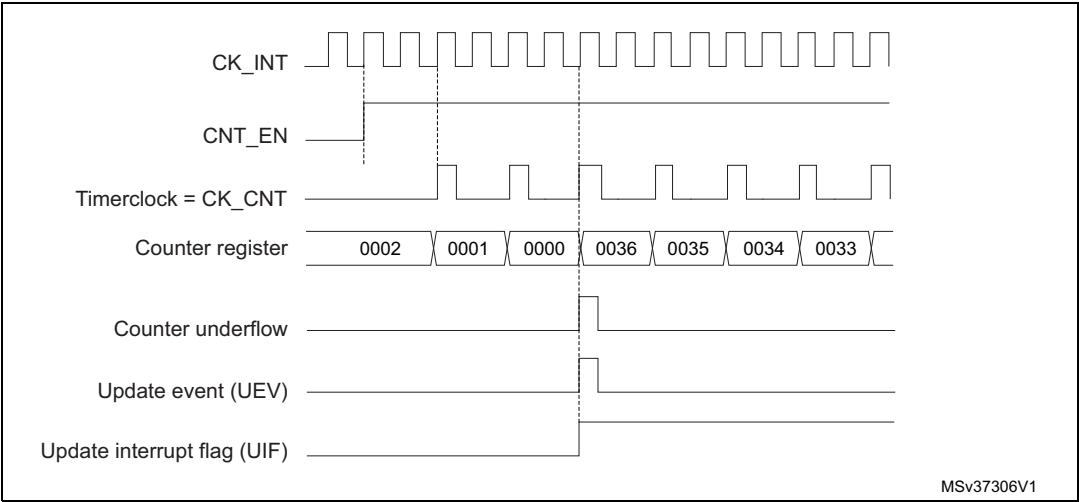


Figure 145. Counter timing diagram, internal clock divided by 4

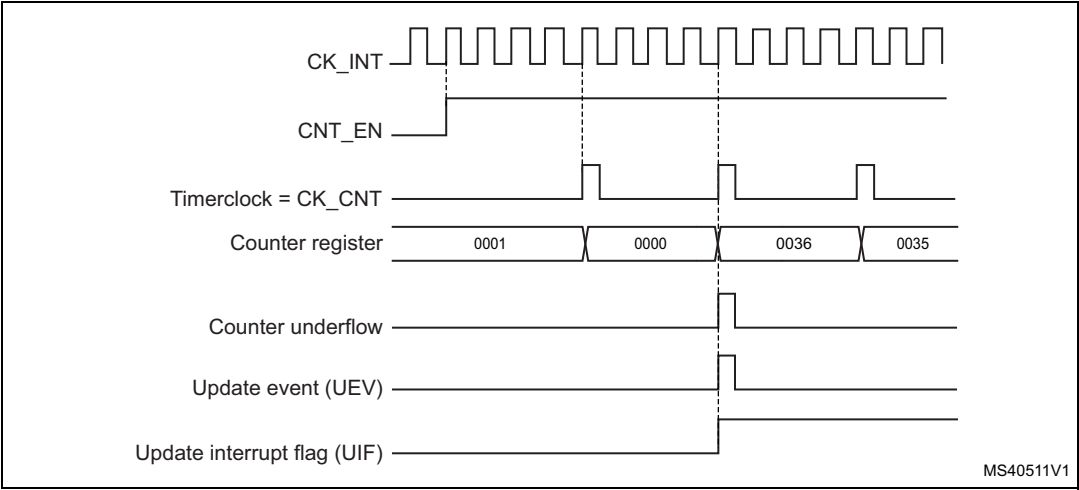


Figure 146. Counter timing diagram, internal clock divided by N

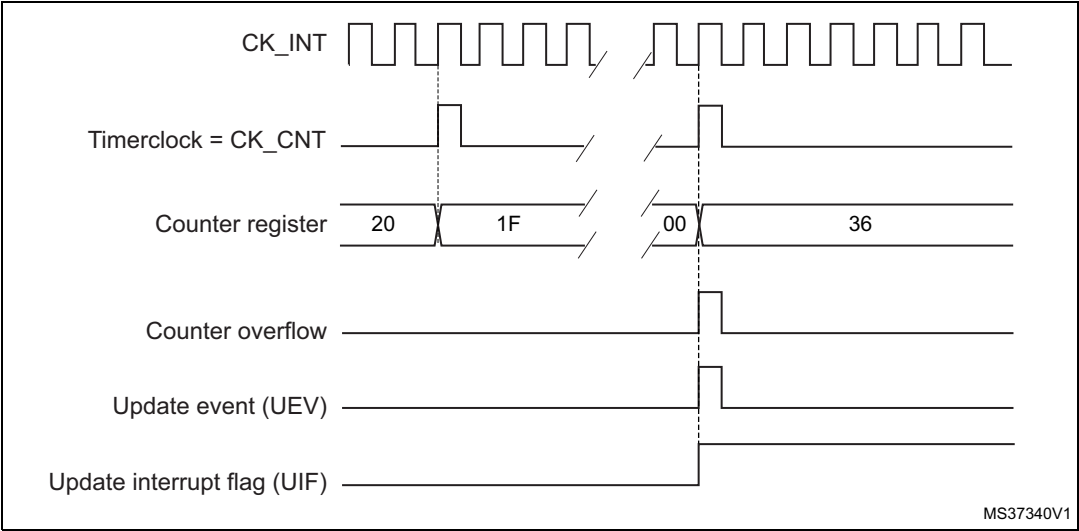
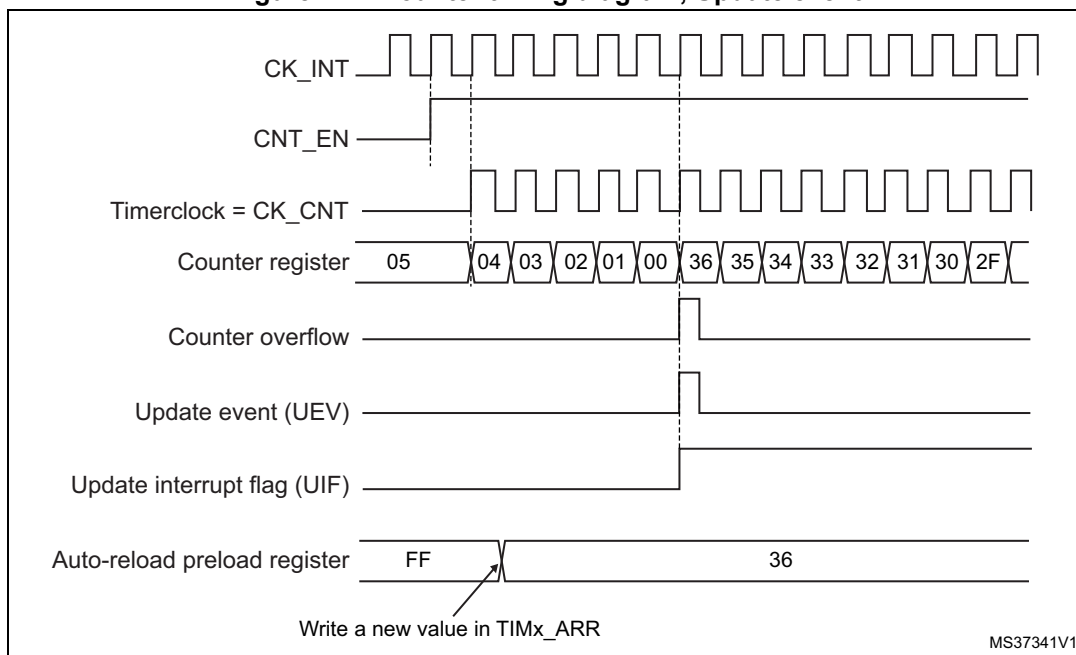


Figure 147. Counter timing diagram, Update event



Center-aligned mode (up/down counting)

In center-aligned mode, the counter counts from 0 to the auto-reload value (content of the TIMx_ARR register) – 1, generates a counter overflow event, then counts from the auto-reload value down to 1 and generates a counter underflow event. Then it restarts counting from 0.

Center-aligned mode is active when the CMS bits in TIMx_CR1 register are not equal to '00'. The Output compare interrupt flag of channels configured in output is set when: the counter counts down (Center aligned mode 1, CMS = "01"), the counter counts up (Center aligned mode 2, CMS = "10") the counter counts up and down (Center aligned mode 3, CMS = "11").

In this mode, the direction bit (DIR from TIMx_CR1 register) cannot be written. It is updated by hardware and gives the current direction of the counter.

The update event can be generated at each counter overflow and at each counter underflow or by setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller) also generates an update event. In this case, the counter restarts counting from 0, as well as the counter of the prescaler.

The UEV update event can be disabled by software by setting the UDIS bit in TIMx_CR1 register. This is to avoid updating the shadow registers while writing new values in the preload registers. Then no update event occurs until the UDIS bit has been written to 0. However, the counter continues counting up and down, based on the current auto-reload value.

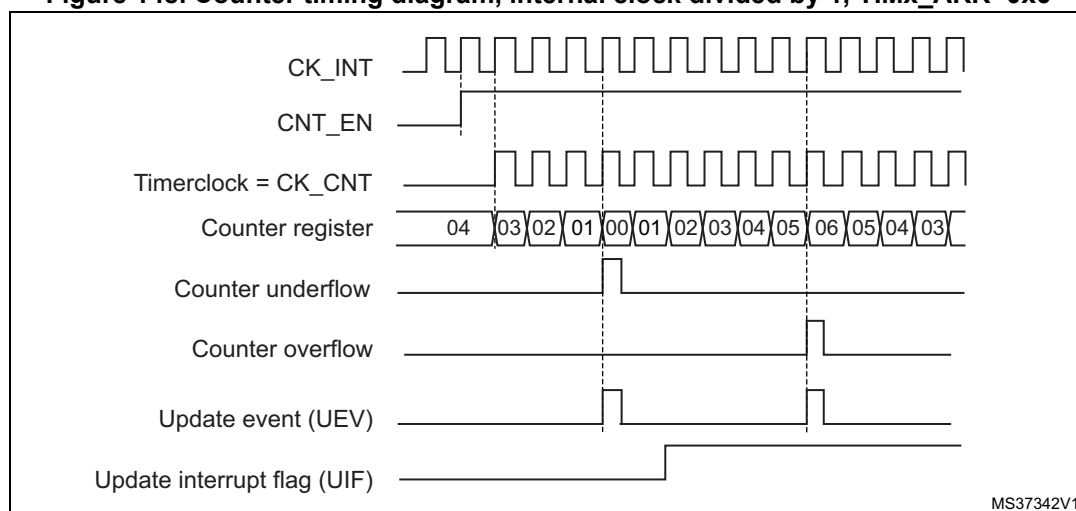
In addition, if the URS bit (update request selection) in TIMx_CR1 register is set, setting the UG bit generates an update event UEV but without setting the UIF flag (thus no interrupt or DMA request is sent). This is to avoid generating both update and capture interrupt when clearing the counter on the capture event.

When an update event occurs, all the registers are updated and the update flag (UIF bit in TIMx_SR register) is set (depending on the URS bit):

- The buffer of the prescaler is reloaded with the preload value (content of the TIMx_PSC register).
- The auto-reload active register is updated with the preload value (content of the TIMx_ARR register). Note that if the update source is a counter overflow, the auto-reload is updated before the counter is reloaded, so that the next period is the expected one (the counter is loaded with the new value).

The following figures show some examples of the counter behavior for different clock frequencies.

Figure 148. Counter timing diagram, internal clock divided by 1, TIMx_ARR=0x6



1. Here, center-aligned mode 1 is used, for more details refer to [Section 18.4.1: TIMx control register 1 \(TIMx_CR1\)](#).

Figure 149. Counter timing diagram, internal clock divided by 2

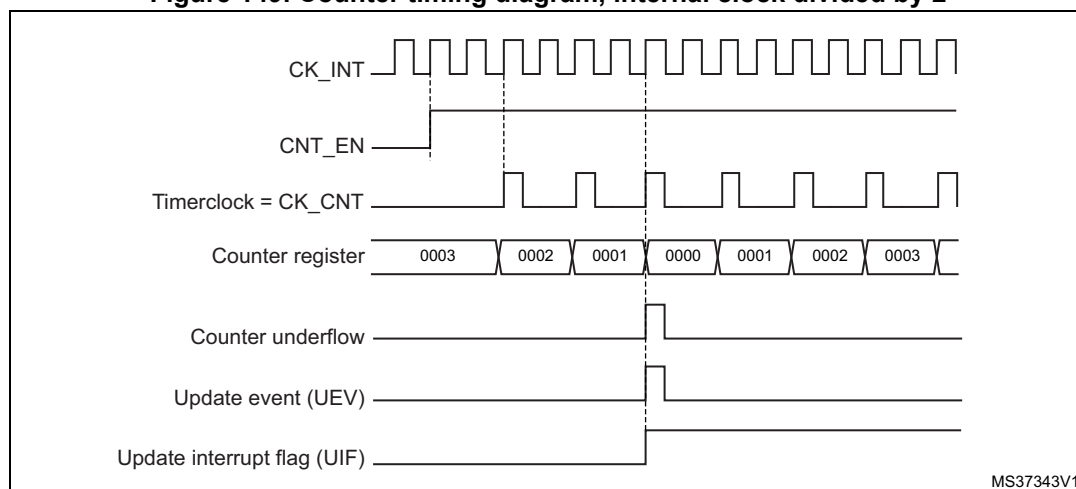
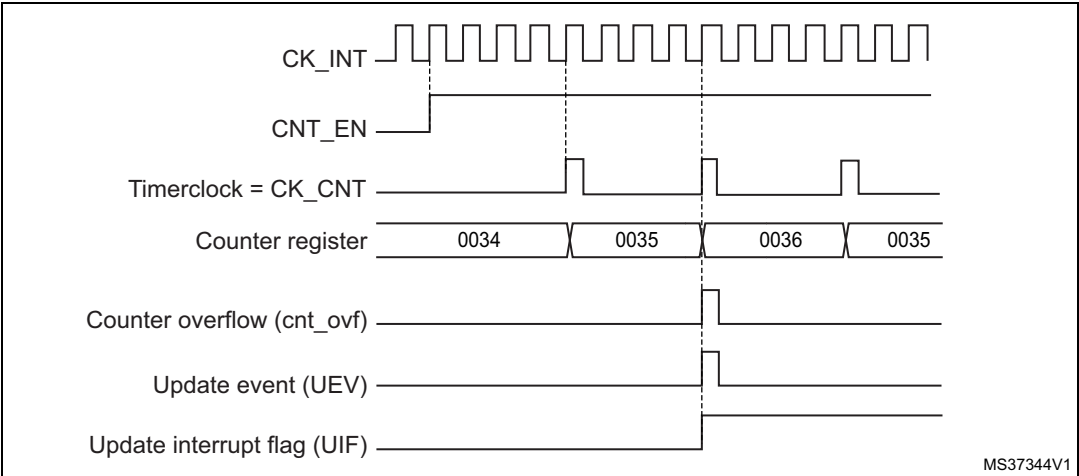


Figure 150. Counter timing diagram, internal clock divided by 4, TIMx_ARR=0x36



1. Center-aligned mode 2 or 3 is used with an UIF on overflow.

Figure 151. Counter timing diagram, internal clock divided by N

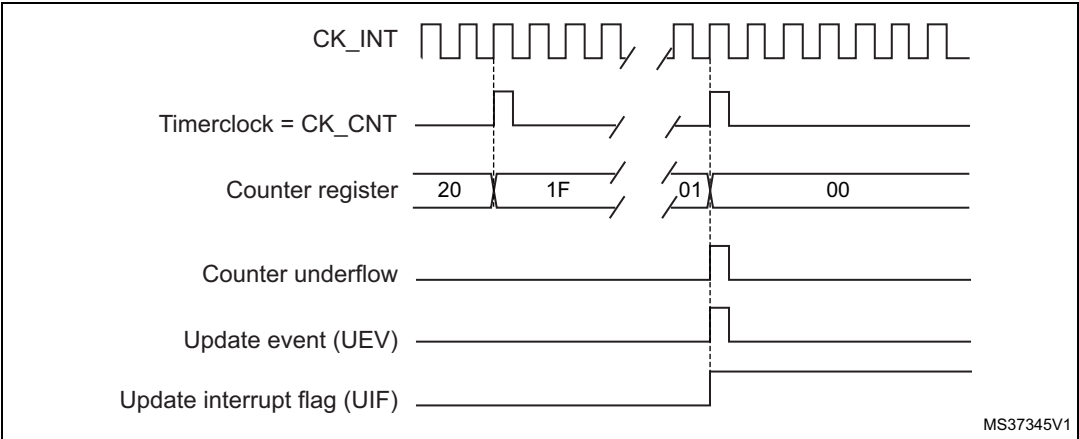


Figure 152. Counter timing diagram, Update event with ARPE=1 (counter underflow)

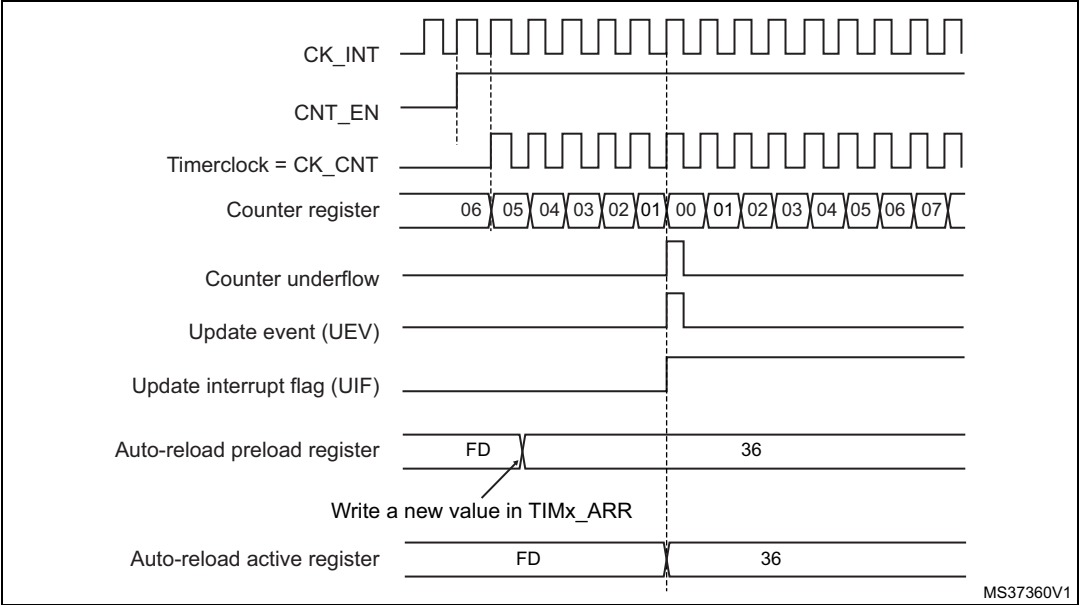
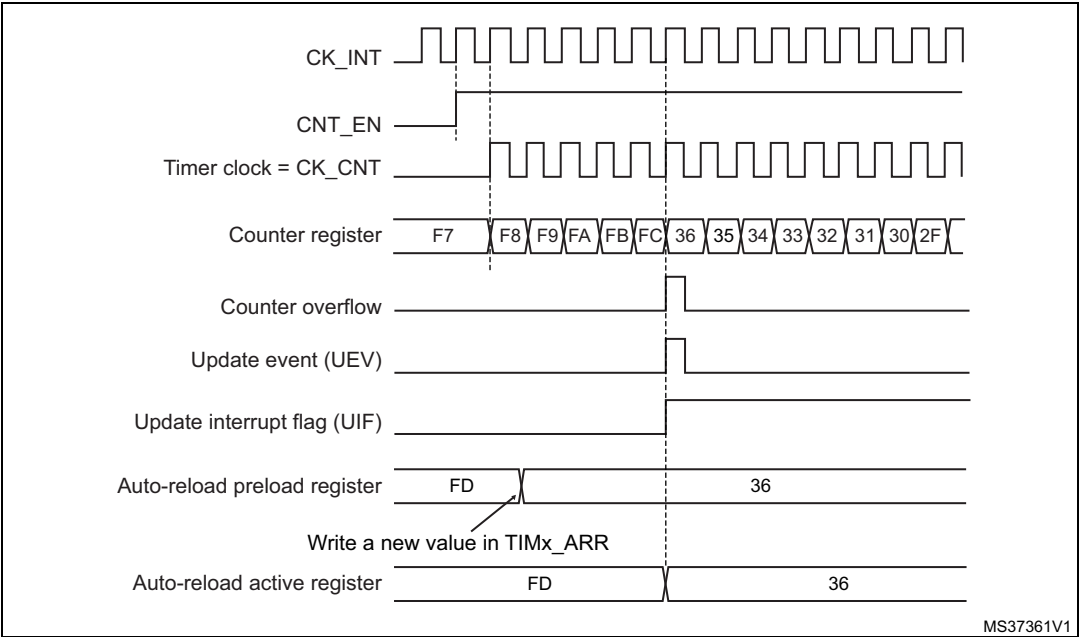


Figure 153. Counter timing diagram, Update event with ARPE=1 (counter overflow)



18.3.3 Clock selection

The counter clock can be provided by the following clock sources:

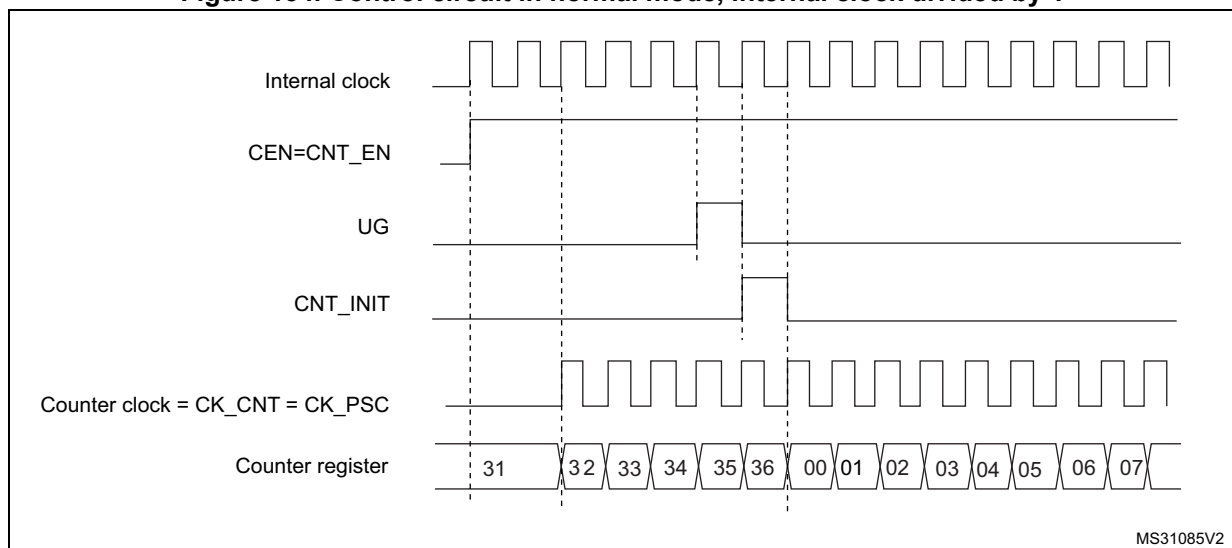
- Internal clock (CK_INT)
- External clock mode1: external input pin (TIx)
- External clock mode2: external trigger input (ETR) available on TIM2, TIM3 and TIM4 only.
- Internal trigger inputs (ITRx): using one timer as prescaler for another timer, for example, Timer3 can be configured to act as a prescaler for Timer 2. Refer to [Using one timer as prescaler for another timer](#) for more details.

Internal clock source (CK_INT)

If the slave mode controller is disabled (SMS=000 in the TIMx_SMCR register), then the CEN, DIR (in the TIMx_CR1 register) and UG bits (in the TIMx_EGR register) are actual control bits and can be changed only by software (except UG which remains cleared automatically). As soon as the CEN bit is written to 1, the prescaler is clocked by the internal clock CK_INT.

[Figure 154](#) shows the behavior of the control circuit and the upcounter in normal mode, without prescaler.

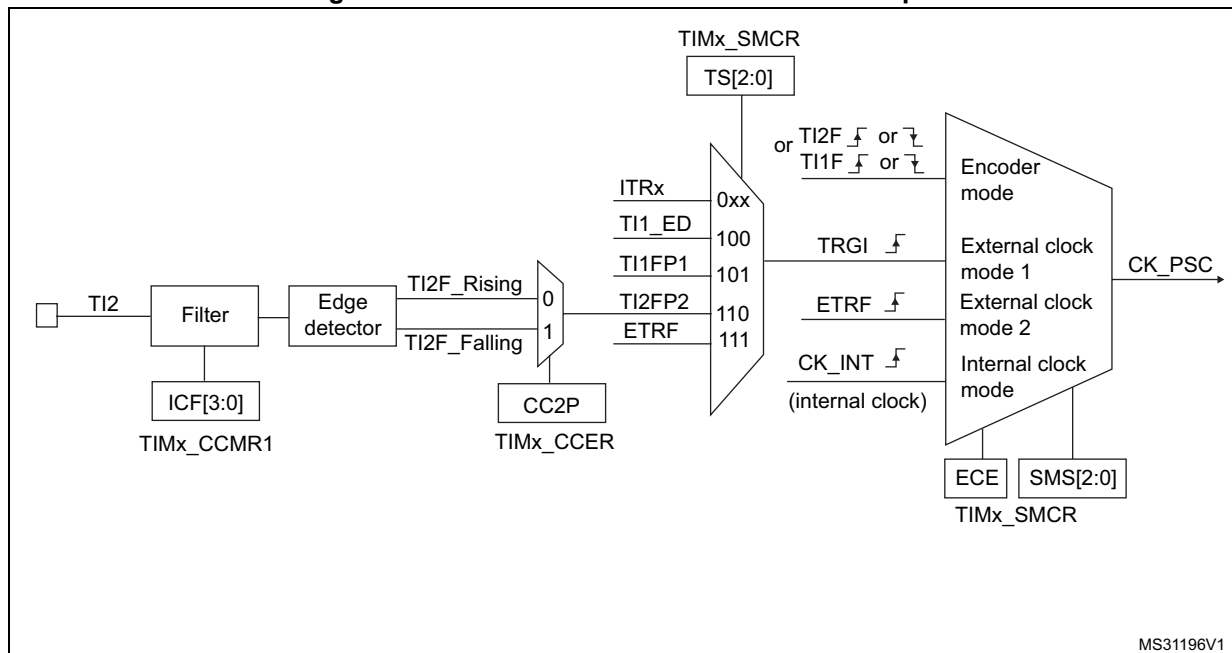
Figure 154. Control circuit in normal mode, internal clock divided by 1



External clock source mode 1

This mode is selected when SMS=111 in the TIMx_SMCR register. The counter can count at each rising or falling edge on a selected input.

Figure 155. TI2 external clock connection example



For example, to configure the upcounter to count in response to a rising edge on the TI2 input, use the following procedure:

1. Configure channel 2 to detect rising edges on the TI2 input by writing CC2S= '01 in the TIMx_CCMR1 register.
2. Configure the input filter duration by writing the IC2F[3:0] bits in the TIMx_CCMR1 register (if no filter is needed, keep IC2F=0000).

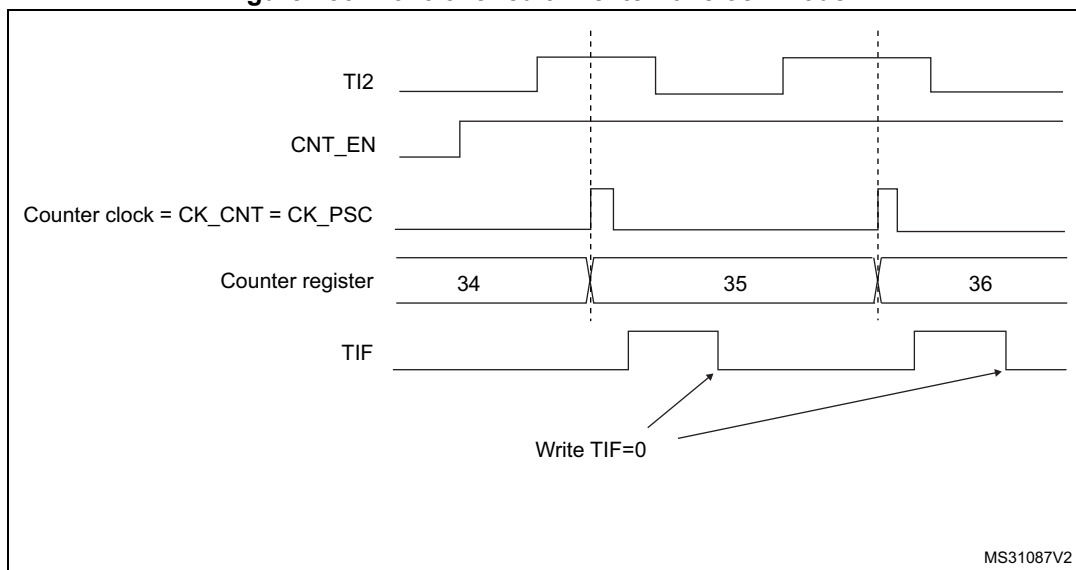
Note: The capture prescaler is not used for triggering, so there's no need to configure it.

3. Select rising edge polarity by writing CC2P=0 and CC2NP=0 in the TIMx_CCER register.
4. Configure the timer in external clock mode 1 by writing SMS=111 in the TIMx_SMCR register.
5. Select TI2 as the input source by writing TS=110 in the TIMx_SMCR register.
6. Enable the counter by writing CEN=1 in the TIMx_CR1 register.

When a rising edge occurs on TI2, the counter counts once and the TIF flag is set.

The delay between the rising edge on TI2 and the actual clock of the counter is due to the resynchronization circuit on TI2 input.

Figure 156. Control circuit in external clock mode 1



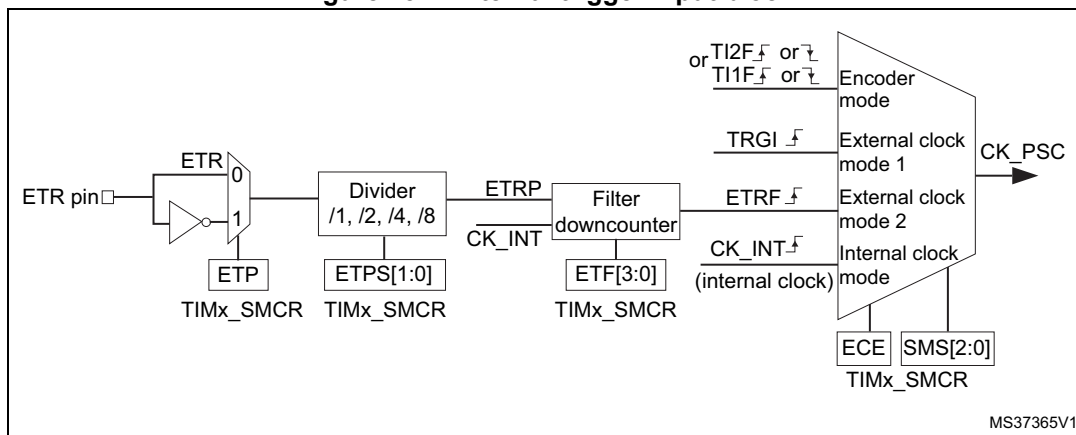
External clock source mode 2

This mode is selected by writing ECE=1 in the TIMx_SMCR register.

The counter can count at each rising or falling edge on the external trigger input ETR.

Figure 157 gives an overview of the external trigger input block.

Figure 157. External trigger input block



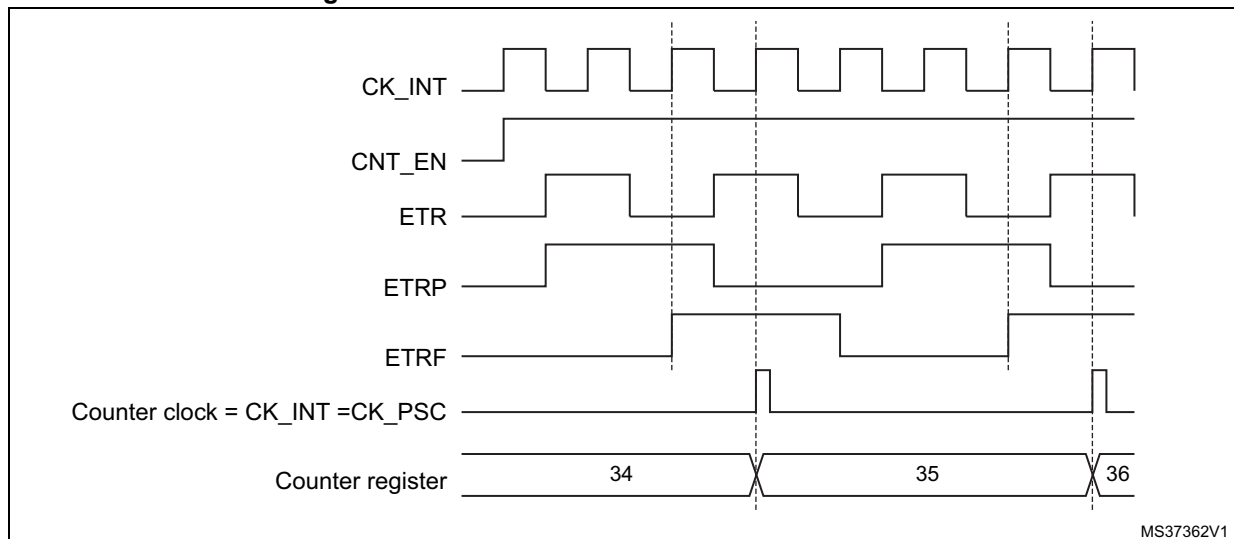
For example, to configure the upcounter to count each 2 rising edges on ETR, use the following procedure:

1. As no filter is needed in this example, write ETF[3:0]=0000 in the TIMx_SMCR register.
2. Set the prescaler by writing ETPS[1:0]=01 in the TIMx_SMCR register
3. Select rising edge detection on the ETR pin by writing ETP=0 in the TIMx_SMCR register
4. Enable external clock mode 2 by writing ECE=1 in the TIMx_SMCR register.
5. Enable the counter by writing CEN=1 in the TIMx_CR1 register.

The counter counts once each 2 ETR rising edges.

The delay between the rising edge on ETR and the actual clock of the counter is due to the resynchronization circuit on the ETRP signal.

Figure 158. Control circuit in external clock mode 2

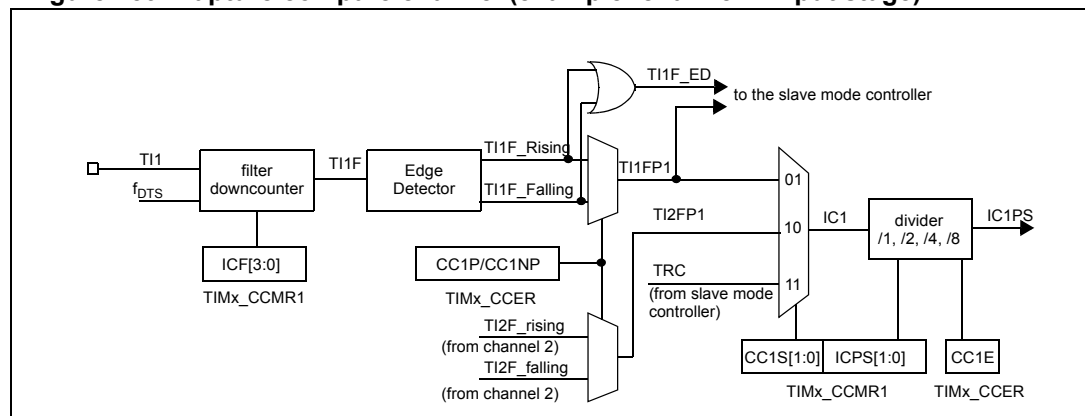


18.3.4 Capture/compare channels

Each Capture/Compare channel (see [Figure 159](#)) is built around a capture/compare register (including a shadow register), an input stage for capture (with digital filter, multiplexing and prescaler) and an output stage (with comparator and output control).

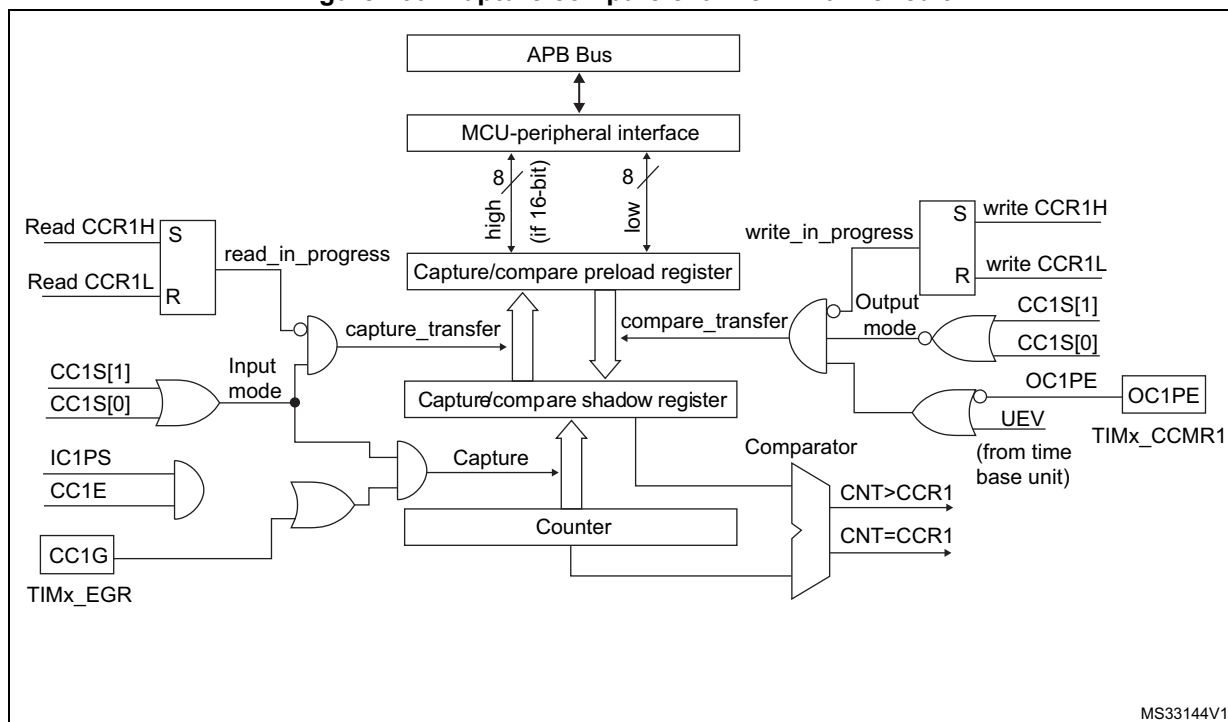
The input stage samples the corresponding TIx input to generate a filtered signal TIxF. Then, an edge detector with polarity selection generates a signal (TIxFPx) which can be used as trigger input by the slave mode controller or as the capture command. It is prescaled before the capture register (ICxPS).

Figure 159. Capture/compare channel (example: channel 1 input stage)



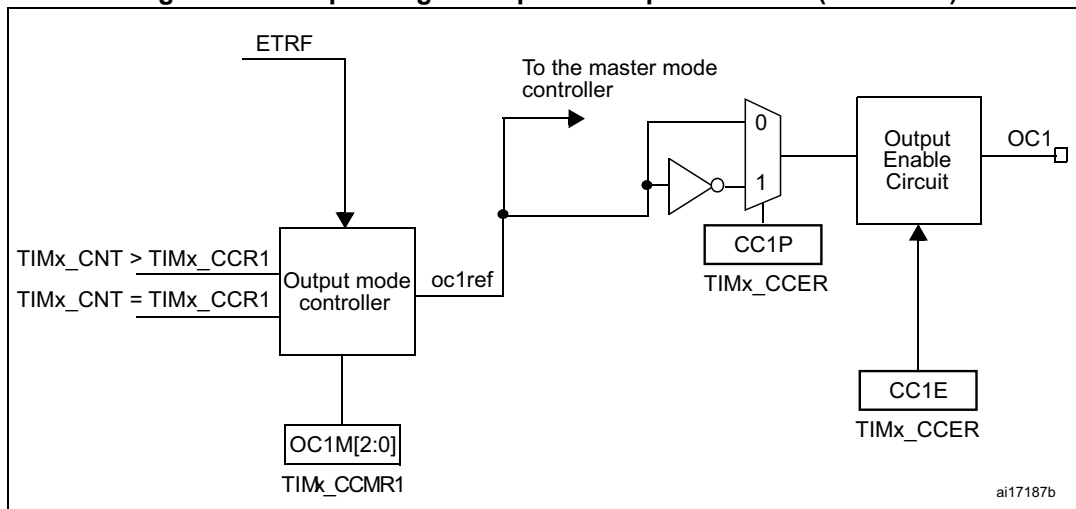
The output stage generates an intermediate waveform which is then used for reference: OCxRef (active high). The polarity acts at the end of the chain.

Figure 160. Capture/compare channel 1 main circuit



MS33144V1

Figure 161. Output stage of capture/compare channel (channel 1)



ai17187b

The capture/compare block is made of one preload register and one shadow register. Write and read always access the preload register.

In capture mode, captures are actually done in the shadow register, which is copied into the preload register.

In compare mode, the content of the preload register is copied into the shadow register which is compared to the counter.

18.3.5 Input capture mode

In Input capture mode, the Capture/Compare registers (TIMx_CCRx) are used to latch the value of the counter after a transition detected by the corresponding ICx signal. When a capture occurs, the corresponding CCxIF flag (TIMx_SR register) is set and an interrupt or a DMA request can be sent if they are enabled. If a capture occurs while the CCxIF flag was already high, then the over-capture flag CCxOF (TIMx_SR register) is set. CCxIF can be cleared by software by writing it to 0 or by reading the captured data stored in the TIMx_CCRx register. CCxOF is cleared when written to 0.

The following example shows how to capture the counter value in TIMx_CCR1 when TI1 input rises. To do this, use the following procedure:

- Select the active input: TIMx_CCR1 must be linked to the TI1 input, so write the CC1S bits to 01 in the TIMx_CCMR1 register. As soon as CC1S becomes different from 00, the channel is configured in input and the TIMx_CCR1 register becomes read-only.
- Program the needed input filter duration with respect to the signal connected to the timer (by programming the ICxF bits in the TIMx_CCMRx register if the input is one of the TIx inputs). Let's imagine that, when toggling, the input signal is not stable during at least five internal clock cycles. We must program a filter duration longer than these five clock cycles. We can validate a transition on TI1 when eight consecutive samples with the new level have been detected (sampled at f_{DTS} frequency). Then write IC1F bits to 0011 in the TIMx_CCMR1 register.
- Select the edge of the active transition on the TI1 channel by writing the CC1P and CC1NP bits to 00 in the TIMx_CCER register (rising edge in this case).
- Program the input prescaler. In our example, we wish the capture to be performed at each valid transition, so the prescaler is disabled (write IC1PS bits to 00 in the TIMx_CCMR1 register).
- Enable capture from the counter into the capture register by setting the CC1E bit in the TIMx_CCER register.
- If needed, enable the related interrupt request by setting the CC1IE bit in the TIMx_DIER register, and/or the DMA request by setting the CC1DE bit in the TIMx_DIER register.

When an input capture occurs:

- The TIMx_CCR1 register gets the value of the counter on the active transition.
- CC1IF flag is set (interrupt flag). CC1OF is also set if at least two consecutive captures occurred whereas the flag was not cleared.
- An interrupt is generated depending on the CC1IE bit.
- A DMA request is generated depending on the CC1DE bit.

In order to handle the overcapture, it is recommended to read the data before the overcapture flag. This is to avoid missing an overcapture which could happen after reading the flag and before reading the data.

Note: IC interrupt and/or DMA requests can be generated by software by setting the corresponding CCxG bit in the TIMx_EGR register.

18.3.6 PWM input mode

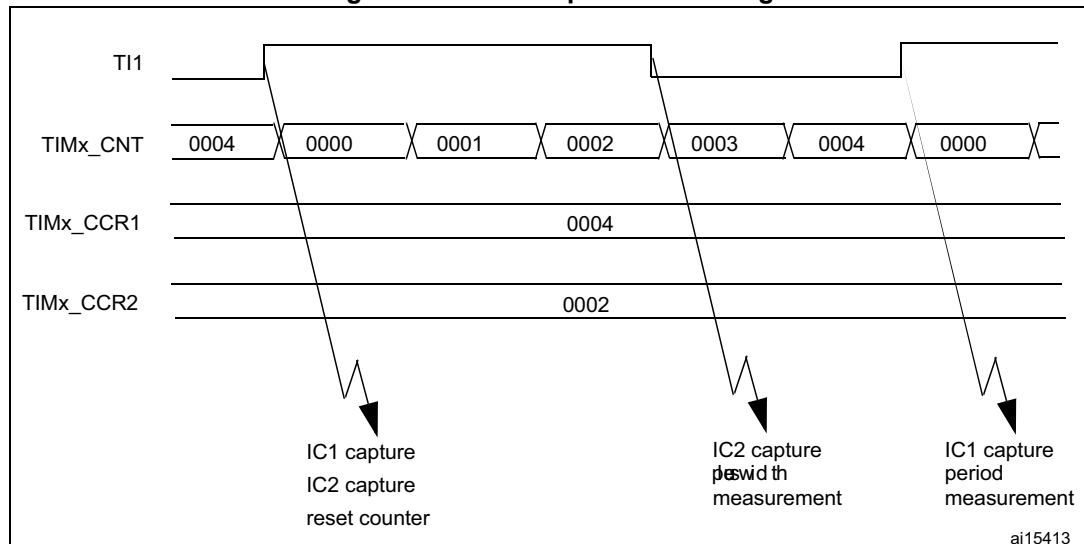
This mode is a particular case of input capture mode. The procedure is the same except:

- Two ICx signals are mapped on the same Tlx input.
- These 2 ICx signals are active on edges with opposite polarity.
- One of the two TlxFP signals is selected as trigger input and the slave mode controller is configured in reset mode.

For example, the user can measure the period (in TIMx_CCR1 register) and the duty cycle (in TIMx_CCR2 register) of the PWM applied on TI1 using the following procedure (depending on CK_INT frequency and prescaler value):

- Select the active input for TIMx_CCR1: write the CC1S bits to 01 in the TIMx_CCMR1 register (TI1 selected).
- Select the active polarity for TI1FP1 (used both for capture in TIMx_CCR1 and counter clear): write the CC1P to '0' and the CC1NP bit to '0' (active on rising edge).
- Select the active input for TIMx_CCR2: write the CC2S bits to 10 in the TIMx_CCMR1 register (TI1 selected).
- Select the active polarity for TI1FP2 (used for capture in TIMx_CCR2): write the CC2P bit to '1' and the CC2NP bit to '0'(active on falling edge).
- Select the valid trigger input: write the TS bits to 101 in the TIMx_SMCR register (TI1FP1 selected).
- Configure the slave mode controller in reset mode: write the SMS bits to 100 in the TIMx_SMCR register.
- Enable the captures: write the CC1E and CC2E bits to '1 in the TIMx_CCER register.

Figure 162. PWM input mode timing



18.3.7 Forced output mode

In output mode (CCxS bits = 00 in the TIMx_CCMRx register), each output compare signal (OCxREF and then OCx) can be forced to active or inactive level directly by software, independently of any comparison between the output compare register and the counter.

To force an output compare signal (ocxref/OCx) to its active level, the user just needs to write 101 in the OCxM bits in the corresponding TIMx_CCMRx register. Thus ocxref is forced high (OCxREF is always active high) and OCx get opposite value to CCxP polarity bit.

e.g.: CCxP=0 (OCx active high) => OCx is forced to high level.

ocxref signal can be forced low by writing the OCxM bits to 100 in the TIMx_CCMRx register.

Anyway, the comparison between the TIMx_CCRx shadow register and the counter is still performed and allows the flag to be set. Interrupt and DMA requests can be sent accordingly. This is described in the next section.

18.3.8 Output compare mode

This function is used to control an output waveform or indicating when a period of time has elapsed.

When a match is found between the capture/compare register and the counter, the output compare function:

- Assigns the corresponding output pin to a programmable value defined by the output compare mode (OCxM bits in the TIMx_CCMRx register) and the output polarity (CCxP bit in the TIMx_CCER register). The output pin can keep its level (OCxM=000), be set active (OCxM=001), be set inactive (OCxM=010) or can toggle (OCxM=011) on match.
- Sets a flag in the interrupt status register (CCxIF bit in the TIMx_SR register).
- Generates an interrupt if the corresponding interrupt mask is set (CCxIE bit in the TIMx_DIER register).
- Sends a DMA request if the corresponding enable bit is set (CCxDE bit in the TIMx_DIER register, CCDS bit in the TIMx_CR2 register for the DMA request selection).

The TIMx_CCRx registers can be programmed with or without preload registers using the OCxPE bit in the TIMx_CCMRx register.

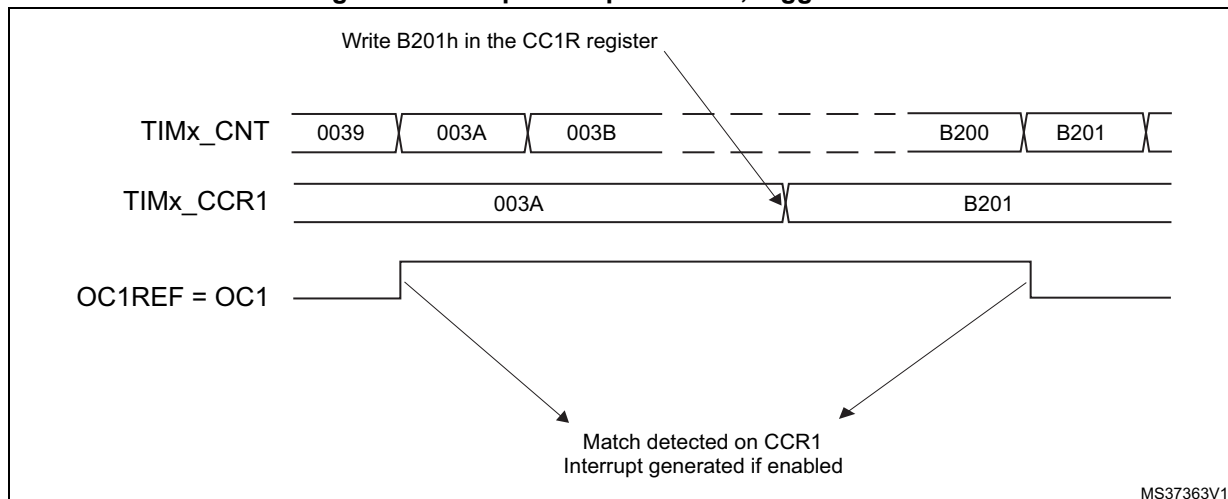
In output compare mode, the update event UEV has no effect on ocxref and OCx output. The timing resolution is one count of the counter. Output compare mode can also be used to output a single pulse (in One-pulse mode).

Procedure:

1. Select the counter clock (internal, external, prescaler).
2. Write the desired data in the TIMx_ARR and TIMx_CCRx registers.
3. Set the CCxIE and/or CCxDE bits if an interrupt and/or a DMA request is to be generated.
4. Select the output mode. For example, the user must write OCxM=011, OCxPE=0, CCxP=0 and CCxE=1 to toggle OCx output pin when CNT matches CCRx, CCRx preload is not used, OCx is enabled and active high.
5. Enable the counter by setting the CEN bit in the TIMx_CR1 register.

The TIMx_CCRx register can be updated at any time by software to control the output waveform, provided that the preload register is not enabled (OCxPE=0, else TIMx_CCRx shadow register is updated only at the next update event UEV). An example is given in [Figure 163](#).

Figure 163. Output compare mode, toggle on OC1



18.3.9 PWM mode

Pulse width modulation mode allows generating a signal with a frequency determined by the value of the TIMx_ARR register and a duty cycle determined by the value of the TIMx_CCRx register.

The PWM mode can be selected independently on each channel (one PWM per OCx output) by writing 110 (PWM mode 1) or '111 (PWM mode 2) in the OCxM bits in the TIMx_CCMRx register. The user must enable the corresponding preload register by setting the OCxPE bit in the TIMx_CCMRx register, and eventually the auto-reload preload register by setting the ARPE bit in the TIMx_CR1 register.

As the preload registers are transferred to the shadow registers only when an update event occurs, before starting the counter, the user has to initialize all the registers by setting the UG bit in the TIMx_EGR register.

OCx polarity is software programmable using the CCxP bit in the TIMx_CCER register. It can be programmed as active high or active low. OCx output is enabled by the CCxE bit in the TIMx_CCER register. Refer to the TIMx_CCERx register description for more details.

In PWM mode (1 or 2), TIMx_CNT and TIMx_CCRx are always compared to determine whether $TIMx_CCRx \leq TIMx_CNT$ or $TIMx_CNT \leq TIMx_CCRx$ (depending on the direction of the counter). However, to comply with the ETRF (OCREF can be cleared by an external event through the ETR signal until the next PWM period), the OCREF signal is asserted only:

- When the result of the comparison changes, or
- When the output compare mode (OCxM bits in TIMx_CCMRx register) switches from the "frozen" configuration (no comparison, OCxM='000) to one of the PWM modes (OCxM='110 or '111).

This forces the PWM by software while the timer is running.

The timer is able to generate PWM in edge-aligned mode or center-aligned mode depending on the CMS bits in the TIMx_CR1 register.

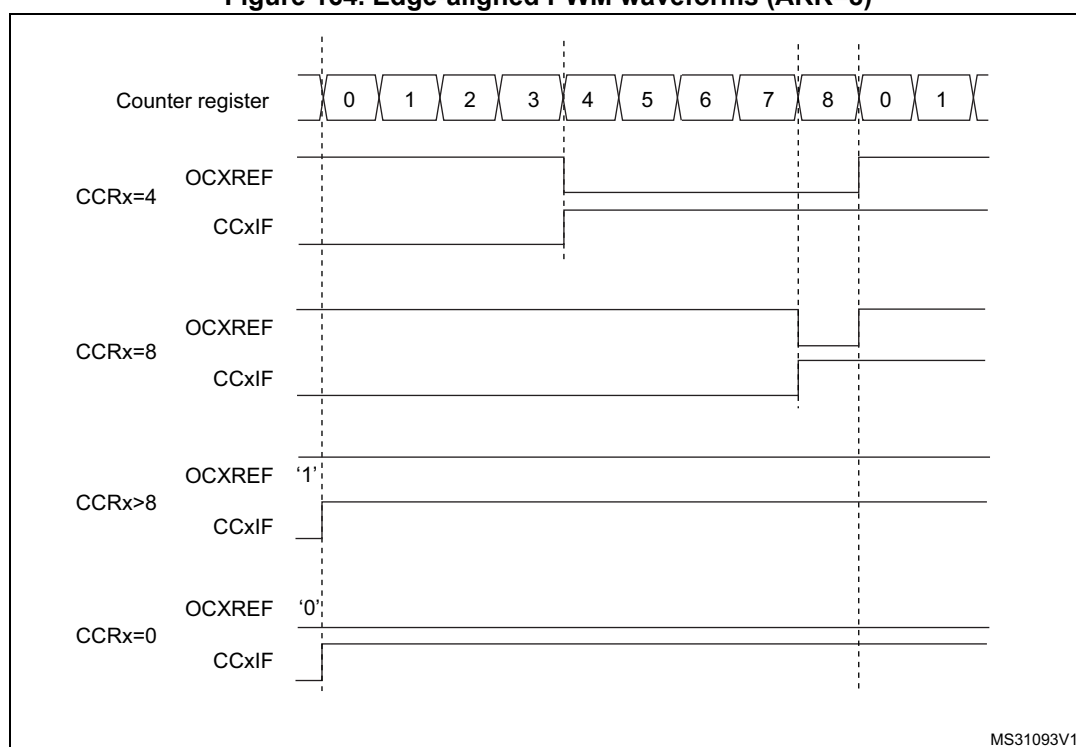
PWM edge-aligned mode

Upcounting configuration

Upcounting is active when the DIR bit in the TIMx_CR1 register is low. Refer to [Upcounting mode](#).

In the following example, we consider PWM mode 1. The reference PWM signal OCxREF is high as long as TIMx_CNT < TIMx_CCRx else it becomes low. If the compare value in TIMx_CCRx is greater than the auto-reload value (in TIMx_ARR) then OCxREF is held at '1'. If the compare value is 0 then OCxREF is held at '0'. [Figure 164](#) shows some edge-aligned PWM waveforms in an example where TIMx_ARR=8.

Figure 164. Edge-aligned PWM waveforms (ARR=8)



Downcounting configuration

Downcounting is active when DIR bit in TIMx_CR1 register is high. Refer to [Downcounting mode](#).

In PWM mode 1, the reference signal ocxref is low as long as TIMx_CNT > TIMx_CCRx else it becomes high. If the compare value in TIMx_CCRx is greater than the auto-reload value in TIMx_ARR, then ocxref is held at '1'. 0% PWM is not possible in this mode.

PWM center-aligned mode

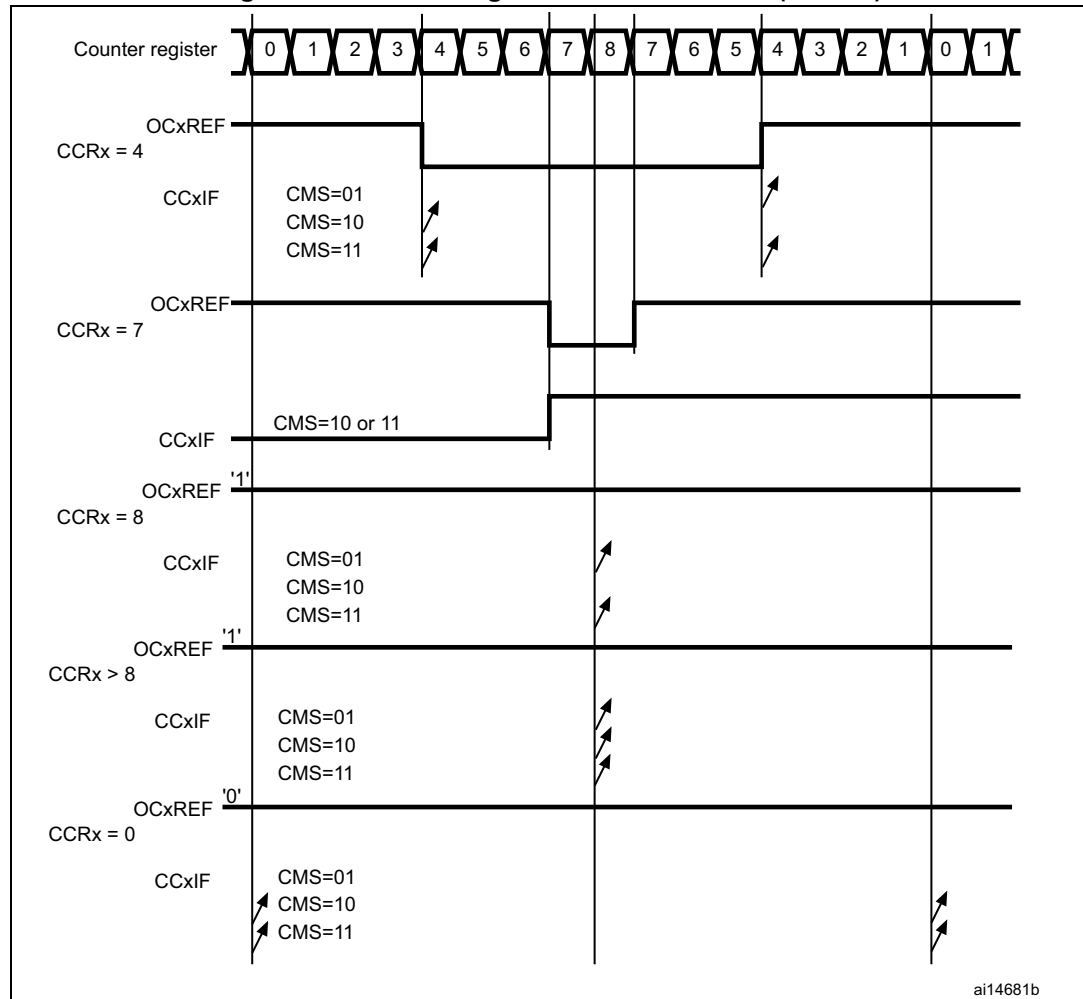
Center-aligned mode is active when the CMS bits in TIMx_CR1 register are different from '00' (all the remaining configurations having the same effect on the ocxref/OCx signals). The compare flag is set when the counter counts up, when it counts down or both when it counts

up and down depending on the CMS bits configuration. The direction bit (DIR) in the TIMx_CR1 register is updated by hardware and must not be changed by software. Refer to [Center-aligned mode \(up/down counting\)](#).

Figure 165 shows some center-aligned PWM waveforms in an example where:

- TIMx_ARR=8,
- PWM mode is the PWM mode 1,
- The flag is set when the counter counts down corresponding to the center-aligned mode 1 selected for CMS=01 in TIMx_CR1 register.

Figure 165. Center-aligned PWM waveforms (ARR=8)



Hints on using center-aligned mode:

- When starting in center-aligned mode, the current up-down configuration is used. It means that the counter counts up or down depending on the value written in the DIR bit

in the TIMx_CR1 register. Moreover, the DIR and CMS bits must not be changed at the same time by the software.

- Writing to the counter while running in center-aligned mode is not recommended as it can lead to unexpected results. In particular:
 - The direction is not updated if the user writes a value in the counter that is greater than the auto-reload value ($TIMx_CNT > TIMx_ARR$). For example, if the counter was counting up, it continues to count up.
 - The direction is updated if the user writes 0 or write the TIMx_ARR value in the counter but no Update Event UEV is generated.
- The safest way to use center-aligned mode is to generate an update by software (setting the UG bit in the TIMx_EGR register) just before starting the counter and not to write the counter while it is running.

18.3.10 One-pulse mode

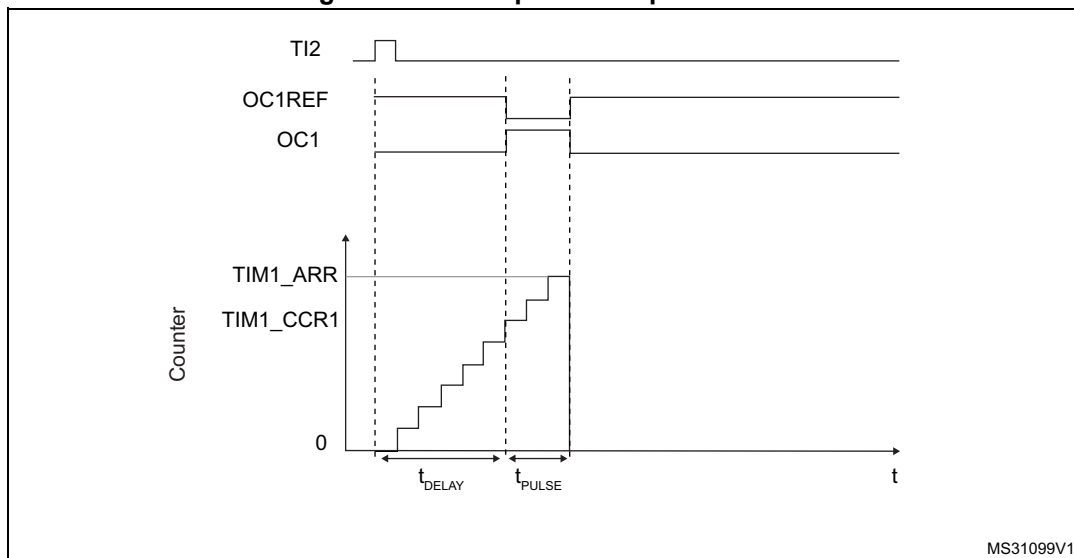
One-pulse mode (OPM) is a particular case of the previous modes. It allows the counter to be started in response to a stimulus and to generate a pulse with a programmable length after a programmable delay.

Starting the counter can be controlled through the slave mode controller. Generating the waveform can be done in output compare mode or PWM mode. Select One-pulse mode by setting the OPM bit in the TIMx_CR1 register. This makes the counter stop automatically at the next update event UEV.

A pulse can be correctly generated only if the compare value is different from the counter initial value. Before starting (when the timer is waiting for the trigger), the configuration must be:

- In upcounting: $CNT < CCRx \leq ARR$ (in particular, $0 < CCRx$),
- In downcounting: $CNT > CCRx$.

Figure 166. Example of one-pulse mode



For example the user may want to generate a positive pulse on OC1 with a length of t_{PULSE} and after a delay of t_{DELAY} as soon as a positive edge is detected on the TI2 input pin.

Let's use TI2FP2 as trigger 1:

- Map TI2FP2 on TI2 by writing CC2S=01 in the TIMx_CCMR1 register.
- TI2FP2 must detect a rising edge, write CC2P=0 and CC2NP='0' in the TIMx_CCER register.
- Configure TI2FP2 as trigger for the slave mode controller (TRGI) by writing TS=110 in the TIMx_SMCR register.
- TI2FP2 is used to start the counter by writing SMS to '110 in the TIMx_SMCR register (trigger mode).

The OPM waveform is defined by writing the compare registers (taking into account the clock frequency and the counter prescaler).

- The t_{DELAY} is defined by the value written in the TIMx_CCR1 register.
- The t_{PULSE} is defined by the difference between the auto-reload value and the compare value (TIMx_ARR - TIMx_CCR + 1).
- Let us say user wants to build a waveform with a transition from '0 to '1 when a compare match occurs and a transition from '1 to '0 when the counter reaches the auto-reload value. To do this enable PWM mode 2 by writing OC1M=111 in the TIMx_CCMR1 register. The user can optionally enable the preload registers by writing OC1PE=1 in the TIMx_CCMR1 register and ARPE in the TIMx_CR1 register. In this case write the compare value in the TIMx_CCR1 register, the auto-reload value in the TIMx_ARR register, generate an update by setting the UG bit and wait for external trigger event on TI2. CC1P is written to '0 in this example.

In our example, the DIR and CMS bits in the TIMx_CR1 register should be low.

User only wants one pulse (Single mode), so write '1 in the OPM bit in the TIMx_CR1 register to stop the counter at the next update event (when the counter rolls over from the auto-reload value back to 0). When OPM bit in the TIMx_CR1 register is set to '0', so the Repetitive mode is selected.

Particular case: OCx fast enable:

In One-pulse mode, the edge detection on Tlx input set the CEN bit which enables the counter. Then the comparison between the counter and the compare value makes the output toggle. But several clock cycles are needed for these operations and it limits the minimum delay $t_{\text{DELAY min}}$ we can get.

To output a waveform with the minimum delay, the user can set the OCxFE bit in the TIMx_CCMRx register. Then OCxRef (and OCx) is forced in response to the stimulus, without taking in account the comparison. Its new level is the same as if a compare match had occurred. OCxFE acts only if the channel is configured in PWM1 or PWM2 mode.

18.3.11 Clearing the OCxREF signal on an external event

The OCxREF signal for a given channel can be driven Low by applying a High level to the ETRF input (OCxCE enable bit of the corresponding TIMx_CCMRx register set to '1'). The OCxREF signal remains Low until the next update event, UEV, occurs.

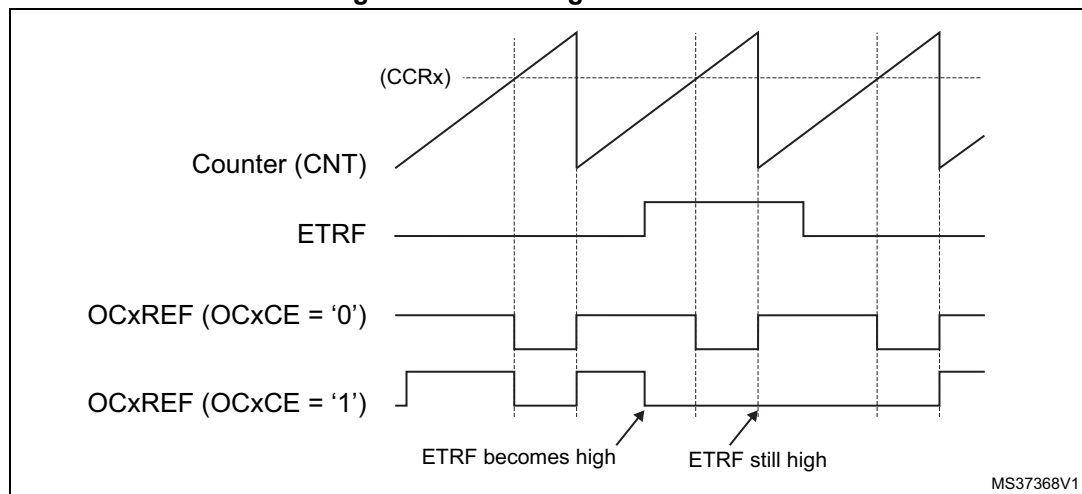
This function can only be used in output compare and PWM modes, and does not work in forced mode.

For example, the ETR signal can be connected to the output of a comparator to be used for current handling. In this case, ETR must be configured as follows:

1. The external trigger prescaler should be kept off: bits ETPS[1:0] in the TIMx_SMCR register are cleared to 00.
2. The external clock mode 2 must be disabled: bit ECE in the TIM1_SMCR register is cleared to 0.
3. The external trigger polarity (ETP) and the external trigger filter (ETF) can be configured according to the application's needs.

Figure 167 shows the behavior of the OCxREF signal when the ETRF input becomes high, for both values of the OCxCE enable bit. In this example, the timer TIMx is programmed in PWM mode.

Figure 167. Clearing TIMx OCxREF



1. In case of a PWM with a 100% duty cycle (if CCRx>ARR), OCxREF is enabled again at the next counter overflow.

18.3.12 Encoder interface mode

To select Encoder Interface mode write SMS='001' in the TIMx_SMCR register if the counter is counting on TI2 edges only, SMS=010 if it is counting on TI1 edges only and SMS=011 if it is counting on both TI1 and TI2 edges.

Select the TI1 and TI2 polarity by programming the CC1P and CC2P bits in the TIMx_CCER register. When needed, program the input filter as well.

The two inputs TI1 and TI2 are used to interface to an incremental encoder. Refer to [Table 98](#). The counter is clocked by each valid transition on TI1FP1 or TI2FP2 (TI1 and TI2 after input filter and polarity selection, TI1FP1=TI1 if not filtered and not inverted, TI2FP2=TI2 if not filtered and not inverted) assuming that it is enabled (CEN bit in TIMx_CR1 register written to '1'). The sequence of transitions of the two inputs is evaluated and generates count pulses as well as the direction signal. Depending on the sequence the counter counts up or down, the DIR bit in the TIMx_CR1 register is modified by hardware accordingly. The DIR bit is calculated at each transition on any input (TI1 or TI2), whatever the counter is counting on TI1 only, TI2 only or both TI1 and TI2.

Encoder interface mode acts simply as an external clock with direction selection. This means that the counter just counts continuously between 0 and the auto-reload value in the TIMx_ARR register (0 to ARR or ARR down to 0 depending on the direction). So the user must configure TIMx_ARR before starting. In the same way, the capture, compare, prescaler, trigger output features continue to work as normal.

In this mode, the counter is modified automatically following the speed and the direction of the incremental encoder and its content, therefore, always represents the encoder's position. The count direction correspond to the rotation direction of the connected sensor. The table summarizes the possible combinations, assuming TI1 and TI2 do not switch at the same time.

Table 98. Counting direction versus encoder signals

Active edge	Level on opposite signal (TI1FP1 for TI2, TI2FP2 for TI1)	TI1FP1 signal		TI2FP2 signal	
		Rising	Falling	Rising	Falling
Counting on TI1 only	High	Down	Up	No Count	No Count
	Low	Up	Down	No Count	No Count
Counting on TI2 only	High	No Count	No Count	Up	Down
	Low	No Count	No Count	Down	Up
Counting on TI1 and TI2	High	Down	Up	Up	Down
	Low	Up	Down	Down	Up

An external incremental encoder can be connected directly to the MCU without external interface logic. However, comparators are normally be used to convert the encoder's differential outputs to digital signals. This greatly increases noise immunity. The third encoder output which indicate the mechanical zero position, may be connected to an external interrupt input and trigger a counter reset.

[Figure 168](#) gives an example of counter operation, showing count signal generation and direction control. It also shows how input jitter is compensated where both edges are selected. This might occur if the sensor is positioned near to one of the switching points. For this example we assume that the configuration is the following:

- CC1S= '01' (TIMx_CCMR1 register, TI1FP1 mapped on TI1)
- CC2S= '01' (TIMx_CCMR1 register, TI2FP2 mapped on TI2)
- CC1P= '0', CC1NP = '0', IC1F = '0000' (TIMx_CCER register, TI1FP1 noninverted, TI1FP1=TI1)
- CC2P= '0', CC2NP = '0', IC2F = '0000' (TIMx_CCER register, TI2FP2 noninverted, TI2FP2=TI2)
- SMS= '011' (TIMx_SMCR register, both inputs are active on both rising and falling edges)
- CEN = 1 (TIMx_CR1 register, Counter is enabled)

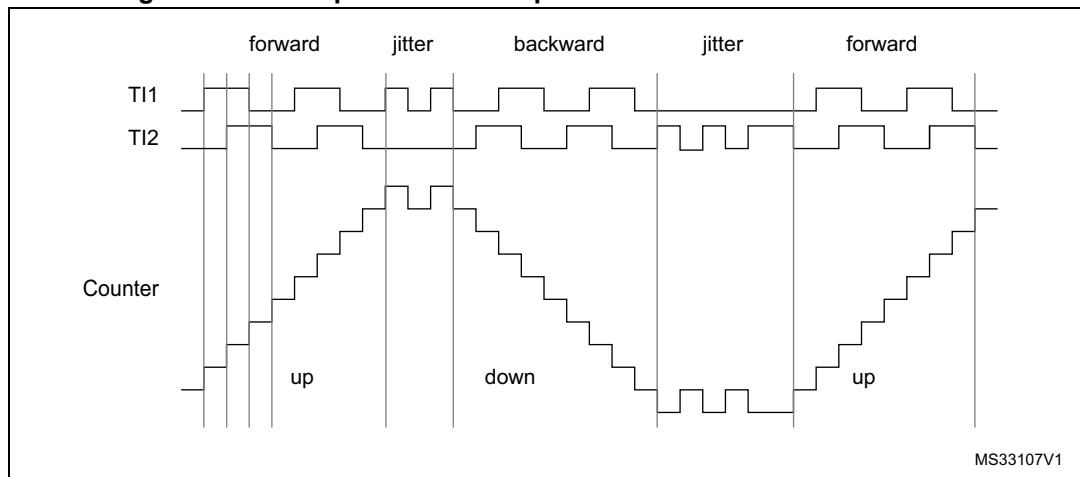
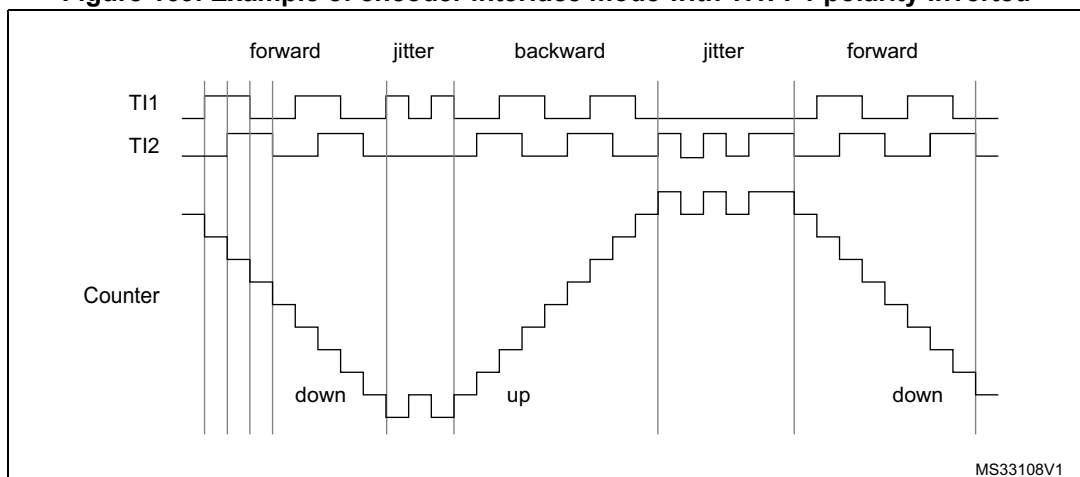
Figure 168. Example of counter operation in encoder interface mode

Figure 169 gives an example of counter behavior when TI1FP1 polarity is inverted (same configuration as above except CC1P=1).

Figure 169. Example of encoder interface mode with TI1FP1 polarity inverted

The timer, when configured in Encoder Interface mode provides information on the sensor's current position. The user can obtain dynamic information (speed, acceleration, deceleration) by measuring the period between two encoder events using a second timer configured in capture mode. The output of the encoder which indicates the mechanical zero can be used for this purpose. Depending on the time between two events, the counter can also be read at regular times. The user can do this by latching the counter value into a third input capture register if available (then the capture signal must be periodic and can be generated by another timer). when available, it is also possible to read its value through a DMA request generated by a Real-Time clock.

18.3.13 Timer input XOR function

The TI1S bit in the TIM_CR2 register, allows the input filter of channel 1 to be connected to the output of a XOR gate, combining the three input pins TIMx_CH1 to TIMx_CH3.

The XOR output can be used with all the timer input functions such as trigger or input capture.

18.3.14 Timers and external trigger synchronization

The TIMx Timers can be synchronized with an external trigger in several modes: Reset mode, Gated mode and Trigger mode.

Slave mode: Reset mode

The counter and its prescaler can be reinitialized in response to an event on a trigger input. Moreover, if the URS bit from the TIMx_CR1 register is low, an update event UEV is generated. Then all the preloaded registers (TIMx_ARR, TIMx_CCRx) are updated.

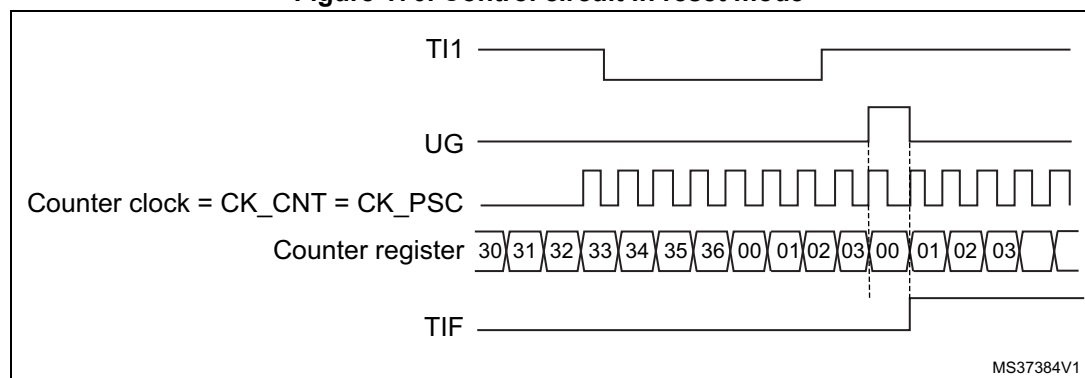
In the following example, the upcounter is cleared in response to a rising edge on TI1 input:

- Configure the channel 1 to detect rising edges on TI1. Configure the input filter duration (in this example, no need of any filter, IC1F=0000 kept). The capture prescaler is not used for triggering, so the user does not need to configure it. The CC1S bits select the input capture source only, CC1S = 01 in the TIMx_CCMR1 register. Write CC1P=0 and CC1NP=0 in TIMx_CCER register to validate the polarity (and detect rising edges only).
- Configure the timer in reset mode by writing SMS=100 in TIMx_SMCR register. Select TI1 as the input source by writing TS=101 in TIMx_SMCR register.
- Start the counter by writing CEN=1 in the TIMx_CR1 register.

The counter starts counting on the internal clock, then behaves normally until TI1 rising edge. When TI1 rises, the counter is cleared and restarts from 0. In the meantime, the trigger flag is set (TIF bit in the TIMx_SR register) and an interrupt request, or a DMA request can be sent if enabled (depending on the TIE and TDE bits in TIMx_DIER register).

[Figure 170](#) shows this behavior when the auto-reload register TIMx_ARR=0x36. The delay between the rising edge on TI1 and the actual reset of the counter is due to the resynchronization circuit on TI1 input.

Figure 170. Control circuit in reset mode



Slave mode: Gated mode

The counter can be enabled depending on the level of a selected input.

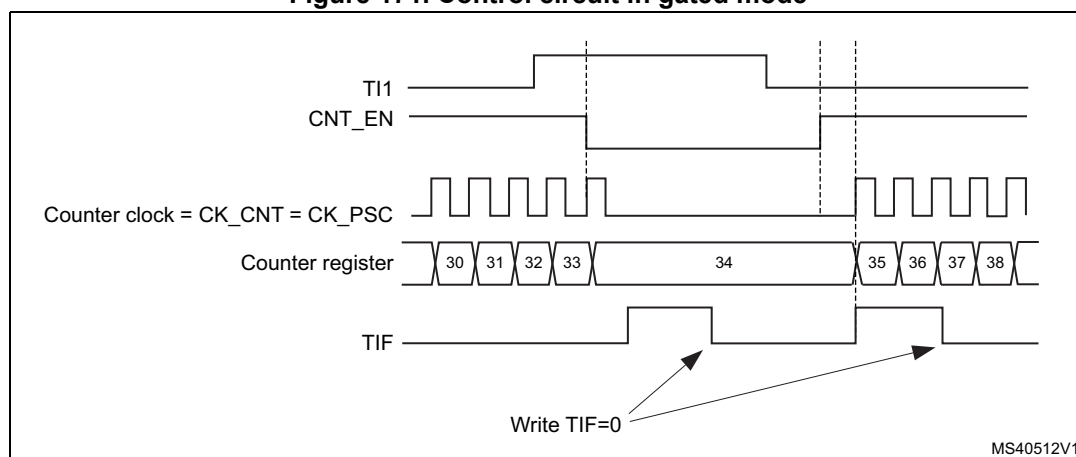
In the following example, the upcounter counts only when TI1 input is low:

- Configure the channel 1 to detect low levels on TI1. Configure the input filter duration (in this example, no need of any filter, IC1F=0000 kept). The capture prescaler is not used for triggering, so the user does not need to configure it. The CC1S bits select the input capture source only, CC1S=01 in TIMx_CCMR1 register. Write CC1P=1 in TIMx_CCER register to validate the polarity (and detect low level only).
- Configure the timer in gated mode by writing SMS=101 in TIMx_SMCR register. Select TI1 as the input source by writing TS=101 in TIMx_SMCR register.
- Enable the counter by writing CEN=1 in the TIMx_CR1 register (in gated mode, the counter does not start if CEN=0, whatever is the trigger input level).

The counter starts counting on the internal clock as long as TI1 is low and stops as soon as TI1 becomes high. The TIF flag in the TIMx_SR register is set both when the counter starts or stops.

The delay between the rising edge on TI1 and the actual stop of the counter is due to the resynchronization circuit on TI1 input.

Figure 171. Control circuit in gated mode



1. The configuration "CCxP=CCxNP=1" (detection of both rising and falling edges) does not have any effect in gated mode because gated mode acts on a level and not on an edge.

Slave mode: Trigger mode

The counter can start in response to an event on a selected input.

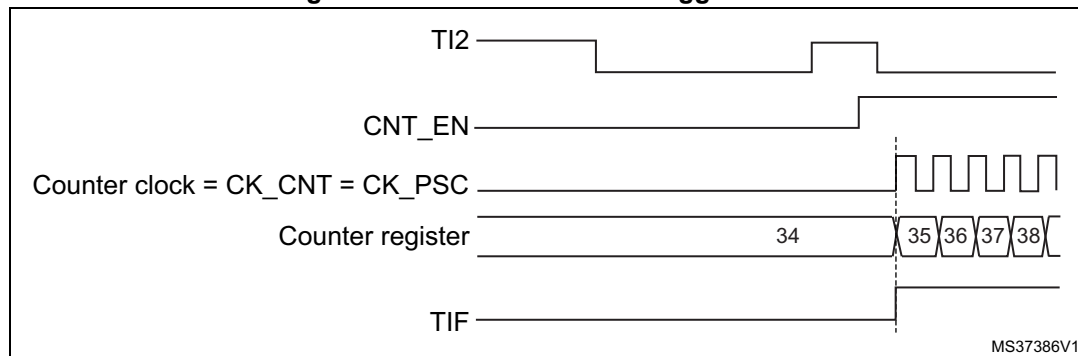
In the following example, the upcounter starts in response to a rising edge on TI2 input:

- Configure the channel 2 to detect rising edges on TI2. Configure the input filter duration (in this example, no need of any filter, IC2F=0000 kept). The capture prescaler is not used for triggering, so the user does not need to configure it. CC2S bits are selecting the input capture source only, CC2S=01 in TIMx_CCMR1 register. Write CC2P=1 in TIMx_CCER register to validate the polarity (and detect low level only).
- Configure the timer in trigger mode by writing SMS=110 in TIMx_SMCR register. Select TI2 as the input source by writing TS=110 in TIMx_SMCR register.

When a rising edge occurs on TI2, the counter starts counting on the internal clock and the TIF flag is set.

The delay between the rising edge on TI2 and the actual start of the counter is due to the resynchronization circuit on TI2 input.

Figure 172. Control circuit in trigger mode



Slave mode: External Clock mode 2 + trigger mode

The external clock mode 2 can be used in addition to another slave mode (except external clock mode 1 and encoder mode). In this case, the ETR signal is used as external clock input, and another input can be selected as trigger input when operating in reset mode, gated mode or trigger mode. It is recommended not to select ETR as TRGI through the TS bits of TIMx_SMCR register.

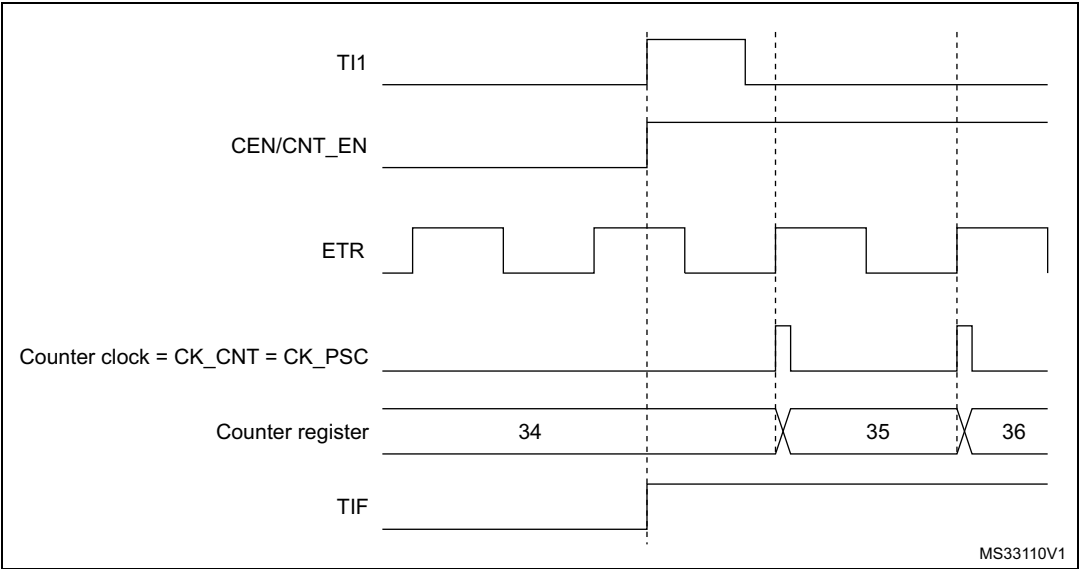
In the following example, the upcounter is incremented at each rising edge of the ETR signal as soon as a rising edge of TI1 occurs:

1. Configure the external trigger input circuit by programming the TIMx_SMCR register as follows:
 - ETF = 0000: no filter
 - ETPS = 00: prescaler disabled
 - ETP = 0: detection of rising edges on ETR and ECE=1 to enable the external clock mode 2.
2. Configure the channel 1 as follows, to detect rising edges on TI1:
 - IC1F = 0000: no filter.
 - The capture prescaler is not used for triggering and does not need to be configured.
 - CC1S = 01 in TIMx_CCMR1 register to select only the input capture source
 - CC1P = 0 in TIMx_CCER register to validate the polarity (and detect rising edge only).
3. Configure the timer in trigger mode by writing SMS=110 in TIMx_SMCR register. Select TI1 as the input source by writing TS=101 in TIMx_SMCR register.

A rising edge on TI1 enables the counter and sets the TIF flag. The counter then counts on ETR rising edges.

The delay between the rising edge of the ETR signal and the actual reset of the counter is due to the resynchronization circuit on ETRP input.

Figure 173. Control circuit in external clock mode 2 + trigger mode



18.3.15 Timer synchronization

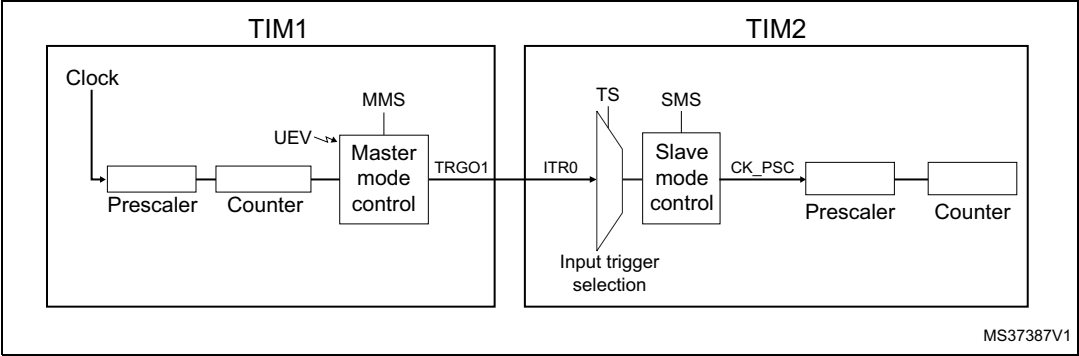
The TIMx timers are linked together internally for timer synchronization or chaining. When one Timer is configured in Master mode, it can reset, start, stop or clock the counter of another Timer configured in Slave mode.

Figure 174 presents an overview of the trigger selection and the master mode selection blocks.

Note: The clock of the slave timer must be enabled prior to receiving events from the master timer, and must not be changed on-the-fly while triggers are received from the master timer.

Using one timer as prescaler for another timer

Figure 174. Master/Slave timer example



For example, the user can configure Timer 1 to act as a prescaler for Timer 2 (see [Figure 174](#)). To do this:

- Configure Timer 1 in master mode so that it outputs a periodic trigger signal on each update event UEV. If you write MMS=010 in the TIM1_CR2 register, a rising edge is output on TRGO1 each time an update event is generated.
- To connect the TRGO1 output of Timer 1 to Timer 2, Timer 2 must be configured in slave mode using ITR0 as internal trigger. You select this through the TS bits in the TIM2_SMCR register (writing TS=000).
- Then you put the slave mode controller in external clock mode 1 (write SMS=111 in the TIM2_SMCR register). This causes Timer 2 to be clocked by the rising edge of the periodic Timer 1 trigger signal (which correspond to the timer 1 counter overflow).
- Finally both timers must be enabled by setting their respective CEN bits (TIMx_CR1 register).

Note: If OCx is selected on Timer 1 as trigger output (MMS=1xx), its rising edge is used to clock the counter of timer 2.

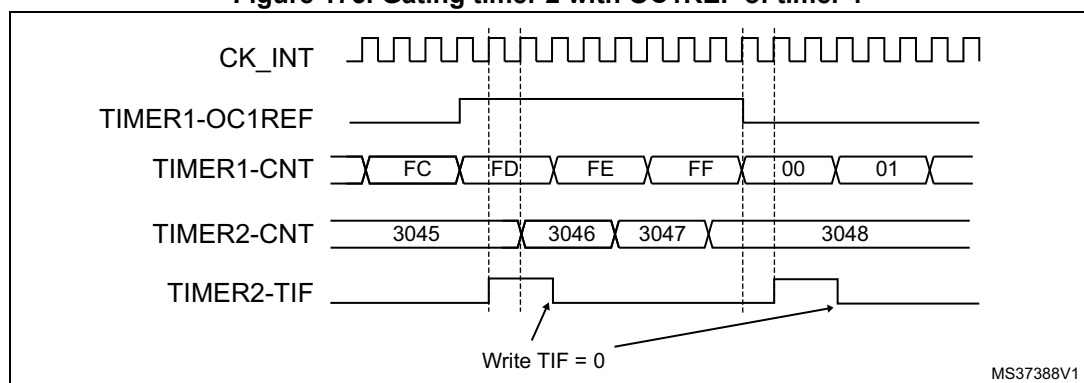
Using one timer to enable another timer

In this example, we control the enable of Timer 2 with the output compare 1 of Timer 1. Refer to [Figure 174](#) for connections. Timer 2 counts on the divided internal clock only when OC1REF of Timer 1 is high. Both counter clock frequencies are divided by 3 by the prescaler compared to CK_INT ($f_{CK_CNT} = f_{CK_INT}/3$).

- Configure Timer 1 master mode to send its Output compare 1 Reference (OC1REF) signal as trigger output (MMS=100 in the TIM1_CR2 register).
- Configure the Timer 1 OC1REF waveform (TIM1_CCMR1 register).
- Configure Timer 2 to get the input trigger from Timer 1 (TS=000 in the TIM2_SMCR register).
- Configure Timer 2 in gated mode (SMS=101 in TIM2_SMCR register).
- Enable Timer 2 by writing '1' in the CEN bit (TIM2_CR1 register).
- Start Timer 1 by writing '1' in the CEN bit (TIM1_CR1 register).

Note: The counter 2 clock is not synchronized with counter 1, this mode only affects the Timer 2 counter enable signal.

Figure 175. Gating timer 2 with OC1REF of timer 1



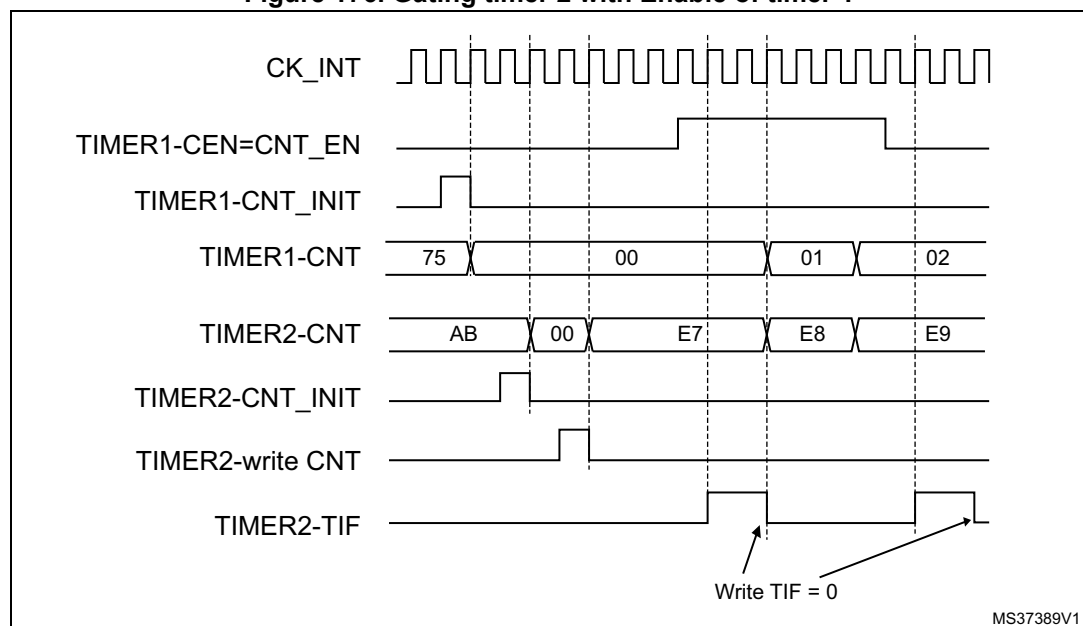
In the example in [Figure 175](#), the Timer 2 counter and prescaler are not initialized before being started. So they start counting from their current value. It is possible to start from a given value by resetting both timers before starting Timer 1. You can then write any value

you want in the timer counters. The timers can easily be reset by software using the UG bit in the TIMx_EGR registers.

In the next example, we synchronize Timer 1 and Timer 2. Timer 1 is the master and starts from 0. Timer 2 is the slave and starts from 0xE7. The prescaler ratio is the same for both timers. Timer 2 stops when Timer 1 is disabled by writing '0 to the CEN bit in the TIM1_CR1 register:

- Configure Timer 1 master mode to send its Output compare 1 Reference (OC1REF) signal as trigger output (MMS=100 in the TIM1_CR2 register).
- Configure the Timer 1 OC1REF waveform (TIM1_CCMR1 register).
- Configure Timer 2 to get the input trigger from Timer 1 (TS=000 in the TIM2_SMCR register).
- Configure Timer 2 in gated mode (SMS=101 in TIM2_SMCR register).
- Reset Timer 1 by writing '1 in UG bit (TIM1_EGR register).
- Reset Timer 2 by writing '1 in UG bit (TIM2_EGR register).
- Initialize Timer 2 to 0xE7 by writing '0xE7' in the timer 2 counter (TIM2_CNT).
- Enable Timer 2 by writing '1 in the CEN bit (TIM2_CR1 register).
- Start Timer 1 by writing '1 in the CEN bit (TIM1_CR1 register).
- Stop Timer 1 by writing '0 in the CEN bit (TIM1_CR1 register).

Figure 176. Gating timer 2 with Enable of timer 1



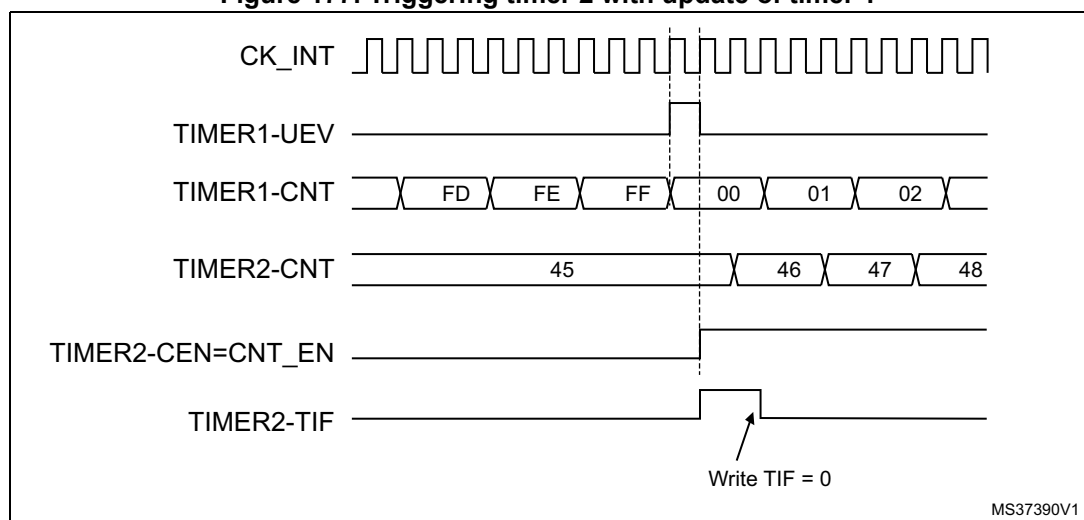
MS37389V1

Using one timer to start another timer

In this example, we set the enable of Timer 2 with the update event of Timer 1. Refer to [Figure 174](#) for connections. Timer 2 starts counting from its current value (which can be nonzero) on the divided internal clock as soon as the update event is generated by Timer 1. When Timer 2 receives the trigger signal its CEN bit is automatically set and the counter counts until we write '0 to the CEN bit in the TIM2_CR1 register. Both counter clock frequencies are divided by 3 by the prescaler compared to CK_INT ($f_{CK_CNT} = f_{CK_INT}/3$).

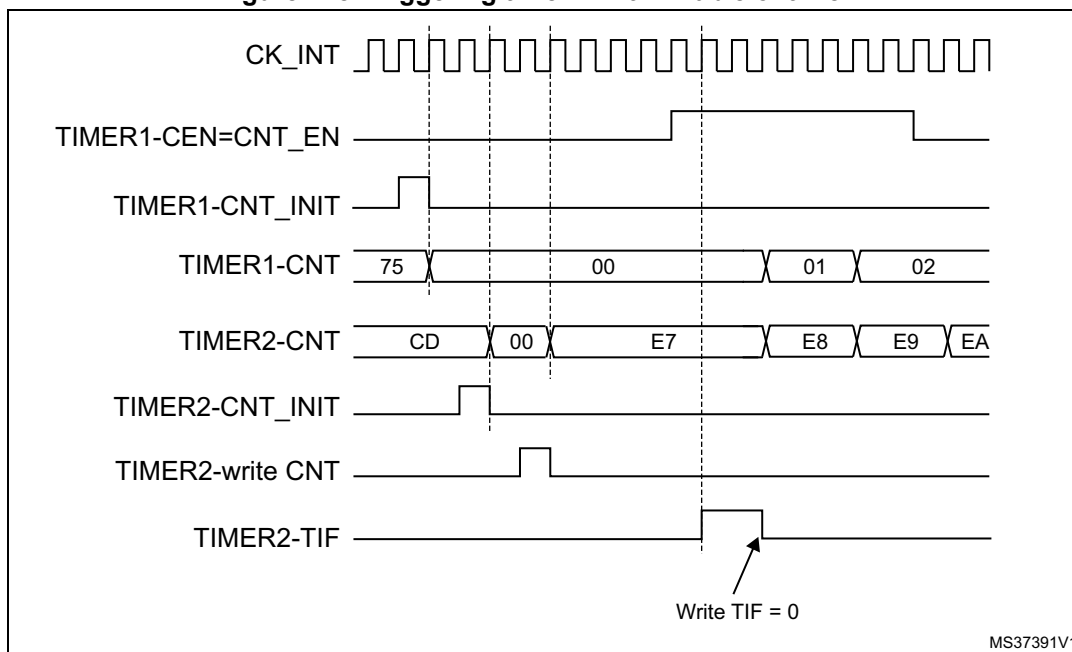
- Configure Timer 1 master mode to send its Update Event (UEV) as trigger output (MMS=010 in the TIM1_CR2 register).
- Configure the Timer 1 period (TIM1_ARR registers).
- Configure Timer 2 to get the input trigger from Timer 1 (TS=000 in the TIM2_SMCR register).
- Configure Timer 2 in trigger mode (SMS=110 in TIM2_SMCR register).
- Start Timer 1 by writing '1 in the CEN bit (TIM1_CR1 register).

Figure 177. Triggering timer 2 with update of timer 1



As in the previous example, the user can initialize both counters before starting counting. [Figure 178](#) shows the behavior with the same configuration as in [Figure 177](#) but in trigger mode instead of gated mode (SMS=110 in the TIM2_SMCR register).

Figure 178. Triggering timer 2 with Enable of timer 1



MS37391V1

Starting 2 timers synchronously in response to an external trigger

In this example, we set the enable of timer 1 when its TI1 input rises, and the enable of Timer 2 with the enable of Timer 1. Refer to [Figure 174](#) for connections. To ensure the counters are aligned, Timer 1 must be configured in Master/Slave mode (slave with respect to TI1, master with respect to Timer 2):

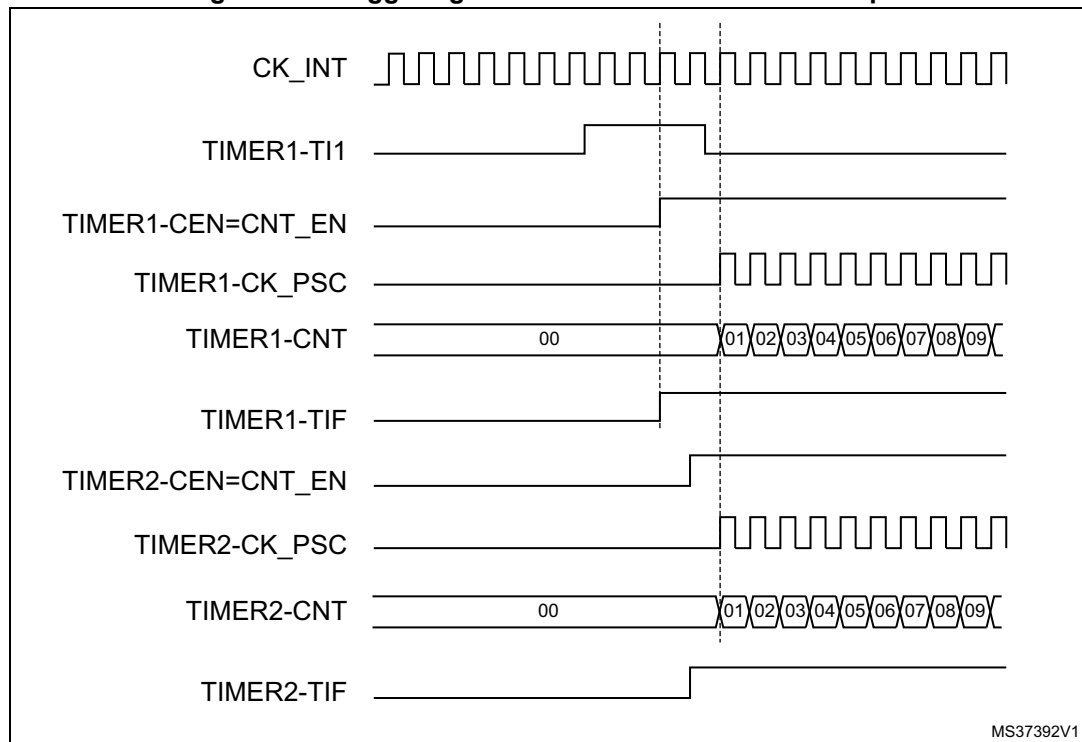
- Configure Timer 1 master mode to send its Enable as trigger output (MMS=001 in the TIM1_CR2 register).
- Configure Timer 1 slave mode to get the input trigger from TI1 (TS=100 in the TIM1_SMCR register).
- Configure Timer 1 in trigger mode (SMS=110 in the TIM1_SMCR register).
- Configure the Timer 1 in Master/Slave mode by writing MSM=1 (TIM1_SMCR register).
- Configure Timer 2 to get the input trigger from Timer 1 (TS=000 in the TIM2_SMCR register).
- Configure Timer 2 in trigger mode (SMS=110 in the TIM2_SMCR register).

When a rising edge occurs on TI1 (Timer 1), both counters start counting synchronously on the internal clock and both TIF flags are set.

Note:

In this example both timers are initialized before starting (by setting their respective UG bits). Both counters start from 0, but you can easily insert an offset between them by writing any of the counter registers (TIMx_CNT). You can see that the master/slave mode insert a delay between CNT_EN and CK_PSC on timer 1.

Figure 179. Triggering timer 1 and 2 with timer 1 TI1 input



18.3.16 Debug mode

When the microcontroller enters debug mode (Cortex[®]-M4 with FPU core - halted), the TIMx counter either continues to work normally or stops, depending on DBG_TIMx_STOP configuration bit in DBGMCU module. For more details, refer to [Section 38.16.2: Debug support for timers, watchdog, bxCAN and I²C](#).

18.4 TIM2 to TIM5 registers

Refer to [Section 2.2](#) for a list of abbreviations used in register descriptions.

The 32-bit peripheral registers have to be written by words (32 bits). All other peripheral registers have to be written by half-words (16 bits) or words (32 bits). Read accesses can be done by bytes (8 bits), half-words (16 bits) or words (32 bits).

18.4.1 TIMx control register 1 (TIMx_CR1)

Address offset: 0x00

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						CKD[1:0]		ARPE	CMS		DIR	OPM	URS	UDIS	CEN
						rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:10 Reserved, must be kept at reset value.

Bits 9:8 **CKD**: Clock division

This bit-field indicates the division ratio between the timer clock (CK_INT) frequency and sampling clock used by the digital filters (ETR, Tlx),

00: $t_{DTS} = t_{CK_INT}$

01: $t_{DTS} = 2 \times t_{CK_INT}$

10: $t_{DTS} = 4 \times t_{CK_INT}$

11: Reserved

Bit 7 **ARPE**: Auto-reload preload enable

0: TIMx_ARR register is not buffered

1: TIMx_ARR register is buffered

Bits 6:5 **CMS**: Center-aligned mode selection

00: Edge-aligned mode. The counter counts up or down depending on the direction bit (DIR).

01: Center-aligned mode 1. The counter counts up and down alternatively. Output compare interrupt flags of channels configured in output (CCxS=00 in TIMx_CCMRx register) are set only when the counter is counting down.

10: Center-aligned mode 2. The counter counts up and down alternatively. Output compare interrupt flags of channels configured in output (CCxS=00 in TIMx_CCMRx register) are set only when the counter is counting up.

11: Center-aligned mode 3. The counter counts up and down alternatively. Output compare interrupt flags of channels configured in output (CCxS=00 in TIMx_CCMRx register) are set both when the counter is counting up or down.

Note: It is not allowed to switch from edge-aligned mode to center-aligned mode as long as the counter is enabled (CEN=1)

Bit 4 **DIR**: Direction

0: Counter used as upcounter

1: Counter used as downcounter

Note: This bit is read only when the timer is configured in Center-aligned mode or Encoder mode.

Bit 3 **OPM**: One-pulse mode

0: Counter is not stopped at update event

1: Counter stops counting at the next update event (clearing the bit CEN)

Bit 2 URS: Update request source

This bit is set and cleared by software to select the UEV event sources.

0: Any of the following events generate an update interrupt or DMA request if enabled.

These events can be:

- Counter overflow/underflow
- Setting the UG bit
- Update generation through the slave mode controller

1: Only counter overflow/underflow generates an update interrupt or DMA request if enabled.

Bit 1 UDIS: Update disable

This bit is set and cleared by software to enable/disable UEV event generation.

0: UEV enabled. The Update (UEV) event is generated by one of the following events:

- Counter overflow/underflow
- Setting the UG bit
- Update generation through the slave mode controller

Buffered registers are then loaded with their preload values.

1: UEV disabled. The Update event is not generated, shadow registers keep their value (ARR, PSC, CCRx). However the counter and the prescaler are reinitialized if the UG bit is set or if a hardware reset is received from the slave mode controller.

Bit 0 CEN: Counter enable

0: Counter disabled

1: Counter enabled

Note: External clock, gated mode and encoder mode can work only if the CEN bit has been previously set by software. However trigger mode can set the CEN bit automatically by hardware.

CEN is cleared automatically in one-pulse mode, when an update event occurs.

18.4.2 TIMx control register 2 (TIMx_CR2)

Address offset: 0x04

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								TI1S	MMS[2:0]			CCDS	Reserved		
								rw	rw	rw	rw	rw			

Bits 15:8 Reserved, must be kept at reset value.

Bit 7 **TI1S**: TI1 selection

0: The TIMx_CH1 pin is connected to TI1 input

1: The TIMx_CH1, CH2 and CH3 pins are connected to the TI1 input (XOR combination)

Bits 6:4 **MMS[2:0]**: Master mode selection

These bits can be used to select the information to be sent in master mode to slave timers for synchronization (TRGO). The combination is as follows:

000: **Reset** - the UG bit from the TIMx_EGR register is used as trigger output (TRGO). If the reset is generated by the trigger input (slave mode controller configured in reset mode) then the signal on TRGO is delayed compared to the actual reset.

001: **Enable** - the Counter enable signal, CNT_EN, is used as trigger output (TRGO). It is useful to start several timers at the same time or to control a window in which a slave timer is enabled. The Counter Enable signal is generated by a logic OR between CEN control bit and the trigger input when configured in gated mode.

When the Counter Enable signal is controlled by the trigger input, there is a delay on TRGO, except if the master/slave mode is selected (see the MSM bit description in TIMx_SMCR register).

010: **Update** - The update event is selected as trigger output (TRGO). For instance a master timer can then be used as a prescaler for a slave timer.

011: **Compare Pulse** - The trigger output send a positive pulse when the CC1IF flag is to be set (even if it was already high), as soon as a capture or a compare match occurred. (TRGO)

100: **Compare** - OC1REF signal is used as trigger output (TRGO)

101: **Compare** - OC2REF signal is used as trigger output (TRGO)

110: **Compare** - OC3REF signal is used as trigger output (TRGO)

111: **Compare** - OC4REF signal is used as trigger output (TRGO)

Note: The clock of the slave timer and ADC must be enabled prior to receiving events from the master timer, and must not be changed on-the-fly while triggers are received from the master timer.

Bit 3 **CCDS**: Capture/compare DMA selection

0: CCx DMA request sent when CCx event occurs

1: CCx DMA requests sent when update event occurs

Bits 2:0 Reserved, must be kept at reset value.

18.4.3 TIMx slave mode control register (TIMx_SMCR)

Address offset: 0x08

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETP	ECE	ETPS[1:0]		ETF[3:0]				MSM	TS[2:0]			Res.	SMS[2:0]		
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw

Bit 15 **ETP**: External trigger polarity

This bit selects whether ETR or $\overline{\text{ETR}}$ is used for trigger operations

0: ETR is noninverted, active at high level or rising edge

1: ETR is inverted, active at low level or falling edge

Bit 14 **ECE**: External clock enable

This bit enables External clock mode 2.

0: External clock mode 2 disabled

1: External clock mode 2 enabled. The counter is clocked by any active edge on the ETRF signal.

1: Setting the ECE bit has the same effect as selecting external clock mode 1 with TRGI connected to ETRF (SMS=111 and TS=111).

2: It is possible to simultaneously use external clock mode 2 with the following slave modes: reset mode, gated mode and trigger mode. Nevertheless, TRGI must not be connected to ETRF in this case (TS bits must not be 111).

3: If external clock mode 1 and external clock mode 2 are enabled at the same time, the external clock input is ETRF.

Bits 13:12 **ETPS**: External trigger prescaler

External trigger signal ETRP frequency must be at most 1/4 of CK_INT frequency. A prescaler can be enabled to reduce ETRP frequency. It is useful when inputting fast external clocks.

00: Prescaler OFF

01: ETRP frequency divided by 2

10: ETRP frequency divided by 4

11: ETRP frequency divided by 8

Bits 11:8 **ETF[3:0]**: External trigger filter

This bit-field then defines the frequency used to sample ETRP signal and the length of the digital filter applied to ETRP. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, sampling is done at f_{DTS}

0001: $f_{\text{SAMPLING}} = f_{\text{CK_INT}}$, N=2

0010: $f_{\text{SAMPLING}} = f_{\text{CK_INT}}$, N=4

0011: $f_{\text{SAMPLING}} = f_{\text{CK_INT}}$, N=8

0100: $f_{\text{SAMPLING}} = f_{\text{DTS}}/2$, N=6

0101: $f_{\text{SAMPLING}} = f_{\text{DTS}}/2$, N=8

0110: $f_{\text{SAMPLING}} = f_{\text{DTS}}/4$, N=6

0111: $f_{\text{SAMPLING}} = f_{\text{DTS}}/4$, N=8

1000: $f_{\text{SAMPLING}} = f_{\text{DTS}}/8$, N=6

1001: $f_{\text{SAMPLING}} = f_{\text{DTS}}/8$, N=8

1010: $f_{\text{SAMPLING}} = f_{\text{DTS}}/16$, N=5

1011: $f_{\text{SAMPLING}} = f_{\text{DTS}}/16$, N=6

1100: $f_{\text{SAMPLING}} = f_{\text{DTS}}/16$, N=8

1101: $f_{\text{SAMPLING}} = f_{\text{DTS}}/32$, N=5

1110: $f_{\text{SAMPLING}} = f_{\text{DTS}}/32$, N=6

1111: $f_{\text{SAMPLING}} = f_{\text{DTS}}/32$, N=8

Bit 7 **MSM**: Master/Slave mode

0: No action

1: The effect of an event on the trigger input (TRGI) is delayed to allow a perfect synchronization between the current timer and its slaves (through TRGO). It is useful if we want to synchronize several timers on a single external event.

Bits 6:4 **TS**: Trigger selection

This bit-field selects the trigger input to be used to synchronize the counter.

000: Internal Trigger 0 (ITR0)

001: Internal Trigger 1 (ITR1).

010: Internal Trigger 2 (ITR2).

011: Internal Trigger 3 (ITR3).

100: TI1 Edge Detector (TI1F_ED)

101: Filtered Timer Input 1 (TI1FP1)

110: Filtered Timer Input 2 (TI2FP2)

111: External Trigger input (ETRF)

See [Table 99: TIMx internal trigger connection on page 635](#) for more details on ITRx meaning for each Timer.*Note: These bits must be changed only when they are not used (e.g. when SMS=000) to avoid wrong edge detections at the transition.*

Bit 3 Reserved, must be kept at reset value.

Bits 2:0 **SMS**: Slave mode selection

When external signals are selected the active edge of the trigger signal (TRGI) is linked to the polarity selected on the external input (see Input Control register and Control register description).

000: Slave mode disabled - if CEN = '1 then the prescaler is clocked directly by the internal clock.

001: Encoder mode 1 - Counter counts up/down on TI1FP1 edge depending on TI2FP2 level.

010: Encoder mode 2 - Counter counts up/down on TI2FP2 edge depending on TI1FP1 level.

011: Encoder mode 3 - Counter counts up/down on both TI1FP1 and TI2FP2 edges depending on the level of the other input.

100: Reset mode - Rising edge of the selected trigger input (TRGI) reinitializes the counter and generates an update of the registers.

101: Gated mode - The counter clock is enabled when the trigger input (TRGI) is high. The counter stops (but is not reset) as soon as the trigger becomes low. Both start and stop of the counter are controlled.

110: Trigger mode - The counter starts at a rising edge of the trigger TRGI (but it is not reset). Only the start of the counter is controlled.

111: External Clock mode 1 - Rising edges of the selected trigger (TRGI) clock the counter.

*Note: The gated mode must not be used if TI1F_ED is selected as the trigger input (TS=100). Indeed, TI1F_ED outputs 1 pulse for each transition on TI1F, whereas the gated mode checks the level of the trigger signal.**The clock of the slave timer must be enabled prior to receiving events from the master timer, and must not be changed on-the-fly while triggers are received from the master timer.*

Table 99. TIMx internal trigger connection

Slave TIM	ITR0 (TS = 000)	ITR1 (TS = 001)	ITR2 (TS = 010)	ITR3 (TS = 011)
TIM2	TIM1_TRGO	TIM8_TRGO	TIM3_TRGO	TIM4_TRGO
TIM3	TIM1_TRGO	TIM2_TRGO	TIM5_TRGO	TIM4_TRGO
TIM4	TIM1_TRGO	TIM2_TRGO	TIM3_TRGO	TIM8_TRGO
TIM5	TIM2_TRGO	TIM3_TRGO	TIM4_TRGO	TIM8_TRGO

18.4.4 TIMx DMA/Interrupt enable register (TIMx_DIER)

Address offset: 0x0C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	TDE	Res	CC4DE	CC3DE	CC2DE	CC1DE	UDE	Res.	TIE	Res	CC4IE	CC3IE	CC2IE	CC1IE	UIE
	rw		rw	rw	rw	rw	rw		rw		rw	rw	rw	rw	rw

Bit 15 Reserved, must be kept at reset value.

Bit 14 **TDE**: Trigger DMA request enable
 0: Trigger DMA request disabled.
 1: Trigger DMA request enabled.

Bit 13 Reserved, always read as 0

Bit 12 **CC4DE**: Capture/Compare 4 DMA request enable
 0: CC4 DMA request disabled.
 1: CC4 DMA request enabled.

Bit 11 **CC3DE**: Capture/Compare 3 DMA request enable
 0: CC3 DMA request disabled.
 1: CC3 DMA request enabled.

Bit 10 **CC2DE**: Capture/Compare 2 DMA request enable
 0: CC2 DMA request disabled.
 1: CC2 DMA request enabled.

Bit 9 **CC1DE**: Capture/Compare 1 DMA request enable
 0: CC1 DMA request disabled.
 1: CC1 DMA request enabled.

Bit 8 **UDE**: Update DMA request enable
 0: Update DMA request disabled.
 1: Update DMA request enabled.

Bit 7 Reserved, must be kept at reset value.

Bit 6 **TIE**: Trigger interrupt enable
 0: Trigger interrupt disabled.
 1: Trigger interrupt enabled.

Bit 5 Reserved, must be kept at reset value.

Bit 4 **CC4IE**: Capture/Compare 4 interrupt enable
 0: CC4 interrupt disabled.
 1: CC4 interrupt enabled.

Bit 3 **CC3IE**: Capture/Compare 3 interrupt enable

0: CC3 interrupt disabled

1: CC3 interrupt enabled

Bit 2 **CC2IE**: Capture/Compare 2 interrupt enable

0: CC2 interrupt disabled

1: CC2 interrupt enabled

Bit 1 **CC1IE**: Capture/Compare 1 interrupt enable

0: CC1 interrupt disabled

1: CC1 interrupt enabled

Bit 0 **UIE**: Update interrupt enable

0: Update interrupt disabled

1: Update interrupt enabled

18.4.5 TIMx status register (TIMx_SR)

Address offset: 0x10

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			CC4OF	CC3OF	CC2OF	CC1OF	Reserved		TIF	Res	CC4IF	CC3IF	CC2IF	CC1IF	UIF
			rc_w0	rc_w0	rc_w0	rc_w0			rc_w0		rc_w0	rc_w0	rc_w0	rc_w0	

Bits 15:13 Reserved, must be kept at reset value.

Bit 12 **CC4OF**: Capture/Compare 4 overcapture flag

refer to CC1OF description

Bit 11 **CC3OF**: Capture/Compare 3 overcapture flag

refer to CC1OF description

Bit 10 **CC2OF**: Capture/compare 2 overcapture flag

refer to CC1OF description

Bit 9 **CC1OF**: Capture/Compare 1 overcapture flag

This flag is set by hardware only when the corresponding channel is configured in input capture mode. It is cleared by software by writing it to '0.

0: No overcapture has been detected

1: The counter value has been captured in TIMx_CCR1 register while CC1IF flag was already set

Bits 8:7 Reserved, must be kept at reset value.

Bit 6 **TIF**: Trigger interrupt flag

This flag is set by hardware on trigger event (active edge detected on TRGI input when the slave mode controller is enabled in all modes but gated mode. It is set when the counter starts or stops when gated mode is selected. It is cleared by software.

0: No trigger event occurred

1: Trigger interrupt pending

Bit 5 Reserved, must be kept at reset value.

Bit 4 **CC4IF**: Capture/Compare 4 interrupt flag

refer to CC1IF description

Bit 3 **CC3IF**: Capture/Compare 3 interrupt flag
refer to CC1IF description

Bit 2 **CC2IF**: Capture/Compare 2 interrupt flag
refer to CC1IF description

Bit 1 **CC1IF**: Capture/compare 1 interrupt flag

If channel CC1 is configured as output:

This flag is set by hardware when the counter matches the compare value, with some exception in center-aligned mode (refer to the CMS bits in the TIMx_CR1 register description). It is cleared by software.

0: No match

1: The content of the counter TIMx_CNT matches the content of the TIMx_CCR1 register.

When the contents of TIMx_CCR1 are greater than the contents of TIMx_ARR, the CC1IF bit goes high on the counter overflow (in upcounting and up/down-counting modes) or underflow (in downcounting mode)

If channel CC1 is configured as input:

This bit is set by hardware on a capture. It is cleared by software or by reading the TIMx_CCR1 register.

0: No input capture occurred

1: The counter value has been captured in TIMx_CCR1 register (An edge has been detected on IC1 which matches the selected polarity)

Bit 0 **UIF**: Update interrupt flag

" This bit is set by hardware on an update event. It is cleared by software.

0: No update occurred.

1: Update interrupt pending. This bit is set by hardware when the registers are updated:

" At overflow or underflow (for TIM2 to TIM5) and if UDIS=0 in the TIMx_CR1 register.

" When CNT is reinitialized by software using the UG bit in TIMx_EGR register, if URS=0 and UDIS=0 in the TIMx_CR1 register.

When CNT is reinitialized by a trigger event (refer to the synchro control register description), if URS=0 and UDIS=0 in the TIMx_CR1 register.

18.4.6 TIMx event generation register (TIMx_EGR)

Address offset: 0x14

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									TG	Res.	CC4G	CC3G	CC2G	CC1G	UG
									w		w	w	w	w	w

Bits 15:7 Reserved, must be kept at reset value.

Bit 6 **TG**: Trigger generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: The TIF flag is set in TIMx_SR register. Related interrupt or DMA transfer can occur if enabled.

Bit 5 Reserved, must be kept at reset value.

Bit 4 **CC4G**: Capture/compare 4 generation

refer to CC1G description

Bit 3 **CC3G**: Capture/compare 3 generation

refer to CC1G description

Bit 2 **CC2G**: Capture/compare 2 generation

refer to CC1G description

Bit 1 **CC1G**: Capture/compare 1 generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: A capture/compare event is generated on channel 1:

If channel CC1 is configured as output:

CC1IF flag is set, Corresponding interrupt or DMA request is sent if enabled.

If channel CC1 is configured as input:

The current value of the counter is captured in TIMx_CCR1 register. The CC1IF flag is set, the corresponding interrupt or DMA request is sent if enabled. The CC1OF flag is set if the CC1IF flag was already high.

Bit 0 **UG**: Update generation

This bit can be set by software, it is automatically cleared by hardware.

0: No action

1: Re-initialize the counter and generates an update of the registers. Note that the prescaler counter is cleared too (anyway the prescaler ratio is not affected). The counter is cleared if the center-aligned mode is selected or if DIR=0 (upcounting), else it takes the auto-reload value (TIMx_ARR) if DIR=1 (downcounting).

18.4.7 TIMx capture/compare mode register 1 (TIMx_CCMR1)

Address offset: 0x18

Reset value: 0x0000

The channels can be used in input (capture mode) or in output (compare mode). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits of this register have a different function in input and in output mode. For a given bit, OCxx describes its function when the channel is configured in output, ICxx describes its function when the channel is configured in input. Take care that the same bit can have a different meaning for the input stage and for the output stage.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC2CE	OC2M[2:0]			OC2PE	OC2FE	CC2S[1:0]		OC1CE	OC1M[2:0]			OC1PE	OC1FE	CC1S[1:0]	
IC2F[3:0]				IC2PSC[1:0]				IC1F[3:0]			IC1PSC[1:0]				
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Output compare mode

Bit 15 **OC2CE**: Output compare 2 clear enable

Bits 14:12 **OC2M[2:0]**: Output compare 2 mode

Bit 11 **OC2PE**: Output compare 2 preload enable

Bit 10 **OC2FE**: Output compare 2 fast enable

Bits 9:8 **CC2S[1:0]**: Capture/Compare 2 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC2 channel is configured as output

01: CC2 channel is configured as input, IC2 is mapped on TI2

10: CC2 channel is configured as input, IC2 is mapped on TI1

11: CC2 channel is configured as input, IC2 is mapped on TRC. This mode is working only if an internal trigger input is selected through the TS bit (TIMx_SMCR register)

Note: CC2S bits are writable only when the channel is OFF (CC2E = 0 in TIMx_CCER).

Bit 7 **OC1CE**: Output compare 1 clear enable

OC1CE: Output compare 1 Clear Enable

0: OC1Ref is not affected by the ETRF input

1: OC1Ref is cleared as soon as a High level is detected on ETRF input

Bits 6:4 **OC1M**: Output compare 1 mode

These bits define the behavior of the output reference signal OC1REF from which OC1 and OC1N are derived. OC1REF is active high whereas OC1 and OC1N active level depends on CC1P and CC1NP bits.

000: Frozen - The comparison between the output compare register TIMx_CCR1 and the counter TIMx_CNT has no effect on the outputs. (this mode is used to generate a timing base).

001: Set channel 1 to active level on match. OC1REF signal is forced high when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).

010: Set channel 1 to inactive level on match. OC1REF signal is forced low when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).

011: Toggle - OC1REF toggles when TIMx_CNT=TIMx_CCR1.

100: Force inactive level - OC1REF is forced low.

101: Force active level - OC1REF is forced high.

110: PWM mode 1 - In upcounting, channel 1 is active as long as TIMx_CNT < TIMx_CCR1 else inactive. In downcounting, channel 1 is inactive (OC1REF=0) as long as TIMx_CNT > TIMx_CCR1 else active (OC1REF=1).

111: PWM mode 2 - In upcounting, channel 1 is inactive as long as TIMx_CNT < TIMx_CCR1 else active. In downcounting, channel 1 is active as long as TIMx_CNT > TIMx_CCR1 else inactive.

Note: In PWM mode 1 or 2, the OCREF level changes only when the result of the comparison changes or when the output compare mode switches from "frozen" mode to "PWM" mode.

Bit 3 **OC1PE**: Output compare 1 preload enable

0: Preload register on TIMx_CCR1 disabled. TIMx_CCR1 can be written at anytime, the new value is taken in account immediately.

1: Preload register on TIMx_CCR1 enabled. Read/Write operations access the preload register. TIMx_CCR1 preload value is loaded in the active register at each update event.

Note: These bits can not be modified as long as LOCK level 3 has been programmed (LOCK bits in TIMx_BDTR register) and CC1S=00 (the channel is configured in output).

Bit 2 **OC1FE**: Output compare 1 fast enable

This bit is used to accelerate the effect of an event on the trigger in input on the CC output.
0: CC1 behaves normally depending on counter and CCR1 values even when the trigger is ON. The minimum delay to activate CC1 output when an edge occurs on the trigger input is 5 clock cycles.

1: An active edge on the trigger input acts like a compare match on CC1 output. Then, OC is set to the compare level independently from the result of the comparison. Delay to sample the trigger input and to activate CC1 output is reduced to 3 clock cycles. OC1FE acts only if the channel is configured in PWM1 or PWM2 mode.

Bits 1:0 **CC1S**: Capture/Compare 1 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output.

01: CC1 channel is configured as input, IC1 is mapped on TI1.

10: CC1 channel is configured as input, IC1 is mapped on TI2.

11: CC1 channel is configured as input, IC1 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC1S bits are writable only when the channel is OFF (CC1E = 0 in TIMx_CCER).

Input capture mode

Bits 15:12 **IC2F**: Input capture 2 filter

Bits 11:10 **IC2PSC[1:0]**: Input capture 2 prescaler

Bits 9:8 **CC2S**: Capture/compare 2 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.
00: CC2 channel is configured as output.

01: CC2 channel is configured as input, IC2 is mapped on TI2.

10: CC2 channel is configured as input, IC2 is mapped on TI1.

11: CC2 channel is configured as input, IC2 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC2S bits are writable only when the channel is OFF (CC2E = 0 in TIMx_CCER).

Bits 7:4 **IC1F**: Input capture 1 filter

This bit-field defines the frequency used to sample TI1 input and the length of the digital filter applied to TI1. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, sampling is done at f_{DTS}

0001: $f_{SAMPLING}=f_{CK_INT}$, N=2

0010: $f_{SAMPLING}=f_{CK_INT}$, N=4

0011: $f_{SAMPLING}=f_{CK_INT}$, N=8

0100: $f_{SAMPLING}=f_{DTS}/2$, N=6

0101: $f_{SAMPLING}=f_{DTS}/2$, N=8

0110: $f_{SAMPLING}=f_{DTS}/4$, N=6

0111: $f_{SAMPLING}=f_{DTS}/4$, N=8

1000: $f_{SAMPLING}=f_{DTS}/8$, N=6

1001: $f_{SAMPLING}=f_{DTS}/8$, N=8

1010: $f_{SAMPLING}=f_{DTS}/16$, N=5

1011: $f_{SAMPLING}=f_{DTS}/16$, N=6

1100: $f_{SAMPLING}=f_{DTS}/16$, N=8

1101: $f_{SAMPLING}=f_{DTS}/32$, N=5

1110: $f_{SAMPLING}=f_{DTS}/32$, N=6

1111: $f_{SAMPLING}=f_{DTS}/32$, N=8

Bits 3:2 **IC1PSC**: Input capture 1 prescaler

This bit-field defines the ratio of the prescaler acting on CC1 input (IC1).

The prescaler is reset as soon as CC1E=0 (TIMx_CCER register).

00: no prescaler, capture is done each time an edge is detected on the capture input

01: capture is done once every 2 events

10: capture is done once every 4 events

11: capture is done once every 8 events

Bits 1:0 **CC1S**: Capture/Compare 1 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output

01: CC1 channel is configured as input, IC1 is mapped on TI1

10: CC1 channel is configured as input, IC1 is mapped on TI2

11: CC1 channel is configured as input, IC1 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC1S bits are writable only when the channel is OFF (CC1E = 0 in TIMx_CCER).

18.4.8 TIMx capture/compare mode register 2 (TIMx_CCMR2)

Address offset: 0x1C

Reset value: 0x0000

Refer to the above CCMR1 register description.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC4CE	OC4M[2:0]			OC4PE	OC4FE	CC4S[1:0]		OC3CE	OC3M[2:0]			OC3PE	OC3FE	CC3S[1:0]	
IC4F[3:0]				IC4PSC[1:0]				IC3F[3:0]			IC3PSC[1:0]				
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Output compare mode

Bit 15 **OC4CE**: Output compare 4 clear enable

Bits 14:12 **OC4M**: Output compare 4 mode

Bit 11 **OC4PE**: Output compare 4 preload enable

Bit 10 **OC4FE**: Output compare 4 fast enable

Bits 9:8 **CC4S**: Capture/Compare 4 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC4 channel is configured as output

01: CC4 channel is configured as input, IC4 is mapped on TI4

10: CC4 channel is configured as input, IC4 is mapped on TI3

11: CC4 channel is configured as input, IC4 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC4S bits are writable only when the channel is OFF (CC4E = 0 in TIMx_CCER).

Bit 7 **OC3CE**: Output compare 3 clear enable

Bits 6:4 **OC3M**: Output compare 3 mode

Bit 3 **OC3PE**: Output compare 3 preload enable

Bit 2 **OC3FE**: Output compare 3 fast enable

Bits 1:0 **CC3S**: Capture/Compare 3 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC3 channel is configured as output

01: CC3 channel is configured as input, IC3 is mapped on TI3

10: CC3 channel is configured as input, IC3 is mapped on TI4

11: CC3 channel is configured as input, IC3 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC3S bits are writable only when the channel is OFF (CC3E = 0 in TIMx_CCER).

Input capture mode

Bits 15:12 **IC4F**: Input capture 4 filter

Bits 11:10 **IC4PSC**: Input capture 4 prescaler

Bits 9:8 **CC4S**: Capture/Compare 4 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC4 channel is configured as output

01: CC4 channel is configured as input, IC4 is mapped on TI4

10: CC4 channel is configured as input, IC4 is mapped on TI3

11: CC4 channel is configured as input, IC4 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC4S bits are writable only when the channel is OFF (CC4E = 0 in TIMx_CCER).

Bits 7:4 **IC3F**: Input capture 3 filter

Bits 3:2 **IC3PSC**: Input capture 3 prescaler

Bits 1:0 **CC3S**: Capture/Compare 3 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC3 channel is configured as output

01: CC3 channel is configured as input, IC3 is mapped on TI3

10: CC3 channel is configured as input, IC3 is mapped on TI4

11: CC3 channel is configured as input, IC3 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC3S bits are writable only when the channel is OFF (CC3E = 0 in TIMx_CCER).

18.4.9 TIMx capture/compare enable register (TIMx_CCER)

Address offset: 0x20

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CC4NP	Res.	CC4P	CC4E	CC3NP	Res.	CC3P	CC3E	CC2NP	Res.	CC2P	CC2E	CC1NP	Res.	CC1P	CC1E
rw		rw	rw	rw		rw	rw	rw		rw	rw	rw		rw	rw

Bit 15 **CC4NP**: Capture/Compare 4 output Polarity.

Refer to CC1NP description

Bit 14 Reserved, must be kept at reset value.

Bit 13 **CC4P**: Capture/Compare 4 output Polarity.

refer to CC1P description

Bit 12 **CC4E**: Capture/Compare 4 output enable.

refer to CC1E description

Bit 11 **CC3NP**: Capture/Compare 3 output Polarity.

refer to CC1NP description

Bit 10 Reserved, must be kept at reset value.

Bit 9 **CC3P**: Capture/Compare 3 output Polarity.

refer to CC1P description

- Bit 8 **CC3E**: *Capture/Compare 3 output enable.*
refer to CC1E description
- Bit 7 **CC2NP**: *Capture/Compare 2 output Polarity.*
refer to CC1NP description
- Bit 6 Reserved, must be kept at reset value.
- Bit 5 **CC2P**: *Capture/Compare 2 output Polarity.*
refer to CC1P description
- Bit 4 **CC2E**: *Capture/Compare 2 output enable.*
refer to CC1E description
- Bit 3 **CC1NP**: *Capture/Compare 1 output Polarity.*
CC1 channel configured as output:
CC1NP must be kept cleared in this case.
CC1 channel configured as input:
This bit is used in conjunction with CC1P to define TI1FP1/TI2FP1 polarity. refer to CC1P description.
- Bit 2 Reserved, must be kept at reset value.
- Bit 1 **CC1P**: *Capture/Compare 1 output Polarity.*
CC1 channel configured as output:
0: OC1 active high
1: OC1 active low
CC1 channel configured as input:
CC1NP/CC1P bits select TI1FP1 and TI2FP1 polarity for trigger or capture operations.
00: noninverted/rising edge
Circuit is sensitive to TIxFP1 rising edge (capture, trigger in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger in gated mode, encoder mode).
01: inverted/falling edge
Circuit is sensitive to TIxFP1 falling edge (capture, trigger in reset, external clock or trigger mode), TIxFP1 is inverted (trigger in gated mode, encoder mode).
10: reserved, do not use this configuration.
11: noninverted/both edges
Circuit is sensitive to both TIxFP1 rising and falling edges (capture, trigger in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger in gated mode). This configuration must not be used for encoder mode.
- Bit 0 **CC1E**: *Capture/Compare 1 output enable.*
CC1 channel configured as output:
0: Off - OC1 is not active
1: On - OC1 signal is output on the corresponding output pin
CC1 channel configured as input:
This bit determines if a capture of the counter value can actually be done into the input capture/compare register 1 (TIMx_CCR1) or not.
0: Capture disabled
1: Capture enabled

Table 100. Output control bit for standard OCx channels

CCxE bit	OCx output state
0	Output Disabled (OCx=0, OCx_EN=0)
1	OCx=OCxREF + Polarity, OCx_EN=1

Note: The state of the external IO pins connected to the standard OCx channels depends on the OCx channel state and the GPIO registers.

18.4.10 TIMx counter (TIMx_CNT)

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CNT[31:16] (depending on timers)															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 **CNT[31:16]**: High counter value (on TIM2 and TIM5).

Bits 15:0 **CNT[15:0]**: Counter value

18.4.11 TIMx prescaler (TIMx_PSC)

Address offset: 0x28

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **PSC[15:0]**: Prescaler value

The counter clock frequency CK_CNT is equal to $f_{CK_PSC} / (PSC[15:0] + 1)$.

PSC contains the value to be loaded in the active prescaler register at each update event (including when the counter is cleared through UG bit of TIMx_EGR register or through trigger controller when configured in “reset mode”).

18.4.12 TIMx auto-reload register (TIMx_ARR)

Address offset: 0x2C

Reset value: 0xFFFF FFFF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ARR[31:16] (depending on timers)															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 **ARR[31:16]**: High auto-reload value (on TIM2 and TIM5).

Bits 15:0 **ARR[15:0]**: Auto-reload value

ARR is the value to be loaded in the actual auto-reload register.

Refer to the [Section 18.3.1: Time-base unit](#) for more details about ARR update and behavior.

The counter is blocked while the auto-reload value is null.

18.4.13 TIMx capture/compare register 1 (TIMx_CCR1)

Address offset: 0x34

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CCR1[31:16] (depending on timers)															
rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1[15:0]															
rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro

Bits 31:16 **CCR1[31:16]**: High Capture/Compare 1 value (on TIM2 and TIM5).

Bits 15:0 **CCR1[15:0]**: Low Capture/Compare 1 value

If channel CC1 is configured as output:

CCR1 is the value to be loaded in the actual capture/compare 1 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR1 register (bit OC1PE). Else the preload value is copied in the active capture/compare 1 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signaled on OC1 output.

If channel CC1 is configured as input:

CCR1 is the counter value transferred by the last input capture 1 event (IC1). The TIMx_CCR1 register is read-only and cannot be programmed.

18.4.14 TIMx capture/compare register 2 (TIMx_CCR2)

Address offset: 0x38

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CCR2[31:16] (depending on timers)															
rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR2[15:0]															
rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro

Bits 31:16 **CCR2[31:16]**: High Capture/Compare 2 value (on TIM2 and TIM5).

Bits 15:0 **CCR2[15:0]**: Low Capture/Compare 2 value

If channel CC2 is configured as output:

CCR2 is the value to be loaded in the actual capture/compare 2 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR register (bit OC2PE). Else the preload value is copied in the active capture/compare 2 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signalled on OC2 output.

If channel CC2 is configured as input:

CCR2 is the counter value transferred by the last input capture 2 event (IC2). The TIMx_CCR2 register is read-only and cannot be programmed.

18.4.15 TIMx capture/compare register 3 (TIMx_CCR3)

Address offset: 0x3C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CCR3[31:16] (depending on timers)															
rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR3[15:0]															
rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro

Bits 31:16 **CCR3[31:16]**: High Capture/Compare 3 value (on TIM2 and TIM5).

Bits 15:0 **CCR3[15:0]**: Low Capture/Compare value

If channel CC3 is configured as output:

CCR3 is the value to be loaded in the actual capture/compare 3 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR register (bit OC3PE). Else the preload value is copied in the active capture/compare 3 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signalled on OC3 output.

If channel CC3 is configured as input:

CCR3 is the counter value transferred by the last input capture 3 event (IC3). The TIMx_CCR3 register is read-only and cannot be programmed.

18.4.16 TIMx capture/compare register 4 (TIMx_CCR4)

Address offset: 0x40

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CCR4[31:16] (depending on timers)															
rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR4[15:0]															
rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro

Bits 31:16 **CCR4[31:16]**: High Capture/Compare 4 value (on TIM2 and TIM5).

Bits 15:0 **CCR4[15:0]**: Low Capture/Compare value

1. if CC4 channel is configured as output (CC4S bits):
CCR4 is the value to be loaded in the actual capture/compare 4 register (preload value). It is loaded permanently if the preload feature is not selected in the TIMx_CCMR register (bit OC4PE). Else the preload value is copied in the active capture/compare 4 register when an update event occurs.
The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signalled on OC4 output.
2. if CC4 channel is configured as input (CC4S bits in TIMx_CCMR4 register):
CCR4 is the counter value transferred by the last input capture 4 event (IC4). The TIMx_CCR4 register is read-only and cannot be programmed.

18.4.17 TIMx DMA control register (TIMx_DCR)

Address offset: 0x48

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			DBL[4:0]					Reserved			DBA[4:0]				
			rw	rw	rw	rw	rw				rw	rw	rw	rw	rw

Bits 15:13 Reserved, must be kept at reset value.

Bits 12:8 **DBL[4:0]**: DMA burst length

This 5-bit vector defines the number of DMA transfers (the timer recognizes a burst transfer when a read or a write access is done to the TIMx_DMAR address).

00000: 1 transfer,
00001: 2 transfers,
00010: 3 transfers,
...
10001: 18 transfers.

Bits 7:5 Reserved, must be kept at reset value.

Bits 4:0 **DBA[4:0]**: DMA base address

This 5-bit vector defines the base-address for DMA transfers (when read/write access are done through the TIMx_DMAR address). DBA is defined as an offset starting from the address of the TIMx_CR1 register.

Example:

00000: TIMx_CR1,
00001: TIMx_CR2,
00010: TIMx_SMCR,
...

Example: Let us consider the following transfer: DBL = 7 transfers & DBA = TIMx_CR1. In this case the transfer is done to/from 7 registers starting from the TIMx_CR1 address.

18.4.18 TIMx DMA address for full transfer (TIMx_DMAR)

Address offset: 0x4C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMAB[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **DMAB[15:0]**: DMA register for burst accesses

A read or write operation to the DMAR register accesses the register located at the address
 $(\text{TIMx_CR1 address}) + (\text{DBA} + \text{DMA index}) \times 4$

where TIMx_CR1 address is the address of the control register 1, DBA is the DMA base address configured in TIMx_DCR register, DMA index is automatically controlled by the DMA transfer, and ranges from 0 to DBL (DBL configured in TIMx_DCR).

Example of how to use the DMA burst feature

In this example the timer DMA burst feature is used to update the contents of the CCRx registers (x = 2, 3, 4) with the DMA transferring half words into the CCRx registers.

This is done in the following steps:

1. Configure the corresponding DMA channel as follows:
 - DMA channel peripheral address is the DMAR register address
 - DMA channel memory address is the address of the buffer in the RAM containing the data to be transferred by DMA into CCRx registers.
 - Number of data to transfer = 3 (See note below).
 - Circular mode disabled.
2. Configure the DCR register by configuring the DBA and DBL bit fields as follows:
 DBL = 3 transfers, DBA = 0xE.
3. Enable the TIMx update DMA request (set the UDE bit in the DIER register).
4. Enable TIMx
5. Enable the DMA channel

Note: *This example is for the case where every CCRx register to be updated once. If every CCRx register is to be updated twice for example, the number of data to transfer should be 6. Let's take the example of a buffer in the RAM containing data1, data2, data3, data4, data5 and data6. The data is transferred to the CCRx registers as follows: on the first update DMA request, data1 is transferred to CCR2, data2 is transferred to CCR3, data3 is transferred to CCR4 and on the second update DMA request, data4 is transferred to CCR2, data5 is transferred to CCR3 and data6 is transferred to CCR4.*

18.4.19 TIM2 option register (TIM2_OR)

Address offset: 0x50

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				ITR1_RMP		Reserved									
				rw	rw										

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:10 **ITR1_RMP**: Internal trigger 1 remap
Set and cleared by software.
00: TIM8_TRGOUT
01: PTP trigger output is connected to TIM2_ITR1
10: OTG FS SOF is connected to the TIM2_ITR1 input
11: OTG HS SOF is connected to the TIM2_ITR1 input

Bits 9:0 Reserved, must be kept at reset value.

18.4.20 TIM5 option register (TIM5_OR)

Address offset: 0x50
Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								TI4_RMP		Reserved					
								rw	rw						

Bits 15:8 Reserved, must be kept at reset value.

Bits 7:6 **TI4_RMP**: Timer Input 4 remap
Set and cleared by software.
00: TIM5 Channel4 is connected to the GPIO: Refer to the Alternate function mapping table in the datasheets.
01: the LSI internal clock is connected to the TIM5_CH4 input for calibration purposes
10: the LSE internal clock is connected to the TIM5_CH4 input for calibration purposes
11: the RTC wake-up interrupt is connected to TIM5_CH4 input for calibration purposes. Wake-up interrupt should be enabled.

Bits 5:0 Reserved, must be kept at reset value.

18.4.21 TIMx register map

TIMx registers are mapped as described in the table below:

Table 101. TIM2 to TIM5 register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x00	TIMx_CR1	Reserved																						CKD [1:0]		ARPE	CMS [1:0]		DIR	OPM	URS	UDIS	CEN	
	0																							0	0	0	0	0	0	0				
0x04	TIMx_CR2	Reserved																						T1S		MMS[2:0]		CCDS	Reserved					
	0																							0	0	0	0							
0x08	TIMx_SMCR	Reserved																	ETP	ECE	ETPS [1:0]		ETF[3:0]		MSM	TS[2:0]		Reserved	SMS[2:0]					
	0																		0	0	0	0	0	0	0	0	0					0	0	0
0x0C	TIMx_DIER	Reserved																	TDE		COMDE	CC4DE	CC3DE	CC2DE	CC1DE	UDE	Reserved	TIE	Reserved	CC4IE	CC3IE	CC2IE	CC1IE	UIE
	0																		0	0	0	0	0	0	0	0		0	0	0	0	0	0	0
0x10	TIMx_SR	Reserved																			CC4OF		CC3OF	CC2OF	CC1OF	Reserved	TIF	Reserved	CC4IF	CC3IF	CC2IF	CC1IF	UIF	
	0																				0	0	0	0	0		0	0	0	0	0	0	0	0
0x14	TIMx_EGR	Reserved																						TG		Reserved	CC4G	CC3G	CC2G	CC1G	UG			
	0																							Reserved	0	0	0	0	0	0	0			
0x18	TIMx_CCMR1 Output Compare mode	Reserved																	OC2CE	OC2M [2:0]		OC2PE	OC2FE	CC2S [1:0]		OC1CE	OC1M [2:0]		OC1PE	OC1FE	CC1S [1:0]			
	0																		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	TIMx_CCMR1 Input Capture mode	Reserved																	IC2F[3:0]		IC2PSC [1:0]		CC2S [1:0]		IC1F[3:0]		IC1PSC [1:0]		CC1S [1:0]					
	0																		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x1C	TIMx_CCMR2 Output Compare mode	Reserved																	OC4CE	OC4M [2:0]		OC4PE	OC4FE	CC4S [1:0]		OC3CE	OC3M [2:0]		OC3PE	OC3FE	CC3S [1:0]			
	0																		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	TIMx_CCMR2 Input Capture mode	Reserved																	IC4F[3:0]		IC4PSC [1:0]		CC4S [1:0]		IC3F[3:0]		IC3PSC [1:0]		CC3S [1:0]					
	0																		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x20	TIMx_CCER	Reserved																	CC4NP	Reserved	CC4P	CC4E	CC3NP	Reserved	CC3P	CC3E	CC2NP	Reserved	CC2P	CC2E	CC1NP	Reserved	CC1P	CC1E
	0																			0	0	0		0	0	0		0	0	0		0	0	
0x24	TIMx_CNT	CNT[31:16] (TIM2 and TIM5 only, reserved on the other timers)																	CNT[15:0]															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x28	TIMx_PSC	Reserved																	PSC[15:0]															
	0																		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 101. TIM2 to TIM5 register map and reset values (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x2C	TIMx_ARR	ARR[31:16] (TIM2 and TIM5 only, reserved on the other timers)																ARR[15:0]															
	Reset value	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
0x30	Reserved																																
0x34	TIMx_CCR1	CCR1[31:16] (TIM2 and TIM5 only, reserved on the other timers)																CCR1[15:0]															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x38	TIMx_CCR2	CCR2[31:16] (TIM2 and TIM5 only, reserved on the other timers)																CCR2[15:0]															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x3C	TIMx_CCR3	CCR3[31:16] (TIM2 and TIM5 only, reserved on the other timers)																CCR3[15:0]															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x40	TIMx_CCR4	CCR4[31:16] (TIM2 and TIM5 only, reserved on the other timers)																CCR4[15:0]															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x44	Reserved																																
0x48	TIMx_DCR	Reserved																DBL[4:0]				Reserved				DBA[4:0]							
	Reset value																	0	0	0	0					0	0	0	0				
0x4C	TIMx_DMAR	Reserved																DMAB[15:0]															
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x50	TIM2_OR	Reserved																Reserved		ITR1_RMP		Reserved											
	Reset value																			0	0												
0x50	TIM5_OR	Reserved																Reserved						IT4_RMP		Reserved							
	Reset value																							0	0								

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.

19 General-purpose timers (TIM9 to TIM14)

This section applies to the whole STM32F4xx family, unless otherwise specified.

19.1 TIM9 to TIM14 introduction

The TIM9 to TIM14 general-purpose timers consist of a 16-bit auto-reload counter driven by a programmable prescaler.

They may be used for a variety of purposes, including measuring the pulse lengths of input signals (input capture) or generating output waveforms (output compare, PWM).

Pulse lengths and waveform periods can be modulated from a few microseconds to several milliseconds using the timer prescaler and the RCC clock controller prescalers.

The TIM9 to TIM14 timers are completely independent, and do not share any resources. They can be synchronized together as described in [Section 19.3.12](#).

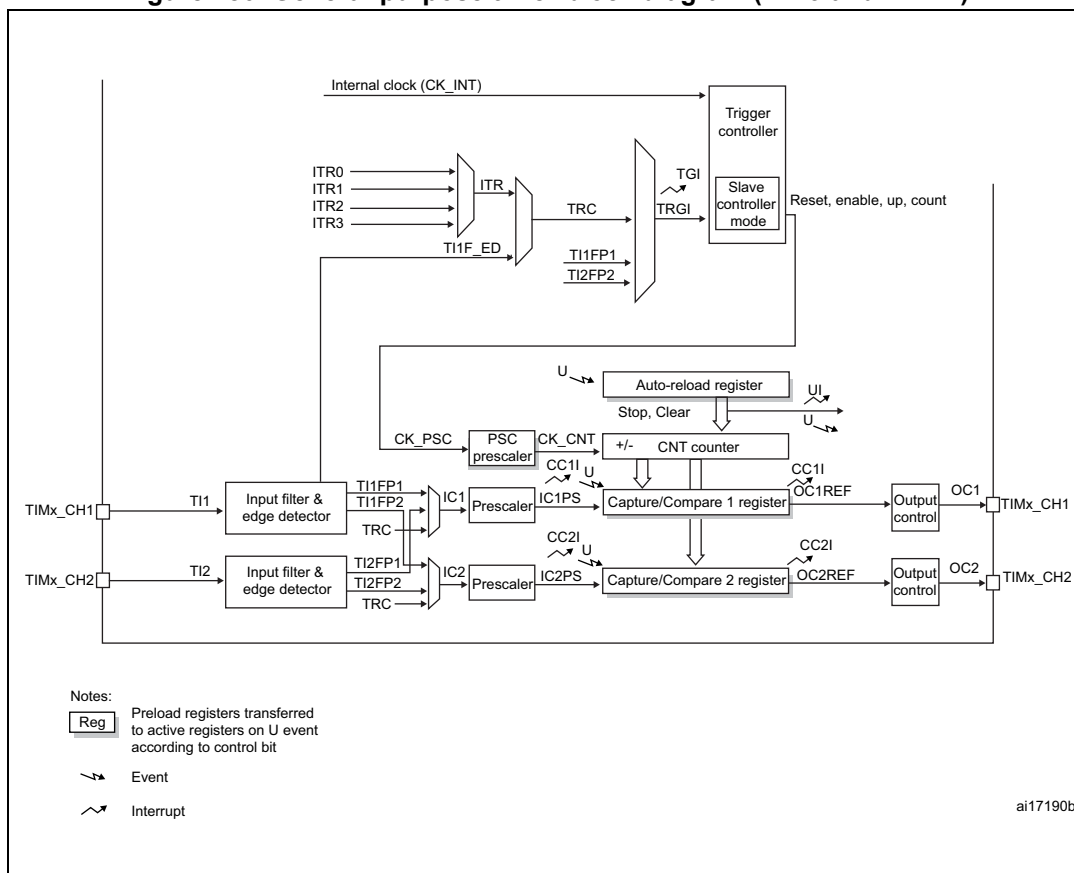
19.2 TIM9 to TIM14 main features

19.2.1 TIM9/TIM12 main features

The features of the TIM9 to TIM14 general-purpose timers include:

- 16-bit auto-reload upcounter
- 16-bit programmable prescaler used to divide the counter clock frequency by any factor between 1 and 65536 (can be changed “on the fly”)
- Up to 2 independent channels for:
 - Input capture
 - Output compare
 - PWM generation (edge-aligned mode)
 - One-pulse mode output
- Synchronization circuit to control the timer with external signals and to interconnect several timers together
- Interrupt generation on the following events:
 - Update: counter overflow, counter initialization (by software or internal trigger)
 - Trigger event (counter start, stop, initialization or count by internal trigger)
 - Input capture
 - Output compare

Figure 180. General-purpose timer block diagram (TIM9 and TIM12)

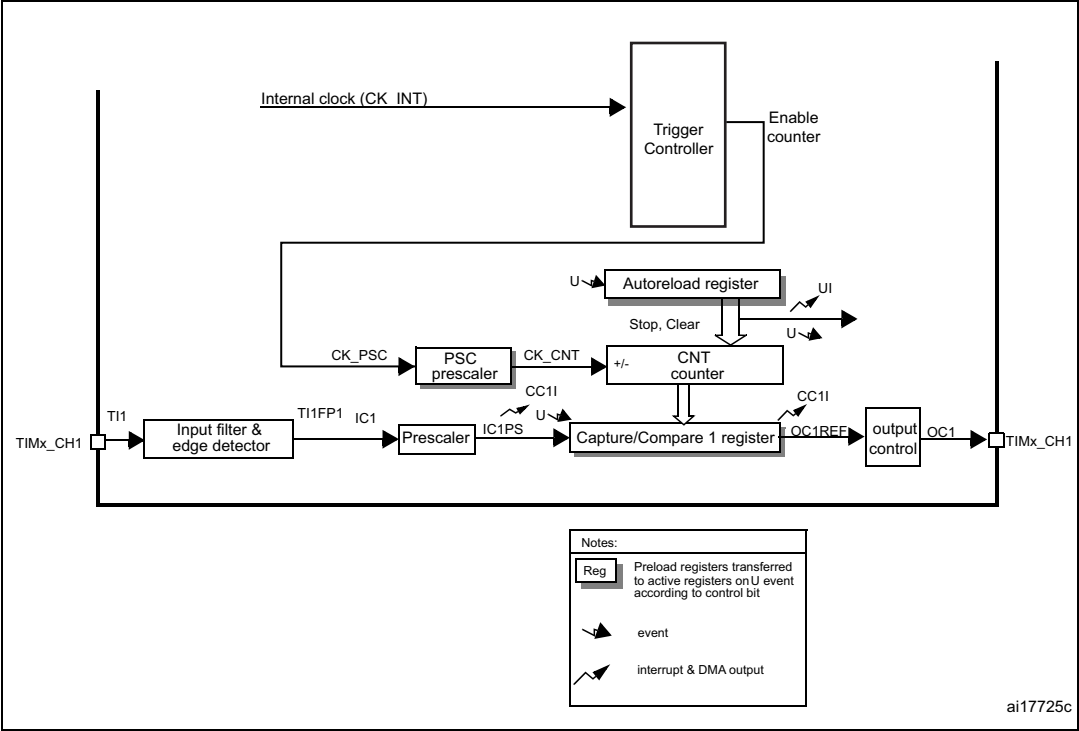


19.2.2 TIM10/TIM11 and TIM13/TIM14 main features

The features of general-purpose timers TIM10/TIM11 and TIM13/TIM14 include:

- 16-bit auto-reload upcounter
- 16-bit programmable prescaler used to divide the counter clock frequency by any factor between 1 and 65536 (can be changed “on the fly”)
- independent channel for:
 - Input capture
 - Output compare
 - PWM generation (edge-aligned mode)
 - One-pulse mode output
- Interrupt generation on the following events:
 - Update: counter overflow, counter initialization (by software)
 - Input capture
 - Output compare

Figure 181. General-purpose timer block diagram (TIM10/11/13/14)



19.3 TIM9 to TIM14 functional description

19.3.1 Time-base unit

The main block of the timer is a 16-bit counter with its related auto-reload register. The counter counts up.

The counter clock can be divided by a prescaler.

The counter, the auto-reload register and the prescaler register can be written or read by software. This is true even when the counter is running.

The time-base unit includes:

- Counter register (TIMx_CNT)
- Prescaler register (TIMx_PSC)
- Auto-reload register (TIMx_ARR)

The auto-reload register is preloaded. Writing to or reading from the auto-reload register accesses the preload register. The content of the preload register are transferred into the shadow register permanently or at each update event (UEV), depending on the auto-reload preload enable bit (ARPE) in TIMx_CR1 register. The update event is sent when the counter reaches the overflow and if the UDIS bit equals 0 in the TIMx_CR1 register. It can also be generated by software. The generation of the update event is described in details for each configuration.

The counter is clocked by the prescaler output CK_CNT, which is enabled only when the counter enable bit (CEN) in TIMx_CR1 register is set (refer also to the slave mode controller description to get more details on counter enabling).

Note that the counter starts counting 1 clock cycle after setting the CEN bit in the TIMx_CR1 register.

Prescaler description

The prescaler can divide the counter clock frequency by any factor between 1 and 65536. It is based on a 16-bit counter controlled through a 16-bit register (in the TIMx_PSC register). It can be changed on the fly as this control register is buffered. The new prescaler ratio is taken into account at the next update event.

[Figure 182](#) and [Figure 183](#) give some examples of the counter behavior when the prescaler ratio is changed on the fly.

Figure 182. Counter timing diagram with prescaler division change from 1 to 2

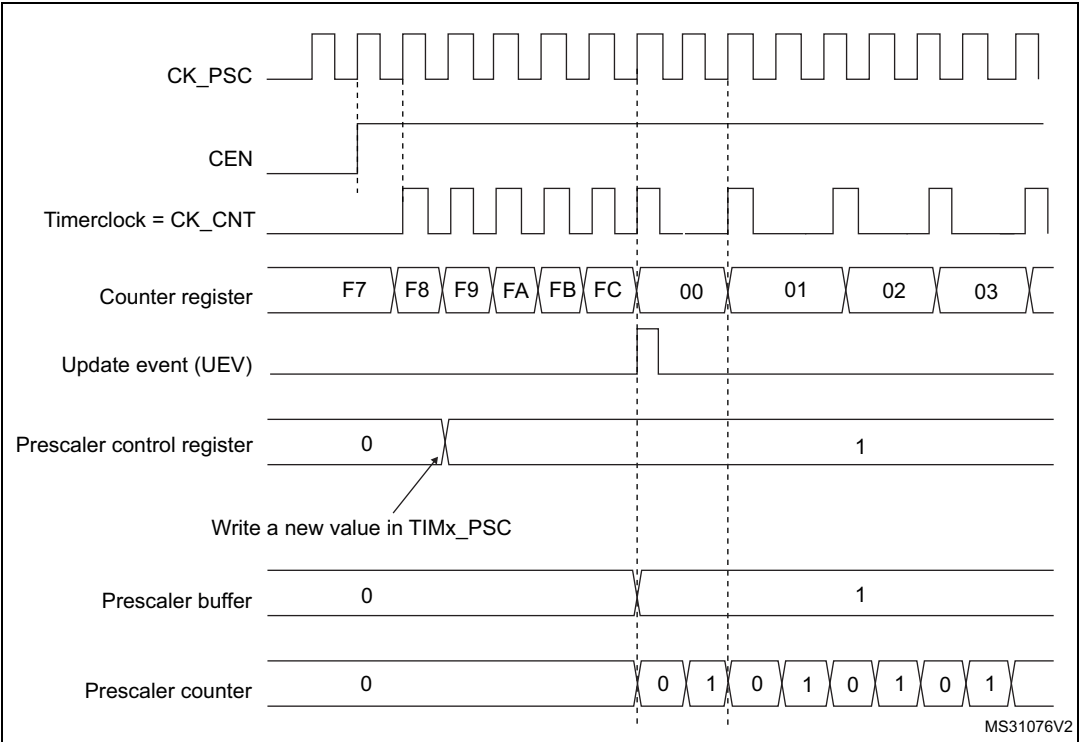
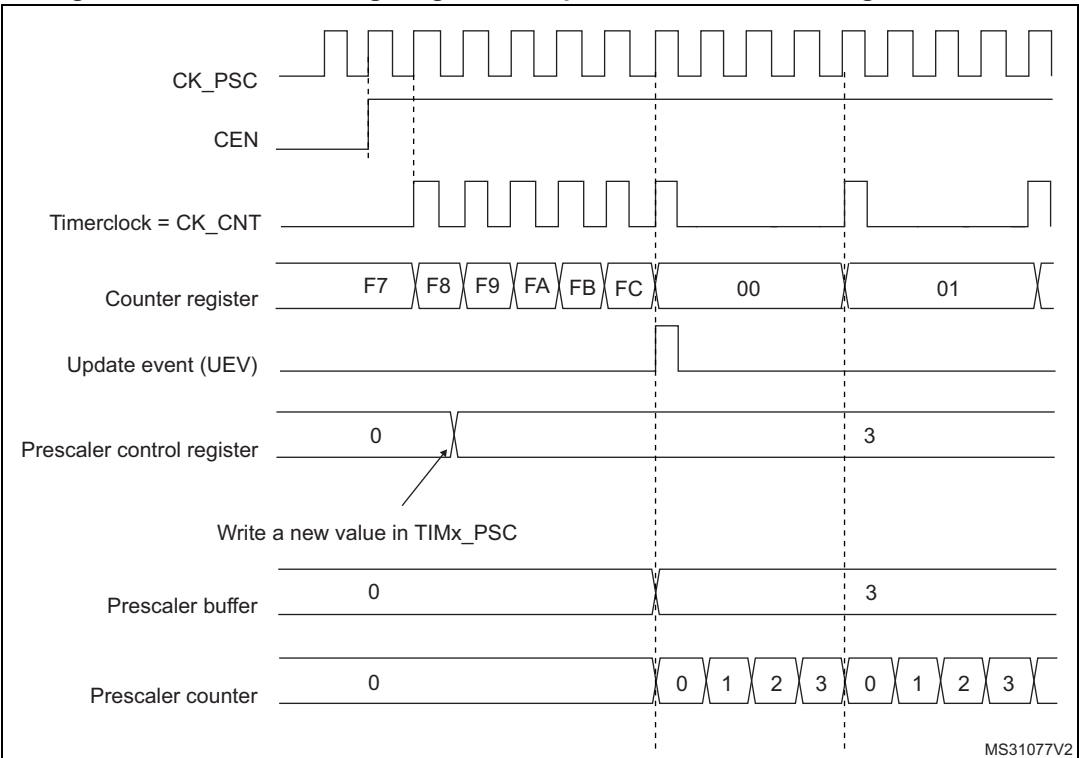


Figure 183. Counter timing diagram with prescaler division change from 1 to 4



19.3.2 Counter modes

Upcounting mode

In upcounting mode, the counter counts from 0 to the auto-reload value (content of the TIMx_ARR register), then restarts from 0 and generates a counter overflow event.

Setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller on TIM9 and TIM12) also generates an update event.

The UEV event can be disabled by software by setting the UDIS bit in the TIMx_CR1 register. This is to avoid updating the shadow registers while writing new values in the preload registers. Then no update event occurs until the UDIS bit has been written to 0. However, the counter restarts from 0, as well as the counter of the prescaler (but the prescale rate does not change). In addition, if the URS bit (update request selection) in TIMx_CR1 register is set, setting the UG bit generates an update event UEV but without setting the UIF flag (thus no interrupt is sent). This is to avoid generating both update and capture interrupts when clearing the counter on the capture event.

When an update event occurs, all the registers are updated and the update flag (UIF bit in TIMx_SR register) is set (depending on the URS bit):

- The auto-reload shadow register is updated with the preload value (TIMx_ARR),
- The buffer of the prescaler is reloaded with the preload value (content of the TIMx_PSC register).

The following figures show some examples of the counter behavior for different clock frequencies when TIMx_ARR=0x36.

Figure 184. Counter timing diagram, internal clock divided by 1

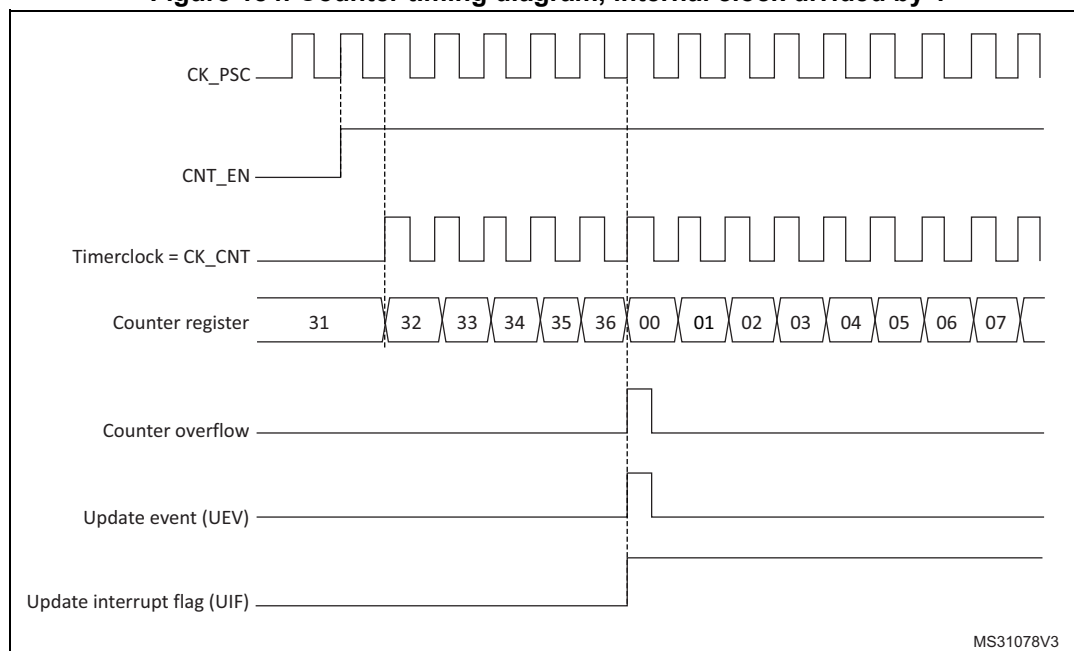


Figure 185. Counter timing diagram, internal clock divided by 2

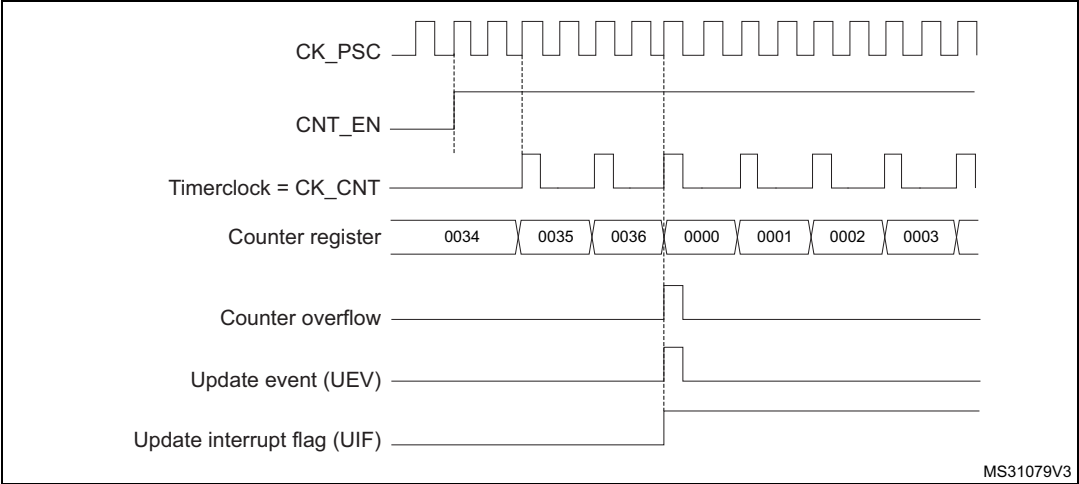


Figure 186. Counter timing diagram, internal clock divided by 4

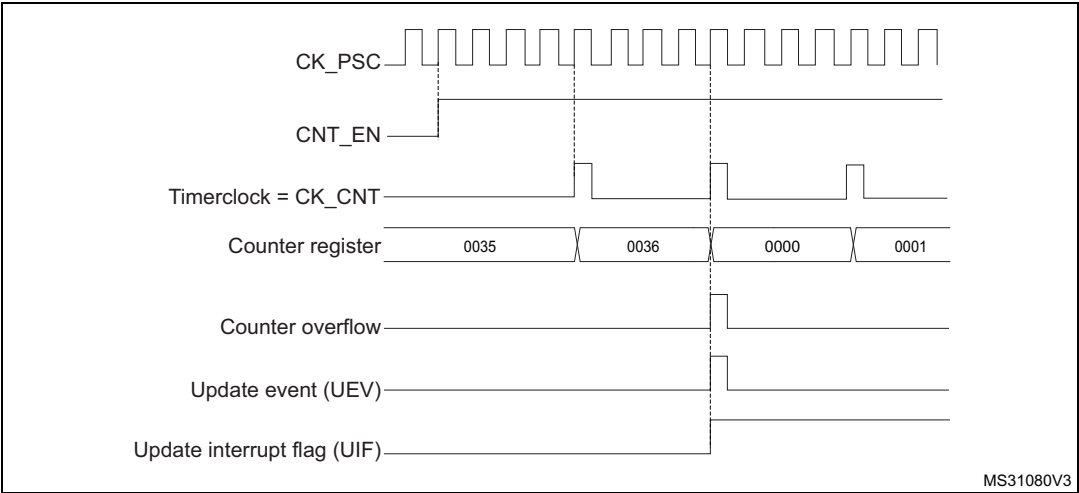


Figure 187. Counter timing diagram, internal clock divided by N

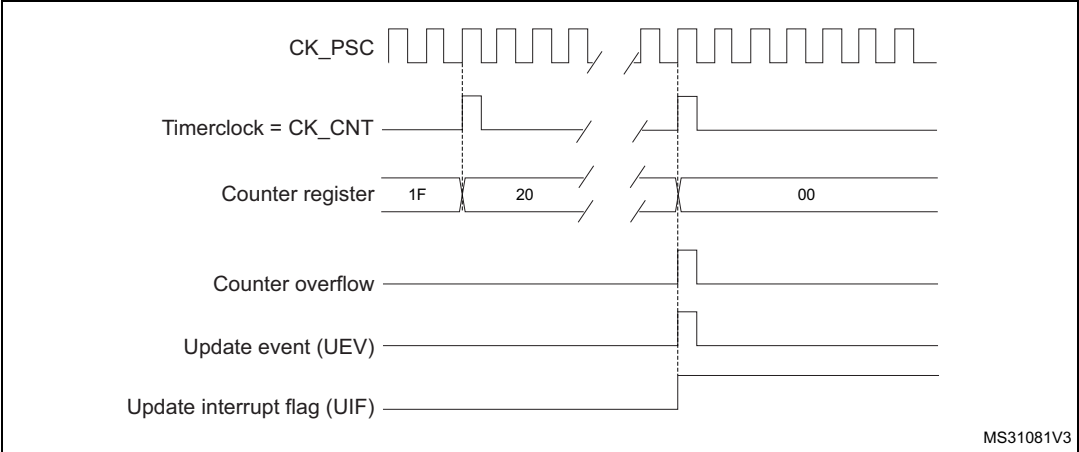


Figure 188. Counter timing diagram, update event when ARPE=0 (TIMx_ARR not preloaded)

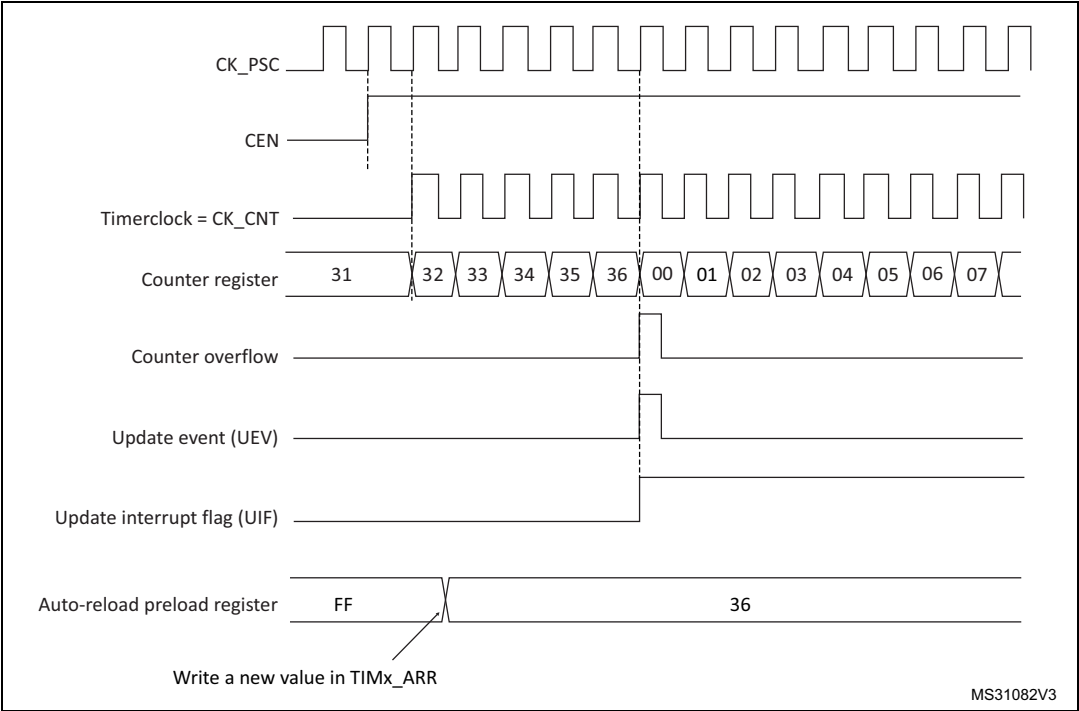
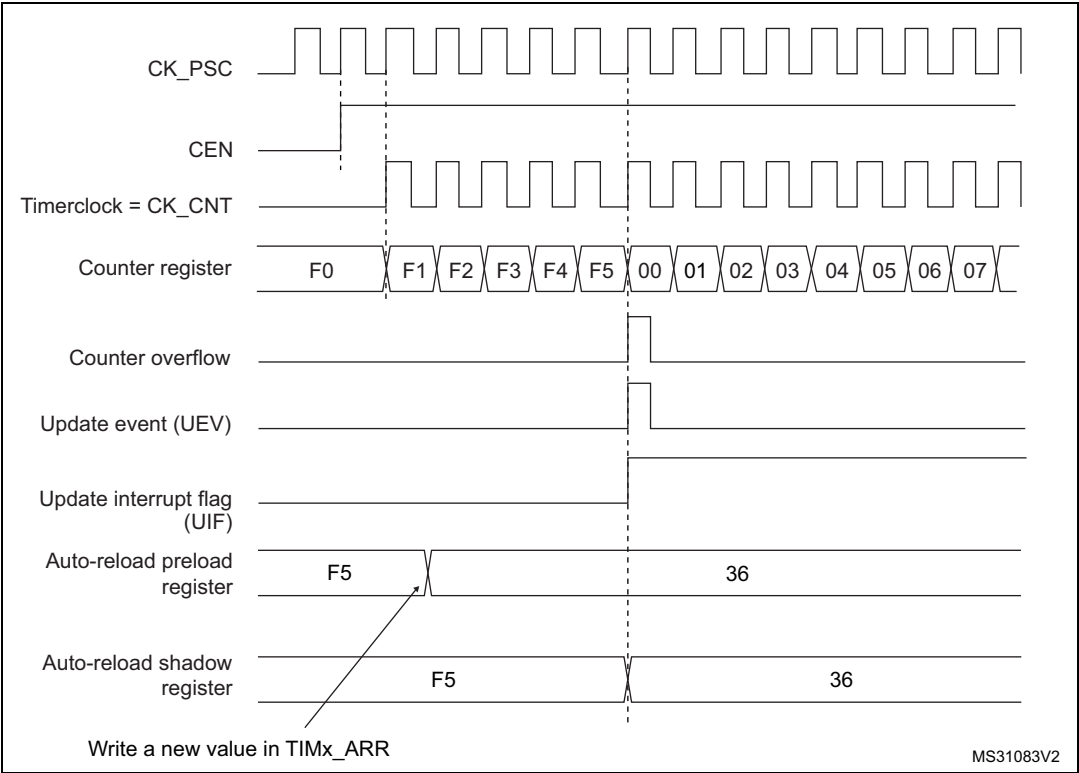


Figure 189. Counter timing diagram, update event when ARPE=1 (TIMx_ARR preloaded)



19.3.3 Clock selection

The counter clock can be provided by the following clock sources:

- Internal clock (CK_INT)
- External clock mode1 (for **TIM9 and TIM12**): external input pin (Tlx)
- Internal trigger inputs (ITRx) (for **TIM9 and TIM12**): connecting the trigger output from another timer. Refer to [Using one timer as prescaler for another timer](#) for more details.

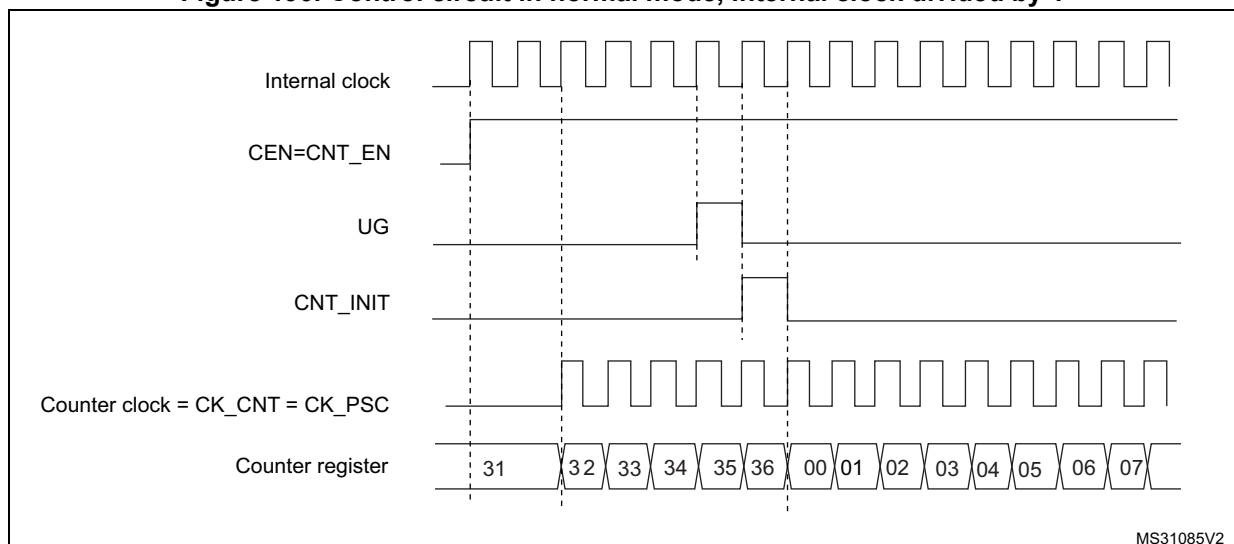
Internal clock source (CK_INT)

The internal clock source is the default clock source for TIM10/TIM11 and TIM13/TIM14.

For TIM9 and TIM12, the internal clock source is selected when the slave mode controller is disabled (SMS='000'). The CEN bit in the TIMx_CR1 register and the UG bit in the TIMx_EGR register are then used as control bits and can be changed only by software (except for UG which remains cleared). As soon as the CEN bit is programmed to 1, the prescaler is clocked by the internal clock CK_INT.

[Figure 190](#) shows the behavior of the control circuit and of the upcounter in normal mode, without prescaler.

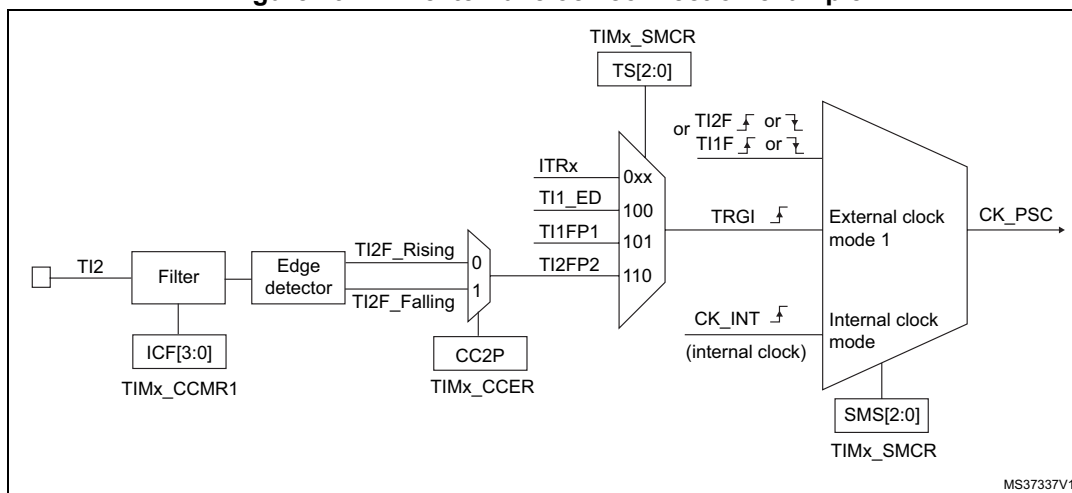
Figure 190. Control circuit in normal mode, internal clock divided by 1



External clock source mode 1 (TIM9 and TIM12)

This mode is selected when SMS='111' in the TIMx_SMCR register. The counter can count at each rising or falling edge on a selected input.

Figure 191. TI2 external clock connection example



For example, to configure the upcounter to count in response to a rising edge on the TI2 input, use the following procedure:

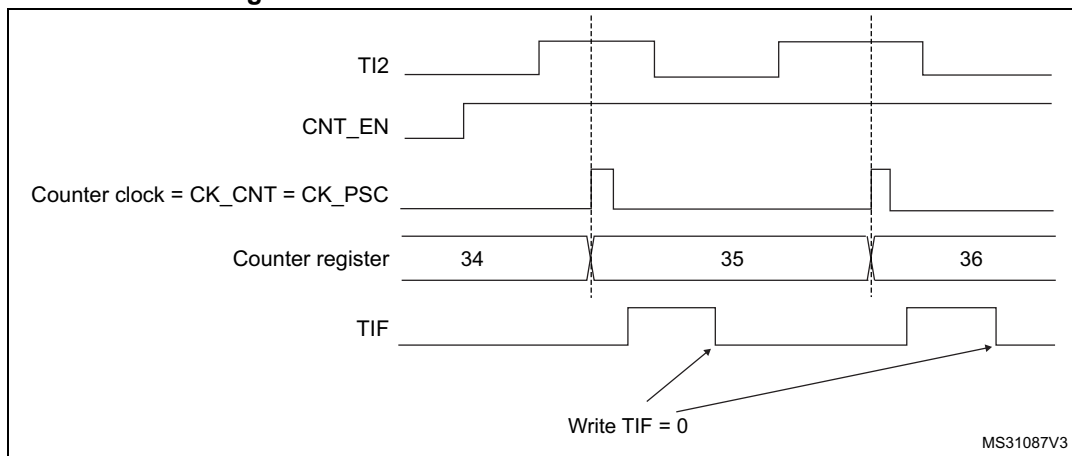
1. Configure channel 2 to detect rising edges on the TI2 input by writing CC2S = '01' in the TIMx_CCMR1 register.
2. Configure the input filter duration by writing the IC2F[3:0] bits in the TIMx_CCMR1 register (if no filter is needed, keep IC2F='0000').
3. Select the rising edge polarity by writing CC2P='0' and CC2NP='0' in the TIMx_CCER register.
4. Configure the timer in external clock mode 1 by writing SMS='111' in the TIMx_SMCR register.
5. Select TI2 as the trigger input source by writing TS='110' in the TIMx_SMCR register.
6. Enable the counter by writing CEN='1' in the TIMx_CR1 register.

Note: The capture prescaler is not used for triggering, so no need to configure it.

When a rising edge occurs on TI2, the counter counts once and the TIF flag is set.

The delay between the rising edge on TI2 and the actual clock of the counter is due to the resynchronization circuit on TI2 input.

Figure 192. Control circuit in external clock mode 1



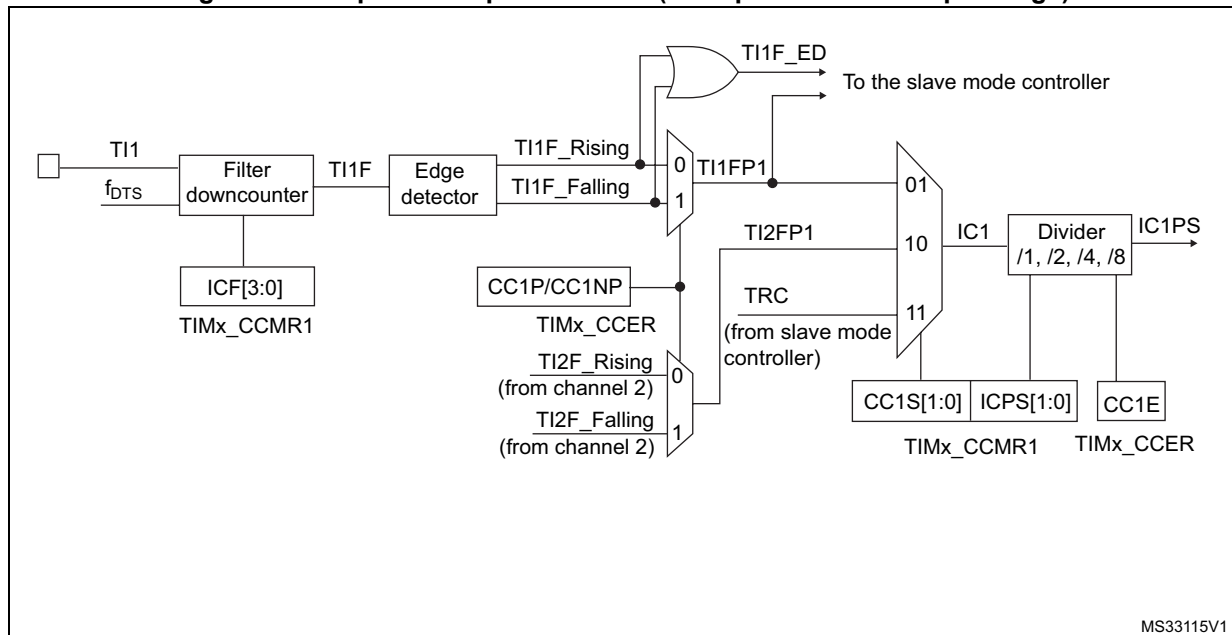
19.3.4 Capture/compare channels

Each Capture/Compare channel is built around a capture/compare register (including a shadow register), a input stage for capture (with digital filter, multiplexing and prescaler) and an output stage (with comparator and output control).

[Figure 193](#) to [Figure 195](#) give an overview of a capture/compare channel.

The input stage samples the corresponding TIx input to generate a filtered signal TIxF. Then, an edge detector with polarity selection generates a signal (TIxFPx) which can be used as trigger input by the slave mode controller or as the capture command. It is prescaled before the capture register (ICxPS).

Figure 193. Capture/compare channel (example: channel 1 input stage)



The output stage generates an intermediate waveform which is then used for reference: OCxRef (active high). The polarity acts at the end of the chain.

Figure 194. Capture/compare channel 1 main circuit

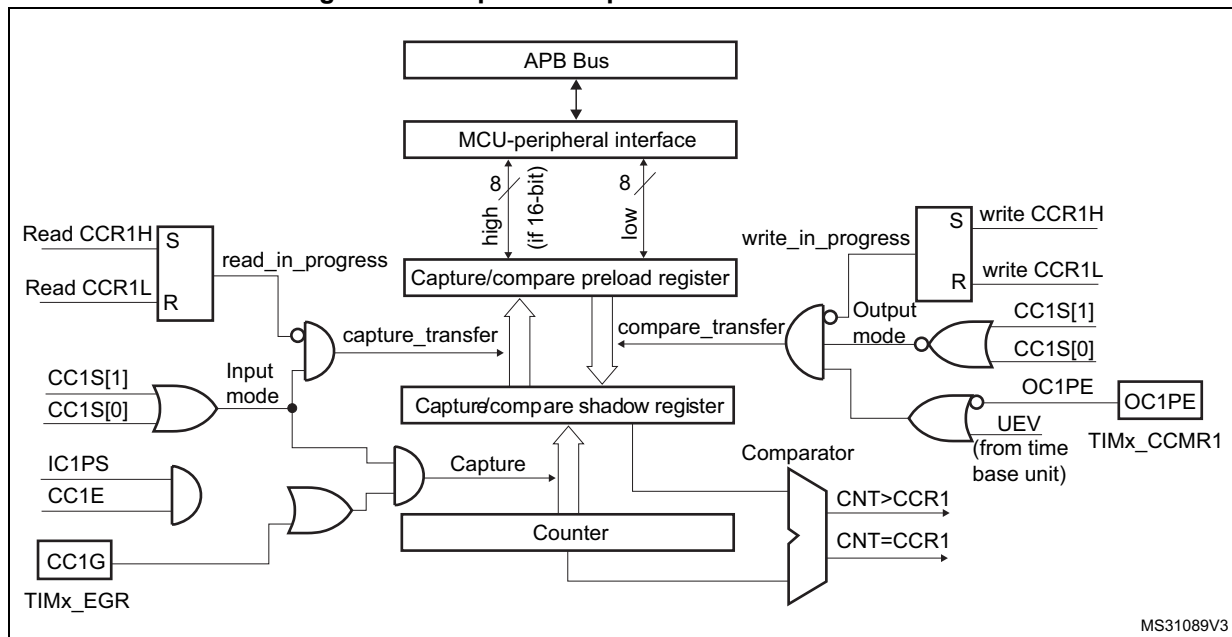
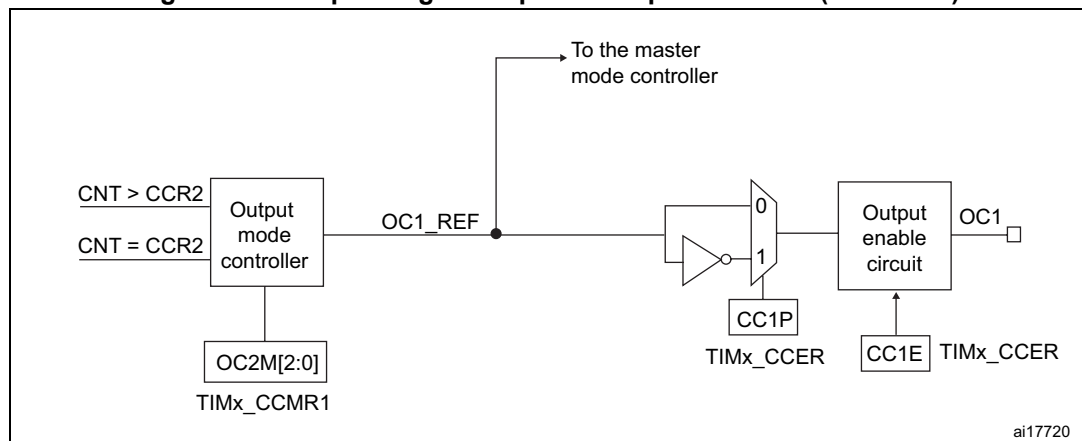


Figure 195. Output stage of capture/compare channel (channel 1)



The capture/compare block is made of one preload register and one shadow register. Write and read always access the preload register.

In capture mode, captures are actually done in the shadow register, which is copied into the preload register.

In compare mode, the content of the preload register is copied into the shadow register which is compared to the counter.

19.3.5 Input capture mode

In Input capture mode, the Capture/Compare registers (TIMx_CCRx) are used to latch the value of the counter after a transition detected by the corresponding ICx signal. When a capture occurs, the corresponding CCXIF flag (TIMx_SR register) is set and an interrupt or a DMA request can be sent if they are enabled. If a capture occurs while the CCxIF flag was already high, then the over-capture flag CCxOF (TIMx_SR register) is set. CCxIF can be

cleared by software by writing it to '0' or by reading the captured data stored in the TIMx_CCRx register. CCxOF is cleared when the user writes it to '0'.

The following example shows how to capture the counter value in TIMx_CCR1 when TI1 input rises. To do this, use the following procedure:

1. Select the active input: TIMx_CCR1 must be linked to the TI1 input, so write the CC1S bits to '01' in the TIMx_CCMR1 register. As soon as CC1S becomes different from '00', the channel is configured in input mode and the TIMx_CCR1 register becomes read-only.
2. Program the needed input filter duration with respect to the signal connected to the timer (by programming the ICxF bits in the TIMx_CCMRx register if the input is one of the TIx inputs). Let us imagine that, when toggling, the input signal is not stable during at least five internal clock cycles. We must program a filter duration longer than these 5 clock cycles. We can validate a transition on TI1 when eight consecutive samples with the new level have been detected (sampled at f_{DTS} frequency). Then write IC1F bits to '0011' in the TIMx_CCMR1 register.
3. Select the edge of the active transition on the TI1 channel by programming CC1P and CC1NP bits to '00' in the TIMx_CCER register (rising edge in this case).
4. Program the input prescaler. In our example, we wish the capture to be performed at each valid transition, so the prescaler is disabled (write IC1PS bits to '00' in the TIMx_CCMR1 register).
5. Enable capture from the counter into the capture register by setting the CC1E bit in the TIMx_CCER register.
6. If needed, enable the related interrupt request by setting the CC1IE bit in the TIMx_DIER register.

When an input capture occurs:

- The TIMx_CCR1 register gets the value of the counter on the active transition.
- CC1IF flag is set (interrupt flag). CC1OF is also set if at least two consecutive captures occurred whereas the flag was not cleared.
- An interrupt is generated depending on the CC1IE bit.

In order to handle the overcapture, it is recommended to read the data before the overcapture flag. This is to avoid missing an overcapture which could happen after reading the flag and before reading the data.

Note: IC interrupt requests can be generated by software by setting the corresponding CCxG bit in the TIMx_EGR register.

19.3.6 PWM input mode (only for TIM9/12)

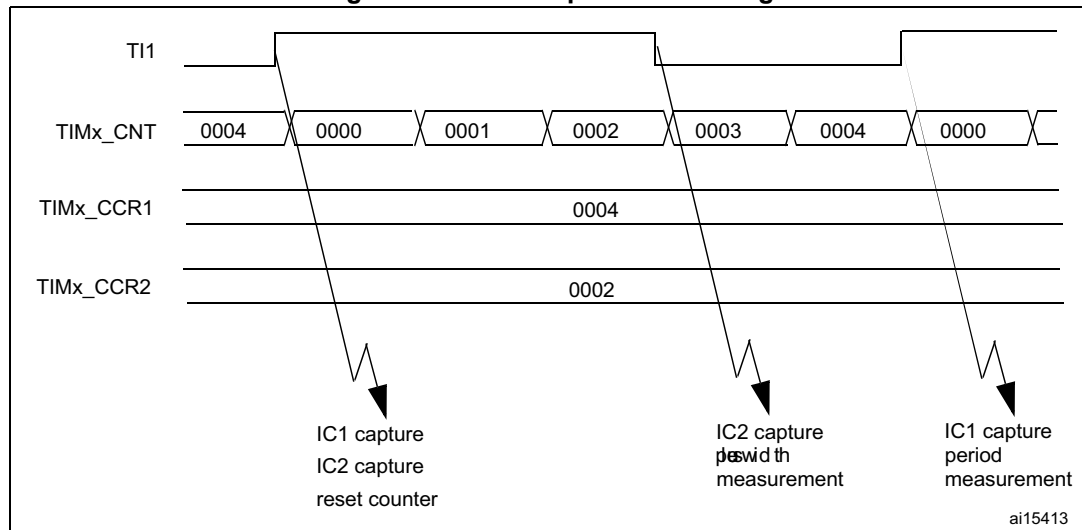
This mode is a particular case of input capture mode. The procedure is the same except:

- Two ICx signals are mapped on the same TIx input.
- These 2 ICx signals are active on edges with opposite polarity.
- One of the two TIxFP signals is selected as trigger input and the slave mode controller is configured in reset mode.

For example, the user can measure the period (in TIMx_CCR1 register) and the duty cycle (in TIMx_CCR2 register) of the PWM applied on TI1 using the following procedure (depending on CK_INT frequency and prescaler value):

1. Select the active input for TIMx_CCR1: write the CC1S bits to '01' in the TIMx_CCMR1 register (TI1 selected).
2. Select the active polarity for TI1FP1 (used both for capture in TIMx_CCR1 and counter clear): program the CC1P and CC1NP bits to '00' (active on rising edge).
3. Select the active input for TIMx_CCR2: write the CC2S bits to '10' in the TIMx_CCMR1 register (TI1 selected).
4. Select the active polarity for TI1FP2 (used for capture in TIMx_CCR2): write the CC2P bit to '1' and the CC2NP bit to '0' (active on falling edge).
5. Select the valid trigger input: write the TS bits to '101' in the TIMx_SMCR register (TI1FP1 selected).
6. Configure the slave mode controller in reset mode: write the SMS bits to '100' in the TIMx_SMCR register.
7. Enable the captures: write the CC1E and CC2E bits to '1' in the TIMx_CCER register.

Figure 196. PWM input mode timing



1. The PWM input mode can be used only with the TIMx_CH1/TIMx_CH2 signals due to the fact that only TI1FP1 and TI2FP2 are connected to the slave mode controller.

19.3.7 Forced output mode

In output mode (CCxS bits = '00' in the TIMx_CCMRx register), each output compare signal (OCxREF and then OCx) can be forced to active or inactive level directly by software, independently of any comparison between the output compare register and the counter.

To force an output compare signal (OCXREF/OCx) to its active level, the user just needs to write '101' in the OCxM bits in the corresponding TIMx_CCMRx register. Thus OCXREF is forced high (OCxREF is always active high) and OCx get opposite value to CCxP polarity bit.

For example: CCxP='0' (OCx active high) => OCx is forced to high level.

The OCxREF signal can be forced low by writing the OCxM bits to '100' in the TIMx_CCMRx register.

Anyway, the comparison between the TIMx_CCRx shadow register and the counter is still performed and allows the flag to be set. Interrupt requests can be sent accordingly. This is described in the output compare mode section below.

19.3.8 Output compare mode

This function is used to control an output waveform or indicating when a period of time has elapsed.

When a match is found between the capture/compare register and the counter, the output compare function:

1. Assigns the corresponding output pin to a programmable value defined by the output compare mode (OCxM bits in the TIMx_CCMRx register) and the output polarity (CCxP bit in the TIMx_CCER register). The output pin can keep its level (OCxM='000'), be set active (OCxM='001'), be set inactive (OCxM='010') or can toggle (OCxM='011') on match.
2. Sets a flag in the interrupt status register (CCxIF bit in the TIMx_SR register).
3. Generates an interrupt if the corresponding interrupt mask is set (CCXIE bit in the TIMx_DIER register).

The TIMx_CCRx registers can be programmed with or without preload registers using the OCxPE bit in the TIMx_CCMRx register.

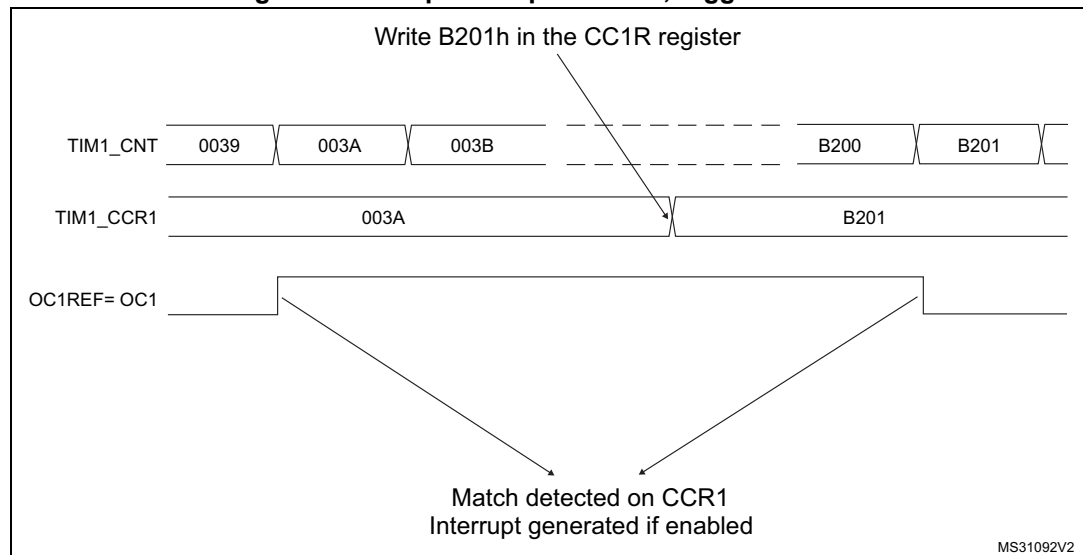
In output compare mode, the update event UEV has no effect on OCxREF and OCx output. The timing resolution is one count of the counter. Output compare mode can also be used to output a single pulse (in One-pulse mode).

Procedure:

1. Select the counter clock (internal, external, prescaler).
2. Write the desired data in the TIMx_ARR and TIMx_CCRx registers.
3. Set the CCxIE bit if an interrupt request is to be generated.
4. Select the output mode. For example:
 - Write OCxM = '011' to toggle OCx output pin when CNT matches CCRx
 - Write OCxPE = '0' to disable preload register
 - Write CCxP = '0' to select active high polarity
 - Write CCxE = '1' to enable the output
5. Enable the counter by setting the CEN bit in the TIMx_CR1 register.

The TIMx_CCRx register can be updated at any time by software to control the output waveform, provided that the preload register is not enabled (OCxPE='0', else TIMx_CCRx shadow register is updated only at the next update event UEV). An example is given in [Figure 197](#).

Figure 197. Output compare mode, toggle on OC1.



19.3.9 PWM mode

Pulse Width Modulation mode allows the user to generate a signal with a frequency determined by the value of the TIMx_ARR register and a duty cycle determined by the value of the TIMx_CCRx register.

The PWM mode can be selected independently on each channel (one PWM per OCx output) by writing '110' (PWM mode 1) or '111' (PWM mode 2) in the OCxM bits in the TIMx_CCMRx register. Enable the corresponding preload register by setting the OCxPE bit in the TIMx_CCMRx register, and eventually the auto-reload preload register by setting the ARPE bit in the TIMx_CR1 register.

As the preload registers are transferred to the shadow registers only when an update event occurs, before starting the counter, the user has to initialize all the registers by setting the UG bit in the TIMx_EGR register.

The OCx polarity is software programmable using the CCxP bit in the TIMx_CCER register. It can be programmed as active high or active low. The OCx output is enabled by the CCxE bit in the TIMx_CCER register. Refer to the TIMx_CCERx register description for more details.

In PWM mode (1 or 2), TIMx_CNT and TIMx_CCRx are always compared to determine whether $TIMx_CNT \leq TIMx_CCRx$.

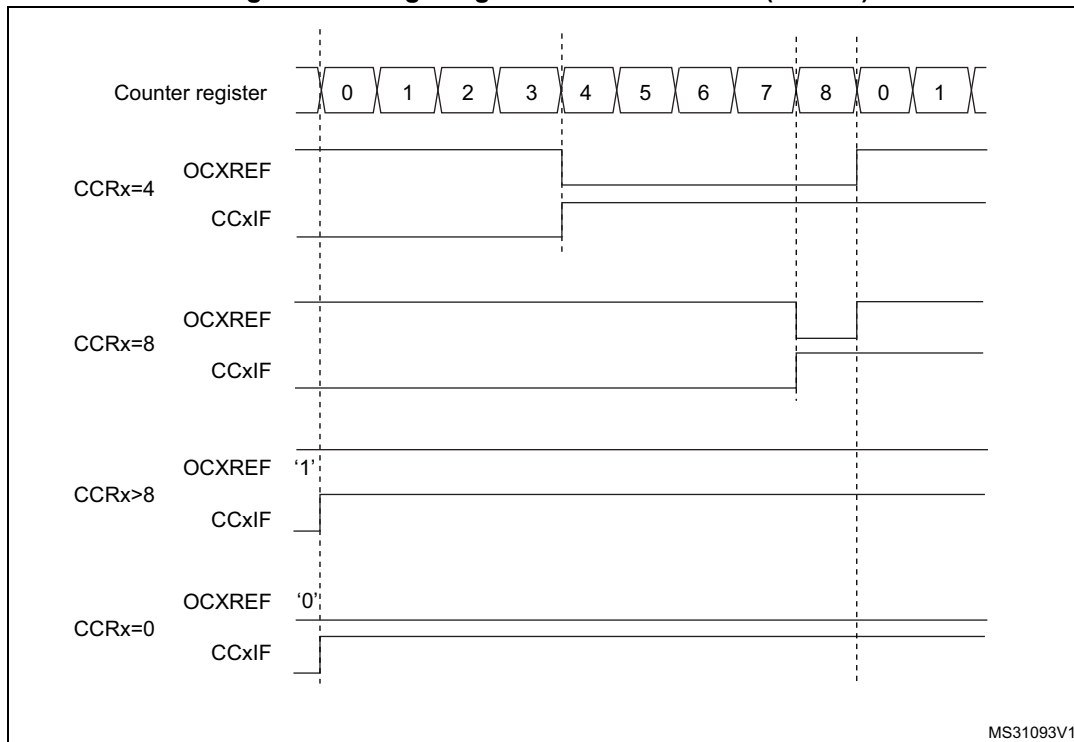
The timer is able to generate PWM in edge-aligned mode only since the counter is upcounting.

PWM edge-aligned mode

In the following example, we consider PWM mode 1. The reference PWM signal OCxREF is high as long as $TIMx_CNT < TIMx_CCRx$ else it becomes low. If the compare value in

TIMx_CCRx is greater than the auto-reload value (in TIMx_ARR) then OCxREF is held at '1'. If the compare value is 0 then OCxRef is held at '0'. [Figure 198](#) shows some edge-aligned PWM waveforms in an example where TIMx_ARR=8.

Figure 198. Edge-aligned PWM waveforms (ARR=8)



19.3.10 One-pulse mode

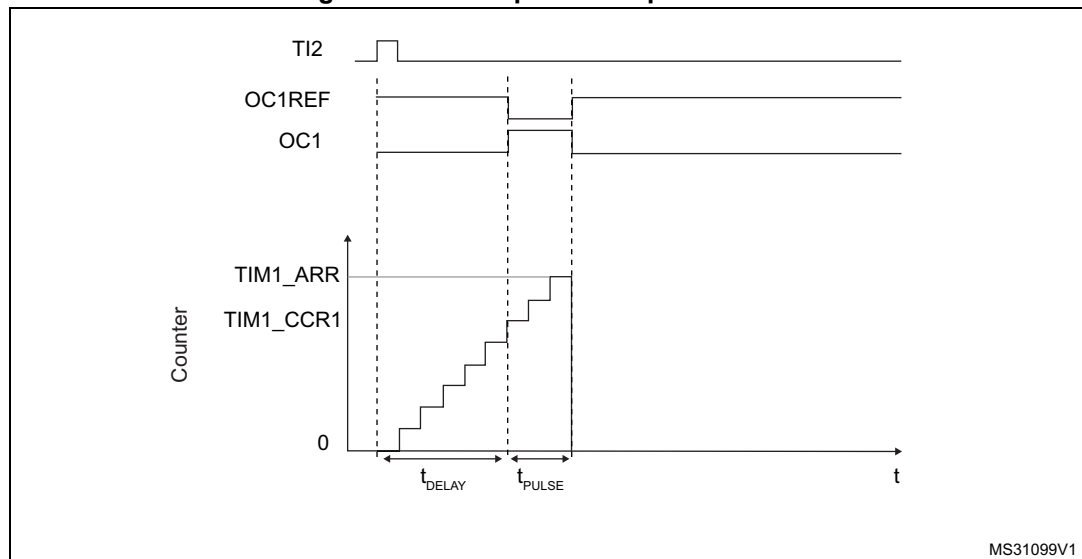
One-pulse mode (OPM) is a particular case of the previous modes. It allows the counter to be started in response to a stimulus and to generate a pulse with a programmable length after a programmable delay.

Starting the counter can be controlled through the slave mode controller. Generating the waveform can be done in output compare mode or PWM mode. Select One-pulse mode by setting the OPM bit in the TIMx_CR1 register. This makes the counter stop automatically at the next update event UEV.

A pulse can be correctly generated only if the compare value is different from the counter initial value. Before starting (when the timer is waiting for the trigger), the configuration must be as follows:

$$CNT < CCRx \leq ARR \text{ (in particular, } 0 < CCRx)$$

Figure 199. Example of one pulse mode.



For example the user may want to generate a positive pulse on OC1 with a length of t_{PULSE} and after a delay of t_{DELAY} as soon as a positive edge is detected on the TI2 input pin.

Use TI2FP2 as trigger 1:

1. Map TI2FP2 to TI2 by writing CC2S='01' in the TIMx_CCMR1 register.
2. TI2FP2 must detect a rising edge, write CC2P='0' and CC2NP = '0' in the TIMx_CCER register.
3. Configure TI2FP2 as trigger for the slave mode controller (TRGI) by writing TS='110' in the TIMx_SMCR register.
4. TI2FP2 is used to start the counter by writing SMS to '110' in the TIMx_SMCR register (trigger mode).

The OPM waveform is defined by writing the compare registers (taking into account the clock frequency and the counter prescaler).

- The t_{DELAY} is defined by the value written in the TIMx_CCR1 register.
- The t_{PULSE} is defined by the difference between the auto-reload value and the compare value (TIMx_ARR - TIMx_CCR1).
- Let us say the user wants to build a waveform with a transition from '0' to '1' when a compare match occurs and a transition from '1' to '0' when the counter reaches the auto-reload value. To do this enable PWM mode 2 by writing OC1M='111' in the TIMx_CCMR1 register. The user can optionally enable the preload registers by writing OC1PE='1' in the TIMx_CCMR1 register and ARPE in the TIMx_CR1 register. In this case the user has to write the compare value in the TIMx_CCR1 register, the auto-reload value in the TIMx_ARR register, generate an update by setting the UG bit and wait for external trigger event on TI2. CC1P is written to '0' in this example.

The user only wants one pulse (Single mode), so write '1' in the OPM bit in the TIMx_CR1 register to stop the counter at the next update event (when the counter rolls over from the auto-reload value back to 0). When OPM bit in the TIMx_CR1 register is set to '0', so the Repetitive mode is selected.

Particular case: OCx fast enable

In One-pulse mode, the edge detection on TIx input set the CEN bit which enables the counter. Then the comparison between the counter and the compare value makes the output toggle. But several clock cycles are needed for these operations and it limits the minimum delay $t_{\text{DELAY min}}$ we can get.

If the user wants to output a waveform with the minimum delay, set the OCxFE bit in the TIMx_CCMRx register. Then OCxRef (and OCx) are forced in response to the stimulus, without taking in account the comparison. Its new level is the same as if a compare match had occurred. OCxFE acts only if the channel is configured in PWM1 or PWM2 mode.

19.3.11 TIM9/12 external trigger synchronization

The TIM9/12 timers can be synchronized with an external trigger in several modes: Reset mode, Gated mode and Trigger mode.

Slave mode: Reset mode

The counter and its prescaler can be reinitialized in response to an event on a trigger input. Moreover, if the URS bit from the TIMx_CR1 register is low, an update event UEV is generated. Then all the preloaded registers (TIMx_ARR, TIMx_CCRx) are updated.

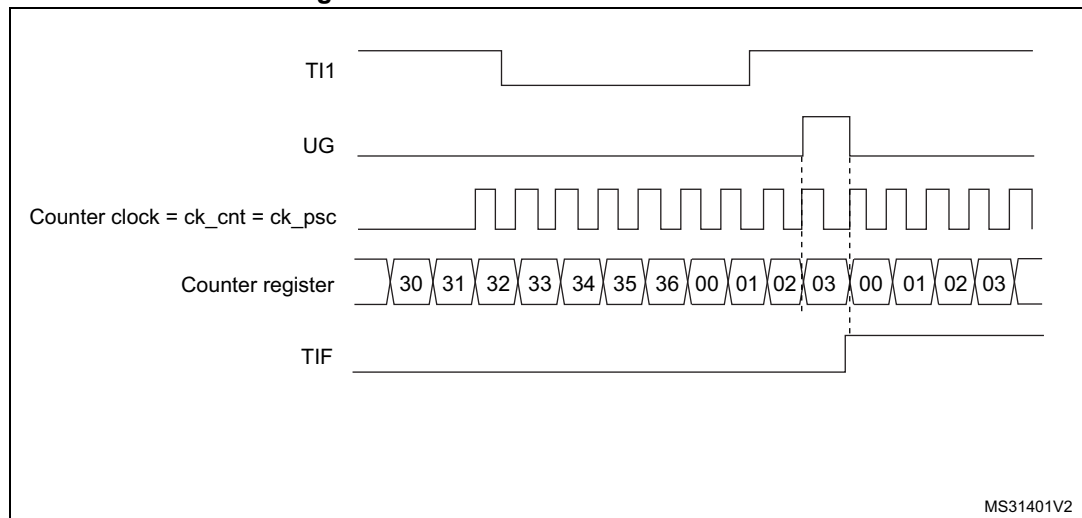
In the following example, the upcounter is cleared in response to a rising edge on TI1 input:

1. Configure the channel 1 to detect rising edges on TI1. Configure the input filter duration (in this example, no need of any filter, IC1F = 0000 kept). The capture prescaler is not used for triggering, so there's no need to configure it. The CC1S bits select the input capture source only, CC1S = '01' in the TIMx_CCMR1 register. Program CC1P and CC1NP to '00' in TIMx_CCER register to validate the polarity (and detect rising edges only).
2. Configure the timer in reset mode by writing SMS='100' in TIMx_SMCR register. Select TI1 as the input source by writing TS='101' in TIMx_SMCR register.
3. Start the counter by writing CEN='1' in the TIMx_CR1 register.

The counter starts counting on the internal clock, then behaves normally until TI1 rising edge. When TI1 rises, the counter is cleared and restarts from 0. In the meantime, the trigger flag is set (TIF bit in the TIMx_SR register) and an interrupt request can be sent if enabled (depending on the TIE bit in TIMx_DIER register).

The following figure shows this behavior when the auto-reload register TIMx_ARR=0x36. The delay between the rising edge on TI1 and the actual reset of the counter is due to the resynchronization circuit on TI1 input.

Figure 200. Control circuit in reset mode



Slave mode: Gated mode

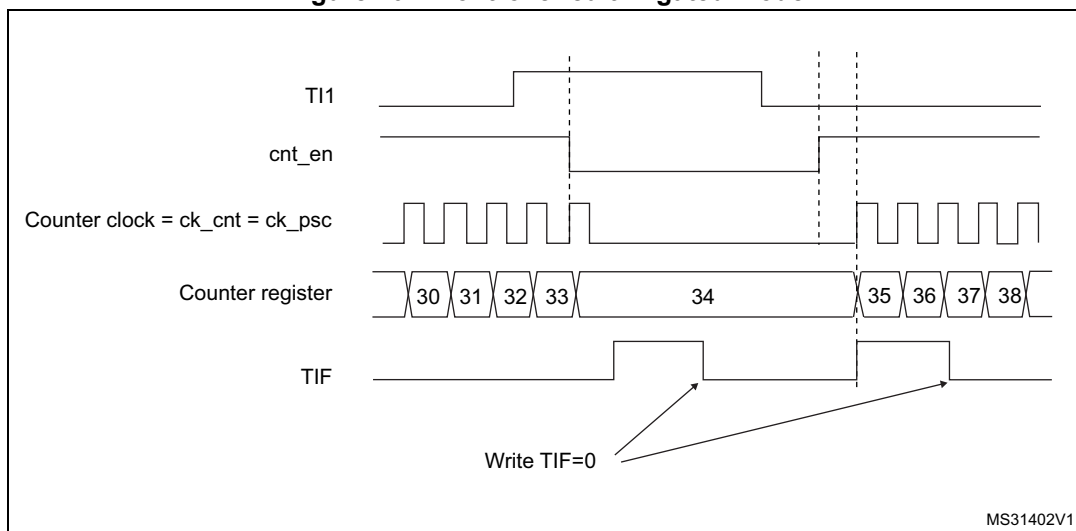
The counter can be enabled depending on the level of a selected input.

In the following example, the upcounter counts only when TI1 input is low:

1. Configure the channel 1 to detect low levels on TI1. Configure the input filter duration (in this example, no need of any filter, IC1F='0000' kept). The capture prescaler is not used for triggering, so there's no need to configure it. The CC1S bits select the input capture source only, CC1S='01' in TIMx_CCMR1 register. Program CC1P='1' and CC1NP='0' in TIMx_CCER register to validate the polarity (and detect low level only).
2. Configure the timer in gated mode by writing SMS='101' in TIMx_SMCR register. Select TI1 as the input source by writing TS='101' in TIMx_SMCR register.
3. Enable the counter by writing CEN='1' in the TIMx_CR1 register (in gated mode, the counter does not start if CEN='0', whatever is the trigger input level).

The counter starts counting on the internal clock as long as TI1 is low and stops as soon as TI1 becomes high. The TIF flag in the TIMx_SR register is set both when the counter starts or stops.

The delay between the rising edge on TI1 and the actual stop of the counter is due to the resynchronization circuit on TI1 input.

Figure 201. Control circuit in gated mode**Slave mode: Trigger mode**

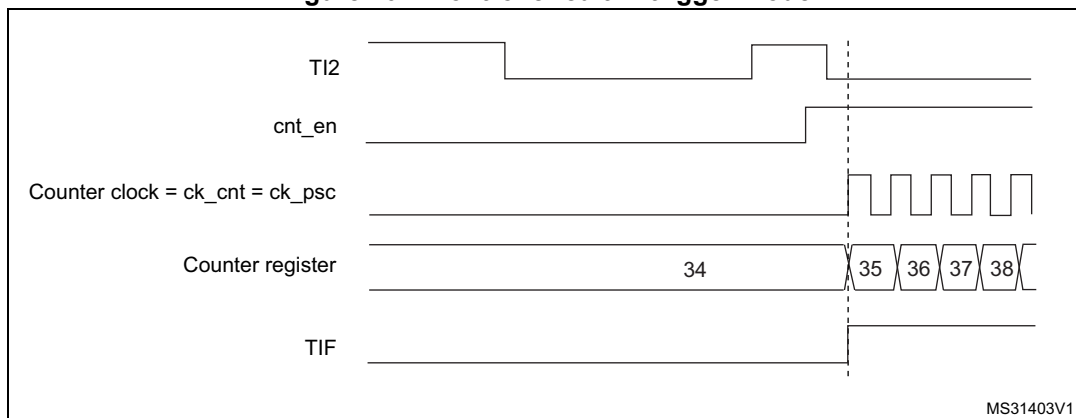
The counter can start in response to an event on a selected input.

In the following example, the upcounter starts in response to a rising edge on TI2 input:

1. Configure the channel 2 to detect rising edges on TI2. Configure the input filter duration (in this example, no need of any filter, IC2F='0000' kept). The capture prescaler is not used for triggering, so there's no need to configure it. The CC2S bits are configured to select the input capture source only, CC2S='01' in TIMx_CCMR1 register. Program CC2P='1' and CC2NP='0' in TIMx_CCER register to validate the polarity (and detect low level only).
2. Configure the timer in trigger mode by writing SMS='110' in TIMx_SMCR register. Select TI2 as the input source by writing TS='110' in TIMx_SMCR register.

When a rising edge occurs on TI2, the counter starts counting on the internal clock and the TIF flag is set.

The delay between the rising edge on TI2 and the actual start of the counter is due to the resynchronization circuit on TI2 input.

Figure 202. Control circuit in trigger mode

19.3.12 Timer synchronization (TIM9/12)

The TIM timers are linked together internally for timer synchronization or chaining. Refer to [Section 18.3.15: Timer synchronization](#) for details.

Note: The clock of the slave timer must be enabled prior to receive events from the master timer, and must not be changed on-the-fly while triggers are received from the master timer.

19.3.13 Debug mode

When the microcontroller enters debug mode (Cortex[®]-M4 with FPU core halted), the TIMx counter either continues to work normally or stops, depending on DBG_TIMx_STOP configuration bit in DBG module. For more details, refer to [Section 38.16.2: Debug support for timers, watchdog, bxCAN and I²C](#).

19.4 TIM9 and TIM12 registers

Refer to [Section 2.2](#) for a list of abbreviations used in register descriptions.

The peripheral registers have to be written by half-words (16 bits) or words (32 bits). Read accesses can be done by bytes (8 bits), half-words (16 bits) or words (32 bits).

19.4.1 TIM9/12 control register 1 (TIMx_CR1)

Address offset: 0x00

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reserved						CKD[1:0]		ARPE	Reserved				OPM	URS	UDIS	CEN
						rw	rw	rw					rw	rw	rw	rw

Bits 15:10 Reserved, must be kept at reset value.

Bits 9:8 **CKD**: Clock division

This bit-field indicates the division ratio between the timer clock (CK_INT) frequency and sampling clock used by the digital filters (Tlx),

00: $t_{DTS} = t_{CK_INT}$

01: $t_{DTS} = 2 \times t_{CK_INT}$

$$10: t_{DTS} = 4 \times t_{CK_INT}$$

11: Reserved

Bit 7 **ARPE**: Auto-reload preload enable

0: TIMx ARR register is not buffered.

1: TIMx ARR register is buffered.

Bits 6:4 Reserved, must be kept at reset value.

Bit 3 **OPM**: One-pulse mode

0: Counter is not stopped on the update event

1: Counter stops counting on the next update event (clearing the CEN bit).

Bit 2 **URS**: Update request source

This bit is set and cleared by software to select the UEV event sources.

0: Any of the following events generates an update interrupt if enabled:

- Counter overflow
- Setting the UG bit

1: Only counter overflow generates an update interrupt if enabled.

Bit 1 **UDIS**: Update disable

This bit is set and cleared by software to enable/disable update event (UEV) generation.

0: UEV enabled. An UEV is generated by one of the following events:

- Counter overflow
- Setting the UG bit

Buffered registers are then loaded with their preload values.

1: UEV disabled. No UEV is generated, shadow registers keep their value (ARR, PSC, CCRx). The counter and the prescaler are reinitialized if the UG bit is set.

Bit 0 **CEN**: Counter enable

0: Counter disabled

1: Counter enabled

CEN is cleared automatically in one-pulse mode, when an update event occurs.

19.4.2 TIM9/12 slave mode control register (TIMx_SMCR)

Address offset: 0x08

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reserved								MSM	TS[2:0]				Res.	SMS[2:0]		
								rw	rw	rw	rw	rw		rw	rw	

Bits 15:8 Reserved, must be kept at reset value.

Bit 7 **MSM**: Master/Slave mode

0: No action

1: The effect of an event on the trigger input (TRGI) is delayed to allow a perfect synchronization between the current timer and its slaves (through TRGO). It is useful in order to synchronize several timers on a single external event.

Bits 6:4 **TS**: Trigger selection

This bit field selects the trigger input to be used to synchronize the counter.

000: Internal Trigger 0 (ITR0)

001: Internal Trigger 1 (ITR1)

010: Internal Trigger 2 (ITR2)

011: Internal Trigger 3 (ITR3)

100: TI1 Edge Detector (TI1F_ED)

101: Filtered Timer Input 1 (TI1FP1)

110: Filtered Timer Input 2 (TI2FP2)

111: Reserved.

See [Table 102](#) for more details on the meaning of ITRx for each timer.

Note: These bits must be changed only when they are not used (e.g. when SMS='000') to avoid wrong edge detections at the transition.

Bit 3 Reserved, must be kept at reset value.

Bits 2:0 **SMS**: Slave mode selection

When external signals are selected, the active edge of the trigger signal (TRGI) is linked to the polarity selected on the external input (see Input control register and Control register descriptions).

000: Slave mode disabled - if CEN = 1 then the prescaler is clocked directly by the internal clock

001: Reserved

010: Reserved

011: Reserved

100: Reset mode - Rising edge of the selected trigger input (TRGI) reinitializes the counter and generates an update of the registers

101: Gated mode - The counter clock is enabled when the trigger input (TRGI) is high. The counter stops (but is not reset) as soon as the trigger becomes low. Counter starts and stops are both controlled

110: Trigger mode - The counter starts on a rising edge of the trigger TRGI (but it is not reset). Only the start of the counter is controlled

111: External clock mode 1 - Rising edges of the selected trigger (TRGI) clock the counter

Note: The Gated mode must not be used if TI1F_ED is selected as the trigger input (TS='100'). Indeed, TI1F_ED outputs 1 pulse for each transition on TI1F, whereas the Gated mode checks the level of the trigger signal.

Note: The clock of the slave timer must be enabled prior to receive events from the master timer, and must not be changed on-the-fly while triggers are received from the master timer.

Table 102. TIMx internal trigger connection

Slave TIM	ITR0 (TS = 000)	ITR1 (TS = 001)	ITR2 (TS = 010)	ITR3 (TS = 011)
TIM9	TIM2_TRGO	TIM3_TRGO	TIM10_OC	TIM11_OC
TIM12	TIM4_TRGO	TIM5_TRGO	TIM13_OC	TIM14_OC

19.4.3 TIM9/12 Interrupt enable register (TIMx_DIER)

Address offset: 0x0C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									TIE	Res			CC2IE	CC1IE	UIE
									rw				rw	rw	rw

Bits 15:7 Reserved, must be kept at reset value.

Bit 6 **TIE**: Trigger interrupt enable

0: Trigger interrupt disabled.

1: Trigger interrupt enabled.

Bit 5:3 Reserved, must be kept at reset value.

Bit 2 **CC2IE**: Capture/Compare 2 interrupt enable

0: CC2 interrupt disabled.

1: CC2 interrupt enabled.

Bit 1 **CC1IE**: Capture/Compare 1 interrupt enable

0: CC1 interrupt disabled.

1: CC1 interrupt enabled.

Bit 0 **UIE**: Update interrupt enable

0: Update interrupt disabled.

1: Update interrupt enabled.

19.4.4 TIM9/12 status register (TIMx_SR)

Address offset: 0x10

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reserved					CC2OF	CC1OF	Reserved			TIF	Reserved			CC2IF	CC1IF	UIF
					rc_w0	rc_w0				rc_w0				rc_w0	rc_w0	rc_w0

Bits 15:11 Reserved, must be kept at reset value.

Bit 10 **CC2OF**: Capture/compare 2 overcapture flag
refer to CC1OF description

Bit 9 **CC1OF**: Capture/Compare 1 overcapture flag

This flag is set by hardware only when the corresponding channel is configured in input capture mode. It is cleared by software by writing it to '0'.

0: No overcapture has been detected.

1: The counter value has been captured in TIMx_CCR1 register while CC1IF flag was already set

Bits 8:7 Reserved, must be kept at reset value.

Bit 6 **TIF**: Trigger interrupt flag

This flag is set by hardware on trigger event (active edge detected on TRGI input when the slave mode controller is enabled in all modes but gated mode. It is set when the counter starts or stops when gated mode is selected. It is cleared by software.

0: No trigger event occurred.

1: Trigger interrupt pending.

Bits 5:3 Reserved, must be kept at reset value.

Bit 2 **CC2IF**: Capture/Compare 2 interrupt flag
refer to CC1IF description

Bit 1 **CC1IF**: Capture/compare 1 interrupt flag

If channel CC1 is configured as output:

This flag is set by hardware when the counter matches the compare value. It is cleared by software.

0: No match.

1: The content of the counter TIMx_CNT matches the content of the TIMx_CCR1 register. When the contents of TIMx_CCR1 are greater than the contents of TIMx_ARR, the CC1IF bit goes high on the counter overflow.

If channel CC1 is configured as input:

This bit is set by hardware on a capture. It is cleared by software or by reading the TIMx_CCR1 register.

0: No input capture occurred.

1: The counter value has been captured in TIMx_CCR1 register (an edge has been detected on IC1 which matches the selected polarity).

Bit 0 UIF: Update interrupt flag

This bit is set by hardware on an update event. It is cleared by software.

0: No update occurred.

1: Update interrupt pending. This bit is set by hardware when the registers are updated:

- At overflow and if UDIS='0' in the TIMx_CR1 register.
- When CNT is reinitialized by software using the UG bit in TIMx_EGR register, if URS='0' and UDIS='0' in the TIMx_CR1 register.
- When CNT is reinitialized by a trigger event (refer to the synchro control register description), if URS='0' and UDIS='0' in the TIMx_CR1 register.

19.4.5 TIM9/12 event generation register (TIMx_EGR)

Address offset: 0x14

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									TG	Reserved			CC2G	CC1G	UG
									w				w	w	w

Bits 15:7 Reserved, must be kept at reset value.

Bit 6 TG: Trigger generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: The TIF flag is set in the TIMx_SR register. Related interrupt can occur if enabled

Bits 5:3 Reserved, must be kept at reset value.

Bit 2 CC2G: Capture/compare 2 generation

refer to CC1G description

Bit 1 CC1G: Capture/compare 1 generation

This bit is set by software to generate an event, it is automatically cleared by hardware.

0: No action

1: A capture/compare event is generated on channel 1:

If channel CC1 is configured as output:

the CC1IF flag is set, the corresponding interrupt is sent if enabled.

If channel CC1 is configured as input:

The current counter value is captured in the TIMx_CCR1 register. The CC1IF flag is set, the corresponding interrupt is sent if enabled. The CC1OF flag is set if the CC1IF flag was already high.

Bit 0 UG: Update generation

This bit can be set by software, it is automatically cleared by hardware.

0: No action

1: Re-initializes the counter and generates an update of the registers. The prescaler counter is also cleared and the prescaler ratio is not affected. The counter is cleared.

19.4.6 TIM9/12 capture/compare mode register 1 (TIMx_CCMR1)

Address offset: 0x18

Reset value: 0x0000

The channels can be used in input (capture mode) or in output (compare mode). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits in this register have different functions in input and output modes. For a given bit, OCxx describes its function when the channel is configured in output mode, ICxx describes its function when the channel is configured in input mode. Take care that the same bit can have different meanings for the input stage and the output stage.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	OC2M[2:0]			OC2PE	OC2FE	CC2S[1:0]		Res.	OC1M[2:0]			OC1PE	OC1FE	CC1S[1:0]	
IC2F[3:0]			IC2PSC[1:0]		IC1F[3:0]			IC1PSC[1:0]							
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Output compare mode

Bit 15 Reserved, must be kept at reset value.

Bits 14:12 **OC2M[2:0]**: Output compare 2 mode

Bit 11 **OC2PE**: Output compare 2 preload enable

Bit 10 **OC2FE**: Output compare 2 fast enable

Bits 9:8 **CC2S[1:0]**: Capture/Compare 2 selection

This bitfield defines the direction of the channel (input/output) as well as the used input.

00: CC2 channel is configured as output

01: CC2 channel is configured as input, IC2 is mapped on TI2

10: CC2 channel is configured as input, IC2 is mapped on TI1

11: CC2 channel is configured as input, IC2 is mapped on TRC. This mode works only if an internal trigger input is selected through the TS bit (TIMx_SMCR register)

Note: The CC2S bits are writable only when the channel is OFF (CC2E = 0 in TIMx_CCER).

Bit 7 Reserved, must be kept at reset value.

Bits 6:4 **OC1M**: Output compare 1 mode

These bits define the behavior of the output reference signal OC1REF from which OC1 and OC1N are derived. OC1REF is active high whereas the active levels of OC1 and OC1N depend on the CC1P and CC1NP bits, respectively.

000: Frozen - The comparison between the output compare register TIMx_CCR1 and the counter TIMx_CNT has no effect on the outputs. (this mode is used to generate a timing base).

001: Set channel 1 to active level on match. The OC1REF signal is forced high when the TIMx_CNT counter matches the capture/compare register 1 (TIMx_CCR1).

010: Set channel 1 to inactive level on match. The OC1REF signal is forced low when the TIMx_CNT counter matches the capture/compare register 1 (TIMx_CCR1).

011: Toggle - OC1REF toggles when TIMx_CNT=TIMx_CCR1

100: Force inactive level - OC1REF is forced low

101: Force active level - OC1REF is forced high

110: PWM mode 1 - In upcounting, channel 1 is active as long as TIMx_CNT < TIMx_CCR1 else it is inactive. In downcounting, channel 1 is inactive (OC1REF=0) as long as TIMx_CNT > TIMx_CCR1, else it is active (OC1REF=1)

111: PWM mode 2 - In upcounting, channel 1 is inactive as long as TIMx_CNT < TIMx_CCR1 else it is active. In downcounting, channel 1 is active as long as TIMx_CNT > TIMx_CCR1 else it is inactive.

Note: In PWM mode 1 or 2, the OCREF level changes only when the result of the comparison changes or when the output compare mode switches from "frozen" mode to "PWM" mode.

Bit 3 **OC1PE**: Output compare 1 preload enable

0: Preload register on TIMx_CCR1 disabled. TIMx_CCR1 can be written at anytime, the new value is taken into account immediately

1: Preload register on TIMx_CCR1 enabled. Read/Write operations access the preload register. TIMx_CCR1 preload value is loaded into the active register at each update event

Bit 2 **OC1FE**: Output compare 1 fast enable

This bit is used to accelerate the effect of an event on the trigger in input on the CC output.
0: CC1 behaves normally depending on the counter and CCR1 values even when the trigger is ON. The minimum delay to activate the CC1 output when an edge occurs on the trigger input is 5 clock cycles

1: An active edge on the trigger input acts like a compare match on the CC1 output. Then, OC is set to the compare level independently of the result of the comparison. Delay to sample the trigger input and to activate CC1 output is reduced to 3 clock cycles. OC1FE acts only if the channel is configured in PWM1 or PWM2 mode.

Bits 1:0 **CC1S**: Capture/Compare 1 selection

This bitfield defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output

01: CC1 channel is configured as input, IC1 is mapped on TI1

10: CC1 channel is configured as input, IC1 is mapped on TI2

11: CC1 channel is configured as input, IC1 is mapped on TRC. This mode works only if an internal trigger input is selected through the TS bit (TIMx_SMCR register)

Note: The CC1S bits are writable only when the channel is OFF (CC1E = 0 in TIMx_CCER).

Input capture mode

Bits 15:12 **IC2F**: Input capture 2 filter

Bits 11:10 **IC2PSC[1:0]**: Input capture 2 prescaler

Bits 9:8 **CC2S**: Capture/compare 2 selection

This bitfield defines the direction of the channel (input/output) as well as the used input.

00: CC2 channel is configured as output

01: CC2 channel is configured as input, IC2 is mapped on TI2

10: CC2 channel is configured as input, IC2 is mapped on TI1

11: CC2 channel is configured as input, IC2 is mapped on TRC. This mode works only if an internal trigger input is selected through the TS bit (TIMx_SMCR register)

Note: The CC2S bits are writable only when the channel is OFF (CC2E = 0 in TIMx_CCER).

Bits 7:4 **IC1F**: Input capture 1 filter

This bitfield defines the frequency used to sample the TI1 input and the length of the digital filter applied to TI1. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, sampling is done at f_{DTS}

0001: $f_{SAMPLING}=f_{CK_INT}$, N=2

0010: $f_{SAMPLING}=f_{CK_INT}$, N=4

0011: $f_{SAMPLING}=f_{CK_INT}$, N=8

0100: $f_{SAMPLING}=f_{DTS}/2$, N=6

0101: $f_{SAMPLING}=f_{DTS}/2$, N=8

0110: $f_{SAMPLING}=f_{DTS}/4$, N=6

0111: $f_{SAMPLING}=f_{DTS}/4$, N=8

1000: $f_{SAMPLING}=f_{DTS}/8$, N=6

1001: $f_{SAMPLING}=f_{DTS}/8$, N=8

1010: $f_{SAMPLING}=f_{DTS}/16$, N=5

1011: $f_{SAMPLING}=f_{DTS}/16$, N=6

1100: $f_{SAMPLING}=f_{DTS}/16$, N=8

1101: $f_{SAMPLING}=f_{DTS}/32$, N=5

1110: $f_{SAMPLING}=f_{DTS}/32$, N=6

1111: $f_{SAMPLING}=f_{DTS}/32$, N=8

Bits 3:2 **IC1PSC**: Input capture 1 prescaler

This bitfield defines the ratio of the prescaler acting on the CC1 input (IC1).

The prescaler is reset as soon as CC1E='0' (TIMx_CCER register).

00: no prescaler, capture is done each time an edge is detected on the capture input

01: capture is done once every 2 events

10: capture is done once every 4 events

11: capture is done once every 8 events

Bits 1:0 **CC1S**: Capture/Compare 1 selection

This bitfield defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output

01: CC1 channel is configured as input, IC1 is mapped on TI1

10: CC1 channel is configured as input, IC1 is mapped on TI2

11: CC1 channel is configured as input, IC1 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: The CC1S bits are writable only when the channel is OFF (CC1E = 0 in TIMx_CCER).

19.4.7 TIM9/12 capture/compare enable register (TIMx_CCER)

Address offset: 0x20

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								CC2NP	Res.	CC2P	CC2E	CC1NP	Res.	CC1P	CC1E
								rw		rw	rw	rw		rw	rw

Bits 15:8 Reserved, must be kept at reset value.

Bit 7 **CC2NP**: Capture/Compare 2 output Polarity
refer to CC1NP description

Bits 6 Reserved, must be kept at reset value.

Bit 5 **CC2P**: Capture/Compare 2 output Polarity
refer to CC1P description

Bit 4 **CC2E**: Capture/Compare 2 output enable
refer to CC1E description

Bit 3 **CC1NP**: Capture/Compare 1 complementary output Polarity
CC1 channel configured as output: CC1NP must be kept cleared
CC1 channel configured as input: CC1NP is used in conjunction with CC1P to define
TI1FP1/TI2FP1 polarity (refer to CC1P description).

Bit 2 Reserved, must be kept at reset value.

Bit 1 **CC1P**: Capture/Compare 1 output Polarity.**CC1 channel configured as output:**

0: OC1 active high.

1: OC1 active low.

CC1 channel configured as input:

CC1NP/CC1P bits select TI1FP1 and TI2FP1 polarity for trigger or capture operations.

00: noninverted/rising edge

Circuit is sensitive to TIxFP1 rising edge (capture, trigger in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger in gated mode, encoder mode).

01: inverted/falling edge

Circuit is sensitive to TIxFP1 falling edge (capture, trigger in reset, external clock or trigger mode), TIxFP1 is inverted (trigger in gated mode, encoder mode).

10: reserved, do not use this configuration.

Note: 11: noninverted/both edges

Circuit is sensitive to both TIxFP1 rising and falling edges (capture, trigger in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger in gated mode). This configuration must not be used for encoder mode.

Bit 0 **CC1E**: Capture/Compare 1 output enable.**CC1 channel configured as output:**

0: Off - OC1 is not active.

1: On - OC1 signal is output on the corresponding output pin.

CC1 channel configured as input:

This bit determines if a capture of the counter value can actually be done into the input capture/compare register 1 (TIMx_CCR1) or not.

0: Capture disabled.

1: Capture enabled.

Table 103. Output control bit for standard OCx channels

CCxE bit	OCx output state
0	Output disabled (OCx='0', OCx_EN='0')
1	OCx=OCxREF + Polarity, OCx_EN='1'

Note: The states of the external I/O pins connected to the standard OCx channels depend on the state of the OCx channel and on the GPIO registers.

19.4.8 TIM9/12 counter (TIMx_CNT)

Address offset: 0x24

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 15:0 **CNT[15:0]**: Counter value

19.4.9 TIM9/12 prescaler (TIMx_PSC)

Address offset: 0x28

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 15:0 **PSC[15:0]**: Prescaler value

The counter clock frequency CK_CNT is equal to $f_{CK_PSC} / (PSC[15:0] + 1)$.

PSC contains the value to be loaded in the active prescaler register at each update event (including when the counter is cleared through UG bit of TIMx_EGR register or through trigger controller when configured in "reset mode").

19.4.10 TIM9/12 auto-reload register (TIMx_ARR)

Address offset: 0x2C

Reset value: 0xFFFF

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 15:0 **ARR[15:0]**: Auto-reload value

ARR is the value to be loaded into the actual auto-reload register.

Refer to the [Section 19.3.1: Time-base unit](#) for more details about ARR update and behavior.

The counter is blocked while the auto-reload value is null.

19.4.11 TIM9/12 capture/compare register 1 (TIMx_CCR1)

Address offset: 0x34

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1[15:0]															
rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro

Bits 15:0 **CCR1[15:0]**: Capture/Compare 1 value**If channel CC1 is configured as output:**

CCR1 is the value to be loaded into the actual capture/compare 1 register (preload value). It is loaded permanently if the preload feature is not selected in the TIMx_CCMR1 register (OC1PE bit). Else the preload value is copied into the active capture/compare 1 register when an update event occurs.

The active capture/compare register contains the value to be compared to the TIMx_CNT counter and signaled on the OC1 output.

If channel CC1 is configured as input:

CCR1 is the counter value transferred by the last input capture 1 event (IC1). The TIMx_CCR1 register is read-only and cannot be programmed.

19.4.12 TIM9/12 capture/compare register 2 (TIMx_CCR2)

Address offset: 0x38

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR2[15:0]															
rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro

Bits 15:0 **CCR2[15:0]**: Capture/Compare 2 value**If channel CC2 is configured as output:**

CCR2 is the value to be loaded into the actual capture/compare 2 register (preload value). It is loaded permanently if the preload feature is not selected in the TIMx_CCMR2 register (OC2PE bit). Else the preload value is copied into the active capture/compare 2 register when an update event occurs.

The active capture/compare register contains the value to be compared to the TIMx_CNT counter and signalled on the OC2 output.

If channel CC2 is configured as input:

CCR2 is the counter value transferred by the last input capture 2 event (IC2). The TIMx_CCR2 register is read-only and cannot be programmed.

19.4.13 TIM9/12 register map

TIM9/12 registers are mapped as 16-bit addressable registers as described below. The reserved memory areas are highlighted in gray in the table.

Table 104. TIM9/12 register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0													
0x00	TIMx_CR1 Reset value	Reserved																						CKD [1:0]		0	0	0	ARPE		Reserved		OPM	URS	UDIS	CEN										
0x08	TIMx_SMCR Reset value	Reserved																						MSM		0	0	0	0	TS[2:0]		Reserved	SMS[2:0]		0	0	0									
0x0C	TIMx_DIER Reset value	Reserved																						TIE		0	Reserved		CC2IE	CC1IE	UIE															
0x10	TIMx_SR Reset value	Reserved																				CC2OF		0	CC1OF		0	Reserved		0	Reserved		CC2IF	CC1IF	UIF											
0x14	TIMx_EGR Reset value	Reserved																						TG		0	Reserved		CC2G	CC1G	UG															
0x18	TIMx_CCMR1 Output compare mode Reset value	Reserved														OC2M [2:0]		0	0	0	OC2PE OC2FE		0	0	CC2S [1:0]		0	0	Reserved		OC1M [2:0]		0	0	0	OC1PE OC1FE		0	0	CC1S [1:0]		0	0			
	TIMx_CCMR1 Input capture mode Reset value	Reserved														IC2F[3:0]				0	0	0	0	IC2PSC [1:0]		0	0	CC2S [1:0]		0	0	IC1F[3:0]				0	0	0	0	IC1PSC [1:0]		0	0	CC1S [1:0]		0
0x1C	Reserved																																													
0x20	TIMx_CCER Reset value	Reserved																						CC2NP Reserved		0	CC2P		0	CC2E		0	CC1NP Reserved		0	CC1P		0	CC1E		0					
0x24	TIMx_CNT Reset value	Reserved														CNT[15:0]																														
0x28	TIMx_PSC Reset value	Reserved														PSC[15:0]																														
0x2C	TIMx_ARR Reset value	Reserved														ARR[15:0]																														
0x30	Reserved																																													
0x34	TIMx_CCR1 Reset value	Reserved														CCR1[15:0]																														

Table 104. TIM9/12 register map and reset values (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0x38	TIMx_CCR2	Reserved																CCR2[15:0]																	
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x3C to 0x4C	Reserved																																		

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.



19.5 TIM10/11/13/14 registers

The peripheral registers have to be written by half-words (16 bits) or words (32 bits). Read accesses can be done by bytes (8 bits), half-words (16 bits) or words (32 bits).

19.5.1 TIM10/11/13/14 control register 1 (TIMx_CR1)

Address offset: 0x00

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reserved						CKD[1:0]		ARPE	Reserved				OPM	URS	UDIS	CEN
						rw	rw	rw					rw	rw	rw	rw

Bits 15:10 Reserved, must be kept at reset value.

Bits 9:8 **CKD**: Clock division

This bit-field indicates the division ratio between the timer clock (CK_INT) frequency and sampling clock used by the digital filters (Tl_x),

00: $t_{DTS} = t_{CK_INT}$

01: $t_{DTS} = 2 \times t_{CK_INT}$

10: $t_{DTS} = 4 \times t_{CK_INT}$

11: Reserved

Bit 7 **ARPE**: Auto-reload preload enable

0: TIMx_ARR register is not buffered

1: TIMx_ARR register is buffered

Bits 6:4 Reserved, must be kept at reset value.

Bit 3 **OPM**: One-pulse mode

0: Counter is not stopped on the update event

1: Counter stops counting on the next update event (clearing the CEN bit).

Bit 2 **URS**: Update request source

This bit is set and cleared by software to select the update interrupt (UEV) sources.

0: Any of the following events generate an UEV if enabled:

- Counter overflow
- Setting the UG bit

1: Only counter overflow generates an UEV if enabled.

Bit 1 **UDIS**: Update disable

This bit is set and cleared by software to enable/disable update interrupt (UEV) event generation.

0: UEV enabled. An UEV is generated by one of the following events:

- Counter overflow
- Setting the UG bit.

Buffered registers are then loaded with their preload values.

1: UEV disabled. No UEV is generated, shadow registers keep their value (ARR, PSC, CCR_x). The counter and the prescaler are reinitialized if the UG bit is set.

Bit 0 **CEN**: Counter enable

0: Counter disabled

1: Counter enabled

19.5.2 TIM10/11/13/14 Interrupt enable register (TIMx_DIER)

Address offset: 0x0C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														CC1IE	UIE
														rw	rw

Bits 15:2 Reserved, must be kept at reset value.

Bit 1 **CC1IE**: Capture/Compare 1 interrupt enable

0: CC1 interrupt disabled

1: CC1 interrupt enabled

Bit 0 **UIE**: Update interrupt enable

0: Update interrupt disabled

1: Update interrupt enabled

19.5.3 TIM10/11/13/14 status register (TIMx_SR)

Address offset: 0x10

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						CC1OF	Reserved							CC1IF	UIF
						rc_w0								rc_w0	rc_w0

Bits 15:10 Reserved, must be kept at reset value.

Bit 9 **CC1OF**: Capture/Compare 1 overcapture flag

This flag is set by hardware only when the corresponding channel is configured in input capture mode. It is cleared by software by writing it to '0'.

0: No overcapture has been detected.

1: The counter value has been captured in TIMx_CCR1 register while CC1IF flag was already set

Bits 8:2 Reserved, must be kept at reset value.

Bit 1 **CC1IF**: Capture/compare 1 interrupt flag**If channel CC1 is configured as output:**

This flag is set by hardware when the counter matches the compare value. It is cleared by software.

0: No match.

1: The content of the counter TIMx_CNT matches the content of the TIMx_CCR1 register.

When the contents of TIMx_CCR1 are greater than the contents of TIMx_ARR, the CC1IF bit goes high on the counter overflow.

If channel CC1 is configured as input:

This bit is set by hardware on a capture. It is cleared by software or by reading the TIMx_CCR1 register.

0: No input capture occurred.

1: The counter value has been captured in TIMx_CCR1 register (an edge has been detected on IC1 which matches the selected polarity).

Bit 0 **UIF**: Update interrupt flag

This bit is set by hardware on an update event. It is cleared by software.

0: No update occurred.

1: Update interrupt pending. This bit is set by hardware when the registers are updated:

- At overflow and if UDIS='0' in the TIMx_CR1 register.
- When CNT is reinitialized by software using the UG bit in TIMx_EGR register, if URS='0' and UDIS='0' in the TIMx_CR1 register.

19.5.4 TIM10/11/13/14 event generation register (TIMx_EGR)

Address offset: 0x14

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														CC1G	UG
														w	w

Bits 15:2 Reserved, must be kept at reset value.

Bit 1 **CC1G**: Capture/compare 1 generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: A capture/compare event is generated on channel 1:

If channel CC1 is configured as output:

CC1IF flag is set, Corresponding interrupt or is sent if enabled.

If channel CC1 is configured as input:

The current value of the counter is captured in TIMx_CCR1 register. The CC1IF flag is set, the corresponding interrupt is sent if enabled. The CC1OF flag is set if the CC1IF flag was already high.

Bit 0 **UG**: Update generation

This bit can be set by software, it is automatically cleared by hardware.

0: No action

1: Re-initialize the counter and generates an update of the registers. Note that the prescaler counter is cleared too (anyway the prescaler ratio is not affected). The counter is cleared.

19.5.5 TIM10/11/13/14 capture/compare mode register 1 (TIMx_CCMR1)

Address offset: 0x18

Reset value: 0x0000

The channels can be used in input (capture mode) or in output (compare mode). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits of this register have a different function in input and in output mode. For a given bit, OCxx describes its function when the channel is configured in output, ICxx describes its function when the channel is configured in input. So take care that the same bit can have a different meaning for the input stage and for the output stage.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									OC1M[2:0]			OC1PE	OC1FE	CC1S[1:0]	
Reserved								IC1F[3:0]			IC1PSC[1:0]				
								r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Output compare mode

Bits 15:7 Reserved, must be kept at reset value.

Bits 6:4 **OC1M**: Output compare 1 mode

These bits define the behavior of the output reference signal OC1REF from which OC1 is derived. OC1REF is active high whereas OC1 active level depends on CC1P bit.

000: Frozen. The comparison between the output compare register TIMx_CCR1 and the counter TIMx_CNT has no effect on the outputs.

001: Set channel 1 to active level on match. OC1REF signal is forced high when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).

010: Set channel 1 to inactive level on match. OC1REF signal is forced low when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).

011: Toggle - OC1REF toggles when TIMx_CNT = TIMx_CCR1.

100: Force inactive level - OC1REF is forced low.

101: Force active level - OC1REF is forced high.

110: PWM mode 1 - Channel 1 is active as long as TIMx_CNT < TIMx_CCR1 else inactive.

111: PWM mode 2 - Channel 1 is inactive as long as TIMx_CNT < TIMx_CCR1 else active.

Note: In PWM mode 1 or 2, the OCREF level changes when the result of the comparison changes or when the output compare mode switches from frozen to PWM mode.

Bit 3 **OC1PE**: Output compare 1 preload enable

0: Preload register on TIMx_CCR1 disabled. TIMx_CCR1 can be written at anytime, the new value is taken in account immediately.

1: Preload register on TIMx_CCR1 enabled. Read/Write operations access the preload register. TIMx_CCR1 preload value is loaded in the active register at each update event.

Bit 2 **OC1FE**: Output compare 1 fast enable

This bit is used to accelerate the effect of an event on the trigger in input on the CC output.
0: CC1 behaves normally depending on counter and CCR1 values even when the trigger is ON. The minimum delay to activate CC1 output when an edge occurs on the trigger input is 5 clock cycles.

1: An active edge on the trigger input acts like a compare match on CC1 output. OC is then set to the compare level independently of the result of the comparison. Delay to sample the trigger input and to activate CC1 output is reduced to 3 clock cycles. OC1FE acts only if the channel is configured in PWM1 or PWM2 mode.

Bits 1:0 **CC1S**: Capture/Compare 1 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.
00: CC1 channel is configured as output.

01: CC1 channel is configured as input, IC1 is mapped on TI1.

10:

11:

Note: CC1S bits are writable only when the channel is OFF (CC1E = 0 in TIMx_CCER).

Input capture mode

Bits 15:8 Reserved, must be kept at reset value.

Bits 7:4 **IC1F**: Input capture 1 filter

This bit-field defines the frequency used to sample TI1 input and the length of the digital filter applied to TI1. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, sampling is done at f_{DTS}

0001: $f_{SAMPLING}=f_{CK_INT}$, N=2

0010: $f_{SAMPLING}=f_{CK_INT}$, N=4

0011: $f_{SAMPLING}=f_{CK_INT}$, N=8

0100: $f_{SAMPLING}=f_{DTS}/2$, N=6

0101: $f_{SAMPLING}=f_{DTS}/2$, N=8

0110: $f_{SAMPLING}=f_{DTS}/4$, N=6

0111: $f_{SAMPLING}=f_{DTS}/4$, N=8

1000: $f_{SAMPLING}=f_{DTS}/8$, N=6

1001: $f_{SAMPLING}=f_{DTS}/8$, N=8

1010: $f_{SAMPLING}=f_{DTS}/16$, N=5

1011: $f_{SAMPLING}=f_{DTS}/16$, N=6

1100: $f_{SAMPLING}=f_{DTS}/16$, N=8

1101: $f_{SAMPLING}=f_{DTS}/32$, N=5

1110: $f_{SAMPLING}=f_{DTS}/32$, N=6

1111: $f_{SAMPLING}=f_{DTS}/32$, N=8

Bits 3:2 **IC1PSC**: Input capture 1 prescaler

This bit-field defines the ratio of the prescaler acting on CC1 input (IC1).

The prescaler is reset as soon as CC1E='0' (TIMx_CCER register).

00: no prescaler, capture is done each time an edge is detected on the capture input

01: capture is done once every 2 events

10: capture is done once every 4 events

11: capture is done once every 8 events

Bits 1:0 **CC1S**: Capture/Compare 1 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output

01: CC1 channel is configured as input, IC1 is mapped on TI1

10: Reserved

11: Reserved

Note: CC1S bits are writable only when the channel is OFF (CC1E = 0 in TIMx_CCER).

19.5.6 TIM10/11/13/14 capture/compare enable register (TIMx_CCER)

Address offset: 0x20

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												CC1NP	Res.	CC1P	CC1E
												rw		rw	rw

Bits 15:4 Reserved, must be kept at reset value.

Bit 3 **CC1NP**: Capture/Compare 1 complementary output Polarity.

CC1 channel configured as output: CC1NP must be kept cleared.

CC1 channel configured as input: CC1NP bit is used in conjunction with CC1P to define TI1FP1 polarity (refer to CC1P description).

Bit 2 Reserved, must be kept at reset value.

Bit 1 **CC1P**: Capture/Compare 1 output Polarity.**CC1 channel configured as output:**

0: OC1 active high

1: OC1 active low

CC1 channel configured as input:

The CC1P bit selects TI1FP1 and TI2FP1 polarity for trigger or capture operations.

00: noninverted/rising edge

Circuit is sensitive to TI1FP1 rising edge (capture mode), TI1FP1 is not inverted.

01: inverted/falling edge

Circuit is sensitive to TI1FP1 falling edge (capture mode), TI1FP1 is inverted.

10: reserved, do not use this configuration.

11: noninverted/both edges

Circuit is sensitive to both TI1FP1 rising and falling edges (capture mode), TI1FP1 is not inverted.

Bit 0 **CC1E**: Capture/Compare 1 output enable.**CC1 channel configured as output:**

0: Off - OC1 is not active

1: On - OC1 signal is output on the corresponding output pin

CC1 channel configured as input:

This bit determines if a capture of the counter value can actually be done into the input capture/compare register 1 (TIMx_CCR1) or not.

0: Capture disabled

1: Capture enabled

Table 105. Output control bit for standard OCx channels

CCxE bit	OCx output state
0	Output Disabled (OCx='0', OCx_EN='0')
1	OCx=OCxREF + Polarity, OCx_EN='1'

Note: The state of the external I/O pins connected to the standard OCx channels depends on the OCx channel state and the GPIO registers.

19.5.7 TIM10/11/13/14 counter (TIMx_CNT)

Address offset: 0x24

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **CNT[15:0]**: Counter value**19.5.8 TIM10/11/13/14 prescaler (TIMx_PSC)**

Address offset: 0x28

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **PSC[15:0]**: Prescaler valueThe counter clock frequency CK_CNT is equal to $f_{CK_PSC} / (PSC[15:0] + 1)$.

PSC contains the value to be loaded in the active prescaler register at each update event (including when the counter is cleared through UG bit of TIMx_EGR register or through trigger controller when configured in "reset mode").

19.5.9 TIM10/11/13/14 auto-reload register (TIMx_ARR)

Address offset: 0x2C

Reset value: 0xFFFF

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **ARR[15:0]**: Auto-reload value

ARR is the value to be loaded in the actual auto-reload register.

Refer to [Section 19.3.1: Time-base unit](#) for more details about ARR update and behavior.

The counter is blocked while the auto-reload value is null.

19.5.10 TIM10/11/13/14 capture/compare register 1 (TIMx_CCR1)

Address offset: 0x34

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1[15:0]															
rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro	rw/ro

Bits 15:0 **CCR1[15:0]**: Capture/Compare 1 value**If channel CC1 is configured as output:**

CCR1 is the value to be loaded in the actual capture/compare 1 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR1 register (bit OC1PE). Else the preload value is copied in the active capture/compare 1 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signaled on OC1 output.

If channel CC1 is configured as input:

CCR1 is the counter value transferred by the last input capture 1 event (IC1). The TIMx_CCR1 register is read-only and cannot be programmed.

19.5.11 TIM11 option register 1 (TIM11_OR)

Address offset: 0x50

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														TIM1_RMP[1:0]	
														rw	

Bits 15:2 Reserved, must be kept at reset value.

Bits 1:0 **TIM1_RMP[1:0]**: TIM11 Input 1 remapping capability

Set and cleared by software.

00,01,11: TIM11 Channel1 is connected to the GPIO (refer to the Alternate function mapping table in the datasheets).

10: HSE_RTC clock (HSE divided by programmable prescaler) is connected to the TIM11_CH1 input for measurement purposes

19.5.12 TIM10/11/13/14 register map

TIMx registers are mapped as 16-bit addressable registers as described in the table below.

Table 106. TIM10/11/13/14 register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0x00	TIMx_CR1	Reserved																						CKD [1:0]		ARPE	Reserve d		OPM	URS	UDIS	CEN			
	Reset value																							0	0								0	0	0
0x08	TIMx_SMCR	Reserved																																	
	Reset value																																		
0x0C	TIMx_DIER	Reserved																														CC1IE	UIE		
	Reset value																																	0	0
0x10	TIMx_SR	Reserved																						CC1OF	Reserved						CC1IF	UIF			
	Reset value																																0	0	
0x14	TIMx_EGR	Reserved																														CC1G	UG		
	Reset value																																	0	0
0x18	TIMx_CCMR1 Output compare mode	Reserved																						OC1M [2:0]			OC1PE	OC1FE	CC1S [1:0]						
	Reset value																							0	0	0					0	0	0		
	TIMx_CCMR1 Input capture mode	Reserved																						IC1F[3:0]			IC1 PSC [1:0]	CC1S [1:0]							
	Reset value																							0	0	0				0	0	0	0	0	
0x1C	Reserved																																		
0x20	TIMx_CCER	Reserved																														CC1NP	Reserved	CC1P	CC1E
	Reset value																																		
0x24	TIMx_CNT	Reserved																CNT[15:0]																	
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x28	TIMx_PSC	Reserved																PSC[15:0]																	
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x2C	TIMx_ARR	Reserved																ARR[15:0]																	
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x30	Reserved																																		
0x34	TIMx_CCR1	Reserved																CCR1[15:0]																	
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x38 to 0x4C	Reserved																																		

Table 106. TIM10/11/13/14 register map and reset values (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x50	TIMx_OR	Reserved																														TIM_RMP	
	Reset value																															0	0

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.



20 Basic timers (TIM6 and TIM7)

This section applies to the whole STM32F4xx family, unless otherwise specified.

20.1 TIM6 and TIM7 introduction

The basic timers TIM6 and TIM7 consist of a 16-bit auto-reload counter driven by a programmable prescaler.

They may be used as generic timers for time-base generation but they are also specifically used to drive the digital-to-analog converter (DAC). In fact, the timers are internally connected to the DAC and are able to drive it through their trigger outputs.

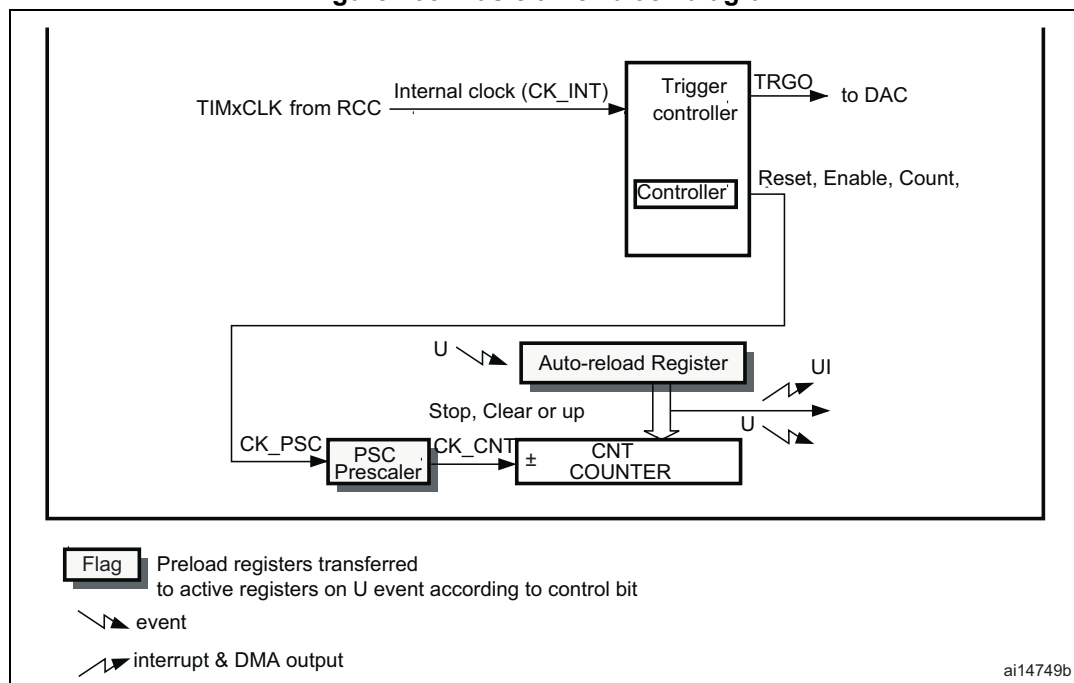
The timers are completely independent, and do not share any resources.

20.2 TIM6 and TIM7 main features

Basic timer (TIM6 and TIM7) features include:

- 16-bit auto-reload upcounter
- 16-bit programmable prescaler used to divide (also “on the fly”) the counter clock frequency by any factor between 1 and 65536
- Synchronization circuit to trigger the DAC
- Interrupt/DMA generation on the update event: counter overflow

Figure 203. Basic timer block diagram



20.3 TIM6 and TIM7 functional description

20.3.1 Time-base unit

The main block of the programmable timer is a 16-bit upcounter with its related auto-reload register. The counter clock can be divided by a prescaler.

The counter, the auto-reload register and the prescaler register can be written or read by software. This is true even when the counter is running.

The time-base unit includes:

- Counter register (TIMx_CNT)
- Prescaler register (TIMx_PSC)
- Auto-Reload register (TIMx_ARR)

The auto-reload register is preloaded. The preload register is accessed each time an attempt is made to write or read the auto-reload register. The contents of the preload register are transferred into the shadow register permanently or at each update event UEV, depending on the auto-reload preload enable bit (ARPE) in the TIMx_CR1 register. The update event is sent when the counter reaches the overflow value and if the UDIS bit equals 0 in the TIMx_CR1 register. It can also be generated by software. The generation of the update event is described in detail for each configuration.

The counter is clocked by the prescaler output CK_CNT, which is enabled only when the counter enable bit (CEN) in the TIMx_CR1 register is set.

Note that the actual counter enable signal CNT_EN is set 1 clock cycle after CEN.

Prescaler description

The prescaler can divide the counter clock frequency by any factor between 1 and 65536. It is based on a 16-bit counter controlled through a 16-bit register (in the TIMx_PSC register). It can be changed on the fly as the TIMx_PSC control register is buffered. The new prescaler ratio is taken into account at the next update event.

[Figure 204](#) and [Figure 205](#) give some examples of the counter behavior when the prescaler ratio is changed on the fly.

Figure 204. Counter timing diagram with prescaler division change from 1 to 2

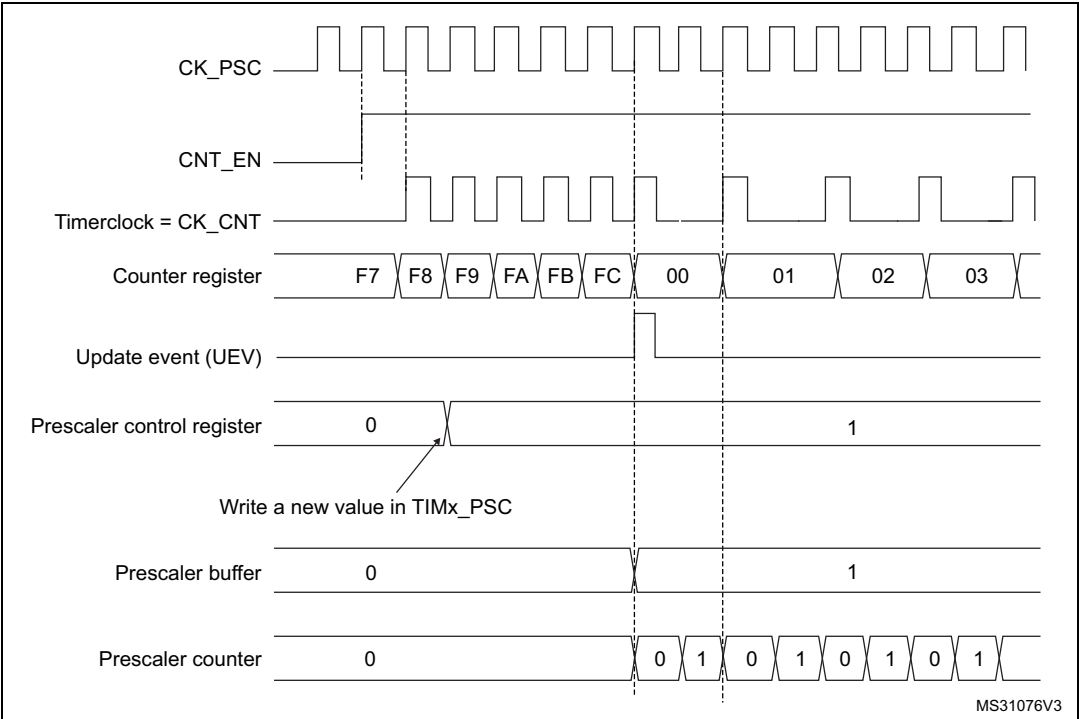
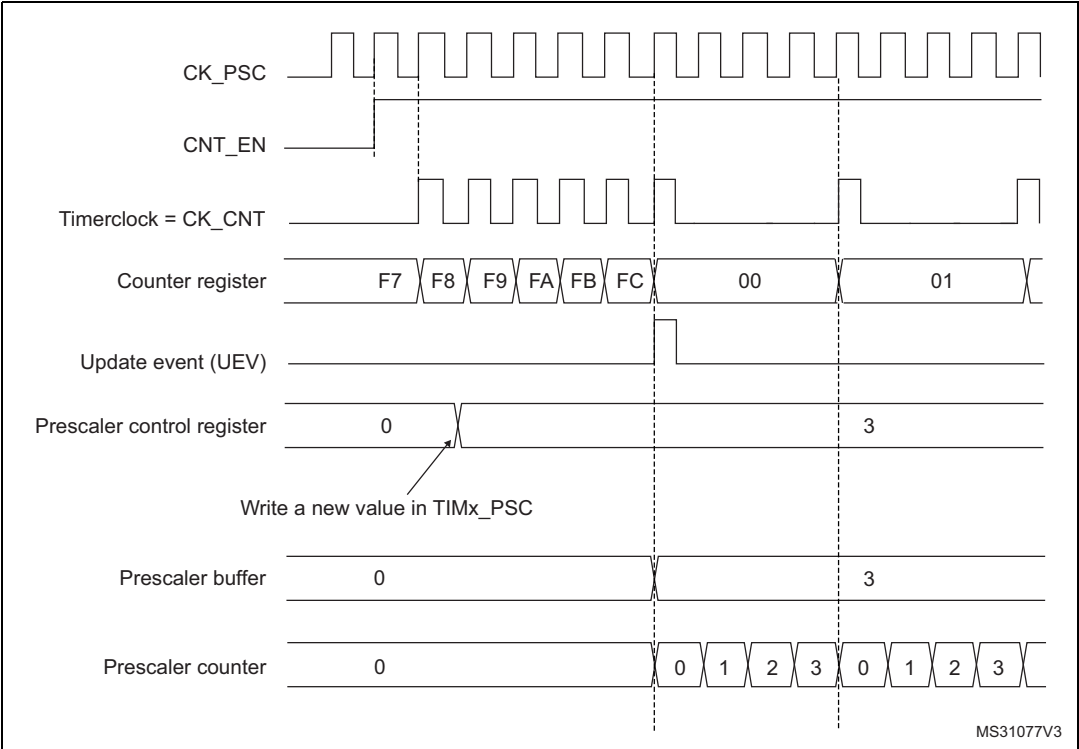


Figure 205. Counter timing diagram with prescaler division change from 1 to 4



20.3.2 Counting mode

The counter counts from 0 to the auto-reload value (contents of the TIMx_ARR register), then restarts from 0 and generates a counter overflow event.

An update event can be generated at each counter overflow or by setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller).

The UEV event can be disabled by software by setting the UDIS bit in the TIMx_CR1 register. This avoids updating the shadow registers while writing new values into the preload registers. In this way, no update event occurs until the UDIS bit has been written to 0, however, the counter and the prescaler counter both restart from 0 (but the prescale rate does not change). In addition, if the URS (update request selection) bit in the TIMx_CR1 register is set, setting the UG bit generates an update event UEV, but the UIF flag is not set (so no interrupt or DMA request is sent).

When an update event occurs, all the registers are updated and the update flag (UIF bit in the TIMx_SR register) is set (depending on the URS bit):

- The buffer of the prescaler is reloaded with the preload value (contents of the TIMx_PSC register)
- The auto-reload shadow register is updated with the preload value (TIMx_ARR)

The following figures show some examples of the counter behavior for different clock frequencies when TIMx_ARR = 0x36.

Figure 206. Counter timing diagram, internal clock divided by 1

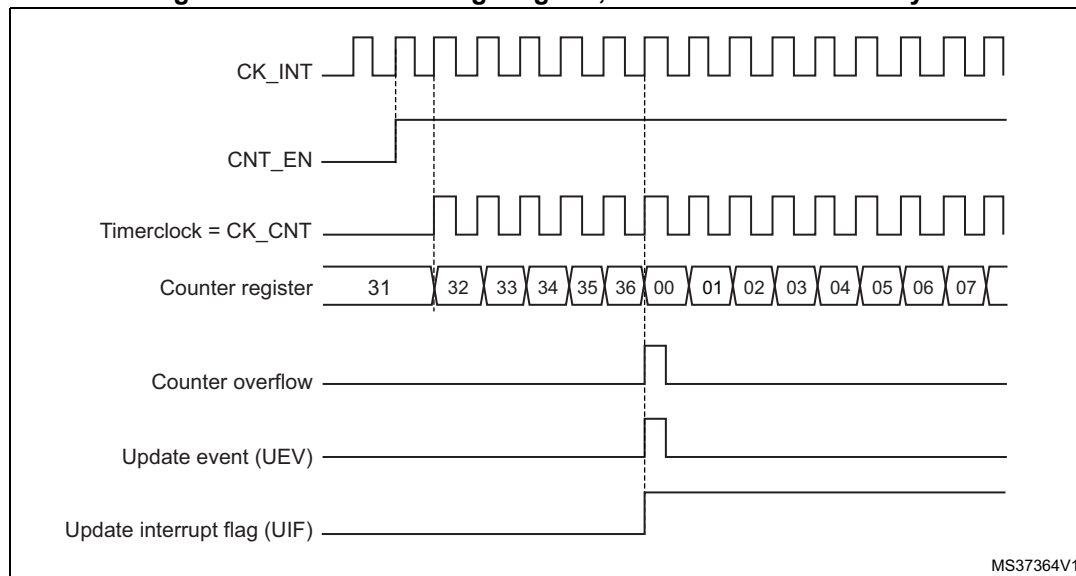


Figure 207. Counter timing diagram, internal clock divided by 2

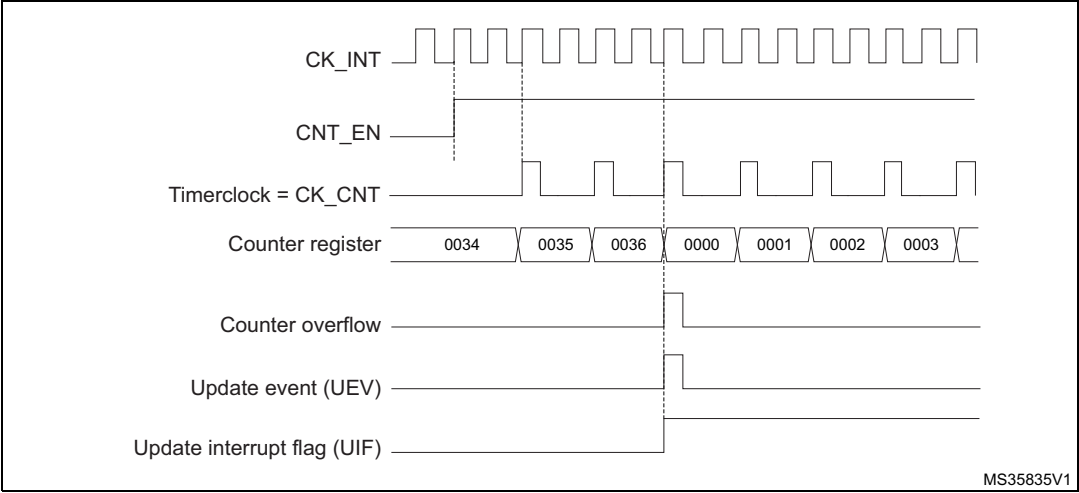


Figure 208. Counter timing diagram, internal clock divided by 4

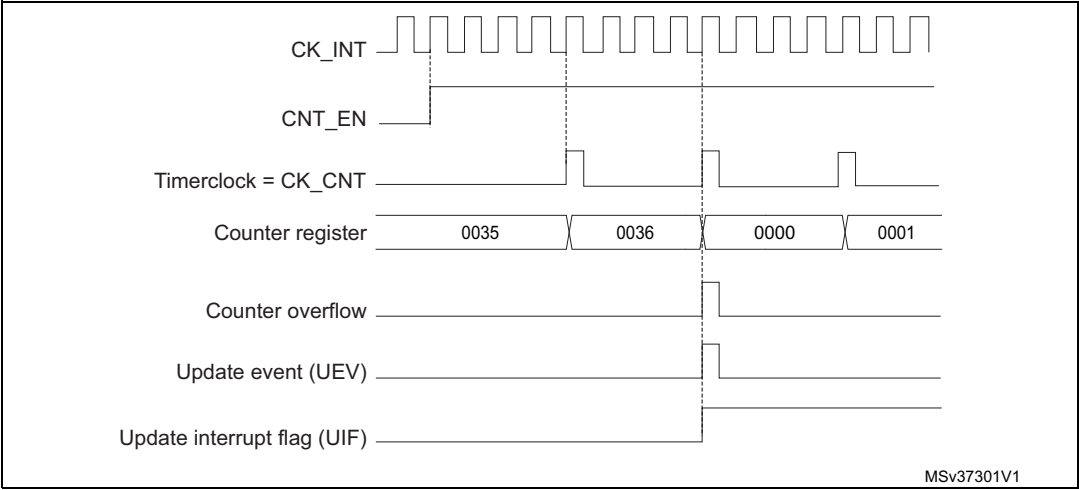


Figure 209. Counter timing diagram, internal clock divided by N

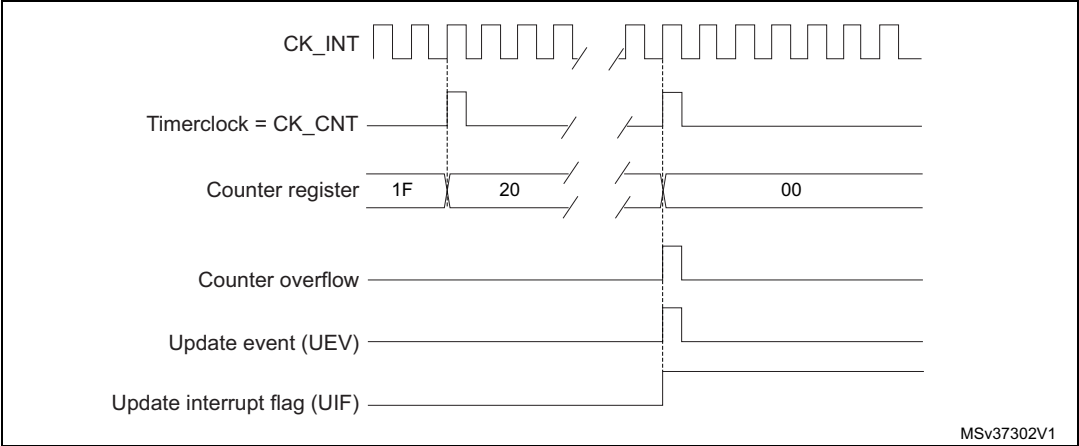
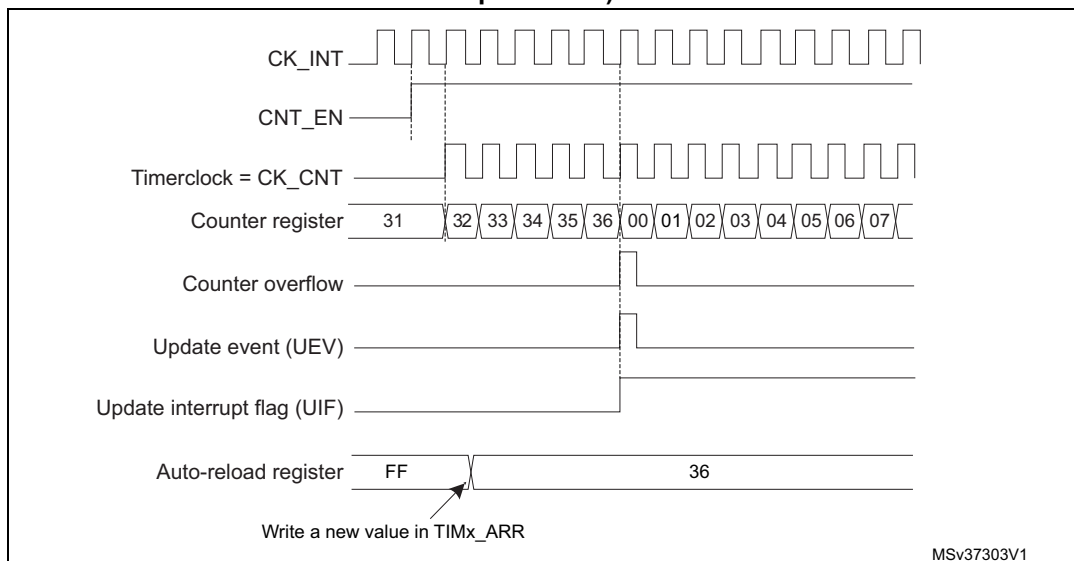
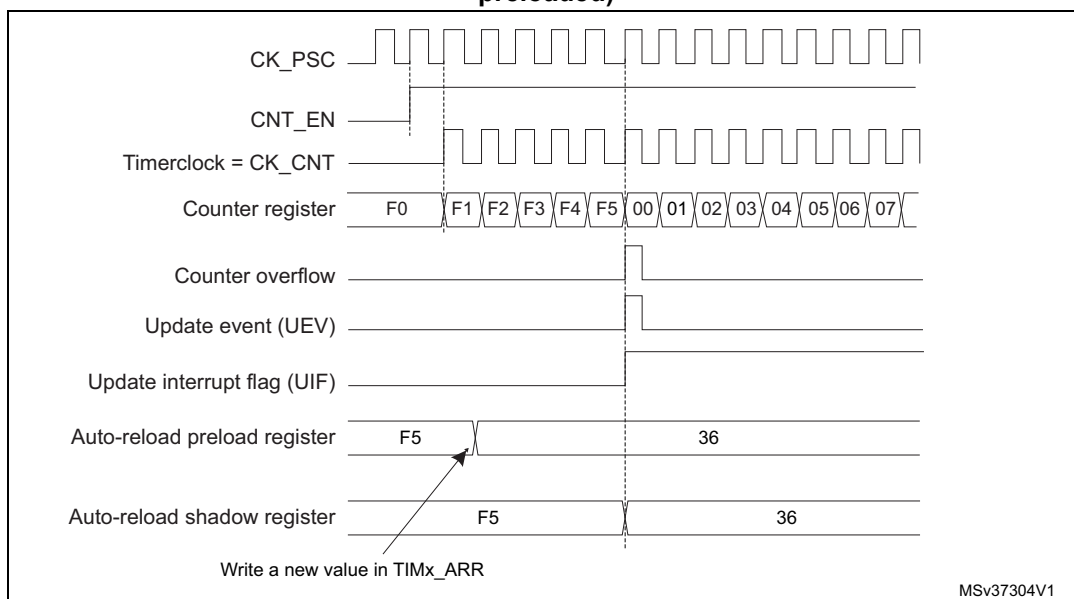


Figure 210. Counter timing diagram, update event when ARPE=0 (TIMx_ARR not preloaded)

MSv37303V1

Figure 211. Counter timing diagram, update event when ARPE=1 (TIMx_ARR preloaded)

MSv37304V1

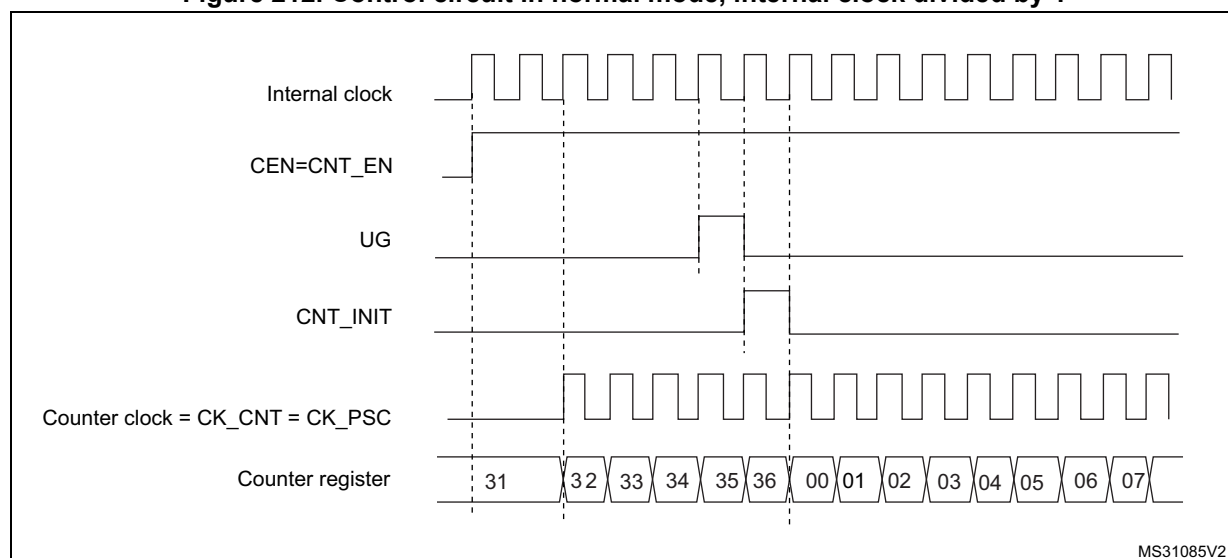
20.3.3 Clock source

The counter clock is provided by the Internal clock (CK_INT) source.

The CEN (in the TIMx_CR1 register) and UG bits (in the TIMx_EGR register) are actual control bits and can be changed only by software (except for UG that remains cleared automatically). As soon as the CEN bit is written to 1, the prescaler is clocked by the internal clock CK_INT.

[Figure 212](#) shows the behavior of the control circuit and the upcounter in normal mode, without prescaler.

Figure 212. Control circuit in normal mode, internal clock divided by 1



20.3.4 Debug mode

When the microcontroller enters the debug mode (Cortex[®]-M4 with FPU core - halted), the TIMx counter either continues to work normally or stops, depending on the DBG_TIMx_STOP configuration bit in the DBG module. For more details, refer to [Section 38.16.2: Debug support for timers, watchdog, bxCAN and I²C](#).

20.4 TIM6 and TIM7 registers

Refer to [Section 2.2](#) for a list of abbreviations used in register descriptions.

The peripheral registers have to be written by half-words (16 bits) or words (32 bits). Read accesses can be done by bytes (8 bits), half-words (16 bits) or words (32 bits).

20.4.1 TIM6 and TIM7 control register 1 (TIMx_CR1)

Address offset: 0x00

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reserved								ARPE	Reserved				OPM	URS	UDIS	CEN
								rw					rw	rw	rw	rw

Bits 15:8 Reserved, must be kept at reset value.

Bit 7 **ARPE**: Auto-reload preload enable

0: TIMx_ARR register is not buffered.

1: TIMx_ARR register is buffered.

Bits 6:4 Reserved, must be kept at reset value.

Bit 3 **OPM**: One-pulse mode

0: Counter is not stopped at update event

1: Counter stops counting at the next update event (clearing the CEN bit).

Bit 2 URS: Update request source

This bit is set and cleared by software to select the UEV event sources.

0: Any of the following events generates an update interrupt or DMA request if enabled.

These events can be:

- Counter overflow/underflow
- Setting the UG bit
- Update generation through the slave mode controller

1: Only counter overflow/underflow generates an update interrupt or DMA request if enabled.

Bit 1 UDIS: Update disable

This bit is set and cleared by software to enable/disable UEV event generation.

0: UEV enabled. The Update (UEV) event is generated by one of the following events:

- Counter overflow/underflow
- Setting the UG bit
- Update generation through the slave mode controller

Buffered registers are then loaded with their preload values.

1: UEV disabled. The Update event is not generated, shadow registers keep their value (ARR, PSC). However the counter and the prescaler are reinitialized if the UG bit is set or if a hardware reset is received from the slave mode controller.

Bit 0 CEN: Counter enable

0: Counter disabled

1: Counter enabled

Note: Gated mode can work only if the CEN bit has been previously set by software. However trigger mode can set the CEN bit automatically by hardware.

CEN is cleared automatically in one-pulse mode, when an update event occurs.

20.4.2 TIM6 and TIM7 control register 2 (TIMx_CR2)

Address offset: 0x04

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									MMS[2:0]			Reserved			
									rw	rw	rw				

Bits 15:7 Reserved, must be kept at reset value.

Bits 6:4 **MMS[2:0]**: Master mode selection

These bits are used to select the information to be sent in master mode to slave timers for synchronization (TRGO). The combination is as follows:

000: **Reset** - the UG bit from the TIMx_EGR register is used as a trigger output (TRGO). If reset is generated by the trigger input (slave mode controller configured in reset mode) then the signal on TRGO is delayed compared to the actual reset.

001: **Enable** - the Counter enable signal, CNT_EN, is used as a trigger output (TRGO). It is useful to start several timers at the same time or to control a window in which a slave timer is enabled. The Counter Enable signal is generated by a logic OR between CEN control bit and the trigger input when configured in gated mode.

When the Counter Enable signal is controlled by the trigger input, there is a delay on TRGO, except if the master/slave mode is selected (see the MSM bit description in the TIMx_SMCR register).

010: **Update** - The update event is selected as a trigger output (TRGO). For instance a master timer can then be used as a prescaler for a slave timer.

Note: The clock of the slave timer and ADC must be enabled prior to receiving events from the master timer, and must not be changed on-the-fly while triggers are received from the master timer.

Bits 3:0 Reserved, must be kept at reset value.

20.4.3 TIM6 and TIM7 DMA/Interrupt enable register (TIMx_DIER)

Address offset: 0x0C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved							UDE	Reserved							UIE
							rw								rw

Bits 15:9 Reserved, must be kept at reset value.

Bit 8 **UDE**: Update DMA request enable

0: Update DMA request disabled.

1: Update DMA request enabled.

Bits 7:1 Reserved, must be kept at reset value.

Bit 0 **UIE**: Update interrupt enable

0: Update interrupt disabled.

1: Update interrupt enabled.

20.4.4 TIM6 and TIM7 status register (TIMx_SR)

Address offset: 0x10

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															UIF
															rc_w0

Bits 15:1 Reserved, must be kept at reset value.

Bit 0 **UIF**: Update interrupt flag

This bit is set by hardware on an update event. It is cleared by software.

0: No update occurred.

1: Update interrupt pending. This bit is set by hardware when the registers are updated:

- At overflow or underflow and if UDIS = 0 in the TIMx_CR1 register.
- When CNT is reinitialized by software using the UG bit in the TIMx_EGR register, if URS = 0 and UDIS = 0 in the TIMx_CR1 register.

20.4.5 TIM6 and TIM7 event generation register (TIMx_EGR)

Address offset: 0x14

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															UG
															w

Bits 15:1 Reserved, must be kept at reset value.

Bit 0 **UG**: Update generation

This bit can be set by software, it is automatically cleared by hardware.

0: No action.

1: Re-initializes the timer counter and generates an update of the registers. Note that the prescaler counter is cleared too (but the prescaler ratio is not affected).

20.4.6 TIM6 and TIM7 counter (TIMx_CNT)

Address offset: 0x24

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **CNT[15:0]**: Counter value

20.4.7 TIM6 and TIM7 prescaler (TIMx_PSC)

Address offset: 0x28

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **PSC[15:0]**: Prescaler value

The counter clock frequency CK_CNT is equal to $f_{CK_PSC} / (PSC[15:0] + 1)$.

PSC contains the value to be loaded in the active prescaler register at each update event (including when the counter is cleared through UG bit of TIMx_EGR register or through trigger controller when configured in "reset mode").

20.4.8 TIM6 and TIM7 auto-reload register (TIMx_ARR)

Address offset: 0x2C

Reset value: 0xFFFF

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **ARR[15:0]**: Auto-reload value

ARR is the value to be loaded into the actual auto-reload register.

Refer to [Section 20.3.1: Time-base unit](#) for more details about ARR update and behavior.

The counter is blocked while the auto-reload value is null.

20.4.9 TIM6 and TIM7 register map

TIMx registers are mapped as 16-bit addressable registers as described in the table below.

Table 107. TIM6 and TIM7 register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	TIMx_CR1	Reserved																							ARPE	Reserved			OPM	URS	UDIS	CEN	
	Reset value																																0
0x04	TIMx_CR2	Reserved																							MMS[2:0]			Reserved					
	Reset value																															0	0
0x08	Reserved																																
0x0C	TIMx_DIER	Reserved																							UDE	Reserved				UIE			
	Reset value																														0		
0x10	TIMx_SR	Reserved																														UIF	
	Reset value																																0
0x14	TIMx_EGR	Reserved																														UG	
	Reset value																																0
0x18	Reserved																																
0x1C	Reserved																																
0x20	Reserved																																
0x24	TIMx_CNT	Reserved																CNT[15:0]															
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x28	TIMx_PSC	Reserved																PSC[15:0]															
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x2C	TIMx_ARR	Reserved																ARR[15:0]															
	Reset value																	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.

21 Independent watchdog (IWDG)

This section applies to the whole STM32F4xx family, unless otherwise specified.

21.1 IWDG introduction

The devices have two embedded watchdog peripherals which offer a combination of high safety level, timing accuracy and flexibility of use. Both watchdog peripherals (Independent and Window) serve to detect and resolve malfunctions due to software failure, and to trigger system reset or an interrupt (window watchdog only) when the counter reaches a given timeout value.

The independent watchdog (IWDG) is clocked by its own dedicated low-speed clock (LSI) and thus stays active even if the main clock fails. The window watchdog (WWDG) clock is prescaled from the APB1 clock and has a configurable time-window that can be programmed to detect abnormally late or early application behavior.

The IWDG is best suited to applications which require the watchdog to run as a totally independent process outside the main application, but have lower timing accuracy constraints. The WWDG is best suited to applications which require the watchdog to react within an accurate timing window. For further information on the window watchdog, refer to [Section 22 on page 716](#).

21.2 IWDG main features

- Free-running downcounter
- clocked from an independent RC oscillator (can operate in Standby and Stop modes)
- Reset (if watchdog activated) when the downcounter value of 0x000 is reached

21.3 IWDG functional description

[Figure 213](#) shows the functional blocks of the independent watchdog module.

When the independent watchdog is started by writing the value 0xCCCC in the Key register (IWDG_KR), the counter starts counting down from the reset value of 0xFFFF. When it reaches the end of count value (0x000) a reset signal is generated (IWDG reset).

Whenever the key value 0xAAAA is written in the IWDG_KR register, the IWDG_RLR value is reloaded in the counter and the watchdog reset is prevented.

21.3.1 Hardware watchdog

If the “Hardware watchdog” feature is enabled through the device option bits, the watchdog is automatically enabled at power-on, and generates a reset unless the Key register is written by the software before the counter reaches end of count.

21.3.2 Register access protection

Write access to the IWDG_PR and IWDG_RLR registers is protected. To modify them, first write the code 0x5555 in the IWDG_KR register. A write access to this register with a

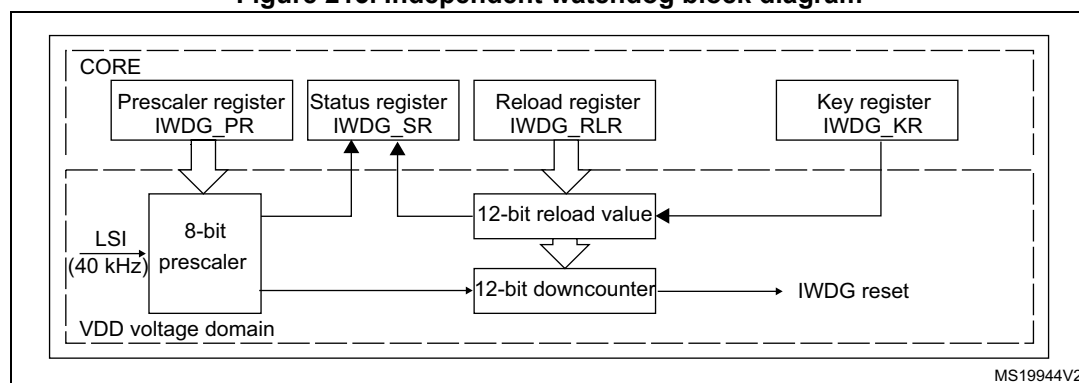
different value breaks the sequence and register access is protected again. This implies that it is the case of the reload operation (writing 0xAAAA).

A status register is available to indicate that an update of the prescaler or the down-counter reload value is on going.

21.3.3 Debug mode

When the microcontroller enters debug mode (Cortex[®]-M4 with FPU core halted), the IWDG counter either continues to work normally or stops, depending on DBG_IWDG_STOP configuration bit in DBG module. For more details, refer to [Section 38.16.2: Debug support for timers, watchdog, bxCAN and I²C](#).

Figure 213. Independent watchdog block diagram



Note: The watchdog function is implemented in the V_{DD} voltage domain, still functional in Stop and Standby modes.

Table 108. Min/max IWDG timeout period (in ms) at 32 kHz (LSI)⁽¹⁾

Prescaler divider	PR[2:0] bits	Min timeout RL[11:0]= 0x000	Max timeout RL[11:0]= 0xFFFF
/4	0	0.125	512
/8	1	0.25	1024
/16	2	0.5	2048
/32	3	1	4096
/64	4	2	8192
/128	5	4	16384
/256	6	8	32768

1. These timings are given for a 32 kHz clock but the microcontroller internal RC frequency can vary. Refer to the LSI oscillator characteristics table in the device datasheet for maximum and minimum values.

21.4 IWDG registers

Refer to [Section 2.2 on page 45](#) for a list of abbreviations used in register descriptions.

The peripheral registers have to be accessed by half-words (16 bits) or words (32 bits).

21.4.1 Key register (IWDG_KR)

Address offset: 0x00

Reset value: 0x0000 0000 (reset by Standby mode)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved																KEY[15:0]															
																w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **KEY[15:0]**: Key value (write only, read 0000h)

These bits must be written by software at regular intervals with the key value AAAAh, otherwise the watchdog generates a reset when the counter reaches 0.

Writing the key value 5555h to enable access to the IWDG_PR and IWDG_RLR registers (see [Section 21.3.2](#))

Writing the key value CCCCh starts the watchdog (except if the hardware watchdog option is selected)

21.4.2 Prescaler register (IWDG_PR)

Address offset: 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved																													PR[2:0]		
																													r/w	r/w	r/w

Bits 31:3 Reserved, must be kept at reset value.

Bits 2:0 **PR[2:0]**: Prescaler divider

These bits are write access protected see [Section 21.3.2](#). They are written by software to select the prescaler divider feeding the counter clock. PVU bit of IWDG_SR must be reset in order to be able to change the prescaler divider.

000: divider /4

001: divider /8

010: divider /16

011: divider /32

100: divider /64

101: divider /128

110: divider /256

111: divider /256

Note: Reading this register returns the prescaler value from the VDD voltage domain. This value may not be up to date/valid if a write operation to this register is ongoing. For this reason the value read from this register is valid only when the PVU bit in the IWDG_SR register is reset.

21.4.3 Reload register (IWDG_RLR)

Address offset: 0x08

Reset value: 0x0000 0FFF (reset by Standby mode)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
Reserved																				RL[11:0]															
																				rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:12 Reserved, must be kept at reset value.

Bits11:0 **RL[11:0]**: Watchdog counter reload value

These bits are write access protected see [Section 21.3.2](#). They are written by software to define the value to be loaded in the watchdog counter each time the value AAAAh is written in the IWDG_KR register. The watchdog counter counts down from this value. The timeout period is a function of this value and the clock prescaler. Refer to [Table 108](#).

The RVU bit in the IWDG_SR register must be reset in order to be able to change the reload value.

Note: Reading this register returns the reload value from the VDD voltage domain. This value may not be up to date/valid if a write operation to this register is ongoing on this register. For this reason the value read from this register is valid only when the RVU bit in the IWDG_SR register is reset.

21.4.4 Status register (IWDG_SR)

Address offset: 0x0C

Reset value: 0x0000 0000 (not reset by Standby mode)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved																												RVU	PVU		
																												r	r		

Bits 31:2 Reserved, must be kept at reset value.

Bit 1 **RVU**: Watchdog counter reload value update

This bit is set by hardware to indicate that an update of the reload value is ongoing. It is reset by hardware when the reload value update operation is completed in the V_{DD} voltage domain (takes up to 5 RC 40 kHz cycles).

Reload value can be updated only when RVU bit is reset.

Bit 0 **PVU**: Watchdog prescaler value update

This bit is set by hardware to indicate that an update of the prescaler value is ongoing. It is reset by hardware when the prescaler update operation is completed in the V_{DD} voltage domain (takes up to 5 RC 40 kHz cycles).

Prescaler value can be updated only when PVU bit is reset.

Note: If several reload values or prescaler values are used by application, it is mandatory to wait until RVU bit is reset before changing the reload value and to wait until PVU bit is reset before changing the prescaler value. However, after updating the prescaler and/or the reload value it is not necessary to wait until RVU or PVU is reset before continuing code execution (even in case of low-power mode entry, the write operation is taken into account and completes)

21.4.5 IWDG register map

The following table gives the IWDG register map and reset values.

Table 109. IWDG register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	IWDG_KR	Reserved																KEY[15:0]															
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x04	IWDG_PR	Reserved																												PR[2:0]			
	Reset value																													0	0	0	
0x08	IWDG_RLR	Reserved																			RL[11:0]												
	Reset value																				1	1	1	1	1	1	1	1	1	1	1	1	1
0x0C	IWDG_SR	Reserved																												RVU		PVU	
	Reset value																													0	0	0	

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.

22 Window watchdog (WWDG)

This section applies to the whole STM32F4xx family, unless otherwise specified.

22.1 WWDG introduction

The window watchdog is used to detect the occurrence of a software fault, usually generated by external interference or by unforeseen logical conditions, which causes the application program to abandon its normal sequence. The watchdog circuit generates an MCU reset on expiry of a programmed time period, unless the program refreshes the contents of the downcounter before the T6 bit becomes cleared. An MCU reset is also generated if the 7-bit downcounter value (in the control register) is refreshed before the downcounter has reached the window register value. This implies that the counter must be refreshed in a limited window.

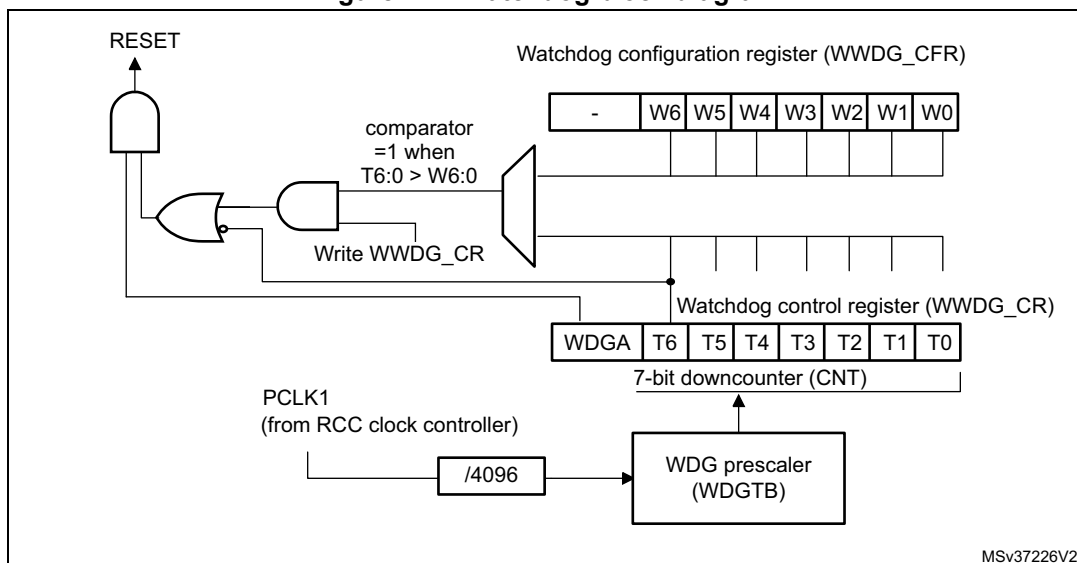
22.2 WWDG main features

- Programmable free-running downcounter
- Conditional reset
 - Reset (if watchdog activated) when the downcounter value becomes less than 0x40
 - Reset (if watchdog activated) if the downcounter is reloaded outside the window (see [Figure 215](#))
- Early wake-up interrupt (EWI): triggered (if enabled and the watchdog activated) when the downcounter is equal to 0x40.

22.3 WWDG functional description

If the watchdog is activated (the WDGA bit is set in the WWDG_CR register) and when the 7-bit downcounter (T[6:0] bits) rolls over from 0x40 to 0x3F (T6 becomes cleared), it initiates a reset. If the software reloads the counter while the counter is greater than the value stored in the window register, then a reset is generated.

Figure 214. Watchdog block diagram



The application program must write in the WWDG_CR register at regular intervals during normal operation to prevent an MCU reset. This operation must occur only when the counter value is lower than the window register value. The value to be stored in the WWDG_CR register must be between 0xFF and 0xC0.

Enabling the watchdog

The watchdog is always disabled after a reset. It is enabled by setting the WDGA bit in the WWDG_CR register, then it cannot be disabled again except by a reset.

Controlling the downcounter

This downcounter is free-running, counting down even if the watchdog is disabled. When the watchdog is enabled, the T6 bit must be set to prevent generating an immediate reset.

The T[5:0] bits contain the number of increments which represents the time delay before the watchdog produces a reset. The timing varies between a minimum and a maximum value due to the unknown status of the prescaler when writing to the WWDG_CR register (see [Figure 215](#)). The Configuration register (WWDG_CFR) contains the high limit of the window: To prevent a reset, the downcounter must be reloaded when its value is lower than the window register value and greater than 0x3F. [Figure 215](#) describes the window watchdog process.

Note: The T6 bit can be used to generate a software reset (the WDGA bit is set and the T6 bit is cleared).

Advanced watchdog interrupt feature

The Early Wake-up Interrupt (EWI) can be used if specific safety operations or data logging must be performed before the actual reset is generated. The EWI interrupt is enabled by setting the EWI bit in the WWDG_CFR register. When the downcounter reaches the value 0x40, an EWI interrupt is generated and the corresponding interrupt service routine (ISR) can be used to trigger specific actions (such as communications or data logging), before resetting the device.

In some applications, the EWI interrupt can be used to manage a software system check and/or system recovery/graceful degradation, without generating a WWDG reset. In this case, the corresponding interrupt service routine (ISR) should reload the WWDG counter to avoid the WWDG reset, then trigger the required actions.

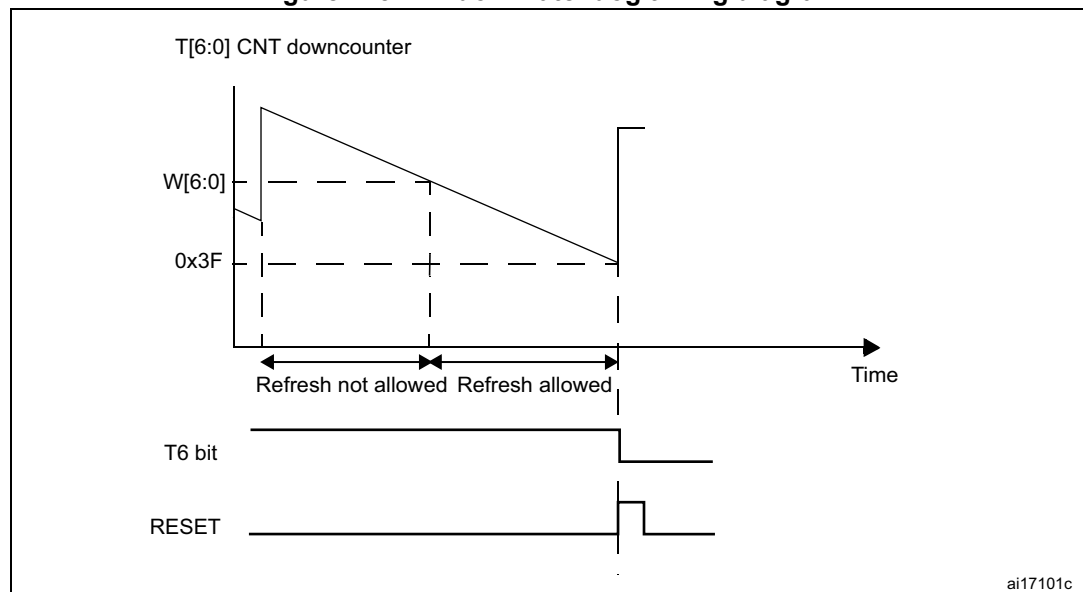
The EWI interrupt is cleared by writing '0' to the EWIF bit in the WWDG_SR register.

Note: *When the EWI interrupt cannot be served (due to a system lock in a higher priority task), the WWDG reset is eventually generated.*

22.4 How to program the watchdog timeout

Warning: When writing to the WWDG_CR register, always write 1 in the T6 bit to avoid generating an immediate reset.

Figure 215. Window watchdog timing diagram



The formula to calculate the WWDG timeout value is given by:

$$t_{\text{WWDG}} = t_{\text{PCLK1}} \times 4096 \times 2^{\text{WDGTB}[1:0]} \times (\text{T}[5:0] + 1) \quad (\text{ms})$$

where:

t_{WWDG} : WWDG timeout

t_{PCLK1} : APB1 clock period measured in ms

4096: value corresponding to internal divider

As an example, let us assume APB1 frequency is equal to 24 MHz, WDG TB[1:0] is set to 3 and T[5:0] is set to 63:

$$t_{\text{WWDG}} = 1 / 24000 \times 4096 \times 2^3 \times (63 + 1) = 21.85\text{ms}$$

Refer to [Table 110](#) for the minimum and maximum values of the t_{WWDG} .

Table 110. Minimum and maximum timeout values at 30 MHz (f_{PCLK1})

Prescaler	WDGTB	Min timeout (μs) T[5:0] = 0x00	Max timeout (ms) T[5:0] = 0x3F
1	0	136.53	8.74
2	1	273.07	17.48
4	2	546.13	34.95
8	3	1092.27	69.91

22.5 Debug mode

When the microcontroller enters debug mode (Cortex[®]-M4 with FPU core halted), the WWDG counter either continues to work normally or stops, depending on DBG_WWDG_STOP configuration bit in DBG module. For more details, refer to [Section 38.16.2: Debug support for timers, watchdog, bxCAN and I²C](#).

22.6 WWDG registers

Refer to [Section 2.2 on page 45](#) for a list of abbreviations used in register descriptions.
 The peripheral registers have to be accessed by half-words (16 bits) or words (32 bits).

22.6.1 Control register (WWDG_CR)

Address offset: 0x00

Reset value: 0x0000 007F

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								WDGA	T[6:0]						
								rs	rw						

Bits 31:8 Reserved, must be kept at reset value.

Bit 7 **WDGA**: Activation bit

This bit is set by software and only cleared by hardware after a reset. When WDGA = 1, the watchdog can generate a reset.
 0: Watchdog disabled
 1: Watchdog enabled

Bits 6:0 **T[6:0]**: 7-bit counter (MSB to LSB)

These bits contain the value of the watchdog counter. It is decremented every $(4096 \times 2^{WDGTB[1:0]})$ PCLK1 cycles. A reset is produced when it rolls over from 0x40 to 0x3F (T6 becomes cleared).

22.6.2 Configuration register (WWDG_CFR)

Address offset: 0x04

Reset value: 0x0000 007F

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						EWI		WDGTB[1:0]		W[6:0]					
						rs		rw		rw					

Bit 31:10 Reserved, must be kept at reset value.

Bit 9 **EWI**: Early wake-up interrupt

When set, an interrupt occurs whenever the counter reaches the value 0x40. This interrupt is only cleared by hardware after a reset.

Bits 8:7 **WDGTB[1:0]**: Timer base

The time base of the prescaler can be modified as follows:

00: CK Counter Clock (PCLK1 div 4096) div 1

01: CK Counter Clock (PCLK1 div 4096) div 2

10: CK Counter Clock (PCLK1 div 4096) div 4

11: CK Counter Clock (PCLK1 div 4096) div 8

Bits 6:0 **W[6:0]**: 7-bit window value

These bits contain the window value to be compared to the downcounter.

22.6.3 Status register (WWDG_SR)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														EWIF	
														rc_w0	

Bits 31:1 Reserved, must be kept at reset value.

Bit 0 **EWIF**: Early wake-up interrupt flag

This bit is set by hardware when the counter has reached the value 0x40. It must be cleared by software by writing '0'. A write of '1' has no effect. This bit is also set if the interrupt is not enabled.

22.6.4 WWDG register map

The following table gives the WWDG register map and reset values.

Table 111. WWDG register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	WWDG_CR	Reserved																								WDGA	T[6:0]						
	Reset value																									0	1	1	1	1	1	1	1
0x04	WWDG_CFR	Reserved																						EWI	WDGTB1	WDGTB0	W[6:0]						
	Reset value																							0	0	0	1	1	1	1	1	1	1
0x08	WWDG_SR	Reserved																															EWIF
	Reset value																																0

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.

23 Cryptographic processor (CRYP)

This section applies to STM32F415/417xx and STM32F43xxx devices.

23.1 CRYP introduction

The cryptographic processor can be used to both encipher and decipher data using the DES, Triple-DES or AES (128, 192, or 256) algorithms. It is a fully compliant implementation of the following standards:

- The data encryption standard (DES) and Triple-DES (TDES) as defined by Federal Information Processing Standards Publication (FIPS PUB 46-3, 1999 October 25). It follows the American National Standards Institute (ANSI) X9.52 standard.
- The advanced encryption standard (AES) as defined by Federal Information Processing Standards Publication (FIPS PUB 197, 2001 November 26)

The CRYP processor performs data encryption and decryption using DES and TDES algorithms in Electronic codebook (ECB) or Cipher block chaining (CBC) mode.

The CRYP peripheral is a 32-bit AHB2 peripheral. It supports DMA transfer for incoming and processed data, and has input and output FIFOs (each 8 words deep).

23.2 CRYP main features

- Suitable for AES, DES and TDES enciphering and deciphering operations
- AES
 - Supports the ECB, CBC, CTR, CCM and GCM chaining algorithms (CCM and GCM are available on STM32F42xxx and STM32F43xxx only)
 - Supports 128-, 192- and 256-bit keys
 - 4 × 32-bit initialization vectors (IV) used in the CBC, CTR, CCM and GCM modes

Table 112. Number of cycles required to process each 128-bit block (STM32F415/417xx)

Algorithm / Key size	ECB	CBC	CTR
128b	14	14	14
192b	16	16	16
256b	18	18	18

Table 113. Number of cycles required to process each 128-bit block (STM32F43xxx)

Algorithm / Key size	ECB	CBC	CTR	GCM				CCM			
				Init	Header	Payload	Tag	Init	Header	Payload	Tag
128b	14	14	14	24	10	14	14	12	14	25	14